

The Emergence of a Decolonial Ecology: Democratic Resistance and Socio-ecological Transformation in Rojava/North and East Syria

DISSERTATION

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Please note that differences of formatting and spelling exist between the chapters due to the differing styles and requirements of the publications. The published chapters, Chapters 3 and 4, appear as they have been published in their respective publications. Some formatting and minor spelling grammatical issues of Chapters 5, 6 and 7 have been altered slightly for this thesis; the text remains substantially identical to that which has been submitted for publication.

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Abstract

This dissertation examines the relation between colonial rule and ecocide in Kurdistan, arguing that colonial domination in Rojava operates through ecocidal coloniality that reorganizes land, water and agrarian infrastructures to dismantle Kurdish socio-economies, degrade environments, and erode Indigenous epistemic and ontological formations. Methodologically, it draws on multi-wave fieldwork, interviews, group discussions and activist-scholar engagement. To make this structure explicit, the thesis proceeds through two interrelated dynamics: the colonial structure and the decolonization struggle.

The first dynamic addresses the colonial formation in Rojava, whose core features have shaped Kurdistan's history. Its first characteristic is an intrinsically antidemocratic and genocidal or assimilatory regime of centralization and control that extends across resources, populations and everyday life. Beyond a political mechanism, it functions as a system of classification and standardization enforcing homogeneity upon a heterogeneous landscape. Through the epistemic violence of modernist scientism and positivist rationality, colonialism has sought to produce a uniform nation, a disciplined society and subjugated subjects. A second defining characteristic is its persistent ecocidal logic. Socio-spatial designs centred on land, water and agriculture were never neutral or merely developmental, but instruments of dispossession and control. Ecological destruction, including deforestation, extractive mining and dam construction, functioned simultaneously as assimilation and elimination, exploitation, and counterinsurgency. The ecocidal vector of colonialism is inseparable from its genocidal structure.

The second dynamic shows how, against this matrix of domination, Rojava's struggle has taken shape as anti-colonial praxis. The thesis contends that what has emerged, particularly in Rojava, is a decolonial ecology articulated through two constitutive lines. The first is democratic resistance, grounded in decentralized and directly democratic formations, conceptualized as rhizomatic democracy and rhizomatic governance. Rhizomatic governance crystallizes the political ontology of democratic autonomy: a horizontal, plural and transversal structure that counters the nation-state's antidemocratic, centralist and homogenizing logics. The second is socio-ecological transformation, in which ecological struggle is not secondary but the ethos of life and resistance. This project reconstructs relations with land, water and life by rejecting

ecocidal developmentalism and cultivating cooperative economies, ecological agriculture, and ethical and aesthetic practices rooted in communal life and traditional and liberatory epistemologies. Ecological struggle thus becomes a medium of ontological repair and decolonial futurity, directly confronting ecocidal coloniality. Taken together, democratic resistance and socio-ecological transformation constitute the ethos of an emergent decolonial ecology in Rojava, marking a holistic alternative to colonial modernity and demonstrating decolonial ecology as both critique and lived praxis.

To uncover these dynamics, the dissertation is organized as a cumulative project that interlinks five articles within a shared conceptual and empirical trajectory. Together, they map the passage from colonial design to decolonial praxis, revealing how theoretical, empirical and methodological strands converge in the structure of the thesis. The thesis architecture mirrors this unfolding. This cumulative dissertation is organized in four sections that move from framing to articles, then to synthesis and conclusion, tracing a path from colonial design (land, water, demographic engineering, schooling) to decolonial praxis and socio-ecological transformation. Section I (Chapters 1–2) establishes the groundwork: Chapter 1 outlines the thesis and introduces core concepts and stakes; Chapter 2 formulates objectives and research questions within colonial and decolonization dynamics, and articulates activist-scholar positionality and methodology. Section II (Chapters 3–7) compiles the five articles, tracing a trajectory from colonial mindset and material infrastructures to the horizon of a political ontology of decolonization and transformation. Section III (Chapters 8–9) provides the integrative synthesis: Chapter 8 consolidates definitions and scope conditions of ecocidal developmentalism and colonial drought, and identifies principal vectors of ecocidal coloniality, including agro-hydro-demographic design and epistemicide as elimination, outlining the operational logic of genocide and ecocide. Chapter 9 traces the passage from resistance to socio-ecological transformation, detailing the reconstruction of ecological agriculture, rhizomatic governance, ecopedagogy and the eco-economy as a coherent decolonial ecology, and concludes with reflections on ongoing challenges and threats. Section IV (Chapter 10) restates the findings, outlines lessons from Rojava, explores limitations and research gaps, and indicates directions for future research opened by the thesis. Ultimately, where colonial structure is ecocidal in its structure, decolonization must be ecological in substance and scope. In Rojava, democratic resistance and socio-ecological transformation together constitute decolonial ecology as a lived and evolving praxis.

Zusammenfassung

In dieser Dissertation wird der Zusammenhang zwischen Kolonialherrschaft und Ökozid in Kurdistan analysiert. Es wird die These vertreten, dass die koloniale Herrschaft in Rojava durch eine „ökozidale Kolonialität“ gekennzeichnet ist. Diese organisiert die Kontrolle über und den Zugang zu Land, Wasser und agrarischen Infrastrukturen, um die kurdische Sozioökonomie zu destabilisieren, Umweltzerstörung zu begünstigen und die indigenen epistemischen und ontologischen Strukturen zu untergraben. Methodisch basiert die Arbeit auf einer Kombination aus Feldforschung, Interviews, Gruppendiskussionen sowie auf der Zusammenarbeit zwischen Aktivist:innen und Wissenschaftler:innen.

Die Grundlage der Dissertation bilden zwei miteinander verflochtene Dynamiken: die Reproduktion der kolonialen Strukturen und der Kampf um eine Entkolonialisierung. Die erste Dynamik bezieht sich auf die koloniale Prozesse in Rojava, die die historische Entwicklung Kurdistans maßgeblich beeinflusst haben. Ein wesentliches Charakteristikum ist ein grundsätzlich antidemokratisches, genozidales oder assimilatorisches Regime der Zentralisierung und Kontrolle, das sowohl Ressourcen, Bevölkerungsgruppen und das tägliche Leben durchdringt. Politisch fungiert das System auch als Mechanismus der Klassifizierung und Standardisierung, durch den eine diverse und heterogene Landschaft homogenisiert wird. Mittels epistemischer Gewalt eines modernistischen Szientismus und dessen positivistischen Rationalität, bestanden die Ziele der kolonialen Herrschaft darin, eine einheitliche Nation, eine disziplinierte Gesellschaft sowie unterworfenen Subjekte zu schaffen. Ein weiteres Kennzeichen kolonialer Macht manifestiert sich in der Logik sozialräumlicher Entwürfen und deren Organisation von Land, Wasser und Landwirtschaft. Diese Entwürfe waren nie neutral, sondern dienten stets der Enteignung und Kontrolle. Ökologische Zerstörungen – wie Entwaldung, Rohstoffabbau und der Bau von Staudämmen – dienten immer gleichzeitig der Assimilation, Eliminierung und Ausbeutung der lokalen Bevölkerung sowie der Bekämpfung von Aufständen. Somit ist der ökologische Aspekt des Kolonialismus eng mit den genozidalen Prozessen verbunden.

Die zweite Dynamik verdeutlicht, wie sich der Widerstand in Rojava im Kontext dieser Herrschaftsmatrix als eine antikoloniale Praxis formiert. Die zentrale These lautet dabei, dass in Rojava eine dekoloniale Ökologie entstanden ist, die zwei konstitutiven Elemente aufweist.

Das erste Element ist der demokratische Widerstand, der auf dezentralisierten und direktdemokratischen Strukturen basiert und als rhizomatische Demokratie bzw. rhizomatische Regierungsführung konzeptualisiert wird. Diese Form der Regierungsführung verkörpert die politische Ontologie der demokratischen Autonomie: eine horizontale, pluralistische und transversale Struktur, die der antidemokratischen, zentralistischen und homogenisierenden Logik des Nationalstaates entgegenwirkt. Das zweite Element betrifft die sozioökologische Transformation, bei der der ökologische Kampf nicht zweitrangig ist, sondern im Zentrum des Lebens und des Widerstands steht. Dabei werden die Beziehungen zu Land, Wasser und Leben neu gestaltet, indem ökozidaler Developmentalismus abgelehnt und kooperative Ökonomien, ökologische Landwirtschaft sowie ethische und ästhetische Praktiken gefördert werden, die in Gemeinschaften, traditionellen sowie befreienden Erkenntnistheorien verwurzelt sind. Der ökologische Kampf wird somit sowohl zu einem Mittel der ontologischen Wiederherstellung als auch der Entwicklung einer dekolonialen Zukunft, die sich aktiv mit der ökologischen Kolonialität auseinandersetzt. Zusammen bilden der demokratische Widerstand und die sozioökologische Transformation den Kern einer sich entwickelnden dekolonialen Ökologie in Rojava. Diese stellt eine allumfassende Alternative zur kolonialen Moderne dar und manifestiert sich sowohl als Kritik als auch als gelebte Praxis.

Zur Untersuchung dieser Prozesse ist die Dissertation als ein kumulatives Forschungsprojekt angelegt. Sie besteht unter anderem aus fünf Artikeln, die innerhalb eines gemeinsamen konzeptionellen und empirischen Rahmens miteinander verknüpft sind. Gemeinsam dokumentieren sie den Übergang vom kolonialen Projekt zur dekolonialen Praxis, wobei theoretische, empirische und methodologische Stränge in der Arbeit zusammenfinden. Die Gliederung der Dissertation spiegelt diese Entwicklung wider und ist in vier Abschnitte unterteilt, die von der Rahmenerzählung, über die Artikel selber bis hin zur Synthese und den Schlussfolgerungen reichen. Diese Struktur zeichnet den Weg vom kolonialen Projekt (Land-, Wasser-, Bevölkerungs- und Schulplanung) zur dekolonialen Praxis und sozioökologischen Transformation nach.

Abschnitt I (Kapitel 1–2) legt die Grundlagen: Kapitel 1 skizziert die Forschungsarbeit, stellt zentrale Konzepte und Herausforderungen vor, beschreibt die eigene Positionierung sowie die Methodik der Arbeit. Kapitel 2 formuliert die Ziele und Forschungsfragen im Kontext der kolonialen und dekolonialen Prozesse. Abschnitt II (Kapitel 3–7) umfasst fünf Artikel und zeichnet den Weg vom kolonialen Denken und den materiellen Infrastrukturen hin zu einer

politischen Ontologie der Dekolonialisierung und Transformation nach. Abschnitt III (Kapitel 8–9) bietet eine Synthese: Kapitel 8 stellt Definitionen und Rahmenbedingungen des „ökologischen Entwicklungswahns“ sowie der „kolonialen Dürre“ vor, identifiziert die Hauptfaktoren der „ökologischen Kolonialität“ – darunter das „agro-hydro-demografische Projekt“ und skizziert die operative Logik von Völkermord und Ökozid. Kapitel 9 zeichnet den Übergang vom Widerstand zu einer sozioökologischen Transformation nach und beschreibt detailliert den Wiederaufbau einer ökologischen Landwirtschaft, die in der Verbindung zwischen einer rhizomatische Regierungsführung mit Ökopädagogik und einer ökologischen Wirtschaftsweise eine kohärente dekoloniale Ökologie darstellt. Der Abschnitt endet mit Überlegungen zu den anhaltenden Herausforderungen und Bedrohungen. Abschnitt IV (Kapitel 10) fasst die Ergebnisse zusammen, skizziert Lehren, die von dem Fallbeispiel Rojava gezogen werden können und zeigt Ansätze für zukünftige Forschungsarbeiten sowie einige Grenzen und Forschungslücken der Arbeit auf.

Als Schlussfolgerung kann festgehalten werden, dass eine Dekolonialisierung immer auch eine ökologische Dekolonialisierung sein muss. Als Gegenentwurf zum hegemonialen Projekt formiert sich in Rojava durch den demokratischen Widerstand und der sozioökologischen Transformation eine dekoloniale Ökologie, die als lebendige und sich entwickelnde Praxis zu verstehen ist.

SECTION I. Introduction

Chapter 1. Introduction

The colonial wound I carry: an accursed mark wavering in the whirlwind of modernity. In the terrified eyes of the Angel of History I behold it and carve the unbowed dignity of healing into my rebellious spirit.¹

This research proceeds as a geophilosophical intervention. Rather than applying readymade theory to Rojava (West Kurdistan / Syrian Kurdistan), it lets Rojava oblige thought to invent, treating territory not as backdrop but as an expressive assemblage whose land, waters, infrastructures and struggles contour concepts. The task is double: to diagnose the historical and ongoing architectures of colonial domination that striate this geography, and to chart the horizons and impasses of decolonial praxis already germinating within it (Walsh, 2020). Geophilosophy here names a change in the conditions of thinking: concepts are not universal forms lowered onto places, but tools forged with territory. Thinking, creating, and transforming with territory is therefore not a stylistic preference; it is a method for disarming the colonial view from nowhere and re-grounding thought in situated, local/Indigenous worlds-from-below. Producing concepts with territory is more than a reversal of colonial epistemology; it is a way of seeing from the locus of the earth that opens a space for world-making (Deleuze & Guattari, 1994; Bonta & Protevi, 2004). This is why, and how, I undertake this work: to craft conceptual tools adequate to the Rojava experience, one that names the designs that destroy and equips the practices that build otherwise, as the project traces their interweaving.

¹ This reference to the “Angel of History” is drawn from Walter Benjamin:

A Klee painting named ‘Angelus Novus’ shows an angel looking as though he is about to move away from something he is fixedly contemplating. His eyes are staring, his mouth is open, his wings are spread. This is how one pictures the angel of history. His face is turned toward the past. Where we perceive a chain of events, he sees one single catastrophe which keeps piling wreckage and hurls it in front of his feet. The angel would like to stay, awaken the dead, and make whole what has been smashed. But a storm is blowing in from Paradise; it has got caught in his wings with such a violence that the angel can no longer close them. The storm irresistibly propels him into the future to which his back is turned, while the pile of debris before him grows skyward. This storm is what we call progress. (Benjamin, 1968, pp. 253–264)

To this end, this cumulative dissertation argues that the colonial order in Rojava Kurdistan is ecocidal: a mode of rule that reorganizes land, water and agrarian infrastructures to dismantle Kurdish socio-economies, degrade environments and erode the cultural-political foundations of Kurdish people. To reveal this structure, this thesis unfolds through the analysis of two interrelated and interconnected dynamics: colonial structure and decolonization struggle (see Table 1). In what follows, I use dynamics as patterned mechanisms under study, that is, multiscale processes composed of identifiable instruments and practices that generate effects over time. A dynamic denotes structured causation rather than isolated events, which is why it is specified here by its reproducibility, institutional mediation and observable outcomes in socio-ecological indicators. Lines are the analytical trajectories that relate these dynamics across chapters and supply the organizing pathways that connect those patterns, establish comparability and carry the cumulative argument forward. Put differently, dynamics are the patterned mechanisms under study, and lines are the organizing axes that relate them across chapters and make their interdependence analytically visible.

Table 1: The framework of this thesis

Ecocidal coloniality (first dynamic: colonial formation)	Decolonial ecology (second dynamic: decolonization struggle)
i) Antidemocratic structure (centralized, centre-periphery structure; demographic and economic engineering; systemic marginalization and assimilation; homogeneous one-world universalism; genocidal and colonial violence; epistemicide)	i) Democratic resistance through rhizomatic democracy and rhizomatic governance (decentralized, transversal, autonomous; pluriversal; affirming multiplicity of society and peoples, diverse socio-ecologies and epistemologies)
ii) Ecocide as intrinsic socio-spatial design and colonial rule of nation-state colonialism (genocide-ecocide nexus; ecocide as a counterinsurgency technique; ecocidal developmentalism; extractivism; monocrop agriculture; colonial drought; biodiversity loss)	ii) Socio-ecological transformation (diverse agroecologies; social/communal economy and eco-economies [ecological balance and social benefit]; developing liberatory epistemologies; enhancing seed varieties and biodiversity)

The first dynamic concerns the formation of ecocidal coloniality in Rojava, whose defining characteristics have consistently determined the historical trajectory of Kurdistan. The first characteristic is its intrinsically antidemocratic and genocidal/assimilatory nature: a regime of centralization and control that extends over resources, populations and every domain of life. More than a political mechanism, this structure functions as a mode of classification and standardization, seeking to impose homogeneity upon a heterogeneous landscape. Through the epistemic violence of modernist scientism and positivist rationality, colonialism has endeavoured to produce a uniform nation, a standardized society, disciplined individuals and a rationalized agricultural system. The consequences have been devastating, eroding and liquidating the plural and colourful social fabric of Kurdistan, and degrading its fauna and flora (Türk & Jongerden, 2024). The second characteristic is the ecocidal logic that has always accompanied colonial domination in Kurdistan. Here, I advance the concept of ecocidal coloniality to identify the structural and inherent link between nation-state colonialism and ecocide, asking whether socio-spatial designs centred on land, water and agriculture operate not as neutral or merely developmental interventions, but as instruments of exploitation and colonial rule in eliminatory ways. Ecological destruction, whether through deforestation, extractive mining or dam construction, is thus examined as a possible modality of rule that operates simultaneously as a mechanism of assimilation and elimination, exploitation and counterinsurgency. The analysis explores how the ecocidal vectors of colonialism articulate with their genocidal structures, without presuming that relation in advance.

The second dynamic is that it is precisely against this matrix of colonial domination that Rojava's struggle has constituted itself as an anti-colonial praxis. The thesis therefore examines how in Kurdistan, and particularly in Rojava, an emergent decolonial ecology has been articulated in practice and theory. Ferdinand claims today's ecological crisis is inseparable from colonial and capitalist histories, formulated as colonial ecology (Ferdinand, 2022, p. 175). Decolonial ecology seeks to dismantle the colonial structures that organize relations among humans and with non-human life, and advances an ecology of struggle tied to political emancipation, social justice, and equity (*ibid.*). Crucially, it proceeds from the premise that colonialism is ongoing, actively shaping contemporary socio-ecological conditions, rather than belonging to a closed past.

This decolonial ecology is articulated through two constitutive lines that together form the foundation of Rojava's praxis. The first line is the line of democratic resistance, grounded in

decentralized and directly democratic formations, which I conceptualize as rhizomatic democracy and rhizomatic governance. Rhizomatic governance crystallizes the political ontology of democratic autonomy in Rojava: a horizontal, plural, and transversal structure that opposes the arborescent logics of the nation-state's first character mentioned above: its antidemocratic, centralist and homogeneous structure. The second line is the project of socio-ecological transformation, in which ecological struggle is not treated as a secondary concern but as the ethos of life and resistance itself. This transformation aims to reconstruct society's relation with land, water and life by rejecting ecocidal coloniality and instead cultivating cooperative economies, ecological agriculture and ethical-aesthetic practices rooted in communal life and traditional and liberatory epistemologies. In this sense, ecological struggle becomes the medium through which ontological repair and decolonial futurity are imagined and enacted. Thus, socio-ecological transformation, which is driven by ecological struggle and paradigm, is a direct response to the second character of colonial structure: ecocidal coloniality. Here, I use ecocidal coloniality to identify the colonial structures that destroy and exploit socio-ecologies simultaneously as a means of colonial rule, as discussed below.

Together, democratic resistance and socio-ecological transformation constitute the ethos of the emergent decolonial ecology in Rojava. They mark a structural and holistic alternative to colonial modernity, revealing how decolonial ecology emerges not only as critique, but as a lived and evolving praxis. Thus, the Rojava Revolution has articulated an ecological paradigm as decolonial praxis, linking eco-economy, women's liberation and rhizomatic democracy/governance. Bringing these lines together, the dissertation positions its contribution as an inquiry that intervenes in political ecology, genocide and ecocide studies, and decolonial theory, by centring a problem: how to place ecology at the centre of anti-colonial struggle and socio-ecological transformation in Rojava. Here, decoloniality does not signify absence or surpassing coloniality. Instead, it points to the "postures, positionings, horizons, projects, and practices of being, thinking, sensing, and doing that resist and re-exist, that transgress and interrupt the colonial matrix of power," arising and pushing forward from the "borders, margins, and [...] cracks," and cultivating the conditions for a decolonial otherwise (Walsh, 2020, p. 607).

1.1. Kurdistan

Kurdistan is a mountainous plateau region that arcs from the Zagros Mountains in the east, across the Mesopotamian plain, towards the Anatolian steppe and Taurus Mountains in the west. Elevation, aspect and basin geometry generate sharp ecological gradients over short distances. Summers are hot and dry, winters cold, with wide diurnal ranges. Many valleys follow a winter growing season, followed by summer drought (McDowall, 2004, pp. 1–8). Agrarian life evolved as an adaptive, diversified system rather than a single-crop regime. For centuries, smallholders combined outfield grains and legumes with intensive household gardens of vegetables, culinary and medicinal herbs, and condiments; arboriculture on hillslopes and terraces provided fruit, vines and olives, and integrated with livestock for manure and traction. Together these practices spread risk, conserved soils, and matched production to seasonal water availability (Butzer, 1994; Pimbert, 2021, pp. 115–117). Historically the region supplied cereals, vegetables from irrigated river belts, and fruit from temperate hillsides and irrigated oases; in the modern era, cotton and tobacco expanded in some basins, while cereals retained a significant share of national output in neighbouring states (McDowall, 2004, pp. 1–8).

Pastoralism formed a second pillar of the economy. Kurdish herders maintained locally adapted sheep and goat populations through village-based flocks and seasonal movements ranging from short-distance transhumance to regional circuits; by the late twentieth century, local breeds still dominated ovine herds in Turkish Kurdistan (Aşkın et al., 1989; Thevenin, 2011; Yadirgi, 2020). Seasonal corridors linked upland pastures to lowland stubble and riparian grazing, occasionally producing friction with settled cultivators at river crossings and village margins (McDowall, 2004). Even as long-distance nomadism contracted, stockbreeding remained central to household livelihoods. The co-production of fields, orchards and herds sustained biodiversity at plot and landscape scales, stabilized soils and water, and embedded cuisine, seasonal labour and social institutions in the rhythms of a demanding but productive mountain-steppe environment (Butzer, 1994; Pimbert, 2021, pp. 115–117).

These socio-economic, socio-cultural and socio-ecological foundations have been repeatedly disrupted by colonial intervention and ecocidal policies, including hydraulic and land socio-spatial design, monocrop agriculture, deforestation, displacement and mining, that have degraded ecosystems and unsettled agrarian-pastoral relations (Harris, 2002; Harris, 2006; Jongerden, 2007; Van Etten et al., 2008; Tejel, 2009; Jongerden, 2010; Harris, 2012; Bilgen,

2017; Bilgen, 2021; Akıncı et al., 2020; Yadirgi, 2020; Dinç et al., 2021; Adalet, 2022; Türk, 2023; Türk & Jongerden, 2024; Dinç & Türk, 2025).

The starting point of this study, therefore, is that Kurdistan is a colony. Following the First World War, the region was partitioned by the colonial designs of Britain and France through the Sykes-Picot Agreement of 1916 and the Treaty of Lausanne of 1923. This settlement divided Kurdistan among four states, namely Türkiye (Bakur / Northern Kurdistan), Iran (Rojhilat / Eastern Kurdistan), Iraq (Başûr / Southern Kurdistan) and Syria (Rojava / Western Kurdistan), and denied the Kurds recognition as a people (Dinc & Türk, 2025) (see Figure 1).

Map of Rojava region and Kurdistan



Figure 1: Map of Rojava region and Kurdistan (Jongerden, 2021, p. 8)

As İsmail Beşikçi (1990) has argued, this geopolitical fragmentation rendered Kurdistan an interstate/international colony of these states, even if it was never formally granted colonial status. In the century that followed the division of Kurdistan, the Kurds were subjected to varying combinations of genocidal policies, including mass massacres, assimilation and denial within the homogenizing nation-building projects of Turkish, Persian and Arab nationalisms (Beşikçi, 1990; Bozarslan, 1996; McDowall, 2004; Öktem, 2005; Jongerden, 2007; Öktem,

2008; Tejel, 2009; Fernandes, 2012; Jongerden, 2016; Türk & Jongerden, 2024). Within each state, Kurdish regions functioned as internal colonies (Duruiz, 2020), a culture-specific domination within national borders (Etkind, 2011, p. 7). Persistent denial of this status has rendered Kurdistan an “invisible international colony” (Goner, 2023, p. 169), a denial reproduced by canonical frames that privilege franchise and settler colonialism (Türk & Jongerden, 2024). If colonialism is a spatial practice of domination enforced through military, economic and cultural means (Sunca 2023), then Kurdistan’s case straddles and exceeds the typologies: extraction, demographic re-engineering and the production of peripheries are pursued without overseas distance, and legitimated as nation-state common-sense (Türk & Jongerden, 2024). Within this broader colonial design, Rojava or Western Kurdistan has emerged as the smallest and arguably most vulnerable part of Kurdistan.

1.2. Rojava

Compared to other parts of Kurdistan, Rojava’s demographic composition, marked by ethnic and religious diversity, smaller population size, and weaker organizational capacity, has limited the possibility of pursuing a traditional nationalist project of state-building. These vulnerabilities have encouraged successive Arab nationalist governments in Syria to intensify policies of assimilation, aiming to dissolve Kurds into a homogeneous Arab nation. From the founding of the independent Syrian state in 1946, Kurds were systematically denied recognition (McDowall 2004; Tejel, 2009; Knap et al., 2016; Aslan, 2023). They were often portrayed not as an Indigenous people but as “immigrants” from Türkiye, and thus stripped of basic civil cultural and political rights. This exclusion was institutionalized with the 1962 special decree in Heseke (al-Hasaka) province, which arbitrarily reclassified approximately 120,000 Kurds (around 20 per cent of Syria’s Kurdish population) as “foreign” (*ajanib*) or “unregistered” (*maktumin*). Those stripped of citizenship lost their legal rights to education, property ownership and political participation (Human Rights Watch, 1996; McDowall 2004; Aslan, 2023; Türk & Jongerden, 2024). This dispossession was further consolidated through the infamous Arab Belt (*al-hizam al-‘arabi*) project of the 1960s and 1970s, which combined demographic engineering, land confiscation and settler colonization along the Syria-Türkiye border. As Türk and Jongerden (2024) note, the Arab Belt was not only an assimilationist policy, but also a counterinsurgency strategy: it sought to eradicate Kurdish presence socio-

culturally and politically, while restructuring and recolonizing Rojava's demographic and economic foundations.

In this context, three interwoven dynamics defined Rojava's colonial condition: assimilation of Kurds into the Arab nation; demographic engineering through settler colonization; and de-development, whereby Rojava was maintained as Syria's agrarian periphery (Türk & Jongerden, 2024). Despite being among the most fertile and resource-rich regions of the country, providing wheat, cotton and crude oil, Rojava was systematically underdeveloped (Ababsa 2015; Knap et al., 2016; Aslan, 2023). Centralized control of land, water and agricultural production imposed monocropping, ecological degradation and dependency. This structure exemplifies what can be conceptualized as nation-state colonialism, a variant of internal colonialism marked by socio-spatial engineering, ecocidal developmentalism and cultural erasure (Türk & Jongerden, 2024). The dissertation treats these twinned moves, assimilation and peripherization, as the ground on which ecocidal coloniality materialized: "development" becomes the vehicle by which ecocide is normalized, and infrastructures become the media through which elimination is designed and made durable. In this regard, the thesis advances the concepts of ecocidal developmentalism and colonial drought to specify how socio-spatial design through dams, irrigation, monocultures and demographic engineering operates as elimination, and thus, how ecological struggle becomes an ontological ground for autonomy.

Given this colonial structure, with its mechanisms and effects, the extraordinary trajectory of the Rojava Revolution assumes even greater significance. It offers both a promising anti-colonial exemplar and practice for other parts of Kurdistan, and novel pathways of resistance at the epistemic and ontological levels that can inspire similar geographies; it also indicates how socio-ecologies might be reorganized transversally and, as praxis, how a socio-ecological transformation could constitute a viable alternative to capitalist modernity.

During its struggle against the Islamic State of Iraq and Syria (ISIS) terror group, Rojava emerged as a pluriverse (Escobar, 2018) and received global recognition. Amid the collapse of the Ba'ath regime's control across much of the country after the start of the Syrian uprising in 2011, Kurds in Rojava declared self-governing autonomy in 2012. The three cantons of Afrin, Kobanê and Cizîrê in north Syria established the Autonomous Democratic Governments (*Rêveberiya Xweseriya Demokratîk*) of Rojava between 2012 and 2017 (Dinc & Türk, 2025).

One of the most important features of this revolution is that it unexpectedly took place in Rojava, the presumed weakest link in Kurdistan, which became a driving force of the Kurdish struggle and a site of extraordinary transformation. Syrian Kurds successfully resisted ISIS at the siege of Kobanê in 2015, and subsequently spearheaded the liberation of Raqqa, the group's de facto capital, in 2017 (Aslan, 2023, pp. 95–98). In 2018, the administration declared the Autonomous Administration of North and East Syria (*Rêveberiya Xweser a Bakur û Rojhilatê Sûriyeyê*, AANES), grounded in radical democracy, women's liberation and ecology. Scholars have noted resonances with Zapatistas in Chiapas-Mexico (Aslan, 2023).

Today, Rojava stands at the heart of the anti-colonial struggle, embodying the revival and empowerment of the Kurds after a century of denial, genocidal and ecocidal violence. Yet, this emancipatory experiment has faced relentless external aggression. In January 2018 Afrin, and in October 2019 Serêkaniyê, were invaded and occupied by the Turkish state and Türkiye-backed jihadist groups. In late 2024, after 13 years of civil war, Hay'at Tahrir al-Sham (HTS), led by Ahmed al-Shara'a, toppled the Assad regime and assumed control of the country. HTS rule has already involved massacres targeting Alawi and Druze communities, and has created acute risks for Kurds and other minorities. From the Arab Belt to the invasions of Afrin and Serêkaniyê, settler-colonial practices, including ethnic cleansing, demographic change, ecocidal socio-spatial design, dispossession and the erasure of Kurdish culture have remained central to the genocidal-ecocidal policies of nation-state colonialism (Grasso, 2024; Human Rights Watch, 2024; Dinc & Türk, 2025).

1.3. Thematic flow and conceptual framework

To uncover ecocidal coloniality in Rojava/Kurdistan, the thesis has been designed through five articles, and through conceptual flows and themes simultaneously. The first arc of the argument proceeds through monoculture agriculture as colonization, and reconstruction of ecological agriculture as decolonization, discussed in Article I, "Decolonization agriculture". Prior to 2012, agrarian modernization under the Ba'athist administration centralized authority, fixed Rojava as a supplier of primary products, and asserted territorial control by distributing land and settling loyalist Arab families. This is not simply economic policy but a socio-spatial design that binds livelihoods, crops and communities to the centre. Reconstruction of ecological agriculture has dismantled this colonial agrarian model and initiated a decolonization of

agriculture. Interview-based and field research shows how decentralized governance, seed institutions and cooperatives have reoriented production around locally administered economies of diversity. The analytical wager here is precise: agrarian restructuring is evaluated not only for output or yields, but for its capacity to undo the socio-spatial logics of dependency and ecological degradation, to recompose land-labour relations, and to anchor autonomy in material life as a decolonization process.

A second arc turns to the genocide-ecocide nexus, as discussed in Article II, “Roots of destruction”. Environmental destruction can constitute a technique of extermination embedded in settler-colonial counterinsurgency (Short, 2010; Crook & Short, 2014; Short, 2016; Dunlap, 2018; Lindgren, 2018; Jaber, 2019; Lynch et al., 2021; Priestley, 2021; Wise, 2021). A comparative political ecology of olive tree destruction in occupied Palestine and Afrin (Rojava) demonstrates how militaries, police, paramilitaries and settlers target essential foundations in order to erode the material and symbolic bases of existence and resistance among stateless and marginalized groups. It shows that ecocide has emerged as a principal method of elimination and exploitation within counterinsurgency strategies.

A third arc follows climate change, drought and the water crisis. Article III, “Resisting colonial drought”, introduced the concept of colonial drought to describe how water scarcity in Rojava is not merely the outcome of climate variability, but is the result of intertwined colonial projects. Internally, the Ba’athist regime engineered hydrological infrastructures and agricultural schemes that overexploited groundwater, depleted aquifers and imposed monocultural production, locking Rojava into ecological degradation. Externally, Türkiye’s damming of the Euphrates and Tigris rivers, central to its Southeast Anatolia Project (GAP), asserted regional hydro-hegemony, reducing crossborder flows and weaponizing water as a tool against Kurdish autonomy (Jongerden, 2010; Harris, 2012; Kibaroglu, 2014; Mueller et al., 2021; Daher, 2022; Tinti, 2023). Together, these dynamics constitute a colonial drought: a condition where water scarcity emerges from the convergence of internal colonial underdevelopment and external neocolonial domination. Conceptually, colonial drought challenges climate-security narratives that depoliticize scarcity by attributing it to naturalized climate change or governance failures, as widely discussed (Gleick, 2014; Kelley et al., 2015; Ash & Obradovich, 2020; Daoudy, 2023). The analytical pivot is not to deny climate variability or neoliberal reforms – both are crucial – but rather to centre how drought becomes colonial in cause and effect: scarcity is

designed and managed; vulnerability is patterned by centre-periphery relations; hydrology is folded into counterinsurgency.

The fourth arc focuses on a distinct but complementary dimension of the colonial structure: epistemicide, analysed in Article IV, “‘Education under a tree’”. Under Ba‘athist rule, education functioned as a key apparatus of assimilation, nationalism and patriarchal control, erasing Kurdish language, history and culture, while imposing a centralized curriculum oriented towards Arabization. After 2012, the AANES sought to dismantle this apparatus and reconstruct education on a radically different basis. The introduction of ecopedagogy – education that integrates ecological principles, communal values and democratic participation – signalled a fundamental reorientation of what it means to learn, teach and reproduce social knowledge. In this framework, education is not only about transmitting skills or knowledge, but also about cultivating subjectivities capable of sustaining a democratic, ecological society. It entwines *jineoloji* (science of women), social ecology and democratic decision-making across academies and councils. Education thus becomes a site of epistemic decolonization, a means of undoing the colonial and patriarchal epistemologies that have long shaped the region, and of fostering new ways of knowing and being that are grounded in communal and ecological relations (Espinosa & Ronan, 2023).

These arcs converge in the ethos through which ecology becomes foundational to decolonization, as discussed in Article V, “Ecological struggle as ethos of decolonization”. If colonialism in Kurdistan has always been ecocidal, then the resistance to it has also necessarily taken ecological form. The Rojava Revolution, which unfolded in the midst of war, displacement and embargo after 2012, was not the sudden creation of new structures from nothing but the crystallization of long histories of struggle, critique, and imagination. Its paradigmatic shift owes much to the intellectual trajectory of the Kurdish revolutionary Abdullah Öcalan and the Kurdish Freedom Movement (KFM), which, from the late 1990s onward, moved away from a classical national liberation framework towards a politics of democratic autonomy, social ecology and women’s liberation. This turn constituted not simply a tactical adaptation but an ontological reorientation: away from the state as the telos of liberation, and towards a non-statist ethos rooted in ecology, communal life and autonomy. Within this ethos, ecology was not an auxiliary concern but a constitutive pillar, alongside democracy and gender liberation (Hunt 2019; Öcalan, 2020; Hunt, 2021; Sustam 2021; Piccardi, 2021; Pimbert 2021; Dirik, 2022; Piccardi & Barca, 2022; Türk, forthcoming).

It is this embrace of ecology as ethos of decolonization that the fifth article of this dissertation analyses, tracing how the KFM and its transnational networks came to foreground ecological struggle as an indispensable dimension of autonomy and liberation. The significance of this shift can be fully understood only against the background of ecocidal coloniality. As Article V argues, the policies of the Ottoman Empire, carried forward by Turkish and Syrian nation-states, systematically target Kurdish socio-ecological foundations through methods such as dams, monocultures, forced displacement and infrastructural projects designed to integrate Kurdistan into a centre-periphery of internal colonialist order. These projects were not merely intended for economic exploitation, but for the dismantling of Kurdish autonomy and the consolidation of colonial control. They sought to eliminate not only Kurdish political resistance but the ecological bases of life (land, water and forests) that sustained it. Against this, the KFM reframed ecology as more than environmental protection: it became the terrain of decolonization itself.

The cumulative design of this dissertation reflects these dynamics. The articles were written as autonomous interventions, yet together they assemble two dynamics: from colonial design, which is embedded in ecocide, to decolonization as a process through democratic principles and ecological struggle. The research therefore claims that ecological struggle is not an add-on; it is the ethos of decolonization and the driving force of socio-ecological transformation in Rojava. Finally, it contends that rhizomatic forms of governance (cooperatives, communes, assemblies, academies and municipalities) are not merely organizational devices, but elements of a political ontology that makes other futures imaginable and liveable through rhizomatic resistance.

1.4. Conceptual framework

Having briefly indicated how each of the five articles of this thesis addresses one or more dimensions of ecocidal coloniality in Rojava and Kurdistan, I now turn to the conceptual framework of ecocidal coloniality. The thesis brings together the literatures on colonialism, coloniality, genocide and ecocide, to assess a specific form of ecocidal violence in which policies of exploitation and elimination crystallize into a structural formation embodied in ecocidal coloniality implemented by the nation-state, and to evaluate their application and consequences in Rojava.

Colonialism is the convergence of socio-spatial and mental designs of domination, a dynamic structure that subordinates peoples and socio-ecologies through political, military, economic and cultural mechanisms, stabilizing itself as a compensatory regime. Its operations reorganize territory, re-code meaning and rewire everyday life so that extraction, control and standardization appear necessary and natural. Within this formation, cultural genocide, massacres and ecocide are not aberrations but intrinsic techniques of invasion and rule, directed at dismantling and reconstituting the essential foundations of peoples and nature. Within this matrix, the literature distinguishes two historical bases: franchise colonialism and settler colonialism (Shoemaker, 2015). The first organizes settlement around resource extraction and labour exploitation for capital accumulation; the second aims to displace Indigenous populations to found a permanent settler polity. Due to this limited understanding, which underestimates the ongoing destructions caused by internal colonization, the colonial status of Kurdistan has been routinely dismissed, rendering it an “invisible international colony” (Goner, 2023, p. 169). This outcome follows from the conceptual and epistemic limits of the dominant frameworks of colonialism (Türk & Jongerden, 2024) and neocolonialism (Sunca, 2023). Coloniality endures beyond formal colonial rule, and organizes power far outside colonial administrations. It persists as an evolving, multidimensional structure embedded in the main arteries of current systems, in both material and immaterial articulations, shaping contemporary life after franchise and settler colonialism. This is not merely a lingering legacy; it is the ongoing restructuring of colonial rule and design into the present (Quijano, 2007; Mignolo, 2011; Mignolo & Walsh, 2018).

In this regard, the genocide-ecocide nexus has emerged as a central principle of coloniality in Kurdistan. This nexus reveals that ecocide is not an externality but a constitutive method of genocide, whereby political, economic and legal architectures converge to dismantle essential foundations (Short, 2010; Crook & Short, 2014; Short, 2016; Jaber, 2019; Lynch et al., 2021; Priestley, 2021; Wise, 2021; Türk, 2023; Dinc & Türk, 2025). Building on this line of inquiry, the literature on neocolonial extractivism argues that exterminatory outcomes follow from historically sedimented formations of power. Dunlap extends this argument by situating developmentalism and extractivism as long-standing colonial forms, clarifying how state-corporate collaborations render ecocidal projects both intelligible and legitimate in public reason, while scaling them to the size of dams, pipelines, wind and solar megaprojects with predictable social and ecological fallout (Dunlap, 2018a; Dunlap, 2018b; Dunlap, 2021; Dunlap, 2023a; Dunlap, 2023b). A further strand demonstrates how these arrangements become

counterinsurgency in practice. Here, ecocidal development, militarized conservation and infrastructure corridors function as technologies of control, degrading environments to undercut subsistence, mobility, knowledge systems and insurgent capacity (Le Billon, 2001, Le Billon, 2008; van Solinge, 2008; White, 2008; Peluso & Vandergeest, 2011; Brock & Dunlap, 2018; Crook et al., 2018; Dunlap, 2018b; Dunlap, 2023b). This reframing relocates environmental destruction from collateral effect to constitutive logic, clarifies how national legal-security regimes organize a new colonial and ecocidal order, and bridges colonialism, developmentalism, environmental security and the genocide-ecocide nexus within a single analytical arc.

To formulate this structure, I advance ecocidal coloniality to designate a distinct configuration of colonial structure in which the domination of people and the destruction of environments are inseparable and co-constitutive. It refers to the historical and contemporary structures through which colonial and nation-state formations secure authority, not only by dispossessing and assimilating populations, but also by systematically degrading, appropriating and reordering socio-ecologies. In this sense, ecocidal coloniality is both a mode of rule and a material condition, in which ecological devastation becomes an instrument of governance, control and exploitation. This concept foregrounds that environmental destruction and exploitation under coloniality are not accidental or collateral, but rather intentional and structural mechanisms of domination. The obliteration of forests, the contamination of water sources, the exhaustion of soils and the forced alteration of agrarian practices are not side effects of development or war, but technologies of colonial order. Ecocidal coloniality therefore describes how the colonial state governs through the weaponization of ecology itself, by producing scarcity, toxicity, marginalization and vulnerability as tools of subjugation. To this end, ecocidal coloniality operates through six intertwined, albeit not exhaustive, dimensions.

First, *legal and political engineering* codifies dispossession and ecological exploitation through emergency decrees, property laws and bureaucratic interventions that dissolve communal rights, criminalize subsistence and confiscate the commons. Second, *ecocidal developmentalism* reorganizes land, water and energy systems to serve metropolitan or settler interests under the guise of modernization. Large-scale dam projects, monocrop agriculture and extractive industries reorient entire landscapes towards export economies, deepening dependency and ecological fragility. Third, *socio-spatial and hydraulic design* converts lands, rivers, aquifers and infrastructures into instruments of power and control. The construction of

dams, the diversion of transboundary flows and the targeted destruction of irrigation systems operate as methods of discipline and demographic engineering, turning hydrology into a weapon of statecraft. A fourth dimension is *militarized ecological violence*. This dimension shows how the environment becomes both battlefield and instrument of elimination within a genocide-ecocide nexus. Scorched-earth operations, crop burning, systemic deforestation, security dams, the bombing of water stations, and the deliberate pollution of land and air demonstrate how the annihilation of ecological life functions as counterinsurgency and cultural genocide. Fifth, *epistemic domination and erasure* manifests through the suppression of traditional and liberatory ecological knowledges and cultures, the imposition of colonial and technocratic expertise, and the replacement of communal stewardship with extractivist management. Finally, *biopolitical marginalization* differentiates life along racialized, ethnic, classed and gendered lines, distributing exposure to drought, disease and toxicity unevenly across populations. This dimension lies beyond the scope of this research, which identifies a research gap for further work on the relationality of pollution, extractivism, waste and colonialism.

Unlike the notion of ecocide, which denotes large-scale environmental destruction often detached from historical context and social relations, ecocidal coloniality makes explicit that such destruction is an instrument of political domination as a structure and legacy. It reveals that the elimination of socio-ecological worlds, through land seizure, water control and the erasure of cultural-ecological practices, is constitutive of colonial rule itself. Similarly, while the concept of coloniality describes the enduring structures of power that persist beyond formal colonialism, ecocidal coloniality identifies their ecological dimension: how sovereignty, accumulation, dispossession, displacement and security are produced through interventions in land, water, energy and infrastructure. Understanding ecocidal coloniality thus clarifies the genocide-ecocide nexus, where the destruction of environments undermines the material and cultural conditions of collective existence. The targeting of livelihoods, seeds and ecosystems is simultaneously an assault on the social fabric, and a strategy of political control. Through this lens, decolonization cannot be limited to political recognition or administrative reform; it must entail a profound socio-ecological transformation that dismantles the infrastructural, epistemic and metabolic foundations of ecocidal coloniality. As an analytical framework, ecocidal coloniality directs attention to the concrete mechanisms through which colonial power operates ecocidal methods and mechanisms: the legal codification of land tenure, the design of dams and pipelines, the distribution of pollution and disease, and the epistemic hierarchies that

legitimize environmental violence. It invites a reading of colonialism not merely as territorial conquest, but as the reconfiguration of ecological relations themselves, the remaking of life's material and symbolic conditions in the service of domination.

From this stance, the dissertation's contribution is threefold. Theoretically, it develops the concept of ecocidal coloniality to reveal nation-state colonialism and its operational logic and structure; specifies ecocidal developmentalism (design-led elimination enacted through 'development') and colonial drought (hydropolitical scarcity produced by colonial rule and its legacies); and develops the genocide-ecocide nexus as a technique of extermination within settler-colonial counterinsurgency. It advances rhizomatic democracy/governance as political ontology of socio-ecological transformation, linking eco-economy, *jineoloji*, communal ownership and pluriversal epistemologies. It also develops ecopedagogy based on Misiaszek (2023) as the reconstruction of traditional and liberatory knowledge synthesis against colonial epistemicide. Empirically, the dissertation offers grounded analysis from North and East Syria / Rojava, focusing on agriculture, water, education and governance, based on interviews, surveys, group discussions and engaged observation, and provides a comparative lens on Bakur and Rojava concerning the ecocidal colonial continuum, situating Rojava within a wider field of ecocidal coloniality. Methodologically, it develops activist research as co-production of knowledge with communities and situated accountability in conflict-affected settings, with data sovereignty, careful anonymization, and an ethic of refusal. It also demonstrates a multi-wave, mixed-method design embedded in movement interlocution and collaboration.

Therefore, this dissertation makes significant interdisciplinary contributions. It advances political ecology by reframing ecological degradation as a structural and intentional instrument of nation-state colonial domination, establishing the concept of ecocidal coloniality as a decolonial analytic. Within decolonial and postcolonial studies, it places ecology at the centre of colonial critique, situating Rojava as a site of epistemic and ontological world-making. It further contributes to Kurdish and Middle Eastern studies by treating Kurdistan not as a peripheral case, but as a geography of theoretical production, where ecological and democratic experiments generate new conceptual vocabularies. In social ecology and political theory, the thesis develops rhizomatic governance as a political ontology that fuses Bookchin's ethics of ecology with Deleuzian political philosophy. It extends emerging debates in genocide-ecocide scholarship by theorizing the genocide-ecocide nexus as a technique of elimination in settler-colonial counterinsurgency, and it intervenes in the fields of climate change and socio-

ecological transformation by conceptualizing colonial drought as the hydropolitical manifestation of ecocidal coloniality.

The dissertation does not present Rojava as a model to be replicated, nor as a utopia free of contradictions, but as a living laboratory of decolonial ecology. It is an insurgent geography where the colonial legacies of dispossession, assimilation and ecocide are confronted by practices of autonomy, ecological struggle and rhizomatic resistance. By situating the Kurdish struggle within broader debates on colonialism, ecology and emancipation, this thesis affirms that the future of Kurdistan, and of the world, depends on our capacity to think and live otherwise (Holloway, 2002): to be faithful to the earth in the Nietzschean sense (Nietzsche, 1966, p. 6), to cultivate multiplicity, and to struggle for decolonial forms of being, knowing and doing.

1.5. Thesis structure

This cumulative thesis is organized in four sections that move from framing, to articles, synthesis, and then conclusion. The architecture mirrors the argumentative unfolding of the project: from colonial design to decolonial praxis and transformation. Section I (Chapters 1 and 2) establishes the groundwork. Chapter 1 sets out the thesis outline and introduces the core concepts and stakes. Chapter 2 formulates the objectives and research questions within colonial and decolonizing dynamics, and articulates the activist-scholar positionality and methodology. The methodological design combines multi-wave fieldwork centred in Cezîrê/Jazira (Dêrik, Tirbespiyê, Hesekê) with comparative lenses on Afrin, Bakur, Palestine and Europe, and employs an abductive, triangulated analytic strategy.

Section II (Chapters 3–7) brings together five articles that trace a trajectory from the colonial mindset and its material infrastructures to the horizon of a political ontology of decolonization and transformation. Chapter 3 (Article I, “Decolonization agriculture”) demonstrates how Ba‘athist agricultural modernization operated as the main tool of nation-state colonialism, including peripherization, settlement and land confiscation, and how, after 2012, AANES initiated a decolonization of agriculture through cooperatives, seed institutions and rhizomatic governance that rerouted production towards locally administered economies of diversity as a decolonization and building economic autonomy. Chapter 4 (Article II, “Roots of destruction”)

Chapter 1: Introduction

develops the genocide-ecocide nexus comparatively, analysing the systematic destruction and appropriation of olive trees in occupied Palestine and Afrin to show ecocide as a deliberate technique of extermination within settler-colonial counterinsurgency, and to map repertoires of resistance. Chapter 5 (Article III, “Resisting colonial drought”) theorizes colonial drought as the hydropolitical outcome of Syrian internal colonialism and Türkiye’s regional hydro-hegemony, reading dams and irrigation systems under large-scale development projects as socio-spatial designs that manufacture scarcity and dependency, and tracing autonomy-oriented responses in governance and practice as a socio-ecological transformation. Chapter 6 (Article IV, “Education under a tree”) treats education as ecological infrastructure for building decolonial minds and epistemologies, tracing the shift from Ba’athist schooling to ecopedagogy embedded in democratic confederalism, and linking *jineoloji*, social ecology, and communal decision-making to repair relations to land, life, nature, and collective self-governance. Chapter 7 (Article V, “Ecological struggle as ethos of decolonization”) synthesizes movement texts and fieldwork from Bakur and Rojava to explain why ecology becomes a foundational pillar of the KFM’s anti-colonial politics (eco-economy, women’s liberation, communal ownership, ecological stewardship) and the ethics and aesthetics of autonomy.

Section III (Chapters 8 and 9) provides the integrative synthesis. Chapter 8 consolidates the definitions and scope conditions of ecocidal articulations, and draws together evidence from across the articles to identify the principal vectors and dimensions of ecocidal coloniality. Chapter 9 traces the passage from resistance to socio-ecological transformation, detailing the reconstruction of ecological agriculture, rhizomatic governance, ecopedagogy and the eco-economy as a coherent decolonial ecology; it concludes with reflections on the challenges and threats Rojava faces. Section IV (Chapter 10) closes by restating the findings, outlining lessons from Rojava, and indicating directions for future research concerning the specific research gaps that the thesis opens.

Chapter 2. Research objectives, positionality and methodology

The design of this research is inseparable from its political and epistemic commitments. To study Rojava merely as an object of analysis would reproduce the extractivist logics of colonial research; instead, this project treats research as a co-constitutive process grounded in solidarity, accountability and reflexivity. The following chapter therefore outlines the framework through which the inquiry has been structured. Its aim is to make explicit how the objectives and research questions are formulated, how the positionality of the researcher shapes and constrains the project, and how methodological choices are ethically and politically embedded.

The first section defines the objectives and research questions guiding this inquiry. At stake is a dual movement: critically analysing how colonial state structures, ranging from Syrian Ba‘athist agrarian modernization to Turkish hydro-hegemonic interventions, have produced socio-ecological crisis; and examining how the AANES, in collaboration with grassroots actors, has reconfigured governance, agriculture and ecological practice to advance a project of democratic autonomy and decolonial transformation. These objectives generate research questions that foreground the intersections of ecology, decolonization and political ontology, asking how Rojava emerges as a pluriversal alternative to statist, patriarchal and capitalist modernity.

The second section turns to critical reflexivity and positionality. Here, the research recognizes that knowledge is always situated, shaped by the social, cultural and political location of the researcher. My position as a Kurdish activist-scholar in exile entails both proximity to and distance from the field. It provides access and responsibility, but also demands continuous vigilance against reproducing colonial or extractivist dynamics. Reflexivity is therefore not an afterthought but a central methodological tool, interrogating how personal history, intergenerational trauma and resistance, political commitments, and activist engagements all inflect the production of knowledge.

The third section sets out the methodological and ethical framework. Rejecting detached, positivist approaches, the project adopts an activist research methodology grounded in solidarity, reciprocity and co-production. It mobilizes a plural set of methods, while treating each not only as instruments of data collection, but as spaces of collective learning and situated

theorization. Ethics here are not reduced to formal protocols but understood as daily practice: negotiating safety, consent and the limits of visibility in a conflict-affected context.

2.1. Objectives and research questions

The world is facing intersecting ecological, economic, social, ethical and political crises, driven predominantly by colonial and neocolonial structures underpinned by capitalist modernity. Rojava represents a radical rupture from capitalist modernity and its associated ecocidal and genocidal logics, both ontologically and epistemologically. Situated within this context of colonial violence, war and revolutionary transformation, this research aims to critically analyse the emergent model developed in Rojava, specifically examining how mechanisms of democratic resistance and socio-ecological transformation have facilitated its emergence as a pluriverse (Escobar, 2018).

The primary objective of this research weaves together two elements. First, it critically examines how colonial state structures, particularly Ba‘athist Syria with its modernization and developmental policies, and Türkiye’s neocolonial hydro-hegemonic interventions, have historically shaped, and continue to influence, the socio-spatial, ecological and demographic configurations of Rojava. It seeks to explain the region’s socio-ecological collapse by showing how intersecting crises, including drought, conflict and ecological degradation, are rooted in long-standing practices of colonial dispossession, demographic engineering and displacement, and are further intensified by climate-induced aridity and more than a decade of civil war.

Second, the research explores the innovative organizational and administrative strategies developed by the AANES, in collaboration with local communities, to confront and transform these multidimensional challenges. The study particularly focuses on grassroots ecological initiatives, communes, people’s assemblies, academies, agriculture system, cooperatives and democratic governance mechanisms, which seek not merely to adapt to crises but to actively foster decolonial and emancipatory socio-ecological transformation. In doing so, the research aims to contribute to broader theoretical and practical debates surrounding ecological struggle, decolonization, democratic resistance, autonomy and socio-ecological transformation in contexts affected by ecocidal coloniality.

This research is grounded in sustained engagement and collaboration with local actors, scholars and activists in Rojava. Through participatory workshops, collaborative fieldwork, and critical dialogues, the following interrelated research questions have been co-developed. The articles that answer these questions are indicated in parentheses.

1. How have colonial state structures, particularly Ba‘athist Syria’s agrarian policies and Türkiye’s hydro-hegemonic strategies, shaped socio-spatial configurations, dismantling Rojava’s ecological, economic and cultural integrity? (Articles I, III and V)

a) In what ways have these colonial mechanisms exacerbated contemporary drought conditions and agricultural collapse? How do these dynamics intersect with climate change and ongoing conflict to reinforce structural vulnerabilities? (Article III)

2. To what extent do contemporary struggles over access to land and water reflect and reproduce colonial power relations, and how are these contested through emergent governance models and practices of ecological and economic autonomy reclamation? (Articles I and V)

3. How has AANES developed organizational and administrative responses to address structural socio-ecological challenges and ecocidal policies rooted in colonial and statist legacies? (Articles I, II and III)

a) What structural relationality exists between coloniality and ecocide in Rojava? Why has ecology emerged as a central ethical and aesthetic dimension of the anti-colonial struggle in the region, and how does this shape processes of organizational and community transformation? (Articles II, III and V)

b) What democratic governance models have been implemented to dismantle centralized and hierarchical state logics, and how do this governance mechanisms integrate diverse ethnic and religious groups, women, and social classes? (Articles I, III and V)

4. Considering Rojava’s revolutionary context, how does its model embody an ontological and epistemological rupture with the colonial state structure, capitalist and patriarchal systems it opposes? (Articles I, III, IV and V)

a) How does the Rojava model transcend mimicry and the replication of colonial mentalities and institutional forms, and through which methodological and epistemic frameworks does it reconfigure socio-ecological terrains? (Articles I, III, IV and V)

b) What role can I, as an activist-researcher, play in advancing the epistemic decolonization embedded in this struggle, actively resisting extractivist knowledge production, and affirming the epistemic agency of communities engaged in resistance? (Chapter 2)

These questions are pursued through an activist research methodology rooted in principles of solidarity, reciprocity and situated accountability, allowing for the co-production of knowledge reflective of the lived realities and aspirations of Rojava's communities. Ultimately, this research seeks to offer insights and pathways for broader struggles towards ecological justice, autonomy and decolonization, emphasizing the transformative potential embedded in grassroots anti-colonial ecological resistance and democratic governance.

2.2. Critical reflexivity and positionality

Critical reflexivity and activist positionality are not peripheral considerations, but foundational pillars of ethical and epistemically just research, particularly within geographies shaped by colonial violence, political struggle, and asymmetrical power relations (Bourke, 2014; England, 1994; Holmes, 2020; Thambinathan & Kinsella, 2021; Kitagawa, 2023). Over recent decades, feminist, Indigenous and decolonial scholars have argued that knowledge is never produced from a neutral standpoint, but is always situated in the researcher's social, historical and affective location (Haraway, 1988; Mullings, 1999; Sultana, 2007; Marsh & Furlong, 2017; Adu-Ampong & Adams, 2020; Holmes, 2020; Kitagawa, 2023). Accordingly, interrogating one's positionality is not a methodological ornament, but a political and ethical necessity, especially within frameworks such as activist research (Hale, 2001; Hale, 2006; Fortier, 2017; Martínez, 2023), solidarity-based inquiry (Gagliano, 2021; Ross et al., 2023) and decolonial epistemologies (Mignolo, 2009; Mignolo & Walsh, 2018).

Through engagement in transformative praxis, the activist-researcher participates in the process of change through reflexivity, solidarity-based inquiry, reciprocity, and situated accountability (Haraway, 1988; Kearns, 2005; Thambinathan & Kinsella, 2021; Ross et al., 2023). Reflexivity must go beyond personal introspection to a structural analysis of how power circulates through research relationships and how academic knowledge production, if unexamined, risks reproducing extractivist and colonial dynamics (Sultana, 2007; Smith, 2012; Tuck & Yang,

2012; Fisher, 2015; Taha, 2018; Kitagawa, 2023). In colonized and contested terrains in the Global South such as Rojava, the stakes are particularly high: scholarly practices that marginalize or appropriate Indigenous knowledges can easily become complicit in the systems of domination they seek to critique (Sultana, 2007; Santos, 2015).

As a Kurdish activist-scholar in exile, I am simultaneously embedded within and accountable to the communities at the heart of my research. Rejecting the ideal of detached objectivity, long critiqued as a mask for dominant epistemic paradigms (Cunliffe & Karunanayake, 2013; Hale, 2001; 2006; Haraway, 1988; Kitagawa, 2023; Mullings, 1999; Ross et al., 2023; Sultana, 2007), this research embraces a model of situated accountability that foregrounds the partiality, solidarity, relationality and ethical embeddedness of all knowledge (co-)production (Haraway, 1988; Kwaymullina, 2016; Rasch & van Drunen, 2017; Maclean et al., 2022; Ross et al., 2023). In a context like Rojava, where epistemic disobedience is not merely a theoretical stance but a mode of survival and political affirmation, method and positionality are inseparable. My positionality, shaped by intergenerational trauma, forced displacement and embodied resistance, is not a subjective bias to be bracketed, but the very condition of possibility for this research. It affirms that knowledge emerges through affective, ethical and political entanglements, and that the credibility of research lies not in false neutrality, but in its fidelity to situated struggle and collective transformation (Chatterton et al., 2007; Collins, 2000). Nevertheless, while my political and cultural affiliations facilitate trust, access and shared experience, they also necessitate continuous vigilance against romanticization, over-identification, and epistemic blind spots, precisely the kinds of risks that critical reflexivity seeks to mitigate (Collins, 2000; Kovach, 2010; Thambinathan & Kinsella, 2021).

Given my long-standing involvement in the Kurdish movement, I occupied a position of relative insider status during fieldwork, which facilitated access to data, documents, institutions and research participants. Yet it would be naïve to assume a fully insider position. My residency in Europe and educational and class background positioned me simultaneously as an outsider, producing a form of epistemic and social distances. Nevertheless, speaking Kurdish, arriving with references from the Kurdish movement and being recognized as an activist-researcher enabled a strong sense of trust and solidarity. Despite widespread poverty, nearly all participants welcomed me warmly, offering tea and homemade food with sincere hospitality. Following the interviews, many explicitly expressed their hope that I would amplify Rojava's

voice to the world and foster international solidarity against Turkish aggression and jihadist militias.

Several mothers, in a deeply affective gesture, kissed my eyes, a cultural act conveying both honour and responsibility. In Kurdish culture, to be kissed on the eyes is a gesture layered with profound tenderness, reverence, love and sacredness. When performed by an elderly woman, it becomes an act of deep honour, an affirmation of worth that transcends mere affection. Such a gesture symbolically welcomes the recipient into the moral fabric of the community, as kin. In its quiet grace, it carries the weight of collective memory, entrusting not only personal regard, but also the transmission of social responsibility and the solemn bequest of a legacy of resistance. It is thus no coincidence that Kurdish elderly women have kissed the eyes of fighters departing to confront ISIS – an intimate act of blessing, defiance and communal farewell.

This was a form of traditional negotiation, unspoken yet powerful, that shaped my positionality and demanded that I take an ethical stance. Participants actively sought to influence and shape the research, asserting that it should offer tangible, positive contributions to Rojava and to their lives. In particular, key informants, including local activist-scholars from various groups, non-governmental organizations (NGOs) and the University of Rojava, as well as high-level political cadres and representatives of AANES, were actively involved in co-designing the research questions, objectives and methodological framework. This structural and organic relationality, operating from a Deleuzian perspective, dissolves the binary between subject and object, researcher and researched. It affirms that every encounter entails a reciprocal process of transformation for both sides. In Gilles Deleuze's philosophy, particularly in his writings on cinema, the relation of perception, affection and action is a dynamic interplay rather than a fixed subject-object dichotomy. Deleuze challenges the traditional view of the subject as a stable, pre-existing agent exerting influence over a passive object. Instead, he emphasizes that action and perception are fundamentally entangled, and that subjectivity itself is continually constituted through encounters with forces, events, and affects. Perception, in this framework, is not a passive intake of external information by a subject from the world outside, but an active, co-constitutive process of interaction and becoming (Deleuze, 1986; Deleuze, 1989).

In Deleuzian terms, I am neither a detached observer nor a pure insider, but rather a transversal node in a co-constitutive process, both affected by and affecting the field through activist-scholar engagement. My engagements were shaped by the socio-political intensities of war,

displacement and resistance, while simultaneously shaping the way knowledge was shared, withheld or transformed. Such a perspective dissolves the subject-object binary that underpins extractive, positivist epistemologies, and aligns instead with an ontology of becoming, wherein knowledge emerges through encounters, affective resonances and mutual transformation. This is especially crucial in a context like Rojava, where research is not merely about interpretation, but participation in a living, resisting collective. Therefore, a static, fixed positionality is not only impossible, it is a myth perpetuated by modern European positivist scientism. Knowledge produced through such a lens tends to reflect a hierarchical relationship in which the privileged (white European) researcher exerts, whether consciously or not, epistemic violence, turning participants into objects of extraction within a hegemonic, extractivist regime of knowledge production. Rejecting this epistemic and methodological violence, and committing instead to placing knowledge at the service of local communities, the oppressed and colonized, and those engaged in resistance, become an ethical and political imperative.

To choose this path is to abandon the positivist rhetoric of impartiality and objectivity, and to foreground one's own location as a researcher, politically and ethically. It requires embracing an epistemology grounded in solidarity, capable of confronting and dismantling colonial and epistemic violence. Therefore, knowledge, and the methods through which it is produced, is inherently partial and subjective. From our choice of topic and research field, to the kind of knowledge we produce, for whom and why, every step is shaped by opaque dynamics and ongoing negotiations. Our educational background, class, ethnic identity, gender, institutional affiliation, and personal stories and lived experiences are embedded in this subjectivity, and are never absent.

Conducting fieldwork in a conflict zone like Syria was a tremendous challenge. Many colleagues and friends, understandably, advised me not to go, or even to change my research topic. I waited for over ten months for an opportunity for safe passage, but not only did the conflict persist, the Semelka border crossing under the control of the Kurdistan Regional Government in Iraq repeatedly communicated that the crossing was unsafe or officially closed. Even during my fieldwork, Turkish attacks, including on civilian infrastructure, were ongoing. My decision to take such a high risk was driven by an ethical and political commitment: I viewed and believed in my research as an integral part of my people's anti-colonial struggle. From this positionality, the notion of being 'objective' and producing 'impartial' knowledge

was never convincing, neither in terms of who it would serve, nor why such a study should be conducted in the first place.

Accordingly, my findings are inevitably situated, interpretive and partial, but they bear witness to a narrative of my people that might otherwise remain silenced: a narrative of ecological struggle, decolonization, land and water, agricultural transformation, epistemic resistance, and the articulation of new political ontologies as a terrain of socio-ecological transformation. These insights not only document a locally grounded praxis, but also illuminate broader dynamics of resistance that seek to construct a world in which many worlds can coexist (Escobar, 2018). Ethical fidelity to the relationships, encounters and experiences that unfolded during fieldwork was a guiding imperative. Remaining attentive to the specificity of place, the complexity of context and the interpretive frameworks that emerged, required a reflexive and accountable mode of knowledge production. Throughout the research, I continually negotiated responsibilities to myself, participants, the political stakes of the work, disciplinary conventions and institutional ethics review frameworks; these obligations became deeply entangled in the embodied, everyday practice of fieldwork.

This sensitivity and positionality stem directly from the entanglement of my roots and personal story with the research I conduct. It is, in itself, the process of becoming that has transformed me into an activist-researcher. I was born in a village in the Dicle (in Kurdish, Pîran) district of Diyarbakir (in Kurdish, Amed), located in Bakur / Northern Kurdistan, southeast Türkiye. My first direct encounter with state violence and colonial policies occurred in 1992 at the age of six, when our family home and farm were burned down and we were forcibly displaced by the Turkish state. This moment, ungraspable at the time, was a profound rupture. It remained a deep silence within my family for years, an unspoken trauma, like a black hole that consumed meaning. During my high school years, I began a committed process of inquiry into Kurdish history and resistance, especially the contemporary guerrilla struggle. I also had intense conversations with my father, himself an activist who had been detained and tortured multiple times. He once told me something I could only fully understand years later: “Your grandfather once told me the story of his own father – he passed on a burden I could never shake off, a wound that never closed.” That sentence haunted me. What was this wound? Why was our family story so heavy, so silenced, yet so present?

The roots stretch deep. My great-grandfather, Mihemedê Muso, was part of a prominent landowning and scholarly family in Pîran. He and his relatives established and led the renowned school (*medrese*) in Qilbin, Pîran. Another branch of the family engaged in silk production and trade with local Armenian partners in Qatug-Sûlbetan, particularly Agop and Xacer, two influential religious and business figures in Pîran. This partnership was formalized through the tradition of *kirîvatî*, a cultural and religious pact that established spiritual kinship and mutual protection, forbade intermarriage, and thus sealed a bond of deep trust and interdependence. This relationship brought considerable prosperity to the family, enabling them to acquire vast tracts of land. This world, however, was shattered by the Armenian Genocide of 1915, during which many of our relatives – considered Armenian by association – and Armenian partners were massacred. The family's lands were confiscated and state-aligned tribal actors were settled there, where they remain to this day. Just a decade later, the newly established Turkish Republic turned its focus to eliminating Kurdish resistance. The 1925 semi-national uprising in Pîran, led by Sheik Seîdê Pîran, was brutally crushed, and many of my family members were again exiled or punished due to their ties to him and the religious institutions he inspired. The *medreses* were closed. After a long silence, the 1980s and 1990s brought a resurgence of Kurdish resistance through the *Partiya Karkerên Kurdistanê* (PKK)'s anti-colonial guerrilla movement, reawakening what my father had called the “unhealed wound.”

It was into this legacy, long before I knew what it meant, that I was born. My coming of age was shaped by both trauma and resistance. My father's activism, his repeated detention and torture by state officials and paramilitary units, became a contemporary manifestation of this inherited trauma. It is into this dense, intergenerational memory that I was born. My journey of self-realization, identity and reconnection with my roots has been both deeply traumatic and profoundly liberating. This shaped my path, as I initially studied civil engineering at Dicle University (2005–2006), but left it to pursue a more critical engagement with society. I became active in student movements, organizing around social and political justice, language rights and cultural identity. This led me to Boğaziçi University's philosophy department (2007–2013), where I encountered new tools to name and confront the colonial wounds of my people intellectually. After completing my philosophy degree, I returned to Amed and worked from 2014 to 2016 in pro-Kurdish local governance institutions and NGOs, including academies, human rights and ecological organizations involved in constructing grassroots autonomy. With the Turkish state's subsequent dismantling of all Kurdish-led municipalities and civil institutions, it became impossible to continue academic or activist work there.

In 2019, I moved to Germany to pursue the master's programme in Human Rights at Friedrich-Alexander University of Erlangen-Nürnberg. This provided theoretical depth for my reflections, enabling me to analyse how neocolonial structures expropriate peasant and Indigenous lands, a concern shaped by my historical and political background. I examined this problem throughout the master's thesis. Subsequently, I began my PhD at the University of Hamburg with a clear aim: to challenge the epistemic narratives that reproduce colonial silencing, and to reconnect knowledge with memory, freedom and justice. The central motivation of my doctoral research has been to serve as a mouthpiece for the silenced stories and voices of both my own past and those of other victims and resisters of colonial violence, voices targeted for erasure, and to construct a dignifying language and aesthetic that restores their presence. Through both conceptual interventions and activist collaborations, I strive to decolonize epistemologies to build pathways for socio-ecological and political transformation as part of the anti-colonial struggle of my people. My lived experience of displacement, loss and resistance directly informs both my theoretical approach and methodological commitments, particularly my use of activist, solidarity-based and decolonial research methods rooted in dialogical and collective knowledge-making.

As an activist-scholar in exile, the colonial violence, massacres, forced displacement, and dispossession experienced by me and my family are a reflection of the broader macrohistory endured by the Kurdish people. In this resistance, where the historical and the present, the personal and the collective, are deeply entangled and mutually constituted, my involvement is both inevitable and existentially meaningful. Had I not situated my own story as part of the larger process of liberation through the anti-colonial struggle of the Kurdish people and other oppressed communities, it would likely have led to a tragic internal collapse, a quiet erasure. In failing to do so, I would have condemned not only myself, but also my family and other people who have been subjected to colonial violence – those killed, displaced and stripped of their lives, cultures and memories – to die once more, to be lost again.

I believe that this is a colonial wound that I embody. A striking moment during my bachelor's degree in philosophy came when reading Deleuze's commentary on Joë Bousquet's words "My wound existed before me; I was born to embody it" (Deleuze, 1990, p. 148). I felt as though someone had put into words what I had carried for years. This statement helped me reconceive my trauma not as a private burden, but as a collective, transgenerational event that exceeds individual experience. In Deleuzian philosophy, an event is not merely something that happens

in time or space; it also contains a surplus, an incorporeal dimension not reducible to physical bodies or empirical facts. Events are effectuated, realized in specific persons and things, but their meaning or intensity surpasses that realization. Each event manifests uniquely in a subject, a process Deleuze calls individuation. This singular embodiment of the event forms part of the subject's becoming. Here, the subject is not the origin of the event but its expression, suggesting a reversal of typical subject-object logic. (Deleuze, 1990, pp. 98–99) Thus, Deleuze's theory of the event dissolves binary distinctions (subject/object, cause/effect, self/world) by positioning the event as both transcendent and immanent, always partially unactualized yet materially inscribed. Accepting the wound not as 'mine' but as the wound, a historical and ontological condition, is not resignation, but an ethical and political affirmation of decolonization and liberation. Geography may be fate, but it is a fate shaped by human agency, and resisting it with joy, solidarity and rootedness can be emancipatory. I choose not to accept victimhood, but to embrace the aesthetics and ethics of resistance and transformation. As Angela Davis emphasizes, ethical responsibility in the face of structural injustice requires not passive neutrality, but active resistance: "In a racist society, it is not enough to be non-racist, we must be anti-racist" (cited in Thambinathan & Kinsella, 2021, p. 6).

It is important to emphasize that my engagement in activism began in 2005 across various domains in Bakur / Northern Kurdistan and Europe, and long preceded the formal contours of this research project. This research added a deliberate emphasis on the co-production of knowledge, a methodological and ethical commitment to producing theory *with* rather than *about* communities in struggle. This culminated in a collaboratively authored article written with three Rojava-based activist-scholars who not only supported the fieldwork, but also actively shaped the analytical process. Most crucially, the central aim of this research is not merely academic. It is part of a broader effort to decolonize existing colonial epistemologies, those that have historically silenced, pathologized or instrumentalized Kurdish knowledges and experiences. In this sense, my work is embedded in the collective struggle for liberation led by the Kurdish people, particularly in Rojava, where a new emancipatory horizon is being constructed against the enduring legacies of colonialism, statism, patriarchy and capitalist modernity.

My direct engagement with Rojava began in 2014, when ISIS launched its brutal assault on Kobanê. I travelled to the border town of Suruç (Urfa province), directly across the border from Kobanê, and participated in humanitarian coordination for approximately three months under

the framework of the Union of Southeastern Anatolia Municipalities, supporting displaced populations by facilitating aid delivery, organizing solidarity networks and ensuring safe resettlement. My involvement has always been rooted in organic engagement and principled solidarity with a deep commitment to understanding, supporting and amplifying the revolutionary struggle of the peoples of Rojava, who, despite immense hardship and the realities of war, have been striving to build an alternative system since 2012. For instance, I served as one of the pioneer organizers of the Ecology Conference, “Ecology in Democratic Modernity: From Resistance to Construction – Perspectives, Possibilities and Challenges”, held on 26–27 April 2024 in Qamishlo (Rojava, Syria). At the request of the Rojava Ecology Council, my dear colleague Joost Jongerden and I invited the participating scholars to rework their talks into academic articles; as co-editors, we are currently compiling these contributions into an edited volume. I am also a co-founder and initiator of the master’s programme in Political Ecology at the University of Rojava (see Appendix A). Through ongoing discussions with local scholars and institutions, such as the Agriculture and Irrigation Council, the Ecology Council, economic and municipal organizations, and the university, we identified the region’s needs. In collaboration with colleagues across Europe who offered their solidarity as guest lecturers, we launched the programme in the winter semester of 2025. I deliver lectures and voluntarily serve as a co-coordinator. Furthermore, I am actively involved in the Kurdistan Ecology Movement in Europe as an activist-scholar. In this role, I have shared findings from my research with Kurdish associations and civil society initiatives across Europe. Over recent years, I have participated in numerous academic and international conferences, from Mexico to Japan and throughout Europe, engaging with universities and grassroots social, political and ecological movements. The goal is not only to exchange knowledge and experiences, but also to foster transnational networks of organic solidarity among movements committed to social, political and environmental justice.

These commitments are not without personal cost. One of the most significant challenges has been the intense workload and constant navigation between academic obligations and activist commitments. Maintaining this dual engagement has left little time for personal or social life, and often resulted in emotional and physical exhaustion. These engagements, while intellectually enriching, also entailed continuous exposure to stories of violence, ecological destruction and the suffering of marginalized communities. The emotional burden of witnessing and listening to these experiences cannot be understated. Despite these challenges, what sustained me was the sense of collective purpose and solidarity shared across movements and

scholarly communities. This work would not be possible without such existential grounding and a shared ethical and aesthetic commitment to justice and freedom. The solidarity and encouragement I have received from academic colleagues were vital in helping me persevere through periods of stress and uncertainty.

2.3. Methodology

This research adopts an activist research methodology deeply informed by critical, decolonial, feminist and Indigenous perspectives, anchored in the recognition that research is never a neutral act and has roots deep within colonial and Eurocentric epistemologies (Hale, 2001; Hale, 2006). Through histories of colonial domination, state violence and epistemic erasure, traditional positivist social science has often functioned as an instrument of control – collecting data as intelligence for colonizers, categorizing and dehumanizing Indigenous/colonized peoples, and appropriating local knowledge for the benefit of external powers (Denzin et al., 2008; Mignolo, 2009; Smith, 2012; Mbembe, 2015; Thambinathan & Kinsella, 2021). As Linda Tuhiwai Smith observes, colonial research has followed a pattern: “They came, they saw, they named, they claimed,” transforming knowledge into a weapon of occupation (Smith, 2012, p. 83). Western scientific paradigms historically discovered, extracted, commodified and distributed knowledge about the Indigenous ‘other’, embedding research within the very structures of dispossession and assimilation it purported to study. Decolonizing epistemologies and dismantling methodological architectures are therefore not optional; they are prerequisites for the liberation of colonized and Indigenous peoples and for producing knowledge that can advance social and ecological justice (Battiste, 2000; Bishop, 2005; Denzin et al., 2008; Hart, 2010; Mbembe, 2015; Santos, 2015; Radcliffe, 2017; Walsh, 2018; Barreiros & Moreira, 2020; McGuire-Adams, 2020; Thambinathan & Kinsella, 2021; Martínez, 2023). Thus, this research methodology adopts a decolonial perspective, articulated strikingly by Audre Lorde: “the master’s tools will never dismantle the master’s house” (1984, p. 112).

To this end, central to activist research is the rejection of traditional, detached, objectivist paradigms, and the adoption of methodologies grounded in political solidarity, ethical relationality and epistemic reciprocity (Routledge, 1996; Hale, 2006; Kindon et al., 2007; Pulido, 2008; Zavala, 2013; Thambinathan & Kinsella, 2021). Participatory action research and Routledge’s (1996) “third space” share strong affinities with activist research, as they

foreground solidarity, mutual learning and the destabilization of power relations; however, activist research radicalizes these commitments. Routledge's intervention opens hybrid, reflexive terrains that unsettle insider/outsider binaries and reconfigure positionality, and activist research builds on this by grounding such moves in an organic decolonial ontology of becoming and a sustained praxis aimed at transformative social change. Its priority is not dialogical negotiation or the instrumental deployment of methods for 'knowledge production', nor the technocratic 'solution' of ostensibly local problems, but an ongoing, situated and accountable participation that dissolves researcher-researched dichotomies and pursues structural transformation. In the same spirit, Antonio Conti argues that "the goal of research is not the interpretation of the world, but the organisation of transformation" (cited in Chatterton et al., 2007, p. 3).

It is in this spirit that my methodology rejects the positivist, detached paradigms of mainstream social science (Hale, 2006; Smith, 2012) in favour of activist research – a praxis-oriented approach grounded in political solidarity, ethical relationality, and epistemic reciprocity (Routledge, 1996; Chatterton et al., 2007; Kindon et al., 2007; Pulido, 2008). Smith's (2012) call to interrogate colonial power relations and epistemic violence aligns with my stance: knowledge production is framed as resistance to colonial extraction and epistemic injustice. My choice of methodology recognizes that how we research is inseparable from what and why we research, and that producing knowledge in Rojava without dismantling colonial epistemic frames risks reproducing the very structures my work seeks to oppose. Drawing on Haraway's (1988) "situated knowledges", this research emphasizes the inherently partial, contextual and politically engaged nature of knowledge production. Situated accountability thus becomes foundational, highlighting the ethical obligation to articulate positionalities transparently and to consider how personal histories, identities, privileges, and commitments shape research practices and outcomes. My positionality as a Kurdish activist-scholar compels me to adopt a methodology that is explicitly accountable to participants, reciprocal with communities in Rojava, and relational. Indigenous and decolonial scholars further underscore accountability and relationality as ethical imperatives in research with historically oppressed and marginalized communities (Battiste, 2000; Smith, 2012; TallBear, 2014; Taha, 2018; Thambinathan & Kinsella, 2021).

Methodologically, this research resists extractivist scholarship and emphasizes collaborative co-production through sustained engagement with local actors in Rojava. It uses participatory

workshops, community dialogues and reflective practice to ensure the work remains ethically accountable to, and genuinely reflective of, community aspirations and lived realities. Methods include key informant and expert interviews, group discussions, semi-structured interviews, engaged and observant fieldwork, participatory workshops, and conference-based dialogue. These plural methods are not only tools of data collection, but also spaces of co-learning and co-production, where participants shape both the content and direction of inquiry. In addition to these qualitative methods, I incorporated a targeted survey designed not for statistical generalization, but as a qualitative instrument to gather narrative accounts and patterns across a broader sample, complementing and triangulating other data sources (see Appendix B). The integration of these methods supports a situated and reflexive approach to knowledge production, grounded in the lived experiences and aspirations of Rojava's communities.

Ultimately, activist research suits this project because it transforms the research process itself into a site of struggle, solidarity, and epistemic resistance. It enables contribution to the Kurdish movement's broader project of decolonizing knowledge: developing alternative epistemologies that refuse colonial hierarchies and centre lived experiences, ecological ethics and political visions. Method here is not an external lens applied to a field, but a co-constitutive process in which scholarly practice and community struggle shape and transform one another.

2.4. Research field and methods

Adopting an activist research methodology requires treating research as a relational, situated practice accountable to communities in Rojava. Methods were chosen to resist extractivism and to enable solidarity, reciprocity and co-learning, combining a rural household-level survey with semi-structured expert interviews across organizational and community arenas, group discussions, and engaged observation in cooperatives, communes and people's assemblies. Although the thesis is cumulative (five articles), its empirical centre of gravity is the Cezîrê/Jazira region of Rojava; comparative components (Afrin, Palestine, Bakur, Europe) serve to contextualize rather than redefine the primary case. Fieldwork concentrated on Dêrik, Tirbespiyê and Heseke across 12 villages (see Figure 2), and unfolded in three waves designed to balance embedded depth with continuity and safety under shifting conditions: an initial online phase (October 2023–March 2024), an in-person phase (26 April–6 June 2024) and a second online phase (October 2024–March 2025).

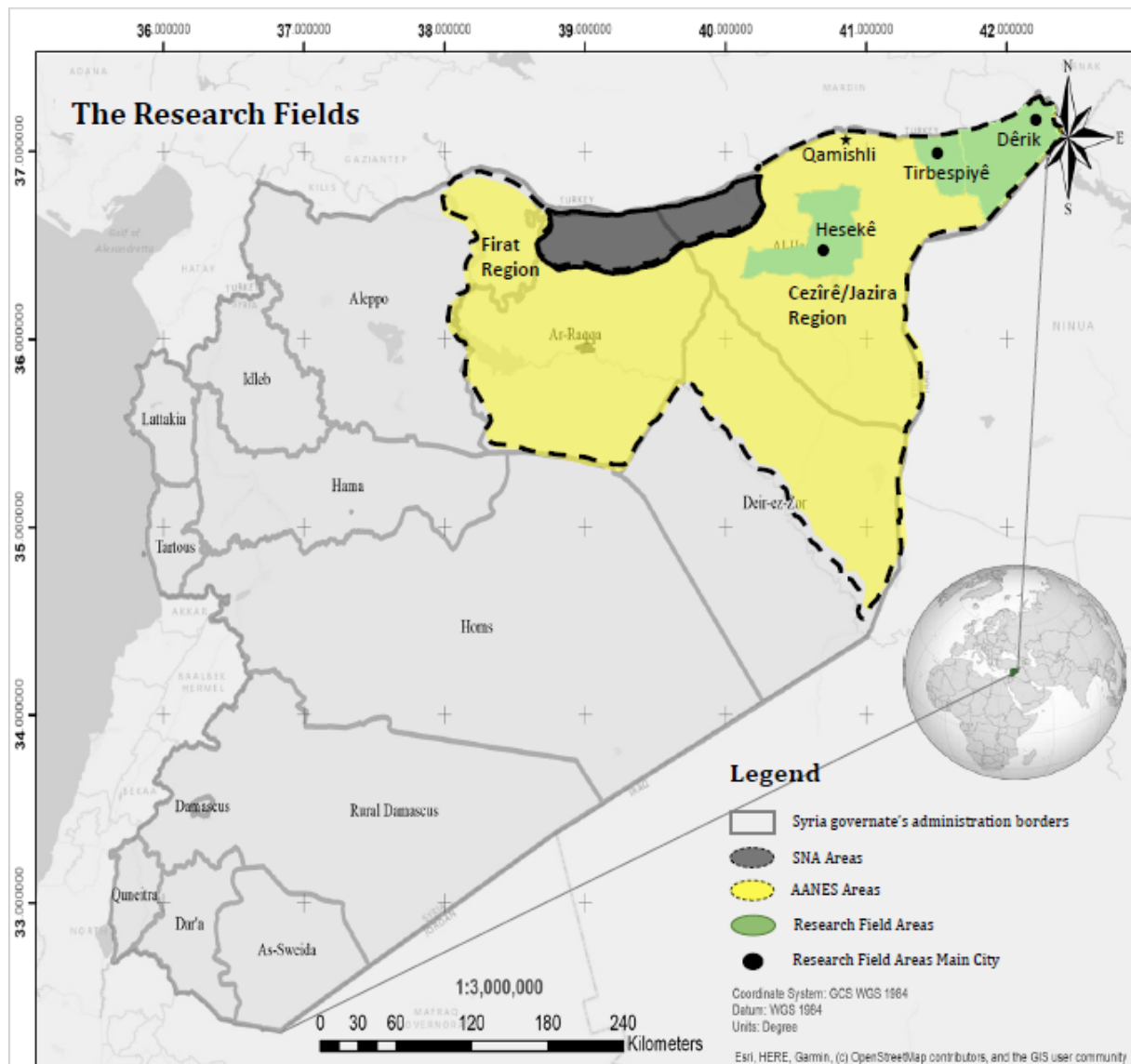


Figure 2: The research fields

The household survey ($n = 50$) captured livelihoods; perceived ownership and management of land and water; and adaptive strategies against climate change, drought and water scarcity, using purposive sampling for maximum variation and political diversity (40 participants active in communes and 10 participants who have not joined the communes because they are neutral towards them or ideologically opposed to the Rojava administration commune system). Intersectional attributes recorded included gender; ethnoreligious identity (Kurds, Yazidis, Arabs, Assyrians – see Table 2a); and livelihood/tenure (farmer-landowner, tenant/sharecropper, landless agricultural worker, pastoralist/herder – see Table 2b).

Tables 2a and 2b: Household survey intersectionality (n = 50)**Table 2a:** Gender and ethnoreligious composition (by district)

District	N	Men	Women	Arabs	Kurds	Yazidis	Assyrians
Dêrik	10	7	3	4	6	0	0
Tirbespiyê	17	11	6	2	10	3	2
Hesekê	23	15	8	7	8	5	3
Total	50	33	17	13	24	8	5

Table 2b: Livelihood and tenure composition (by district)

District	Farmer-landowner	Tenant/sharecropper	Landless agri-worker	Pastoralist/herder	Subtotal
Dêrik	4	4	2	0	10
Tirbespiyê	5	4	6	2	17
Hesekê	6	5	12	0	23
Total	15	13	20	2	50

Instruments were administered primarily in Kurdish and Arabic, and district allocations were fixed at Dêrik 10, Tirbespiyê 17 and Hesekê 23 (see Appendix B).

In-depth interview targeted communes, cooperatives and co-chairs of assemblies, as well as pioneers and workers; municipal and regional ecology, agriculture and economy cadres; women's economy initiatives; research institutes; and movement actors. In-depth interviews in Rojava centred on commune and cooperative practice and governance, as well as reconstructing ecological agriculture as a key element of economic autonomy (Dêrik 6; Tirbespiyê 8; Hesekê 7; total 21). These were complemented by further (in-person) interviews with high-ranking cadres and leaders, and by extensive online interviews before and after the field period. Community representative (either a religious authority or an individual elected by the community to serve in local or general administrative bodies of the AANES) interviews covered Sunni/Muslim, Alevi, Assyrian and Armenian Christian, Zoroastrian (Zerdestî), and Yazidi leaders. Group discussions were held with the Mesopotamia Ecology Movement board (Bakur), the International Commune of Make Rojava Green Again, women's leadership in Jinwar, and a multi-institutional session in Dêrik; a pedagogy-focused article drew additionally on 12 written (email) interviews. Across the project, the totals are: online 28, in-person 32, written 12, group discussions 4, survey 50. A consolidated list of expert interviews and the identification rules appear in Appendix C.

Analysis proceeded abductively and thematically across interviews, survey narratives, observation notes and documents, iterating with activist and decolonial research concepts. Transcripts and notes were coded manually in spreadsheets using a hierarchical codebook and reflexive memoing. The research's rigour rests on triangulation (methods, sites and time), dialogic checks in group settings, thick description and a documented audit trail. Limitations linked to conflict access, infrastructural disruptions (e.g. the Alouk water station under Turkish-backed jihadist control), and modality differences between online and in-person work were mitigated through the multi-wave design and reflexive practice.

2.5. Further considerations

2.5.1. Praxis

I entered this work as a Kurdish activist-scholar in exile, accountable to the peoples and institutions of Rojava and guided by a decolonial ethic: research must be relational, reciprocal and careful with what it makes visible. That commitment shaped every aspect of the work, from design and access, to analysis, and what I have refused to publish. I began with a deliberately plural design that spoke to both governance and everyday life: semi-structured interviews with representatives of AANES across agriculture, economy, women's movements, ecology, education and political bodies, conducted alongside in-depth household interviews and surveys with peasants, farmers and workers. The aim was to trace the frictions between discourse and practice, to assess the interaction between the administration and the communities, and to measure the reach and limits of direct participation in decision-making. Equally important was a comparative lens at community scale. I conducted interviews with leading political cadres of the AANES and local community members to examine the internal functioning of democratic mechanisms, the everyday textures of minority self-governance, and the extent to which the Democratic Nation paradigm – through communes, cooperatives and people's assemblies – enables different communities to govern together and apart without erasure. Accordingly, I sought to understand how, together, these actors and organizational frameworks challenge colonial, statist, patriarchal and capitalist structures by building democratic governance and enabling socio-ecological transformation.

Despite months of preparatory exchanges, the first two weeks in the field revealed that my initial framing, centred on the nexus of climate change, drought and conflict, and on whether the Rojava model could counter the legacies of state-centric, capitalist policies which I studied in the literature, was superficial, and risked obscuring the colonial and historical dimensions of the current multilayered crisis. Thus, my first research design was far from able to delve into these layers: colonial legacies sedimented into infrastructure, the temporalities of civil war, neocolonial interventions, and ecocidal policies misrecognized as administrative common sense. The five articles that emerged follow those entanglements across scales and scenes, showing both how colonial-genocidal-ecocidal designs shape socio-spatial life and how Rojava's institutions attempt plural, decolonial ecological alternatives. This was not merely a tactical adjustment; it marked a shift from theorizing about, to co-theorizing with. To this end, collective research methods and data collection were transformed into co-production. Together with five colleagues from four European countries and three collaborators who participated in fieldwork in Rojava, we jointly produced four articles. This approach not only deepened my understanding of the data and the research topic, but also transformed the process into a more powerful form of knowledge production by fostering collective thinking with experts whose skills and insights were essential for addressing complex, specialized aspects of the research. Had I pursued an individualistic, closed and non-collaborative approach, I doubt the quality and depth of these outputs would have reached their current level.

The co-design of the research was not merely decorative. I discussed and negotiated research questions, sites and security arrangements with local activists, scholars and organizational actors before and during fieldwork. As a concrete obligation of reciprocity, one bachelor's and two master's students, two lecturers from the University of Rojava, and three colleagues from the Ecology Academy joined the data collection. We co-designed research instruments (interview guides, observation protocols and survey questionnaires), practised interviewing, observation and survey administration together, and debriefed each evening with ethics at the centre of our discussions rather than as an afterthought. By the end of fieldwork, they were leading substantial components of the research; later, the master's students told me that the methods they had learned became directly useful for their own theses. This capacity-building was not a by-product of the project; it was part of the method.

Methods in practice became dialogical, situated and slow. Formal interviews and surveys were complemented by engaged observation inside communes, cooperatives and people's

assemblies, and of daily work routines. In sensitive settings, I often set aside visible notetaking, choosing instead to write detailed diaries each night. The most important learning came through unstructured exchanges with veteran organizers who carried the movement's archive in their bodies. Many phenomena which I had observed but not fully understood became legible only through those long conversations. When appropriate, I later sought permission to include non-sensitive details, and invited interlocutors to review drafts; some consented, some declined, and their "no's" remained no. Beyond the Cezîrê region, I traced the genocide-ecocide nexus in Afrin, under the Turkish state and its proxy jihadists, where access was necessarily limited to refugee camps hosting forcibly displaced residents. There, spontaneous conversations and a handful of interviews made grief and endurance legible, particularly around the symbolic and material centrality of olive trees. Key informants actively engaged in resisting colonial and ecocidal violence provided the temporal depth I could not gain in situ. Across Kurdistan and Europe, experiences at workshops, seminars and conferences became primary data rather than mere background: they were public spaces where strategy was argued, meanings contested and futures rehearsed. Triangulating those movement publics with interviews and observation deepened validity and situated local findings within a translocal ecology of struggle.

My long-standing engagement in Kurdish ecological movements opened doors to spaces rarely accessible to outside researchers. Leading cadres, some with 20 or 30 years in the movement, shared institutional histories, dilemmas and strategy, including sensitive knowledge that clarified why certain choices were made and which risks could not be taken. Access, however, was not entitlement. In keeping with Indigenous and decolonial commitments to data sovereignty, much of what I learned does not appear in the thesis and will not circulate beyond responsible interlocutors.

A small number of responsible colleagues in Rojava read my work before academic peer review; names, locations and roles were anonymized, unless an official explicitly wished to be named; and I returned non-sensitive outputs in Kurdish and Arabic. Published articles have been translated into Kurdish by the Ecology Academy and Rojava University, and incorporated into their educational materials. The sequence, local review before academic review, was a deliberate epistemic and ethical positioning: the first obligation is to those who live with the consequences of the knowledge produced.

2.5.2. Ethics

Ethical procedures followed a consent-and-care model appropriate to conflict-affected settings: all participants were verbally briefed and gave informed consent with the option to refuse questions or withdraw at any time; anonymization uses pseudonyms and role descriptors with the masking of potentially identifying site details; and all materials were encrypted with access controls and de-identification keys stored separately. All materials were handled under EU data-protection principles: explicit consent, data minimization, pseudonymization, encryption at rest and in transit, access controls, and limited retention with separate storage of reidentification keys (see Appendix D). Ethics were daily practice rather than paperwork. Meetings were scheduled with care for safety; the most sensitive lines of questioning were avoided where risk outweighed benefit, such as not discussing the Syrian government's Arab Belt policy with Arabs who had originally moved to the region as settlers. These procedures were part of a broader ethic of refusal: refusing to endanger participants for "data," refusing to publish what should not circulate, and refusing to render communities legible to hostile power.

There were moments when methodological ideals met the realities of fear and asymmetry. This occurred, for example, in two settler Arab villages established under the Arab Belt policy after the 1970s. In the first village, my Kurdish identity and perceived proximity to AANES clearly shaped responses towards socially desirable approval of the Rojava administration and its decision-making mechanisms. In the second village, I adopted a bounded, situational concealment: colleagues from the Ecology Academy acted as interpreters and guides, and introduced me as an Italian researcher unconnected to local authority. Only then did participants voice candid criticisms of the AANES, reject the prospect of returning land to displaced Kurds and assert hereditary possession. In this case, I adopted bounded deception (a brief, situational non-disclosure of political identity) to reduce social desirability bias and mitigate risk to participants. This decision follows ethical guidance for research in conflict-affected settings (Wood, 2006; Sriram et al., 2009) and the literature on tightly constrained covert methods under exceptional conditions (Spicker, 2011; Calvey, 2017; Marzano, 2021). I record and defend this limited deception here because, under conditions of asymmetrical power and fear, it surfaced a strand of the story that would otherwise have remained unsaid. All such material communicated below is fully anonymized, restricted to analytic use, and never fed into administrative decision-making.

Analysis proceeded abductively and dialogically. I wrote memos every night, tested interpretations in subsequent interviews and assemblies, and conducted member-style checks with ecology, economy, agriculture and cooperative actors and cadres. To this end, I shared non-sensitive observations directly with high-ranking responsible cadres in their domains, even when such reflections did not appear in the thesis, precisely because public circulation would have caused harm, while private return could support ongoing problem-solving. In this sense, analysis and accountability were braided: interpretation was not a private act, but a loop of proposition, response, correction and consent.

2.5.3. The aesthetics of becoming

All of this changed the research project, and it changed me. I arrived with a question about the drought-conflict nexus and governance; I left with a slower, more granular understanding of deep socio-ecological transformation by decolonial ecology's principles and policies as praxis: its achievements, blockages, contradictions and openings (Ferdinand, 2022). Trust did not pre-exist; it was made through shared labour, careful listening by sensitive minds, frank disagreement, timely return of work and the humility to keep learning. That trust, extended by households, cooperatives, students and responsible authorities, made this dissertation possible. It also binds me beyond it: to guard what must remain quiet, keep returning knowledge in forms that are useful locally, and insist that method here is not a set of techniques, but a relationship unfolding in time, from plan to practice, from extraction to co-labour, from certainty to inquiry, and from 'field site' to a commons of learning.

My positionality as an activist-scholar thus does not merely inform, but is integral to, my research praxis, ensuring that methodological choices explicitly support and are accountable to the socio-political and ecological struggles of Rojava's communities. As a Kurdish activist-researcher, my work is also part of a broader struggle for epistemic sovereignty. Kurdish knowledge systems, histories and ontologies have long been suppressed under the violent weight of Turkish, Syrian, Iraqi and Iranian nation-state colonialism. The very act of writing, theorizing and publishing as a Kurd and from a Kurdish perspective is already a defiant act against epistemic erasure as an Indigenous aesthetics. Situated within broader conversations on feminist, Indigenous and decolonial methodologies, this activist research offers a contribution to scholarship committed to ecological justice, autonomy and transformative social change. It

is not merely a study of the Rojava Revolution's ecological and democratic praxis, but an embedded, co-constitutive process of knowledge-making with and for a people engaged in a transformative struggle against coloniality, patriarchy and capitalist extractivism. To what extent I have succeeded is not for me to judge. But I can say that this journey, spanning from 2005 until today, has become for me a decolonial, existential and collective process of becoming and liberation. It is an ethical and political ontology of life-making with my people, and with all others who struggle for justice, freedom and dignity.

SECTION II. ARTICLES

Chapter 3. Article I: Decolonization agriculture: Challenging colonization through the reconstruction of agriculture in Western Kurdistan (Rojava)

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Decolonisation agriculture: challenging colonisation through the reconstruction of agriculture in Western Kurdistan (Rojava)

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ABSTRACT

This article examines the colonial homogenising policies of the Syrian Ba'ath regime and the subsequent decolonisation processes that led to the emergence of Rojava as a pluriverse. In 1963, the Ba'ath regime implemented nation-state colonialism in the predominantly Kurdish region, using agricultural modernisation as a tool for its colonisation efforts. This modernisation bolstered the central state, perpetuated the underdevelopment of the region as a periphery, and asserted control through the settlement and distribution of land to Arab families loyal to the regime. Following the regime's collapse in Rojava in 2012, the communities that comprise the Autonomous Administration of North and East Syria (AANES) dismantled the colonial agricultural system. They developed a decentralised governance and agrarian development approach, referred to here as the decolonisation of agriculture. Based on interviews and fieldwork in the region, this article explores the interplay between agricultural development and colonial politics, as well as the critical role of agriculture in the broader struggle for decolonisation. We conclude that in the anti-colonial struggle, people and the rhizomatic governance structures they develop challenge colonial submission to the central state, exploring life beyond the nation-state, which is crucial for a decolonial shift.

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Introduction

Prior to the implosion of the Syrian central state with the onset of the Syrian Civil War, Rojava¹ functioned as a colonial periphery under the Syrian government. However, Rojava emerged from this status through a process of political and economic decolonisation. In this article, we explore this decolonisation process through the lens of the political economy of agriculture, set against the backdrop of the colonial policies of the Syrian Ba'ath regime, which has been in power since 1963. We examine these policies as a form of nation-state colonialism, using the modernisation of agriculture as a key tool for the colonisation of peoples and places. This is followed by a discussion of the decolonisation processes since 2012, through

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which Rojava has been reconstituted as a pluriverse (Escobar 2018) with a decentralised governance structure.

Coloniality has been defined as the darker side of modernity, as it emerged as a structure of power built on western ideas of civilisation (Mignolo 2011). Though mostly discussed in terms of relation of the west to other parts of the world, we discuss the politics of the Syrian Ba'ath regime as colonial. We argue that its coloniality emerged as the darker side of agricultural modernisation. Promising progress and productivity, its agricultural modernisation has been a colonial vehicle resting on three main practices. First, agricultural modernisation played a role in bolstering the development of the central state in the region. Second, agricultural modernisation transformed the West Kurdistan region into a supplier of primary (agricultural) products, thereby underdeveloping the region as a periphery of the Syrian central state in the service of its national economy. Third, the distribution of land to and the settlement in Kurdistan of Arab families perceived as loyal to the regime aimed to assert control over the territory and its inhabitants. While these practices exhibited elements of franchise and settler colonialism, collectively, they not only targeted the extraction of wealth from Kurdistan or demographic change for the benefit of the greater Arab nation but also aimed at assimilating the Kurds and Kurdistan into that nation. Thus, the article discusses the agricultural development of West Kurdistan as organised by the administration in Damascus in terms of a disguised form of internal colonial politics. After the collapse of this Ba'ath regime in Rojava in 2012, the colonial agricultural system was dismantled by the newly instituted Autonomous Administration of North and East Syria (AANES). Unlike the pursuit of a nation-state as a model for political restructuring (Goeman 2008), AANES aimed to develop an interconnected network of locally administrated communities. This article thus seeks to explore two key aspects of the process outlined: (1) the interplay between agricultural development and nation-state colonial politics and (2) the Kurdish movement's practice in Rojava/AANES of decolonising agriculture.²

Data was gathered through online interviews in 2023 and fieldwork conducted from 26 April to 6 June 2024. In total, 18 online interviews were conducted. Fourteen interviewees (six women, eight men) were involved in cooperatives. They included co-chairs and spokespersons (one was a women's movement representative/manager). The other four people with whom we conducted in-depth interviews were two political activists, one environmental activist, and a lecturer in North and East Syria (NES). Following the online interviews, fieldwork was conducted in NES in two rural areas in the period April to June 2024: one near Dêrik and the other around Tirbespiyê. The fieldwork involved administering questionnaires and conducting interviews. In Dêrik, questionnaires were completed by 10 participants in three villages (7 men, 3 women; 4 Arabs, 6 Kurds). In Tirbespiyê, questionnaires were completed in five villages by 17 participants (11 men, 6 women; 2 Assyrians, 3 Yazidis, 2 Arabs, 10 Kurds). Although the questionnaires involved a limited number of participants, the outcomes provide an impression of what are considered contentious issues. In Dêrik, a group discussion was held with various local officials, including the co-chairpersons of the Democratic Union Party (*Partiya Yekîtiya Demokrat*, PYD), the head of the ecology department of the Dêrik municipality, the co-chair of the Dêrik agricultural committee, a staff member responsible for cooperatives, and the head of the Seed Institution (*Saziya Tov*). In addition, two agricultural cooperatives in Dêrik were visited. In Tirbespiyê, visits were also made to a cooperative farm and a fertiliser cooperative, where six interviews were conducted with the co-chairs of

the cooperatives. The interviews and questionnaires were mostly conducted in Kurdish and Arabic.³

In the first part of the article, we discuss the question of colonialism and how colonialism in Kurdistan distinguishes itself from franchise and settler forms of colonialism. In order to understand the form colonialism takes in Kurdistan, we introduce the terms *international* and *internal* colony. In the second part of the article, we discuss how agrarian development or modernisation policies operated as a vehicle for the colonisation of Kurdistan. In the third part, we discuss the features of decolonising agriculture in Kurdistan. The main conclusion we draw is that the rhizomatic agricultural restructuring in AANES disrupted the Syrian state's colonial design of West Kurdistan and its monocultural policies directed at the land, crops and people. Instead, it created locally administered economies of diversity.

The colonial status of Kurdistan

Generally, colonialism is described as a spatial practice of dominance entailing the subjugation of one group or people by another. This domination is enforced by a political entity through military, economic and cultural means (Sunca 2023a, 72). Operating within this matrix of subjugation, various kinds of colonialism can be distinguished, which historically have their roots in two basic forms: franchise colonialism and settler colonialism (Shoemaker 2015).⁴ In franchise colonialism, the colonial power is interested in the exploitation of the colony, and settlement is organised around the extraction of resources and the exploitation of labour for processes of capital accumulation. In settler colonialism, the objective is to displace the Indigenous population, making a permanent home for the settlers in the colonised area.

In the twentieth century, Marxist political thought inspired liberation struggles for decolonisation. The status of Kurdistan as a colony has been a fierce issue of debate within the revolutionary left in the region since the 1970s. The Marxist left generally recognised the existence of a Kurdish nation and its status as oppressed but denied the colonial status of Kurdistan for two reasons. First, although the Kurds could be considered an oppressed nation, the status of a colony, it was argued, had to be determined by analysing the status of the oppressing country. As colonialism was a phenomenon that occurred in the historical stage of imperialism, the countries occupying Kurdistan could not be considered imperialist, so the status of Kurdistan could not be that of a colony. Second, since the economy of Kurdistan was integrated into the economy of the occupying countries, one could not speak of a separate economy that was exploited. Hence, the main contradiction was not between coloniser and colonised, but between capital and labour, regardless of ethnocultural or national identity (Vali 2020; Jongerden 2024).

The lack of recognition of Kurdistan as a colony extends to much of the Marxian academic work. The world-system and centre–periphery Marxian discourse that became dominant in the period after the Second World War understood colonialism within a world system characterised by unequal economic relations between states and thus restricted its analysis of colonial pasts and neocolonial continuities to the inter-state level (Sunca 2023a). This approach made it difficult to recognise and analyse intra-state colonialism: the nationalist identity politics of states as colonial projects. Though numerous thinkers have considered how the exertion of power in post-colonial settings, frequently manifested through authoritarian governance, sustains a cycle of colonial oppression and violence, this analysis has

predominantly focused on the actions of past Western colonisers. It has overlooked the examination of such dynamics as a characteristic of internal colonisation perpetuated by a new hegemonic regime (Mbembe 2001).

It was the Turkish sociologist İsmail Beşikçi who introduced a novel approach to the question of the colonial status of Kurdistan. He argued that colonial projects in Kurdistan have derived their logic from (1) its location within (and not outside) the colonial states (Turkey, Iran, Iraq and Syria) that have occupied a divided Kurdistan and, in relation to that, (2) the submission to an aggressive, nascent nationalism according to which the strength of the state was (and continues to be) determined by the extent to which the land and citizens over which it claims sovereignty comprise a single cultural identity (Beşikçi 1990, 2014). Beşikçi introduced the concept of 'international colony' to get at the status of Kurdistan as colonised by (four) states that had emerged during the course of the twentieth century from the Ottoman and Persian Empires. 'International' here does not refer to a relation between two entities (only), a dominating political entity (the coloniser) and a dominated political entity (the colonised), but (also) to the division of Kurdistan across the four different internationally recognised states.

As a colony of multiple states, Kurdistan became subject to homogenising nation-building projects that targeted geographies and peoples (Öktem 2005, 2008; Tejel 2009a, 2009b; Jongerden 2022). Beşikçi described extensively how Kurdistan, relocated within the borders of these states, became subjected to concerted assimilation efforts aiming at the dissolution of Kurdistan and the Kurds (Beşikçi 1990, 1991, 2014). The partitioned territories effectively became 'internal colonies' (Duruiz 2020), referring to the set of practices through which the central state erased the occupied lands as parts-to-be of their nation-states. In other words, internal colonisation abandons the idea of a spatial distance⁵ between the coloniser and the colonised, connoting instead a culture-specific domination within state borders (Etkind 2011, 7).

Pan-Arabism and colonisation by agricultural means

In Syria, the Kurdish issue emerged as a colonial one within the context of a changing political landscape, primarily as a result of the collapse of the Ottoman Empire and the formation of the nation-states of Turkey (1923), Iraq (1932) and Syria (1946). Spread across south-eastern Europe, North Africa and the Middle East, the Ottomans and their vassals had variously administered an empire of peoples and assemblage of systems of direct and indirect rule. They had lacked both the means and any need to mould, shape or otherwise create a unified population. By the time Syria was carved out from the remains of the empire, however – first as a French Mandate (in 1920) before gaining political independence – nationalism had become a vibrant and multifaceted political imaginary, a powerful colonial platform through which to realise political and territorial claims. The main objective of the state's internal colonialism, therefore, was the production of a 'nation' within the allotted borders. Military occupation and political control were quickly followed by economic means as the main vehicle for the colonisation of Kurdistan (Soleimani and Mohammadpour 2020; Yadirgi 2020). While Syria had gained its independence as a result of a struggle against France, it became itself a colonising state. Though presenting the self-image of a post-colonial state, it exerted a colonial politics over Kurdistan and the Kurds.

Exclusionary attitudes towards the Kurds began shortly after Syria's establishment as an independent state in 1946, and they took on a structural form with a population census in 1962 that resulted in the exclusion from citizenship of Kurds on the basis of their categorisation as 'foreign' (*ajanib*) or 'unregistered' (*maktumin*) persons. As a result, approximately 120,000 Kurds, some 20% of the Kurdish population in Syria, were stripped of their citizenship and lost their statutory (civil and political) rights, including the rights to receive an education, to own property, and to participate in elections (HRW 1996; McDowall 2004). The situation worsened when the Arab nationalist Ba'ath party, founded in 1943, gained control of the state in 1963 and promised to develop an Arab nation in Syria. Agricultural modernisation was to be one of the major vehicles driving the creation of this Arab nation, functioning as a means for the colonisation of Kurdistan through three intertwined projects: the modernisation of agricultural production, development of economic dependency relations, and instigation of a demographic politics.

First, the modernisation of agriculture under the Ba'ath functioned as a mechanism for the expansion of the state in rural areas (Schad 2001). From the late nineteenth century, a shift towards land privatisation led to the formation of large estates and specialised crop production, supported by urban investors from cities like Aleppo and Damascus. This shift caused tensions between smallholders and landlords, and between villagers and urban elites. The Ba'ath Party capitalised on this rural resentment, developing a strong support base in rural areas and among the rural poor. The envisioned transformation of the agrarian economy was to be accomplished through the establishment of large-scale state-run irrigated farms, as small-scale agriculture was considered inefficient (Hinnebusch et al. 2011). Advised by Soviet engineers, the Ba'ath regime embarked on several massive irrigation and land reclamation projects in the Kurdish regions (Springborg 1981). The Tabqa Dam was the hydrological centrepiece of this modernisation project. It aimed to transform the small-scale irrigation and dry-land farming in the Euphrates basin into large-scale irrigated agriculture for the production of 'strategic crops', mainly wheat, barley and cotton (Hinnebusch et al. 2011).

As political and technological assemblies that combine hydro-electrical and agricultural interventions with a centralised bureaucracy, dams were conceived and utilised as material agents of the central administration and colonial state power (Scott 1998; Mitchell 2002; Harris 2012; Akıncı et al. 2020). To paraphrase Tilly (1975): states build dams, and dams build states. Centrally planned and supervised by technocrats, agricultural production was regulated through medium- and long-term sector planning that involved a comprehensive annual plan with targets for key crops, areas for planting specified by crop, and instructions about the crops that farmers should grow (Hinnebusch et al. 2011, 5–6). Colonial practices came with high environmental degradation but poor returns. The ecological costs of these schemes were particularly high; soil quality was diminished due to the monocropping and extensive use of water and fertilisers, water quality was reduced as a result of salinisation and pollution,⁶ and water availability decreased because of overexploitation and resource mismanagement, along with lower rainfall and upstream dam construction in Turkey (Haddad 2011; De Châtel 2014; Daoudy 2020).

The second way in which agrarian development enabled colonisation involved the creation of economic dependencies. The Ba'ath's economic politics had resembled the model developed by the economist Friedrich List, whose idea of a national economy replicated imperial centre–periphery relations domestically and developed an imperial colonial politics

to the level of the nation-state (Ince 2016, 380). The creation of added value through conversion of the produce into consumer products became planned in the western part of the country and typically turned the Kurdish regions into providers of resources for the centre. So, under the Ba'ath regime, West Kurdistan mainly produced primary products for processing, which was predominantly situated outside the region. Thus, Rojava became the wheat basket of Syria and produced most of its cotton, yet it was dependent for supplies of flour, bulgur wheat, pasta, and textiles on cities outside the region, such as Hama, Homs, Aleppo and Damascus. This unequal economic development, creating a centre and periphery, is referred to by various authors as a process of internal colonisation by 'de-development' (Roy 1995; Soleimani and Mohammadpour 2020, 742; Yadirgi 2020).

Third, agricultural modernisation was co-constructed with a demographic policy of Arabisation, a form of settler colonialism instrumental in the nation-state colonialism aimed at subordinating Kurdistan and dissolving it into the Arab nation. However, the Ba'athists considered any cultural integration of the Kurds into a Syrian identity to be 'delicate' at best and a 'malignant tumor' at worst (Tejel 2009b, 64). The issue was addressed through an 'Arab Belt' policy, which aimed to replace the Kurdish population of 332 villages with Arab settlers in a 280-km-long, 15-km-wide stretch of well-cultivated land running along the border with Turkey (Tejel 2009b). Extending 375 km in total to the Iraq border, the Arab Belt idea entailed (1) the displacement of the Kurds from their lands, (2) a divide-and-rule policy within the Kurdish community, (3) Arab settlement of the Kurdish areas, and (4) the establishment of an Arab *cordon sanitaire* along the border with Turkey with (5) collective farms for the Arab settlers (McDowall 2004, 475). Land was expropriated for this from Kurdish farmers (Selby 2005), and in the early 1970s, following the construction of the Tabqa dam for the irrigation of land, Arab families were armed and settled in 'model' farms in the heart of the Kurdish region (Tejel 2009b, 62).⁷ A total of 41 model farms were established, and though the Arab Belt project was suspended in 1976, the model farms that formed the core of the belt were not dismantled and displaced Kurds were not allowed to return to their land, effectively keeping the belt project in place (HRW 2009).

Implosion and transition

The Ba'ath regime's colonial policies in West Kurdistan unravelled when its authority and presence there imploded. On the night of 18–19 July 2012, the local People's Protection Units (*Yekîneyên Parastina Gel*, YPG) seized control of Kobanî, a town with a population of 50,000 near the border with Turkey. This event, known as the 'Kobanî Revolution', swiftly spread to other cities and towns in the Kurdistan region, leading to the formation of the Rojava Autonomous Administration (Knapp et al. 2016, 54–55) – the region under autonomous administration adopted the name 'Autonomous Administration of North and East Syria' (*Rêveberiya Xweser a Bakur û Rojhilatê Sûriyeyê*) in 2018.

When the central state lost its control over the Rojava/AANES region in 2012, the agricultural modernisation and rural development policy in Syria co-constructed through a policy of Arabisation and dispossession was halted. Instead, a network of locally administered communities emerged. This not only interrupted the policies of dispossession but undertook the development of a new politics, involving a revolution in both the administration of the territory and its cultural and agricultural practices. Food provisioning for regional markets was made a key feature in the new economy along with the stimulation of family farms and

cooperatives and the development of local processing and manufacturing capacities (Jongerden 2021; Aslan 2023).⁸ This would form the basis for regional food sovereignty (Yarkin 2015).

A rich body of academic literature has been produced in recent years about the revolution in Rojava. Much of it situates the radical transformation of governance structures in the region in the context of the ideological and political transformation of the Kurdish movement inspired by the ideas of Abdullah Öcalan (Sunca 2020; Cemgil 2021; Gerber and Brincat 2021; Martin 2021). This work explores how the movement has transitioned from a party that aimed at the establishment of a state as the means for decolonisation and societal transformation towards the twin concepts of democratic autonomy and democratic confederalism, which emphasise the right to self-administration and the form of a confederation of autonomous assemblies that it takes.

Öcalan's work expressing ideological and political reorientation was written while he was in detention and preparing his defence against allegations of treason. Around the turn of the millennium, having previously proclaimed the establishment of an independent state as the sole legitimate political objective of a national liberation movement and following a critical examination of liberation struggles worldwide, Öcalan began to question whether independence really *should* be conceived and implemented through the construction of a nation-state (Jongerden 2016). Thence, taking the state out of the equation for measuring self-determination, the idea of an ideological and political framework based on the notion of self-government within a stateless society was developed (Gunes 2019; Jongerden 2019).

Rather than leading a state-formation struggle and then becoming its new elite, the party-movement would be centred on a strengthening of society through a form of self-administration. This followed in a radical way the idea that post-colonial states should not be imitations of European states (Fanon 1963). Not only should they not just become a different state, they should not become a state as such at all (Karasu 2009); instead, they should formulate an alternative perspective on modernity (Yarkin 2015). Öcalan named his proposed alternative 'democratic modernity', which was made up of democratic autonomy⁹ and democratic confederalism¹⁰ as the constitutive pillars of every democratic nation or community (Öcalan 2020). Relatedly, many have discussed Öcalan's theoretical engagement with post-anarchist and post-Marxist scholars – particularly Murray Bookchin (Jongerden 2019), but also Antonio Negri and Michael Hardt (Dinc 2020), Samir Amin (Grubacic 2019, 1075), and Fernand Braudel and Immanuel Wallerstein (Aslan 2022). While some have focused on the post-capitalist turn of the movement inspired by Öcalan (Aslan 2023), others have emphasised the feminist dimension (Tax 2016; Dirik 2022; Piccardi and Barca 2022), the ecological dimension (Hunt 2019) or both (Aslan 2022; Piccardi 2022). Yet much of the literature emphasises the anti-capitalist struggle over the anti-colonial struggle of the Kurdish movement. While some works do mention how the politics of the Kurdish movement addresses colonialism (Yarkin 2015; Martin 2021; Piccardi and Barca 2022; Sunca 2023a, 2023b), the focus tends to be placed on overcoming the problem of capitalist development. Although the Kurdish movement aims to develop a social economy as an alternative to the capitalist economy, the main contradiction is between a colonial state and its modernisation politics, on the one hand, and, on the other, the constitution of Kurds and Kurdistan as identities through which these are to be dissolved.

Decolonisation and agriculture

We have argued that agricultural modernisation was a colonial project in three respects. It bolstered central state development in the region through projects around which the central state's authority was built; it made West Kurdistan a supplier of primary agricultural products, which marginalised Rojava, positioning it as a peripheral supplier to the Syrian central state, rather than allowing it to develop its own economy; and it involved the distribution of land to Arab families perceived as loyal to the regime in order to exert control over the territory and its inhabitants with the long-term objective of dissolving the geography and the people into the Arab nation. In the new governance struggle that emerged after 2012, these three pillars of the colonisation project were addressed in three different ways. These involved the parallel, opposing development of (1) a rhizomatic democracy as an alternative to the central state, (2) diverse agroecologies in response to the de-development of peripheralisation, and (3) a communal economy against Arab nationalist identity politics. These are discussed below.

(1) Rhizomatic democracy

When the central state's control in the region collapsed with the war, a new governance model was established. In line with Öcalan's revised thinking, the governance structure has been organised into a network of councils, with communes or local communities organised in assemblies, serving as the fundamental units. Each commune comprises an executive committee and various working committees, including those focused on economics (or agriculture), education, and women's affairs. The commune functions as both a decision-making body and an executive institution. Its coordination is overseen by two co-chairs and members representing the committees. Communes are grouped into neighbourhood councils, which, in turn, are organised into district, city and regional councils, collectively forming the Autonomous Administration. While the names of the councils may have changed over the years, their fundamental principle remains the same: the establishment of a network governance model in which decision-making authority is decentralised to communities rather than centralised in a bureaucracy, ensuring the active participation of all members of the community.

This new political model is mostly defined negatively, as 'non-state' or 'non-statist', but it can also be referred to positively, as a rhizomatic democracy. The concept of rhizome comes from the book *A Thousand Plateaus* by Gilles Deleuze and Félix Guattari (Deleuze and Guattari 1993), where the metaphor of the rhizome is employed to describe a way of thinking about structure and organisation that contrasts with centralised and hierarchical models. Instead of a vertical branching structure, rhizomes have horizontally oriented stem systems that lack a central organising point. Deleuze and Guattari argue that this offers a superior model because the non-centralised, non-hierarchical structuring enables heterogeneous connections between parts (Somers-Hall 2013, 2).

We borrow the concept of the rhizome here to describe how the agrarian economy was reorganised in NES. This emphasises not only the importance of horizontal relations but also the immanent potential for new rhizome formation from its parts and how, in every social organisation, forms of verticality are produced. Thus, although the rhizome metaphor emphasises the 'scaling out', we recognise that there are also forms of 'scaling up' (Garcia-Arias, Tornel, and Flores Gutiérrez 2024). This is the logic of democratic confederalism.

The primary *nodes* in the agrarian structure are the *village communes*. All households are entitled to participate in the commune. The commune is a space for deliberation and decision-making. Every village commune has an agricultural committee where matters related to production, crop rotation, and water management and distribution are determined. However, villages are not spatial envelopes, each separated from places outside; decisions that have an impact beyond the commune or problems that cannot be solved by one commune alone are discussed in a broader context. In the first instance, this is the district or neighbourhood council – the *Bajarok* – in which 40–60 village communes are clustered. Several *Bajaroks* make up a city council – *Bajar* – which combine to create a canton, which themselves make up AANES. It is according to this logic of clustering that the territory is governed, with the rhizome not only spreading out through a horizontal logic but also rising, following a (bottom-up) vertical logic. To support this network of councils, Agricultural Centers for Consultancy and Services (*Îrşadi* or *Erşadi*, meaning ‘consultancy’) have been established, which provide various forms of advice, support and resources related to veterinary needs, licensing, irrigation, seed distribution, diesel subsidies and pest control, while the Agriculture Development Agencies (*Sirketa Pêxistina Candini* or *Sirket*) are responsible for coordinating technical and infrastructural innovations. The region inherited a centralised water and energy infrastructure controlled by Turkey and Syria. To address the vulnerability of this centrally planned system – exacerbated by Turkey’s regular energy supply cut-offs through drone attacks and its withholding of water from upstream dams to disrupt production – a decentralised system has been introduced.

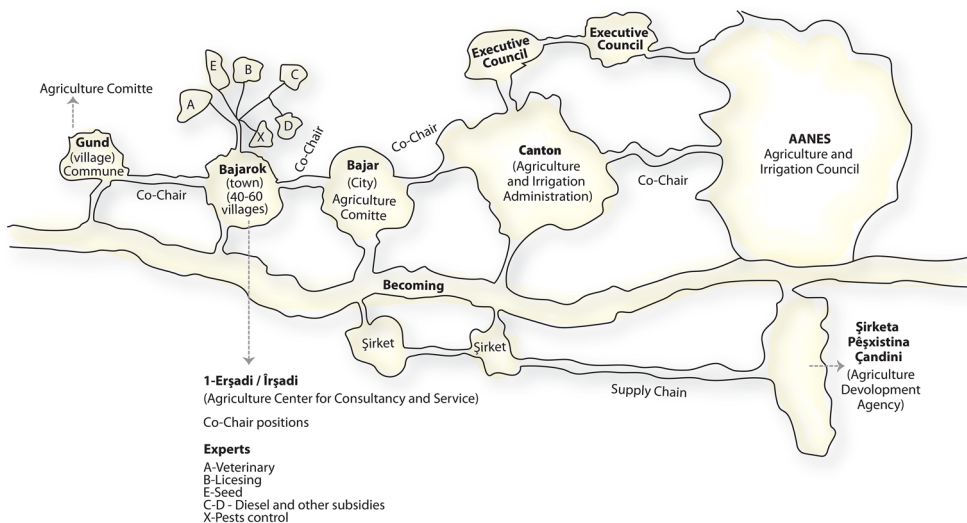


Figure 1. Map of the administrative network in Rojava/Autonomous Administration of North and East Syria (AANES).

This system includes solar panels for energy and water management techniques such as drip irrigation and water harvesting (Figure 2).

The decentralised organisational structure proliferates in diverse directions, and is supported by the AANES Agriculture and Irrigation Council (*Desteya Candini*) which encompasses several autonomous units and institutions, including the Public Land Administration (*Rêveberiya Erdên Giştî*), Water Institution (*Saziya Avê*), and Seed Institution (*Saziya Tov*). For instance, the Public Land Administration is tasked with distributing land and has allocated some 25,000 acres to agricultural cooperatives and 44,000 acres to women's cooperatives. The latter are the product of and enable further new rhizomatic structures. As an autonomous movement, the women's organisation *Kongreya-Star* formed its own governance structure, which includes the establishment of women's cooperatives, which, in turn, form a network referred to as the Women's Economy (*Aboriya Jin*), but which we will not discuss here.

The establishment of a cooperative system does face various difficulties, and this highlights a difference between the cadres in the movement and the peasants. Cadres attribute the difficulty of expanding cooperatives to the people's ingrained statist and individualistic habits, while many peasants argue that unequal contributions (the problem of free riders) and abuse of the cooperative system for personal gain or allocation to family members (corruption/nepotism) are major issues. Consequently, the Kurdish movement's concept of communalising natural resources, particularly land and water, has faced resistance from peasants. Instead of communalisation, they demand land reform and increased individual agricultural subsidies for diesel, fertilisers and electricity. This situation highlights a critical vulnerability within the movement: a significant tension exists between the ideological

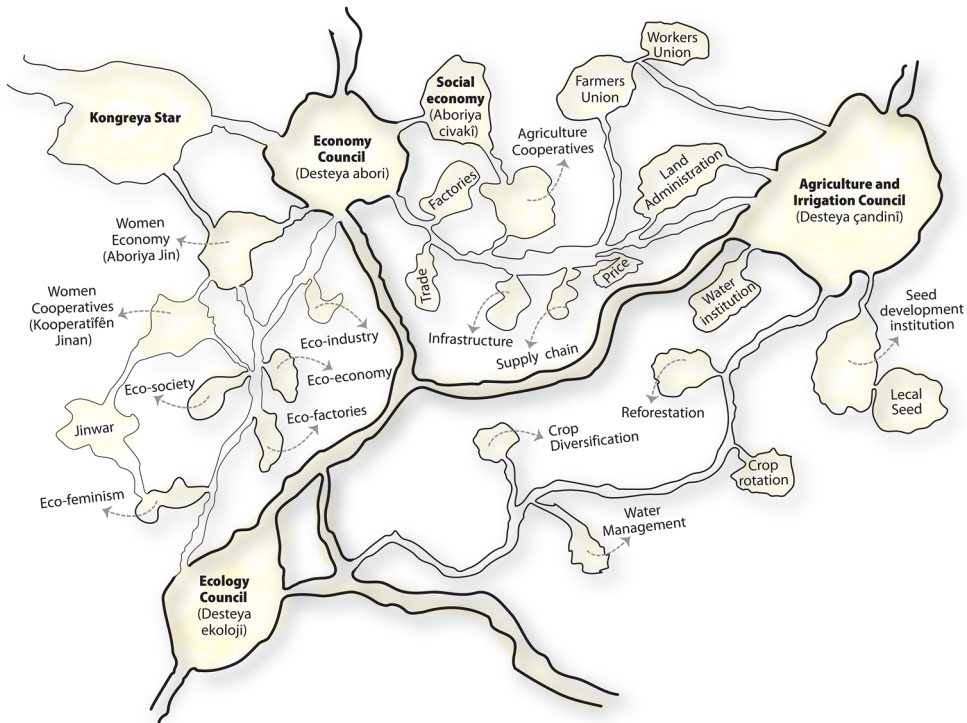


Figure 2. Confederal relationality of organisations in agriculture.

ideals of the politically motivated and the desires of a large part of the farmers and villagers. When asked whether communal use through cooperatives or individual distribution to the peasants would be better, nearly 70% of respondents preferred individual distribution to the peasants, while 24% believed that a well-established cooperative system would be more beneficial.

As Jan Douwe van der Ploeg (Ploeg 2013) shows in his treatise on the agrarian question in the twenty-first century, the radical left has long been divided over this issue of peasant production. Historically, the controversy was epitomised in the debates between Vladimir Lenin and Alexander Chayanov in early twentieth-century Russia and the Soviet Union, reflecting differing views on land reform and the role of peasants in society. Lenin and most of the Bolsheviks advocated for the nationalisation of land and the replacement of individual cultivation with cooperative cultivation. In contrast, Chayanov, representing the *Narodniks*, believed in the emancipation of peasants rather than their imposed dissolution from above. In 1917, Chayanov argued that the socialisation of land cannot be realised through 'enlightened absolutism' but is to 'result from the involvement of local and democratically elected councils'; yet this is not necessarily in contradiction with peasant production (Ploeg 2013, 109).

The question of how to implement land reform – through Leninist depeasantisation or Chayanovian peasantisation – featured in developments and discussions worldwide throughout the twentieth century, from Italy and Portugal to Mozambique, Angola and Guinea-Bissau, from Mexico and Cuba to Viet Nam, China and Japan – and today, in Rojava. From our interviews and discussions with cadres, it became apparent that the Kurdish movement faces a challenging paradox. Excessive idealisation of the cooperative system in combination with resistance from peasants makes practical implementation difficult. This may foster another issue: the emergence of a bureaucratic elite or managerial class seeking to lead the cooperative movement.

(2) Diverse agroecologies

To date, approximately 500,000 hectares of land formerly under the regime's control has been communalised as public land (*Erdên Giştî*). The Public Land Administration had allocated approximately 80% of this land to agricultural cooperatives, women's economy and cooperatives, families, forests/agroforestry, parks, associations, and camps for Internally Displaced Persons (IDP) as outlined in their report dated 20 December 2023.¹¹ Contrary to the old monocropping approach, crop diversification is now a cornerstone of agricultural policy. This encompasses a variety of vegetables and pulses, as well as agroforestry crops such as olives, pistachio nuts, figs and grapes (vineyards).¹² Chickpeas, lentils and beans are now being cultivated, none of which were previously grown, collectively occupying 25% of the cooperative's agricultural land. Cotton production has decreased to 10%, while vegetables now cover 5% of the cultivated area. This diversification initiative was further strengthened by the planting of several hundred thousand fruit-bearing trees. (Aslan 2023, 151). Through the Seed Institution, AANES focuses on selecting drought-resistant and locally adapted seeds, catering to both agroforestry and crop diversification. Relatedly, AANES has dedicated significant investment and resources to rainwater harvesting for water-efficient irrigation utilising the 'drip' method.¹³ This comprises a major departure from the old large-scale, centralised and monopolistic water-management practices, including the dams and irrigation channels in Syria and Turkey, along with the associated water politics.

The Cezire canton, in the eastern part of Rojava, aims to satisfy at least 50% of the region's vegetable needs through the Khabur project, which involves vegetable cooperatives, and it offers products directly to the public at fair prices through a cooperative (MED). Its fruit production is also integral to the agroforestry practised; approximately 138,000 fruit-bearing trees have been planted.¹⁴ Ecological regeneration and agroeconomic reconstruction efforts are reinforced by crop rotation, organic composting, recycling/reusing practices, and the localisation of production, processing, and supply chains through some 500–600 agricultural cooperatives and other economic, ecological and agricultural coordination bodies in addition to the committees and assemblies.

The regime's colonial periphery–centre design was not only directed at exploitation in order to build its Arab national economy but was also a political engineering mechanism that de-developed the Kurdish region. In this design, none of the cities in the Kurdistan region had processing capacities, so an AANES priority is to develop processing infrastructure.¹⁵ Mehran Ahmed, Co-Chair of the Agriculture and Irrigation Council, illustrates this situation thus:

We took all 30,000 tons of cotton from the farmers in NES last year and removed the seeds, but we could not process and sell it Another example, our farmers have concluded that they should plant sunflowers, but after the harvest of our crops, how will we get oil from the products?¹⁶

The Autonomous Administration has managed to establish production facilities. Over the last decade, AANES has developed wheat, flour, bulgur, pasta and lentil processing factories along with an olive oil production facility, almost to the level of self-sufficiency, while the Economy and Agriculture Councils are currently engaged in building sugar, sunflower, soybean and cotton processing and textile production factories over the next five years.¹⁷ The AANES social/communal economy approach – ie through agricultural cooperatives – has made significant progress in terms of dairy products. For instance, the Zozan plant has the capacity to process six tons of milk daily (for cheese, butter, yogurt and buttermilk).¹⁸ It is also noteworthy that Rojava is now able to meet its red meat demand (mostly lamb/mutton and beef), although it is not yet self-sufficient in white meat (poultry, primarily chicken).¹⁹ This model of developing production and processing capacities for the provisioning of food in local markets not only addresses the colonial de-development policy but also creates the foundation for an approach emphasising social and communal values (use value, social benefit), and democratic and ecologic principles (deconstruction of all forms of domination, including gender, class, and nature) (Öcalan 2020).

(3) Demography of the communal economy

The third vehicle for colonialism involved the application of a demographic policy that aimed to assert control over the territory and its inhabitants: the distribution of land to and settlement in West Kurdistan of Arab families perceived as loyal to the regime. The response of the Autonomous Administration to the Arab Belt project was notably measured. Instead of adopting a nationalist stance that would displace Arab families settled in the region decades ago and redistribute the land to Kurdish locals, the Autonomous Administration chose a different path. They did not seize the land of former Arab settlers, not only to avoid sparking ethnic conflict between Arabs and Kurds but also in recognition that some families had lived

in Rojava for several generations and had become locals themselves. Moreover, a fundamental principle of the Autonomous Administration is the idea of a democratic nation, which, unlike the concept of a nation-state, is based not on creating a singular identity but on fostering a society where multiple identities coexist. Therefore, rather than taking land from those settled under the Arab Belt project and redistributing it among the Kurds, the land policy of the autonomous communities has focused on distributing state-owned land to those in need. However, three-quarters of Kurdish respondents to our questionnaire strongly criticised the Autonomous Administration on this issue, viewing it as a policy of de-Kurdification that should be reversed. One-fifth of the respondents considered this a political issue that is difficult to resolve, while the remaining 5% refrained from giving an answer. Nonetheless, all respondents agreed that those dispossessed should be compensated. Interestingly, Arabs who were resettled in the region under the Arab Belt policy also believe the project was wrong. However, they argue that the regime, not they themselves, was responsible for this injustice, implying that they should remain on the land.

While the demographic policies in the past are a highly sensitive issue, demographic change as a result of current developments is an issue of concern. Important drivers of demographic change are the worsening economic conditions due to drought, the war in Syria and Turkish interventions, and an embargo imposed by Turkey and its Kurdish partner in the Kurdistan Region in Iraq were cited as the most important causes of emigration (65%), especially a dramatic increase in emigration after the Turkish military occupation of Afrin and Serêkaniyê. Among Kurds, Assyrians and Yazidis, almost every household reported having one or more members who had emigrated. Among Arabs, just 1 in 10 households reported this. When asked for an explanation of this difference, the answer given by Semo, a 74-year-old Yazidi farmer, ran thus:

No one feels safe except the Arabs. Whether it is ISIS (Islamic State of Iraq and Syria), Turkey, or the regime, if they come again, we are the first ones they'll slaughter. Nobody touches the Arabs. Look at Afrin and Serêkaniyê; Kurds, Yazidis, Assyrians, and Armenians have been massacred and driven out.²⁰

Discussion

This article has delved into the intricate dynamics of Kurdistan's dual colonial status in the context of four nation-states: Turkey, Iran, Iraq and Syria. Occupied and colonised by these, it is an international colony; fragmented within the nation-state borders, it has been made into four internal colonies, subject to dissolution under the influence of hegemonic national identities. This is the logic of the colonialism to which the Kurds and Kurdistan were subjected. Focusing specifically on Syria and Rojava, the article has examined agricultural development as a tool of the central state for internal colonial politics, extending beyond the standard franchise and settler ideas of colonialisation as resource extraction and population resettlement. It has looked at how agricultural modernisation became a vehicle for state-building, economic peripheralisation and demographic engineering with the aim of dissolving Kurdistan within the new nation-state.

Jan Yasin Sunca (2023a, 86) has argued that to 'genuinely achieve decolonization', it is necessary to 'dismantle all structures that perpetuate originally colonial hierarchies' and instead 'establish a system of direct, radical, and democratic self-rule for communities'. This

echoes the words of Frantz Fanon, who wrote in *Wretched of the Earth* (Fanon 1963, 315) 'Let us not pay tribute to Europe by creating states, institutions, and societies which draw their inspiration from her' since humanity was 'waiting for something from us other than such an imitation'. The recent history of the decolonisation struggle in Rojava is such a search for something new. It does not aspire to establish a new state – the traditional route of national(ist) independence – but rather focuses on challenging the authority of the intrinsically colonising state itself in which power is hierarchically ordered and centralised. That is, the movement aims for the delimitation of state power through the development of self-governance. In the spirit of decolonisation, the movement has sought new directions (Fanon 1963).

These new directions involve a double movement of a new governance structure in which the agrarian economy has been and is being reorganised. The aim of this governance structure is to empower communities to manage their affairs in relation to others, a kind of distributed or network governance that we have characterised, following Deleuze and Guattari (1993), as *rhizomatic*. We have shown how, in this network, decolonisation became equated with a decentralisation and (re)diversification of agriculture. This also implied a similar situation with regard to production infrastructure, including the development of a decentralised resource mobilisation (eg energy, water) and processing industry (eg wheat, cotton, vegetable oils). It is in and through this diversification that the colonial past as a legacy of political submission to a centre, economic dependence, and demographic engineering has been revealed and challenged. We have also shown how this rhizomatic network makes clusters – a grouping of assemblies in districts, cities and cantons forming the Autonomous Administration of North and East Syria, that comprises both Rojava or West Kurdistan and parts of Syria predominantly inhabited by Arabs.

This decolonial, transformative shift occurred in the Rojava/AANES region from 2012 on, when the regional authority of Damascus collapsed with the beginning of the civil war. In this process, local communal councils emerged as key actors in the reorganisation of political and economic affairs. However, we have also pointed towards two important challenges. The first is the question of legacies and the impossibility of going back to a forgone past. Particularly difficult is the issue of how to deal with the demographic engineering (Arab Belt) employed in the region in the 1970s. While a considerable number of Kurds want this policy to be reversed and the land reclaimed, many Arab families have now lived in the region for several generations. A retropolitics that would restore a past in the present stands in contradiction to the principle of building a common future, yet the Autonomous Administration is also aware that this would trigger ethnic tensions between Kurds and Arabs. This, in turn, reveals the current operation of political power.

Relatedly, in grappling with the complexities of establishing a cooperative system within the Kurdish movement, we pointed to the agrarian question that underlies choices currently being made. While some emphasise the development of communal forms such as cooperatives, others believe this development should start with peasants' fields. Here, we drew parallels to historical debates on peasant production, particularly as exemplified in the ideological schism between Lenin and Chayanov in early twentieth-century Russia/USSR. At the level of practice, the Kurdish movement finds itself ensnared in a paradox, as an overemphasis on the cooperative model clashes with peasant resistance. A major challenge is overcoming the dichotomy between family farms and communal economic principles, allowing them to coexist without merging land and resources into cooperatives.

Conclusion

We have argued that there is a wealth of scholarship on franchise and settler colonialism, two historical forms of colonialism in the modern world, yet little has been written about the colonial projects of and within nation-states (Bhabba 1994; Bhambra 2018). While the primary objective of franchise colonialism is the appropriation of resources, and that of settler colonialism is the appropriation of land, nation-state colonialism, we have argued, has as its main aim the destruction of ethnocultural difference through dissolution in a hegemonic national identity. Ultimately, the Ba'ath's internal colonialism was oriented towards the erasure of Kurdistan and the Kurds.

In much of the last and present centuries, Kurdistan has been subjected to a project of nation-state colonialism. That is, the states of Turkey, Iran, Iraq and Syria have occupied Kurdistan with the objective of ridding their national imaginary of the Kurds and Kurdistan. In this article we have focused on agricultural modernisation as such a vehicle. The state-led modernisation of agriculture (agrarian developmentalism) in Syria performed this function. Making the area a producer of agricultural products – the state's primary supplier of these goods – while locating their processing for the manufacture of secondary goods – the addition of value through production – outside of the region, the state (re)produced Rojava as an ecologically degraded periphery of the state, a form of colonisation by de-development.

The implosion of the state in the Kurdistan region offered an opportunity for the establishment of a new governance model. An alternative development approach based on distributed responsibilities within a network of autonomous communities was pursued. In this article, we have referred to this in the context of an anti-colonial struggle as people and governance structures (the councils, committees, cooperatives, etc.) have addressed the conventional (modernist) logic and power relations of political and economic submission to the central state. The development of this new, rhizomatic governance structure stands in the best tradition of post-colonial thought as it entails the creation of distinct conditions of life and living beyond the nation-state, which is perceived as an inappropriate means through which to effect a decolonial shift (Paradies 2020; Walsh 2020).

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Notes

1. 'Rojava' means 'west' in Kurdish and stands for 'West Kurdistan'. In Syria, West Kurdistan is divided into the governates of Afrin and Al-Hasakah, previously the Ottoman province of Al-Jazira; historically, it is part of the region of Upper Mesopotamia.
2. The article does not address Turkey's politics of occupation and resettlement or the dynamics of the war economy in Rojava. In 2018, Turkey invaded and occupied Afrin in the south-west of Rojava; a year later, it invaded and occupied a strip of land in north-east Rojava. On 24 September 2019, during the 74th session of the United Nations General Assembly, Turkey's President Erdoğan presented plans to settle (Arab) Syrian refugees in much of Rojava – a manifestation of Turkey's settler colonialism politics in the parts of Kurdistan occupied by Syria. Meanwhile, the war economy in Rojava had highly detrimental effects on the environment and agriculture through the conditions in which oil was produced.
3. Informed consent was obtained from all participants prior to their involvement in the research. Participants were verbally briefed on the research and its objectives. They were informed that participation was voluntary and that they had the option to refrain from answering questions, decline to complete the questionnaire, or withdraw from the study at any time.
4. These emerged, along with colonial epistemologies, in the fifteenth and sixteenth centuries and, with developmentalism as a form of colonialism, in the twentieth century.
5. This 'salt-water' thesis of colonialism defines it as a relationship between a Western metropole and an overseas territory, which has led to the neglect of colonial projects within states (Nelson 2009).
6. Among other reasons, due to deposits of sewage, agricultural chemicals, and the discharges of factories, tanneries and oil refineries.
7. A recent version of this type of idea appeared with the Turkish plan presented to the United Nations in 2019 by President Erdoğan to replace the Kurdish population on the Syrian side of the Turkish–Syrian border with Syrian Arabs (moving the Kurds to the desert in the south). See also <https://foreignpolicy.com/2019/12/18/turkey-pitches-plan-settle-1-million-refugees-northern-syria-erdogan-kurds/>.
8. Interviews: Leyla Saruxan and Mehran Ahmed, Co-Chairs of the Agriculture and Irrigation Council, 21 November 2023; Karker Ismail, Co-Chair of the Office of Community Projects and Cooperatives of Cezire Region, 21 November 2023; Emin, Union of Kurdistan Communities (*Koma Civakên Kurdistan*, KCK) Economy Committee and Economy Council of AANES, 17 May 2024.
9. A major thrust of democratic autonomy is the delimitation of state power though the acceptance of the principle of self-governance. This involves the creation of an institutional framework of citizen assemblies, through which communities develop capacities to manage their own affairs.
10. Democratic confederalism concerns the relational form citizenship takes, namely the strengthening of local administrative capacities through the organisation of councils at the hamlet/village, neighbourhood, district, town/city and region levels. Presenting his perspective on autonomy and confederalism, Öcalan aligned with the radical eco-anarchist Murray Bookchin, who emphasised comprehending autonomy and confederalism as interrelated concepts. In this framework, the economy is organised through the confederal councils.

11. Unpublished report provided by the Public Land Administration (Rêveberiya Erdên Giştî) on 18 December 2023.
12. Leyla Saruxan and Mehran Ahmed, 21 November 2023.
13. This involves a slow supply of piped water directly to the stem/root area of individual plants, which minimises evaporation and runoff, thus maximising water usage.
14. Karker Ismail, 21 November 2023.
15. Leyla Saruxan, 21 November 2023.
16. Mehran Ahmed, 21 November 2023.
17. Leyla Saruxan, Mehran Ahmed, 21 November 2023.
18. Leyla Saruxan, 21 November 2023.
19. Interview and reports, Dr Ahmed Al-Mohamed, Livestock Departments, Office of Agricultural Affairs of AANES, 19 November 2023.
20. Interview, Til Xatûn village/Tirbespiyê, 5 April 2024. Semo's family village and land were confiscated by the regime in 1974, and Arabs were settled there.

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**Chapter 4. Article II: Roots of destruction: Exploring the
genocide-ecocide nexus through the destruction of olive trees in
occupied Palestine and Rojava**

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Roots of destruction: exploring the genocide-ecocide nexus through the destruction of olive trees in occupied Palestine and Rojava

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ABSTRACT

Ecological degradation not only affects local and global environments but also threatens the cultural survival of affected societies, exemplifying the phenomenon known as the genocide–ecocide nexus. This article presents the first comparative case study of olive tree destruction in occupied Palestine and Afrin canton of Rojava, highlighting how ecocide serves as a deliberate strategy of cultural and physical extermination in these regions, home to two of the world’s largest stateless groups. Juxtaposing these cases reveals that environmental destruction and genocidal practices are connected and function as tools of systemic oppression. The study also bridges critical genocide studies and the genocide–ecocide nexus with field political ecology and critical perspectives on environmental security, offering insights into links between conflict/violence and the environment. Our empirical analysis has three parts. First, we examine common mechanisms of olive tree destruction and appropriation in Palestine and Afrin – damaging, uprooting, poisoning, cutting, burning, stealing, and looting—by Israel and Turkey. Second, we explore resistance strategies by Palestinians and Kurds in response to ecological violence. We conclude by discussing similarities and differences between the two cases and problematising the international legal and political systems that fail to address the destruction inflicted on stateless and marginalised groups.

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Introduction

The olive tree represents our roots in the land. [...] I remember how my grandfather and my father used to take care of them. I vividly remember, as a small child, sitting and playing under these trees. They are a symbol for us, representing our identity in the land, said one person.

Another remarked:

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We are a people who have created ourselves with nature, trees and gardens. Everything, from our culture and food to our art and tradition, has been shaped by agriculture and the olive tree. [...] The material and spiritual link between the people and nature is very strong. We see nature as the basis and subject of our own lives.

Both individuals expressed feelings of pain, frustration and longing for their olive groves because they were prevented from living on their land and harvesting olives. Despite the similarities in their remarks about the importance of olive trees for their identity and their experiences and feelings of being displaced from their land and olive groves, these two individuals did not share a common national identity or geography. The first one was a Palestinian from the West Bank and the second was a Kurd from the Afrin canton of Rojava (Western Kurdistan) in northern Syria.

Kurds and Palestinians share many common experiences, such as being stateless nations and engaging in long struggles against nation-states – such as Israel, Turkey, and Syria – that occupy their territories and subject them to systemic oppression through settler colonial practices. They mobilise to defend their territories, ecosystems, and ways of life, often framing their struggles as fights for sovereignty, autonomy, or statehood, even if the latter is not a shared goal across all factions. Kurds and Palestinians also share the experience of environmental destruction at the hands of the occupying forces. Environmental (or ecological) degradation not only has an impact on local and global environments but also threatens the cultural survival of affected societies. Scholars identify this modality as the genocide-ecocide nexus (Crook & Short, 2014), which is inherently derived from the existing political, economic and legal structures. Short (2010, 2016) makes a significant contribution to critical genocide studies by reinterpreting genocide through the lens of Raphael Lemkin's original formulation, thereby highlighting the links between colonialism, ecocide, and cultural genocide. This literature argues that ecocide can constitute a method of genocide, and points to capitalism and colonialism as structural drivers of genocide and ecocide, emphasising how resource extraction and global development policies devastate ecosystems and local communities (Crook et al., 2018; Crook & Short, 2014). Further extending this analysis, Dunlap identifies developmentalism and extractivism as forms of colonialism that date back to around six centuries ago, and explains the collaboration between states and corporations in normalising and justifying ecocidal practices through large-scale projects, which inherently result in exterminating outcomes (Dunlap, 2018b, 2021, 2023b, 2023a). Others have further demonstrated that environmental destruction, including ecocidal development and extractivist projects, can serve as a counterinsurgency technique (Brock & Dunlap, 2018; Crook et al., 2018; Crook & Short, 2014; Dunlap, 2018b, 2023b; Le Billon, 2001, 2008; Peluso & Vandergeest, 2011; van Solinge, 2008; White, 2008). The corporate agendas and structural imperatives of the capitalist mode of production are major drivers of environmental degradation and ecocide in the lands of marginalised groups (Crook & Short, 2021b, 2021a), which also leads to cultural destruction. We see examples of this destruction around the globe, such as in different parts of Kurdistan in Turkey, Syria, Iran, and Kurdistan Region of Iraq (Ahram, 2023; Dinc, 2021, 2022; Dinc et al., 2021; Eklund et al., 2021; Eklund & Dinc, 2024; Etten et al., 2008; Harris, 2012; Hunt, 2019, 2021; Jongerden, 2010; Türk, 2023; Türk & Jongerden, 2024), Iraq (Priestley, 2021), Palestine (al-Butmeh et al., 2019; Jaber, 2019; Molavi, 2024; Rashed & Short, 2016; Twite, 2009), Sudan (Shaw, 2011; Wise, 2021), Peru (Dunlap, 2019), Mexico

(Dunlap, 2018a, 2018b), Sri Lanka (Prakash & Short, 2016), Nepal (Baral & Heinen, 2005), Guatemala (Jonas, 2009), and Nigeria (Lynch et al., 2021) to name a few.

This article examines olive tree destruction with a comparative case study of occupied Palestine and Afrin canton of Rojava (formerly a part of the Autonomous Administration of North and East Syria, NES, hereafter Afrin). It presents the first comparison of the olive tree destruction in Palestine and Afrin, highlighting how ecocide serves as a deliberate strategy of cultural and physical extermination in these regions, which are home to two of the world's largest stateless groups. Juxtaposing these cases reveals that environmental destruction and genocidal practices are connected, and they function as tools of systemic oppression in both cases. The study also bridges critical genocide studies and the genocide-ecocide nexus with field political ecology and critical perspectives on environmental security, offering insights into the links between conflict/violence and the environment (Dunlap, 2021, 2022; Kosek, 2006; Le Billon, 2001, 2008, 2014; Le Billon & Duffy, 2018; Martinez-Alier, 2002, 2021; Neumann, 2009; Peluso & Vandergeest, 2011; Peluso & Watts, 2001; Roy & Hanaček, 2023; Vogler, 2024; Zalik, 2011).

We argue that the ongoing, systemic ecological destruction in Palestine and Afrin constitutes a form of ecologically induced genocide. This is carried out by state actors – including militaries, police, paramilitary forces – as well as settlers, as part of a broader settler-colonial counterinsurgency strategy. We distinguish this from ecological harm that arises as a by-product of capitalist expansion or economic exploitation. In the cases we examine, ecological destruction is not incidental but deliberate – targeting specific lifeworlds, such as olive groves, as a means of eroding the foundations of existence and resistance among stateless and marginalised groups. This strategy seeks to dismantle both the material and symbolic foundations of anti-colonial resistance by targeting the environment as a means of control, dispossession, and erasure. These actions are embedded within the wider apparatus of genocidal state violence. Recognising the environment's role within colonial structures and practices – often overlooked in political ecology and environmental security frameworks – enables a more comprehensive understanding of ecological destruction as a deliberate and systemic tool of colonial domination. Put differently, while ecocide can occur as a by-product of economic exploitation, our analysis focuses on how certain nation-states in the Middle East deliberately operationalise environmental destruction as a counterinsurgency strategy – targeting land, livelihoods, and cultural identity to suppress resistance and displace communities.

Adopting a qualitative approach, we draw on document analysis, key informant interviews, and field observations. The dataset comprises 12 documents (six each from Palestine and Afrin), published between 2012 and 2024 by local and international NGOs, as well as intergovernmental organisations, each selected based on their relevance to the destruction and appropriation of olive trees. We selected these 12 documents not merely for their thematic relevance, but because they originate from civil society and intergovernmental organisations that offer grounded, contemporaneous, and often underrepresented perspectives on the destruction and appropriation of olive trees. These documents provide direct, context-specific accounts – frequently based on field data, testimonies, and on-the-ground monitoring. Their inclusion enables us to trace how the structural dynamics identified in the literature manifest in real time across Palestine and Afrin, offering insights into the lived realities and institutional responses that have unfolded over the past decade. Additional information is gathered from news

outlets and the websites of environmental organisations active in Palestine and Afrin. We conducted four semi-structured interviews¹ with environmental activists and experts – one in Palestine and three in Afrin – focusing on environmental degradation, the symbolic significance of olive trees, and resistance strategies. Field notes and travel diaries from visits to Palestine in 2023 and Rojava in 2024 further contextualise our findings through direct observations and informal conversations.

The remainder of this article is structured as follows. We begin with a background of the genocide-ecocide nexus and introduce the cases of Palestine and Afrin where this nexus unfolds. This is followed by an empirical analysis presented in three sections. The first examines common mechanisms of olive tree destruction and appropriation in Palestine and Afrin, which include damaging, uprooting, poisoning, cutting, burning, stealing olive trees, looting olive trees and their products, among other settler-colonial mechanisms used by Israel and Turkey. The second section explores the resistance strategies of the Palestinians and Kurds who are subjected to this violence. The final section summarises the key similarities and differences between the two cases and problematises the international legal and political system, highlighting its failure to address and overcome the destruction imposed on these stateless and marginalised groups in the Middle East.

The genocide-ecocide nexus

The concept of genocide was coined in 1944 by Raphael Lemkin, who combined the words ‘genos’ (Greek, meaning kin, tribe, race) and ‘cide’ (Latin, meaning destruction, killing) (United Nations Office on Genocide Prevention and the Responsibility to Protect, N/A). Lemkin identified genocide as ‘the destruction of a nation or of an ethnic group [...] a coordinated plan of different actions aiming at *the destruction of essential foundations of the life of national groups*, with the aim of annihilating the groups themselves’ (Mennecke, 2012, p. 145, emphasis added). Lemkin (1944) explains that genocide has two phases. In the first phase, the national pattern of the oppressed group is destroyed, while in the second phase, the national pattern of the oppressor is imposed on the oppressed group. Lemkin writes, ‘This imposition, in turn, may be made upon the oppressed population, which is allowed to remain, or upon the territory alone, after removal of the population and the colonization of the area by the oppressor’s own nationals’ (Crook & Short, 2014; Lemkin, 1944, p. 79). This demonstrates the validity of the understanding that the invasion of indigenous lands is not a single incident or an event, but a structure, ‘an organizing principle of settler colonial society’ (Wolfe, 2006, p. 388). Indeed, Moses argues that in coining the term of genocide, Lemkin was aware of the inherently colonial and imperial nature of the act (Moses, 2010, p. 9).

Colonialism, as ‘a specific form of rule’, is based on a convergence of policies and practices imposed by a dominant country on other countries in order to exploit and control economic resources through a variety of strategies, including genocidal mass killings, cultural assimilation, forced displacement and resettlement, and ecocidal practices (Moses, 2010, p. 22). Others, such as Jean Paul Sartre, Robert Davis, and Mark Zannis have recognised the links between colonisation, capitalism, cultural genocide and environmental genocide as early as the 1960s (Galligan SJ, 2022, p. 1007; Moses, 2010, p. 31). Although

genocide became a crime with the adoption of the Genocide Convention in 1948, the accepted definition of the term had been ‘stripped of its coherence and conceptual integrity’ which denied the relevance of the crime in colonial and settler-colonial contexts (Short, 2017, p. 285).

The concept of ecocide was introduced in 1970, 26 years after the concept of genocide, by Arthur W. Galston, who combined ‘cide’ with ‘eco’, meaning ecosystem. Richard A. Falk’s draft Ecocide Convention in the 1970s, which would have criminalised ecocide, received significant support but was eventually dropped from the agenda of the UN Sub-Commission on Prevention of Discrimination and later from the draft documents of the UN International Law Commission (Short, 2017). Since 2010, the Eradicating Ecocide campaign, launched by Polly Higgins, has been campaigning for ecocide to be included in the Rome Statute of the International Criminal Court as the fifth crime against peace (the first four being genocide, crimes against humanity, war crimes and the crime of aggression). According to Higgins and her colleagues, ecocide is ‘the extensive damage to, destruction of or loss of ecosystem(s) of a given territory, whether by human agency or by other causes, to such an extent that peaceful enjoyment by the inhabitants of that territory has been severely diminished’ (Higgins et al., 2013, p. 257). Ecocide entails the large-scale destruction, degradation and annihilation of ecosystems in ways that rupture the fundamental relational ontology between humans, their socio-cultural fabric, cosmologies, and non-human entities. It is not merely environmental harm but a systemic assault on the web of life, dismantling the deep interconnections that sustain ecological and cultural existence. The extermination of ecosystems is thus an extermination of identities, erasing histories, displacing peoples, and collapsing the conditions for life itself (Crook et al., 2018; Crook & Short, 2014; Dunlap, 2018b, 2021; Lindgren, 2018; Short, 2016).

The genocide-ecocide nexus was first theorised and introduced by Crook and Short in 2014, who suggested that genocide studies should draw on the traditions of political ecology and environmental sociology in order to uncover the common drivers and connections between ecocide and genocidal social death (Crook & Short, 2014). Their theorisation of the genocide-ecocide nexus with a Marxist ecological perspective is significant in showing how ecocide becomes a deliberate act of economic appropriation in the hands of capitalist and colonialist regimes. The genocidal structure of ecocide lies in its ability to dismantle the relationship between communities and their environment, even ‘without specific “intent to destroy” but [as something that] could occur through sporadic and uncoordinated action or be a byproduct of an incompatible expansionist economic system’ (Short, 2016, p. 34). With such a perspective, Crook and Short (2014) conceptualise the experiences of indigenous peoples, ethnic, and religious groups of dispossession, forced displacement, cultural assimilation, poverty and inequality as a result of the destruction of fundamental elements of these communities.

The genocide-ecocide definition suggests that genocide need not be physical or limited to the act of mass killing; rather, it aims to destroy the essential foundations of the targeted groups through various techniques, including deprivation of livelihoods that undermine the life, existence and resistance of the population as part of a strategy of extermination (Dunlap, 2018b, p. 557). Forcible dispossession and alienation from their lands damage the ecosystems that shape the material and spiritual existence of targeted populations, leading to their ‘social death’ and, consequently, genocide (Crook &

Short, 2014, p. 313). The loss of livelihoods for peoples with ‘territory bounded culture’ also means the loss of social and cultural fabric, resulting in ‘community-wide trauma and dysfunction’ (Abed, 2006, p. 326). The nexus of genocide and ecocide thus demonstrates how marginalised communities can be doubly victimised, not only by physical death but also by social death.

The two cases we focus on in this paper, Palestine and Afrin, share similar structures of genocidal violence and ecocidal destruction carried out by two settler colonial states, Israel and Turkey. Given the long and complex histories of both cases, the following section provides only a brief overview of the two cases, focusing on the structures of settler colonialism and the intertwined practices of ethnic cleansing/genocide and environmental destruction under these powers. As Crook and Short (2014, p. 313, emphasis in original) note, ‘indigenous peoples fighting to retain or regain their lands, they are fighting for their *life as distinct peoples* since, for them, their spirituality and cultural vitality is based in and on and with their lands’. This shared struggle forms the foundation of our comparative analysis.

The genocide-ecocide nexus in Palestine

Palestine has a long history under the rule of the Ottoman Empire until the early 1900s, and then became a British Mandate until 1948. With the Nakba in 1948, the State of Israel had been established through a systematic ethnic cleansing of Palestinians, destruction of their villages and urban areas alike, and a continued harassment through looting their houses, confiscating their fields, desecrating their holy places, and violating the basic human rights of Palestinians (Pappe, 2006). After the establishment of Israel, Palestine was divided into the West Bank and Gaza Strip, which were ruled by Jordan and Egypt, respectively. In 1967, the West Bank and Gaza Strip were occupied by Israel. Palestine is therefore an occupied country, including areas that make up the state of Israel, as well as the West Bank and Gaza Strip. While the Gaza Strip is under Israeli blockade, the West Bank is divided into three areas, with Area A under the Palestinian Authority (PA) control, Area B under joint PA and Israeli-Palestinian security control, and Area C under full Israeli control. Since October 2023, the Gaza Strip has been under full-scale invasion by the Israeli army, carrying out an act of genocide, and violence against Palestinians in the West Bank by the Israeli state and settlers has since intensified, according to various reports (Human Rights Watch, 2024b; UNOCHA, 2024). Recent studies have shown the war-induced damage to agricultural lands in Gaza after October 2023 (UNOCHA, 2024; Yin et al., 2025).

There is a vast literature discussing Israel’s occupation of Palestine through the lens of settler colonialism (Dana & Jarbawi, 2017; Francesco Amoroso & Richter-Devroe, 2019) and genocide (Rashed & Short, 2016; Shaw, 2010). In addition to physical, political, economic, and cultural genocide (Abdullah, 2019; Rashed & Short, 2016), scholars have referred to the concepts such as memocide (Rashed et al., 2014) and urbicide (Abujidi, 2014; Graham, 2002) in Palestine. The ecocidal practices of genocide are identified as actions that have ‘a major role in the degradation of agricultural and rangelands, deforestation, desertification, depletion of water resources and the degradation of water quality, leading to far-reaching social, economic and political implications in Palestine’ (Rashed & Short, 2016, p. 81). In addition to the climate-related challenges, reports

show how restrictions on movement and access hinder economic growth and self-governance, while maladaptive practices like settler farming and the destruction of Palestinian olive groves exacerbate environmental challenges (UNDP, 2013, pp. 68–69).

While water politics has been widely studied (Beltrán & Kallis, 2018; Selby, 2013; Trotter, 2007), there is also a significant body of research on tree landscapes in Palestine (Braverman, 2009a, 2014; Fields, 2010; Molavi, 2024; Samar Nazer & Khasib, 2019; Selby et al., 2022). Deforestation has been discussed as an act carried out by Israel, which acts as a major contributor to climate change by increasing carbon loss and damaging the ecosystem (Rashed & Short, 2016). Scholars have shown that olive groves also have an important biodiversity value as well, which is important to conserve. Yet they face numerous threats, including destruction carried out by Israel and Israeli settlers (Awad & Attum, 2017). According to reports, an estimated 1.5 million trees were uprooted by Israel in a 7-year period between 2000 and 2007, the majority of which were olive trees (Rashed & Short, 2016). According to other reports, the number of trees that have been uprooted in Palestine since 1967 is around 2.5 million and includes mainly olive trees, but also fruit and palm trees (Nazzal, 2019). In Gaza alone, the ‘buffer zone’ between Israel and Gaza ‘extends 600 metres beyond the 1967 Green Line [and] encompasses up to 30 per cent of agricultural land’ (Nazzal, 2019, pp. 90–91, see also Molavi, 2024).

Braverman’s (2009a, 2009b) pioneering study on the politics of trees in Israel and Palestine shows how trees become political tools for claiming lands and national identities. Braverman points to the binary juxtaposition of olive and pine trees through bureaucratic and legal frameworks and pine planting campaigns by Israel (and Israeli settlers), while uprooting olive trees that have symbolic and economic significance to Palestinians. Another important point raised by Braverman (2009b, p. 172) concerns Article 78, and her ‘(post)colonial reading’ of it. Article 78, which originated during the Ottoman rule to encourage agriculture in distant lands to ensure the generation of taxes, has been reinterpreted under the Israeli occupation, which decided to apply Article 78 in a different, negative way that would allow its land-capture (Sluglett & Farouk-Sluglett, 1996, p. 226; cited in Braverman, 2009b, p. 172). The land grab is justified by Israel’s self-perception as the modern transformers who would lead development, while devaluing Palestinians as ‘traditional’, which Braverman (2009b, p. 174) argues ‘fits into what some postcolonial scholars have identified as a central feature of the colonial state: its bifurcation into two forms of power that function under a single hegemonic authority’. Other legal tactics used by Israel include the Absentee Property Law and the Land Law, as well as other methods such as seizing land for ‘security’ reasons, or buying lands from the Palestinians (Jaber, 2019).

Today, the lands and olive groves that still belong to Palestinians are not exempt from Israel’s regulatory tactics aimed at detaching Palestinians from their lands as a counterinsurgency technique. In order to be able to access their lands, Palestinians still need to show a number of certificates, in addition to having ‘the ability to decipher countless directives, and a comprehensive engagement with Israel’s military permit and gate bureaucracy’ (Braverman, 2009a). According to the United Nations sources, 39% and 42% of requests for permits to access olive groves behind the wall were denied in 2010 and 2011 respectively (Nazzal, 2019). The existing literature and reports reveal how settler colonialism and its mechanisms ‘attempt to destroy the

Indigenous built and natural landscape' in Palestine (Jaber, 2019, p. 150), which should be further understood as an important example of how the ecocide-genocide nexus operates.

The genocide-ecocide nexus in Afrin

Kurdistan refers to a geographical region that covers a large area of Mesopotamia, encompassing diverse landscapes from the Taurus Mountains in the west to the Zagros Mountains in the east. It was divided into four states (Turkey in the north, Iran in the east, Iraq in the south and Syria in the west) by French and British colonial designs after the World War I, the Sykes – Picot Agreement (1916), and Treaty of Lausanne (1923). This colonial design has made Kurdistan an international colony (Beşikçi, 1990a) of the aforementioned states, although Kurdistan has never formally had the status of a colony. As a result, the Kurds have been subjected to various acts and policies of assimilation and elimination within the homogeneous nation-building processes carried out by the dominant Turkish, Persian or Arab national identities. The case of Kurdistan as a colony (Öcalan, 2011, p. 96) has been systematically denied, making it an 'invisible international colony' (Goner, 2023, p. 169). This situation is derived from the conceptual and epistemic limitations of colonialism (franchise colonialism and settler colonialism) (Türk & Jongerden, 2024) and neo-colonialism (Sunca, 2023).

Afrin is located in the far west of Kurdistan, which was officially a district in the Aleppo Governorate in northwestern Syria. The Syrian state inherited French rule and its colonial mentality/nation-state model in 1946 and Pan-Arab nationalism became the main policy and foundation of the state when the Bathist Party came to power in 1963. Through a special decree (No. 93) in 1962, some 120,000 Kurds were stripped of their citizenship and lost their basic political and economic rights, including land and property (McDowall, 2004, p. 474). The next step was much more aggressive and has had a detrimental effect on the socio-economic, political and cultural structure of the Kurds. The Arab Belt (*al-hizam al-arabi*) was not only aimed at eradicating the Kurdish presence in the socio-cultural and political spheres, but above all it was an example of political engineering involving settler colonisation and the shaping of Rojava's demographic structure and economic resources as a counterinsurgency strategy (Türk & Jongerden, 2024). The plan was implemented in 1973 in the Jazire region, the heart of Rojava, which has favourable land and water resources for agriculture and rich oil fields. As a result of the Arab Belt, at least 60,000 Kurdish families were forcibly displaced, although the plan envisaged the displacement of 140,000 Kurds (Aslan, 2023). Thus, 'agricultural modernisation was co-constructed with a demographic policy of Arabisation, a form of settler colonialism instrumental to state colonialism aimed at subordinating and dissolving Kurdistan into the Arab nation' (Türk & Jongerden, 2024, p. 6).

Afrin became a self-governing canton under the Kurdish rule in 2012, following the political vacuum created by the Syrian uprisings in 2011. Afrin Canton, together with the Kobane and Cezire Cantons, established the Autonomous Democratic Governments (*Rêveberiya Xweseriya Demokratîk*) of Rojava between 2012 and 2017 (Aslan, 2023, p. 92). In January 2018, Afrin was invaded by the Turkish state and Turkey-backed jihadist groups. Since then, Afrin has remained under occupation, which is in line with

Turkey's colonial structure (Beşikçi, 1990a, 1990b; Bozarslan, 2008; Jongerden, 2009). The case of Afrin is an important example of a dual colonial status. After being an internal colony of Syria for decades, Afrin was occupied by Turkey and its allies in 2018, making it an internal and international colony. Despite alarming reports and news coverage, the systematic environmental degradation in Afrin through the destruction and appropriation of olive trees does not receive enough attention in mainstream media and academia.

Shared patterns of destruction and resistance in occupied Palestine and Afrin

We identify both shared patterns and notable differences in the destruction of olive trees in Palestine and Afrin. In both contexts, olive tree destruction functions as a (settler-)colonial counterinsurgency strategy, aiming to sever cultural ties, undermine livelihoods, and impose social death. Despite contextual distinctions, Palestinians and Kurds exhibit similar emotional and cultural attachments to their land, alongside enduring forms of resistance. Table 1 summarises these patterns and divergences, which are further explored in the concluding reflections.

Olive tree destruction in occupied Palestine and Afrin

In this section, we analyse the main themes that emerge from the selected documents on the destruction of olive trees in Palestine and Afrin, which often reveal the common

Table 1. Summary of findings.

		Palestine	Afrin
Peoples' resistance	Shared meanings around olive trees	Identity, culture, history, peace	
	Shared emotions around olive tree destruction	Pain, sorrow, anger, grief, attachment, longing, persistence	
	Collective action	Environmental activism, organisations	
		Several tree-plantation campaigns	No tree-plantation campaigns
Settler-colonial mechanisms	Olive tree destruction & appropriation	Since 1967, carried out by Israel and settlers	Since 2018, carried out by Turkey and its allies
	Walls	Separation Wall, approximately 700 km	Emerging walls, approximately 5 km
	Settlements	Jewish settlements/colonies around Palestinians	Turkish and Arab settlements around Kurds
	Legal accountability	Legal framework exists, but it is not effectively applied	No accountability under Turkish-backed Syrian National Army
The international legal and political system	International law	No legally binding international environmental criminal law Palestinians and Kurds constitute the two largest stateless nations. Palestinian Authority has limited sovereignty. Autonomous Administration of North and East Syria is a de-facto autonomy	
	State-centric structures		
	Securitisation of environmental destruction	Viewed as a security threat by Israel	Viewed as a security threat by Turkey

strategies used by the occupying forces and the settler-colonial systems to destroy the physical, social, cultural, economic and environmental well-being of the occupied peoples, thereby imposing social death on these groups.

Olive tree destruction in Palestine

There are two main approaches to olive tree destruction in Palestine. First, they focus on the symbolic and economic/material significance of olive trees. Second, they discuss the mechanisms and scope of olive tree destruction, providing evidence from the ground.

The symbolic and economic significance of olive trees

Olive trees in Palestine are seen as ‘more than just a tree’, as Palestinians are described as ‘proud of their olive trees’ and have tended to them with ‘care’ and ‘appreciation’ for generations (MIFTAH, 2012, p. 1). ‘The Palestinian olive’ is described as sacred and part of Palestinian culture (The Land Research Center – Al Haq, 2019). In addition to the importance of olive trees in Palestinian history, culture and identity, olive trees in Palestine are also highlighted for their economic importance (Joint Advocacy Initiative, 2022). Due to their resistance to drought and poor soil conditions, olive trees also ‘represent Palestinian resistance and resilience’² (MIFTAH, 2012). As a cultural and economic resource, olive trees tie Palestinians to their land and provide them with a vital source of income. According to the JAI report (Joint Advocacy Initiative, 2022, p. 3), ‘olive fruit comprises the income of 80,000 Palestinian families’. However, due to various means of destroying olive trees and preventing Palestinian farmers from harvesting their olives (The Land Research Center – Al Haq, 2019), Palestinian farmers suffer significant economic losses, ‘resulting in a direct financial setback’ (UNOCHA, 2024).

The destruction of olive trees and hindered olive harvesting

The destruction of olive trees is reported as one of the policies and practices, among others such as water-intensive livestock farming subsidised by settlers, that ‘frustrate the development of Palestinian resilience to climate hazards’ (UNDP, 2013, p. xi). Reportedly, ‘more than a million olive trees and hundreds of thousands of fruit trees have been destroyed on Palestinian land’ in the period following the Israeli occupation of Palestine in 1967 (Joint Advocacy Initiative, 2022, p. 3). The reports list various mechanisms of olive tree destruction, including burning, bulldozing, uprooting, crushing, stealing, spraying with hate graffiti, drowning Palestinian land with sewage from the hilltop settlements. Some reports use the term ‘vandalise’ to describe the destruction of olive trees in Palestine (UNOCHA, 2024). The reports also visualise this destruction with photographs taken on the ground as well as figures, maps, and charts, including tables with detailed information on the perpetrators (e.g. different settlements and colonies with their names, the Israeli army, authorities and companies), the means of destruction (e.g. burning, uprooting, dumping sewage water) together with other information such as the total number of trees damaged in different governorates (e.g. Jerusalem, Bethlehem, Hebron, Ramallah) (The Land Research Center – Al Haq, 2019).

In addition to the material destruction of olive trees, the reports discuss how Palestinian farmers are prevented from harvesting their olive trees. Olive trees in Area C (which covers over 60% of the West Bank) and the ‘seam zone’ between the Green Line and the

Separation Wall are becoming particularly difficult for Palestinian farmers to access, as they need to obtain permits from the Israeli authorities to cultivate or harvest their land (UNOCHA, 2024). In addition to difficulties in obtaining permits, Palestinians face new restrictions such as ‘earth mounds, closed road gates and cement blocks’ erected by Israel to prevent access (UNOCHA, 2024, p. 5). Palestinian farmers reportedly ‘became targets as they harvest their olives’ (MIFTAH, 2012, p. 2). In addition to the possibility of being harmed by the ‘warning shots’, Palestinian farmers are also exposed to attacks by Israeli settlers and the symbolic punishment by Israeli courts, otherwise known as ‘price tags’ – retaliatory acts of violence and vandalism carried out by extremist settlers, who protest the evacuation and demolition of illegal settlements in the West Bank (MIFTAH, 2012, p. 2). The documents indicate that the aggressors in Israel can be the Israeli state (e.g. its armed forces and courts) as well as Israelis living in settlements and colonies surrounding Palestinian lands.

The deliberate destruction of olive trees and the hindrance of olive harvesting are not only attacks on the economic stability of Palestinian families but also a counterinsurgency strategy aimed at eroding their cultural and social fabric. By targeting such a vital resource, the deep-rooted connection between Palestinians and their land is significantly damaged. This tactic aligns with the genocide-ecocide nexus, where environmental destruction is used as a tool to undermine and ultimately annihilate the identity and existence of a people. The loss of olive trees, therefore, represents a direct economic blow and a symbolic assault on Palestinian resilience and heritage, contributing to the broader agenda of dispossession, marginalisation, and social death.

While some instances of land seizure and environmental degradation may appear to stem from economic motives – such as expanding agricultural production or market access – we argue that the systematic and patterned nature of these actions, their alignment with military operations, and their targeting of culturally significant resources point to a broader counterinsurgency strategy. This is not merely about land acquisition for farming, but about weakening the socio-ecological foundations of Palestinian life, identity, and existence. The destruction of olive trees often coincides with military incursions, settlement expansion, and legal mechanisms of dispossession, suggesting a coordinated strategy aimed at suppressing resistance and asserting settler-colonial control. The documents we analyse provide evidence of this intentionality, highlighting how environmental targeting is embedded within broader practices of surveillance, displacement, and spatial domination.

Olive tree destruction in Afrin

Following the invasion of Afrin by Turkey and its allied forces in 2018, documents began to emerge about the demographic changes in Kurdish-dominated areas that went hand in hand with the destruction and appropriation of olive trees. We have identified three main themes that emerge from these reports. First, they discuss the aims of demographic change in Afrin, which includes violence against mainly Kurds, but also other peoples and their lands. Second, they show the scope of destruction and appropriation of olive trees in Afrin, often based on testimony from local residents. Finally, they indicate how the systematic destruction of olive trees has produced conditions of social death, through empirical consequences and the erasure of Afrin’s land-based cultural fabric – an ecology of life that has been cultivated and sustained over centuries.

Demographic change by the Turkish forces and its allies

According to numerous reports, the Kurdish residents who remained in Afrin after Turkey's occupation were subjected to abduction and arrest, harassment and torture under the rule of the Turkish-backed Syrian National Army (SNA) (UN Human Rights Council, 2024). Not only Sunni and Ezidi (Yazidi) Kurds, but also Christians and other minority communities who were 'perceived to have ties to the SDF [Syrian Democratic Forces] and the Autonomous Administration of Northeast Syria' were targeted by the occupying forces (Human Rights Watch, 2024a, p. 3). Such changes were implemented through various mechanisms, including 'killing, arbitrary arrest, forced disappearance, torture and ill-treatment, looting and property confiscations, coercion of Kurdish civilians to abandon their homes, and hampering their returns to their home towns' These documents also include figures on displaced people from Afrin, including the city centre and other neighbourhoods such as Jindires, Rajo, Bulbul, and villages like Deir Sawwan and Eska, among others. Adding up these figures, more than 130,000 people have been settled in Afrin from 2018 to 2021 (SOHR, 2021).

In addition to the mechanisms mentioned above, some documents allege the construction of separation walls (SOHR, 2021). In 2019, Turkey reportedly built walls around the villages of Kimar, Barad, Mariamin and Jalbara in Afrin, followed by another separation wall built in 2022, reportedly more than 2 km long and 4 m high, from the National Hospital in the city of Azaz in northwestern Syria (Human Rights Organisation Afrin, 2022). According to the reports, during the construction of this wall in 2022, 1,600 olive trees were uprooted and a further 24,000 olive trees from the province were annexed (Human Rights Organisation Afrin, 2023, p. 42). Although not to the extent seen in Palestine, Turkey has been building cement walls in Afrin, which people in Afrin refer to as the 'Israeli Wall'.³

The reports also suggest that Turkey is carrying out housing projects as 'model settlements' in occupied Afrin on land covering thousands of dunams that they receive 'for free' (Human Rights Organisation Afrin, 2022). These housing projects are reportedly built in collaboration with various Arab and/or Islamic associations and organisations, such as the Turkish White Hands Association, Al-Isan Society for Turkish Relief, Giving Without Borders Qatari Association, Kuwaiti International Charitable Society for Development, Living with Dignity Association (Palestine) and Wafaa al-Muhsiin (linked to the Palestinian Muslim Brotherhood movement) (Human Rights Organisation Afrin, 2022). There were also media reports in 2024 of the Turkish government's settlement of Hamas militias in occupied Afrin, suggesting an attempt at demographic engineering in the formerly Kurdish-dominated region (ANF, 2023; North Press Agency, 2023; The Syria Report, 2023). It is argued that

the demographic fabric of the Afrin region is apparently changing, especially after IDPs from other areas of Syria (families from Ghouta, rural Damascus, northern Hama and Idlib) were made to settle in the region following the displacement of its predominantly Kurdish population amid widespread suppression of Kurdish culture. (PÊL – Civil Waves et al., 2023, p. 28)

Reports also suggest that the majority of people who benefit from these resettlements are 'Arab and Turkmen militia fighters' (Human Rights Organisation Afrin, 2023).

Other strategies include the display of Turkish flags and portraits of Turkish leaders (e.g. Recep Tayyip Erdoğan, Mustafa Kemal Atatürk) in public spaces, the Turkification of street and shop names and the effective removal of Kurdish names from public spaces

and academic curricula, and the destruction of archaeological sites and cultural treasures (Human Rights Organisation Afrin, 2023; Human Rights Watch, 2024a).

Showing the scope of tree destruction and appropriation in Afrin

According to reports ‘the inhabitants of Afrin depended mainly on agriculture’, and in particular on the cultivation of almost 18 million trees, which annually produced ‘more than 270 thousand tons of olives, equivalent to about 55 thousand tons of olive oil’ (Human Rights Organisation Afrin, 2023, p. 36). After Turkey’s occupation of Afrin, this olive oil was ‘turned into a source of income for the armed groups and their Turkish masters’ (Human Rights Organisation Afrin, 2023, p. 38). The occupying forces reportedly appropriated most of the olive farms in Afrin, forcing local farmers to pay fees to get their olives back from the occupying militias, as well as royalties and taxes (e.g. ‘tree protection tax’) (Human Rights Organisation Afrin, 2023, p. 55). The farmers were also reportedly forced to sell their crops cheaply to Turkish merchants who would process, bottle, and brand the olive oil to be exported to Europe, the United States and the Gulf States (Human Rights Organisation Afrin, 2023; Human Rights Watch, 2024a). This was accepted and further defended by Turkey’s minister of agriculture at the time, Bekir Pekdemirli, who argued that ‘the revenues would otherwise fall into the hands of the PKK’ (Human Rights Watch, 2024a, p. 13; Petti, 2020). The United Nations Human Rights Council report describes how some Syrian National Army factions, particularly the Suleiman Shah Brigade, confiscated parts of Kurdish farmers’ olive harvests, demanding payments as ‘taxes’ that often exceeded the value of the crops (UN Human Rights Council, 2024, p. 14).

Some reports suggest that over 300,000 olive trees have been uprooted in Afrin since the occupation (Human Rights Organisation Afrin, 2023), while other news reports claim even higher figures (North Press Agency, 2020). As a result of this destruction, villagers have reportedly lost their property, including several hundred to several thousand olive trees, as well as other agricultural land and houses (Human Rights Organisation Afrin, 2023). The extensive eyewitness accounts collected on the ground reveal various means of destruction, such as the cutting down, uprooting and burning of olive trees, as well as the practice of confiscating goods and plundering olive groves and vineyards (PÊL – Civil Waves et al., 2023). As a result, olives, which were ‘the main source of income for the people of Afrin’ before the Turkish occupation, have been replaced by income from ‘small businesses and aid, savings, and remittances from family members living in the West’ (Rojava Information Center, 2023).

Social death and cultural erasure: targeting the land-bound cultural fabric of community

The destruction of olive trees in Afrin, following Turkey’s military occupation in 2018, represents a paradigmatic case of ecocide-as-counterinsurgency. Beyond material damage, these acts target the symbolic, economic, and cultural lifeworlds of Kurdish and other minority communities, causing ‘social death’. The olive tree is not merely a crop but a carrier of ritual, memory, and land-bound identity. Its systematic eradication constitutes an attack on the embodied, place-based modes of life that sustain communal cohesion and cultural continuity.

First, ecocide induces the loss of economic autonomy. Olive cultivation – especially the production of olive oil and soap – was the primary livelihood for Afrin’s population.

Reports and our key informants confirm that destruction of olive trees severed communities from their subsistence base. Former olive farmers now depend on remittances, precarious small businesses, or humanitarian aid. This collapse of local autonomy has turned a once self-sustaining agricultural region into a space of dependency, marked by the disappearance of cooperative economies and women-led production networks. Second, the destruction dissolves ritual and seasonal knowledge embedded in olive cultivation. In Afrin, olive harvests were communal and celebratory – rooted in intergenerational rituals, oral histories, and spiritual customs, which connected people to seasons/time, collective identity and land-based socio-economic articulations. These traditions encoded a deep ecological literacy, grounded in local calendars and embodied practices. With the destruction or confiscation of groves, these rituals are no longer enacted, and their cultural relevance is undercut. Harvests no longer mark seasonal rhythms; instead, olive oil has become a militarised commodity, absorbed into settler and state economies. In this sense, ecocide functions as epistemicide: a silencing of the cultural temporalities and meanings that once structured social life. Third, the rupture is generational. Centuries-old trees, once tended collectively by families, are now replaced by walls, militarised checkpoints, and gated settlements. Youth are disconnected from traditional knowledge, having neither the trees to learn from nor the elders to teach them. As land is confiscated and families displaced, the embodied practices of cultivation, care, and stewardship are lost. This break in intergenerational transmission is not incidental; it is central to the colonial logic of erasure. By disrupting the continuity of place-based knowledge and identity, the occupation ensures the cultural disintegration of the community.

Taken together, these mechanisms show that the destruction of olive trees in Afrin cannot be interpreted as collateral damage or economic exploitation alone. It is a mode of ecological warfare – one that produces social death through the annihilation of material sustenance, symbolic meaning, and historical continuity. The targeted eradication of olive groves constitutes both an act of ecocide and a strategy of cultural genocide, engineered to erase a people by severing them from the very landscapes that define who they are.

Resistance in response to olive tree destruction in Occupied Palestine and Afrin

Our analysis of the reports, together with interviews with key informants and observations in Palestine and Afrin, reveals a range of emotions that lead to motivation and determination to resist. These include a sense of belonging and attachment to their land, a sense of collective history and identity, a longing for peace (a peaceful past and/or future), pain and grief over their losses. In this section we discuss these different emotions in Palestine and Afrin, and show how they lead to an important form of individual and collective resistance through tree planting campaigns and organisations. As such, we aim to further highlight the social death experienced by the peoples in Palestine and Afrin as a result of ecocidal violence, while also emphasising their resistance and strategies to counter it.

Resistance in Palestine

Palestinians have cultivated olive trees for many generations, giving them a sense of belonging, rootedness, togetherness and pride (Joint Advocacy Initiative, 2022).

However, the destruction of their olive trees and the prevention of access to their land under Israeli rule has left Palestinians with feelings of anger, pain and sadness. Our key informant in Palestine says:

Talking about the psychological impact of all this [destruction of olive trees], it is very painful because every Palestinian sees the land that they inherited from their ancestors, played on as a child, and saw their father planting trees, and now they cannot access it. This naturally leads to feelings of frustration, helplessness and anger. Despite this, Palestinians try to resist the occupation, as we see in the news, but we are shot at, chased and imprisoned.⁴

In the cross section of these feelings, there is also a strong will to resist the loss, destruction and death that Palestinians are subjected to. As Mohamed says: 'Despite the occupation and the difficult circumstances in which we live, we still cling to our land!'.⁵ This was evident in peoples' descriptions about holding on to their lands in Ramallah and Bethlehem. A young man we met in his small garden, with the broken bricks of what was once a small shack, explained the ongoing struggle and personal hardship involved in holding on to his property. His house was destroyed and his trees damaged several times, but he said he would never let anyone take his land, no matter what.⁶ In another area in the West Bank, showing the young trees with thin, broken branches, another man said: 'This is the third time this has happened here, the settlers do it, especially at night'.⁷ He told about how they keep planting new olive saplings on their land to show that they had no intention of leaving.

A key example of resistance strategy is the determination to remain in their lands are the olive tree planting campaigns, as explained by a young Palestinian activist, who said

Our resistance strategy is to be here, to farm, to be present. A rather passive form of resistance. [...] So the main goal is to protect, to own the land, to keep the attachment to our land. Not because olive trees bring us money.⁸

There are many organisations involved in olive tree planting campaigns, such as the Joint Advocacy Initiative (JAI), the Land of Canaan Foundation and the Palestine Fair Trade Association (PFTA), the Arab Group for the Protection of Nature (APN) and others, supported by partner organisations in other parts of the world (e.g. Europe, North America, Asia). Olive tree planting campaigns have several objectives, including supporting Palestinian farmers, rehabilitating land and protecting local seeds. These campaigns also emphasise their support for wider Palestinian resilience and resistance. The US Campaign for Palestinian Rights uses the concept of 'rooting resistance', inviting supporters to 'root resistance by planting an olive tree today!' (US Campaign for Palestinian Rights, n.d.). The Million Tree Campaign (MTC) led by the APN also highlights the resistance to Israel's policy of destroying olive trees, with the slogan 'They Uproot a Tree ... We Replant Ten' (Arab Group For The Protection of Nature, 2014).

Olive tree plantation and protection in Palestine has 'become a method of nonviolent resistance' (Joint Advocacy Initiative, 2022, p. 3). The organisation has reportedly planted nearly 250,000 olive trees in its first twenty years, with a significant increase in numbers in 2021 and 2022 (Joint Advocacy Initiative, 2022). Islamic Relief Canada also reports on a tree-planting project in Palestine that it launched in November 2021 with a Palestinian partner organisation called the General Union of Palestinian Peasants

and Cooperatives (GUPPAC), which was expected to last until February 2023 and result in the planting of more than 40,000 olive tree saplings (Islamic Relief Canada, 2022).

Resistance in Afrin

What has happened in Afrin since its occupation is often be described in terms of the loss of family members, a sense of rootedness, belonging, and identity.

These olive trees, they are like my family. ... These trees are 50, 60, sometimes 100 years old. Across Afrin, they're [the occupying forces] cutting them down, they take the logs, and they sell them in Idlib [under the control of Syrian opposition forces], and no one can say a word in protest

says a man from Afrin, who was quoted in the Human Rights Watch report (Human Rights Watch, 2024a, pp. 52–53). Similar feelings were expressed by the key informants we interviewed. Azad speaks of the emotional and spiritual bond between the people of Afrin and the olive trees, and draws a clear link between the forced displacement from their land and the killing when he says 'Forced migration is tantamount to killing individuals in various forms'.⁹ Those who are forcibly displaced within Syria or elsewhere (e.g. Europe) often experience feelings of isolation, worthlessness, helplessness, anger, deep sadness, and social death as a result of being separated from their families and land.¹⁰ Another key-informant, Baran, explains how people, especially the elderly, have been very negatively affected by being cut off from their olive trees:

People know that when trees are cut down and gardens burned, part of their own deep-rooted historical relationships and social connections that have shaped the area for centuries are severed, destroyed. That's one aspect. We feel it. There is such a feeling among the people of Afrin, especially among the elderly, it's very strong, very visible ... So the destruction of the olive trees is also seen as their own destruction, which is not wrong ... So the relationship between human beings and nature is a whole; if you cut off one of its limbs, you're essentially cutting off humanity itself.¹¹

Despite these brutal feelings, our key informants in Afrin stress their strong determination to fight against these policies. One example of this is the gardening activities that people from Afrin carry out in their new settlements in other parts of North and East Syria. This is an important form of resistance in which the victims become actors, planting trees and bringing an ecological experience, awareness and culture to their new settlements. Another example is their determination to continue their struggle despite endless military offensives. Azad says that despite the drone attacks and military offensives of the Turkish armed forces, they are determined to create a confederation that will work on ecology and environmental movements in all four parts of Kurdistan.¹²

Although tree planting campaigns in Afrin are not yet on the same scale as in Palestine, there are attempts at reforestation by the Ecology Council, Economy Council, and Agriculture and Irrigation Council under the NES administration. However, it should be noted that the NES authorities are no longer in power in Afrin. There are tree planting campaigns in other parts of the North and East Syria involving displaced people from Afrin, who have greatly contributed to the agroforestry activities in other parts of Rojava they had to move to after Afrin's occupation.¹³ One news article reports that IDPs from Afrin 'continue to plant olive trees [...] as an expression of their attachment

to their occupied land' (ANHA, 2020). The same news article quotes a man, who says: 'Every time the mercenaries cut a tree, we plant 100 trees instead. Everyone resists in their own way; we resist by planting the seed' (ANHA, 2020). The EUTCC-Kurd-Akad report (Güven, 2023) underlines that 'reforestation is seen as a sustainable strategy, and trees are planted in the municipalities of Derik and Qamişlo' where thousands of trees have been reportedly planted, riverbeds were revived, and tree nurseries were started. The report goes on to say that organisations such as Make Rojava Green Again and the University of Rojava continue to act as key actors.

Comparative discussion and concluding reflections

While both Palestine and Afrin exemplify how environmental destruction is weaponised within a settler-colonial counterinsurgency framework, important differences shape the nature and implications of this violence. In Palestine, the destruction of olive trees has unfolded over decades, with a broader scope in terms of frequency, scale, and infrastructural impact – such as the construction of the separation wall. The perpetrators are clearly identifiable: the Israeli state and settler populations, operating within a legal framework that, while formally established, is selectively enforced. In contrast, the targeting of olive groves in Afrin is more recent, emerging in the aftermath of Turkey's military occupation and carried out by Turkish forces in coordination with allied armed groups. The absence of a stable legal regime – amid the ongoing political transition in Syria, marked by efforts toward constitutional reform under the newly declared transitional government in March 2025 – complicates accountability mechanisms. Moreover, while Palestinian collective action around olive cultivation and replanting is long-standing and highly organised, similar forms of ecological resistance are far less visible in Afrin. These differences underscore the need for a context-sensitive analysis of how ecocidal violence operates across distinct geographies, while still being anchored in shared logics of domination, erasure, and control.

There are also numerous similarities between the cases of Palestine and Afrin, particularly in how communities respond to the destruction of olive trees. Both Palestinians and Kurds experience profound emotional distress – marked by grief, anger, and helplessness – yet demonstrate remarkable resilience through acts of ecological resistance. Palestinians engage in replanting campaigns to maintain their connection to the land, while displaced people from Afrin continue to plant trees and preserve cultural practices. These responses reflect not only deep emotional and cultural ties to the land but also a shared determination to resist ongoing colonial violence. In both contexts, olive tree destruction functions as a tool of (settler-)colonial counterinsurgency, imposing social death by severing cultural bonds, degrading the environment, and disrupting social cohesion.

Finally, the cases of Palestine and Afrin underscore the limitations of the international legal and political system in addressing environmental harm in highly securitised and militarised contexts. Stateless nations face significant barriers in seeking justice and recognition, revealing the urgent need to decolonise legal frameworks and to acknowledge the agency of non-state actors in confronting the unfolding genocide – ecocide nexus. In particular, the comparison of olive tree destruction in Palestine and Afrin brings to light three interrelated issues that help explain why the international system struggles to

effectively address environmental degradation in such contexts. Our first point concerns the transboundary nature of ecocide, which requires a change in international law. The existing legal frameworks have significant gaps, including restrictive and ambiguous definitions of impermissible environmental damage, unclear protections for the environment as a civilian object, and issues with the proportionality principle, which weighs military advantage against environmental harm (Bothe et al., 2010). Since the 1970s, scholars, legal experts, and activists have been engaged in a long-running campaign for the recognition of ecocide under the Rome Statute, which would allow ecological destruction to be prosecuted under the jurisdiction of the International Criminal Court (ICC) (Jones, 2022). To date, severe environmental damage during war has only been criminalised as a war crime and related human rights violations, but not as a stand-alone element (Higgins et al., 2013). While making ecocide an international crime isn't a definitive solution that would stop states from committing this crime, it can help break down the hegemonic power structures that allow environmental destruction to continue without repercussions.

The second point relates to the state-centric structure of the existing legal system. Stateless groups such as the Palestinians and the Kurds cannot, under the existing legal structure, apply to the ICC for genocidal or ecocidal acts against their environments/ecosystems committed by states or corporations or their proxies. Recently, South Africa filed a complaint with the International Court of Justice (ICJ) against Israel for violating the Genocide Convention in Gaza. On 26 January 2024, the International Court of Justice ordered provisional measures and later ruled Israel's presence in Palestinian territories unlawful, demanding its end (International Court of Justice, 2024). In 2024, the European Centre for Constitutional and Human Rights and partners filed a criminal complaint with the German Federal Prosecutor against Turkey for crimes in Afrin (ECCHR, 2024). On 5–6 February 2025, the Permanent Peoples' Tribunal in Brussels preliminarily determined Turkey's actions in Afrin/Rojava since 2018 as ethnic cleansing, war crimes, crimes against humanity, and environmental destruction (54th SESSION OF THE PERMANENT PEOPLES TRIBUNAL ROJAVA vs TURKEY, 2025). Despite the importance of these legal actions, it is important to note that neither the Palestinian Authority nor the NES have legal state status, which prevents them from seeking justice in these state-centred structures that welcome occupying states, including Israel and Turkey.

The third issue concerns the lack of international attention and support for local environmental/justice movements. One of the key reasons for the lack (or insufficiency) of international NGO support for environmental protection in Palestine and Afrin is the securitisation and militarisation of these conflicts. The cases of Palestine and Afrin highlight the importance of olive tree planting campaigns and their dependence on international allies and supporters to succeed. Our key informants also highlighted that environmental organisations face a number of challenges in their work, including susceptibility to persecution, the lack of an 'online payment system', lack of international partnerships,¹⁴ and difficulties in reaching out to various NGOs working with the United Nations or the European Union.¹⁵ A common feeling among our key informants was that they were motivated to work harder and to develop relationships between societies in order to overcome these challenges.

In ancient times, planting olive trees was believed to ward off evil spirits and bring peace. In the face of the destruction witnessed in Palestine and Afrin, such beliefs may seem difficult to sustain. Yet, the resilience of Palestinians, Kurds, and other marginalised

and Indigenous peoples around the world – against relentless stigmatisation, criminalisation, and genocidal – ecocidal violence – offers a different kind of hope: that by holding on to their roots and their olive trees, they may one day cultivate the long-awaited freedom and peace their lands deserve.

Notes

1. Informed oral consent was obtained, and pseudonyms are used to protect participants' identities.
2. Also known as *sumoud* that means steadfast perseverance.
3. Baran, personal communication by Necmettin Türk, April 30, 2024.
4. Mohamed, personal communication by Pinar Dinc, October 27, 2023.
5. Mohamed, personal communication by Pinar Dinc, October 27, 2023.
6. Pinar Dinc, Travel Diary, January 2, 2023.
7. Pinar Dinc, Travel Diary, January 2, 2023.
8. Pinar Dinc, Travel Diary, January 2, 2023.
9. Azad, personal communication by Necmettin Türk, May 5, 2024.
10. Necmettin Türk, Travel Diary, May 24, 2024.
11. Baran, personal communication by Necmettin Türk, April 30, 2024.
12. Azad, personal communication by Necmettin Türk, May 5, 2024.
13. Necmettin Türk, Travel Diary, May 24, 2024.
14. Mohamed, personal communication by Pinar Dinc, October 27, 2023.
15. Baran, personal communication by Necmettin Türk, April 30, 2024.

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Chapter 5. Article III: Resisting colonial drought: Water crisis and socio-ecological transformation in Rojava/North and East Syria

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5.1. Abstract

This article introduces the concept of colonial drought to analyse the protracted water crisis in Rojava, framing it as the outcome of intersecting forces: internal colonialism under the Ba‘ath regime and Türkiye’s regional neo-colonial ambitions rooted in hydro-hegemony. Drawing on critical political ecology, the article critiques dominant narratives that attribute Syria’s rural collapse primarily to climate-induced drought or neoliberal governance failures. Instead, it foregrounds how Rojava was systematically underdeveloped as an extractive periphery, with both surface and groundwater resources overexploited through monocultural agrarian strategies. Simultaneously, it explores how Türkiye’s control over transboundary waters has functioned as a geopolitical weapon against Kurdish autonomy. In response to these layered crises, the research examines how Rojava has, since 2012, reimagined governance around principles of direct democracy, ecology, social economy, and women’s liberation. It aims for a structural and holistic socio-ecological transformation, which is conceptualized as the ontology of socio-ecological transformation, grounded in rhizomatic governance—a non-hierarchical, relational, and pluriversal mode of organizing life.

Keywords: Water crisis, colonial drought, internal colonialism, climate change, Rojava, socio-ecological transformation, rhizomatic governance

5.2. Introduction

This article critically investigates the structural drivers of the protracted water crisis in Rojava (North and East Syria), conceptualizing it as a colonized yet insurgent geography—marginalized through historical processes of territorial control and resource extraction, yet reconstituted as a site of ecological resistance and autonomous governance. Drawing on the conceptual lens of internal colonialism demonstrates how the region’s marginalization was institutionally entrenched through state-engineered hydrological and socio-spatial designs. Simultaneously, it scrutinizes Türkiye’s transboundary water regime and regional hydropolitical strategies, revealing how water control has functioned as a key geopolitical mechanism for suppressing Kurdish claims to autonomy. Both the Syrian Ba‘ath regime (Barnes, 2009; Tinti, 2023; Türk & Jongerden, 2024) and Turkish state (Harris, 2002, 2006, 2012; Jongerden, 2007, 2010; Bilgen, 2017, 2021; Akinci et al., 2020; Türk, 2023) have constructed extensive dam networks, irrigation systems, and agricultural megaprojects. These socio-spatial configurations were not neutral instruments of national development, but colonial infrastructures aimed at consolidating state-building and economic centralization while Kurdish regions have been underdeveloped as peripheries by colonial dispossession, displacing, and assimilating. Water, in this context, emerges not as a mere resource, but as a weaponized medium of domination and territorial control. To conceptualize these intersecting dynamics, this article advances the framework of *colonial drought*.

Extensive academic debate has addressed the links between climate change, drought, migration, and conflict in Syria—particularly claims that prolonged drought catalysed agricultural collapse and social unrest leading to the 2011 uprising (Gleick, 2014; Kelley et al., 2015; Ash & Obradovich, 2020; Werrell et al., 2015; Trigo et al., 2010). More recent scholarship, however, shifts attention to structural economic reform, neo liberalization, and resource mismanagement as more decisive causal factors (Selby et al., 2017; Selby, 2018; Daoudy, 2020a; Ababsa, 2015; Mohr, 2021; Fröhlich, 2016; Eklund & Thompson, 2017; Hoffmann, 2018; Eklund et al., 2022; Dinc & Eklund, 2024). While these studies provide valuable insights into the socio-economic collapse of rural Syria, they often neglect the specific ecological degradation and historical

targeting of the Haseke/Cezîrê (Jazira) region—an area known for its agricultural fertility and water abundance. By reducing the crisis to issues of governance failure, neoliberal policy, or climate stress, much of this literature depoliticizes the crisis by failing to account for its colonial origins. Although several scholars have critically engaged with these dynamics, literature directly addressing internal colonialism and hydro-hegemonic dimensions in the Syrian context remains extremely limited. To date, only a few contributions, notably by Barnes (2009), Ababsa (2015), Selby (2018), Hoffmann (2018), Türk and Jongerden (2024) and Türk (forthcoming 2025), have explored some aspects of this perspective. As this research contends, the root of the collapse lies in the Ba‘ath regime’s treatment of Rojava as an extractive periphery—deliberately underdeveloped within a centre–periphery logic of internal colonialism. Through monocultural agrarian exploitation and excessive extraction of underground and surface water resources, the Syrian state catalysed ecological degradation, soil exhaustion, and the depletion of vital aquifers. Türkiye’s assertion of regional hydro-hegemony—embodied in its expansive damming of the Euphrates and Tigris rivers—has compounded the devastation wrought by Syrian internal colonialism. We frame this dual mechanism—Syrian internal colonialism and Turkish neo-colonial hydro-political domination—as colonial drought.

Unpacking these dynamics demands not only critical theoretical analysis but also a method rooted in solidarity and grounded resistance. This research is anchored in a solidarity-based, engaged methodology informed by critical political ecology and decolonial theory. The lead author’s long-standing participation in ecological movements across Kurdistan and Europe—through organizing, conferences, and direct collaboration—has shaped both the theoretical framework and field practices. This positionality fostered a commitment to ethical co-production of knowledge with local communities, activists, and scholars. Methodologically, the study employs qualitative and ethnographic tools including engaged observation, structured field surveys, in-depth interviews, group discussions, and cooperative monitoring across three strategic districts: Dêrik, Tirbespiyê, and Heseke. Fieldwork conducted between April and June 2024 included 17 deep interviews and 50 surveys in 12 villages, complemented by 24 expert interviews conducted remotely throughout 2023. Surveys captured local perceptions of drought, colonial legacies, and resource management, while interviews explored the structural, ideological, and practical barriers to socio-ecological transformation amid protracted conflict. This methodology adopts relational, reciprocal, and decolonial modes of inquiry. Through collective analysis, dialogical reflection, and embedded engagement, it articulates an ethical and grounded research practice aligned with the radical political aspirations of the Rojava

Revolution. The project also sought to enable horizontal learning between the University of Hamburg (Germany) and the University of Rojava (Rojava/Syria), challenging the epistemic asymmetries that often structure North–South academic partnerships (Schmidt & Neuburger, 2017).

The article proceeds in two parts. The first interrogates the historical formation and structural logics of colonial drought, examining its socio-economic, ecological, political, and cultural effects. The second explores how, despite sustained marginalization, conflict, and blockade since 2011, Rojava/the Autonomous Administration of North and East Syria (AANES) has emerged since 2012 as a pluriverse. Rooted in direct democracy, ecology, social economy, and gender liberation, Rojava has forged an emancipatory political ontology. This transformation unfolds amid global ecological crises deepened by climate change and colonial extractivism. In contrast to technocratic, adaptation- or mitigation-focused paradigms, the Rojava model calls for a fundamental rethinking of the relationship between society and nature through a new ontological and epistemological praxis. It enacts not only alternative forms of knowing and organizing, but also a novel aesthetic of ecological coexistence. The central claim of this article is that Rojava’s socio-ecological transformation constitutes a decolonial project at both ontic and epistemic levels. It offers new ways of being, knowing, and doing for human and nonhuman communities alike. We conceptualize this as the ontology of socio-ecological transformation, emerging through *rhizomatic governance*—a non-hierarchical, relational, and pluralistic form of life organization within the pluriverse.

5.3. Water, climate-change and colonial drought in Rojava

*What in water did Bloom, waterlover, drawer of water, watercarrier returning
to the range, admire?*

*Its universality: its democratic equality and constancy... its vastness in the
ocean...unplumbed profundity...the restlessness of its waves and surface
particles...the independence of its units: its hydrostatic quiescence in calm: its
hydrokinetic turgidity in neap and spring tides: its subsidence after devastation:
its sterility... its climatic and commercial significance: its capacity to dissolve
and hold in solution all soluble substances... its alluvial deposits: its weight and*

volume and density: its vehicular ramifications in continental lake... its secrecy in springs, and latent humidity... the simplicity of its composition... its healing virtues: its buoyancy... persevering... properties for cleansing, quenching thirst and fire, nourishing vegetation: its infallibility as paradigm and paragon: its metamorphoses as vapour, mist, cloud, rain, sleet, snow, hail: its strength in rigid hydrants: its variety of forms... solidity... docility in working hydraulic millwheels, turbines, dynamos, electric power stations... its utility in canals, rivers...its submarine fauna and flora...

(Joyce, 2000, pp. 783–785)

In the “Ithaca” episode of *Ulysses*, James Joyce presents water not merely as a natural element, but as a symbolically charged and multi-dimensional force of life, transformation, and knowledge. Joyce depicts water as simultaneously purifying and destructive, static and dynamic, generative and fatal. He emphasizes water’s democratic quality, its tendency to seek its own level. With its capacity to dissolve and contain all soluble substances, water assumes the role of a kind of cosmic memory. Ultimately, it emerges not simply as a natural phenomenon, but as a powerful metaphor situated at the intersection of cosmology, epistemology, and ontology (Joyce, 2000, pp. 783-785). The remarkably vivid portrayal of water in a literary novelist’s work might seem surprising at first. However, considering water’s indispensable role in the existence of both human and non-human entities, its central place within the social sciences is far from unexpected, especially in political ecology area. This symbolic complexity provides a fitting entry point for examining water through the lens of political ecology, which emphasizes water as a site through which power, domination, and resistance are enacted and contested (Swyngedouw, 2009; Bakker, 2012, 2013a, 2013b; Schmitt, 2022). Political ecology critically interrogates the socio-political and colonial structures embedding water within material practices and symbolic discourses. The commodification of water under capitalist systems, alongside community-led resistance that frames water as an existential struggle, continues to shape ideological frameworks, discourses, and policy formations. In this regard, the concept of “modern water,” articulated by Linton (2010), frames water as an abstract, quantifiable commodity within capitalist societies. This abstraction facilitates water’s commodification, privatization, and extraction, exacerbating inequalities and conflicts such as water grabbing (Bakker, 2012, 2013a, 2013b; Franco et al., 2013; Schmitt, 2019). Water has been reconstituted as one of the core commodities underpinning contemporary neo-colonial structures. This shift illuminates how control over

water resources has historically been central to maintaining colonial and authoritarian state power (Scott, 1998; Mitchell, 2002; Whitehead et al., 2007; Harris, 2002, 2012; Wittfogel, 1957; Schmitt, 2017; Estes, 2019; Hartwig et al., 2022; Swyngedouw, 2023). Hydraulic infrastructure, such as dams and irrigation canals, symbolizes state authority and facilitates socio-economic domination, demographic manipulation, and resource extraction—clearly illustrated by Kurdish regions subjected to water control by occupying states, namely Türkiye, Syria and Iran (Harris, 2002, 2006, 2012; Jongerden, 2010; Bilgen, 2017, 2021; Akinci et al., 2020; Türk, 2023; Bezahler, 2024; Wiktor-Mach et al., 2024), Iran (Hassaniyan & Sohrabi, 2022; Hassaniyan, 2024) and Syria (Barnes, 2009; Daoudy, 2020b; Tinti, 2023; Türk & Jongerden, 2024). This chapter investigates the structure and impacts of colonial drought on the quantity and quality of water in Rojava.

5.3.1. Water in Rojava: Access amid climate change, conflict, and pollution

Syria's water crisis cannot be understood without reference to its underlying climatic vulnerability, the uneven hydro-geographies of its territory, and decades of environmental degradation rooted in both natural and political processes. Rojava lies primarily in the semi-arid to arid zones (Aydin-Kandemir & Yıldız, 2022, p. 3; FAO, 2018). Temperature extremes are frequent—hot, dry summers with average highs above 40°C—contributing to intense evapotranspiration rates and seasonal water stress (Aydin-Kandemir & Yıldız, 2022, p. 2; Water Directory of Hasaka, 2024, p. 1). Climatically, the region receives annual rainfall between 250 and 400 mm, which is barely sufficient for rainfed agriculture and highly vulnerable to drought cycles (Aydin-Kandemir & Yıldız, 2022, p. 3; Baba et al., 2021, p. 1). Water availability in Rojava relies on three main sources: surface water (especially the Euphrates-Tigris basin), groundwater (artesian and shallow wells), and seasonal rainfall. The Euphrates alone provides nearly 85 per cent of Syria's surface water supply (Baba et al., 2021, p. 4), and groundwater supports around 60 per cent of irrigated lands nationwide (ibid., p. 7). However, both sources have deteriorated sharply in quantity and quality. Groundwater over-extraction is particularly acute. In NES, aquifer levels have declined to their lowest point in 20 years (Baba et al., 2021, p. 14; REACH, 2023, p. 1), with an average annual drop of few meters. As of 2021, groundwater accounted for 75 per cent of irrigation in Jazire and Hasakah, and 213,248 private wells had been drilled across the region—most unregulated and unsustainable (Aydin-Kandemir & Yıldız, 2022, pp. 5-6). Water tables have collapsed due to years of drought, and intensified

pumping. Recharge has become insufficient even in wet years, resulting in a structural groundwater deficit (Aydin-Kandemir & Yıldız, 2022; Baba et al., 2021).

The quality of water is no less dire. Groundwater quality across Rojava has been critically undermined by overexploitation, pollution, and salinization (see Figure 3). Scientific surveys indicate that up to 43 per cent of Syria's groundwater contains elevated levels of SO_4^{2-} and NaCl , especially in Hasakeh and the Euphrates basin, where TDS levels reach 2500 to 3500 mg/l—far exceeding WHO potable water standards (Aydin-Kandemir & Yıldız, 2022, p. 6; Baba et al., 2021, p. 7). Salinization has been exacerbated by declining recharge due to drought and upstream damming by Türkiye.

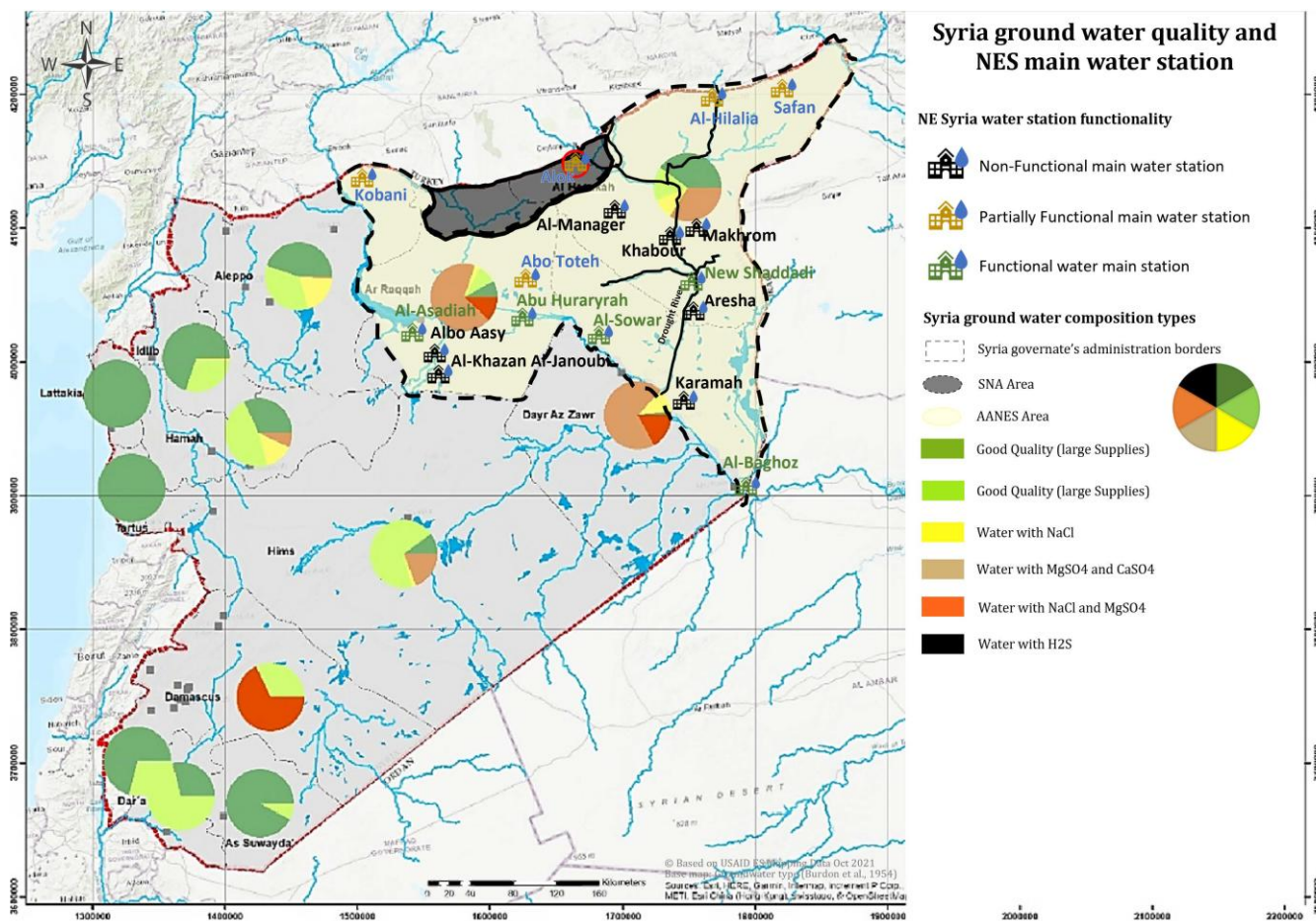


Figure 3: Syria ground water quality and NES main water stations (based on Burdon et al., 1954)

The Euphrates River itself has suffered massive flow reductions. Following the 1987 bilateral agreement with Syria, Türkiye was to release 500 m^3/s across the border. However, since 2019,

the flows released by Türkiye have dropped dramatically, reaching as low as 160–214 m³/s (Water Directory of Hasaka, 2024, p. 2; Daher, 2022, p. 13). This decline has brought Tabqa and Tishrin dams close to dead level, crippled hydroelectric output, and disrupted water delivery to over 5.5 million people in NES. Since Türkiye's 2019 occupation of Serekaniye, the Alouk water station—serving nearly one million people in Hasakah—has been intermittently cut off, leaving the city in disaster conditions till today (Daher, 2022, p. 13; Daoudy, 2023, p. 15; Daoudy, 2020b, p. 1362; REACH, 2023, p. 13; Water Directory of Hasaka, 2024, p. 13). Due to high salinity, Hasaka cannot rely on local groundwater; within a radius of more than 50 km around the city, the water is chemically unsafe (Water Directory of Hasaka, 2024, p. 11). As a result, the population now depends on trucked water, which is insufficient and exorbitantly expensive, reaching 7–10 per cent of household income (Water Directory of Hasaka, 2024, p.10; REACH, 2023, p. 13). Public health consequences are also profound. Outbreaks of diarrhea, leishmaniasis, hepatitis A, and cholera have surged, especially in IDP camps and areas relying on contaminated sources (Water Directory of Hasaka, 2024, p. 13; Daher, 2022, pp. 16–17; Mueller et al., 2021, p. 18).

The Turkish state's manipulation of transboundary water is not just infrastructural or environmental—it is disciplinary, seeking to fragment Kurdish autonomy and disrupt democratic governance models in Rojava. The weaponization of water, combined with the impacts of climate change and institutional collapse, threatens to render Rojava ecologically uninhabitable and politically unstable. Water is not only scarce but increasingly toxic, expensive, and contested. The declining quantity and quality of water in Rojava—manifested in desiccated rivers, salinized aquifers, and polluted canals—is both a driver and a symptom of a broader ecological collapse. Understanding this conjuncture is critical to any meaningful conception of colonial drought. It is not only the absence of rain or the effects of climate change; it is also the systemic degradation, destruction, and weaponization of life-sustaining hydrological systems by water politics and colonial structures.

5.3.2. From colonial drought to climate change: Water crisis and socio-economic breakdown

Although drought initially evokes meteorological and hydrological phenomena, it is, in fact, deeply embedded within complex socio-economic and political structures. Drought simultaneously shapes—and is shaped by—these broader configurations (Schmitt, 2017).

While meteorological droughts have historically occurred as natural events, the contemporary manifestation of drought, intensified by climate change, exhibits increased frequency and severity. Syria ranks among the states most acutely impacted by climate change-induced drought, a phenomenon widely recognized as a significant driver of internal displacement and political unrest, culminating in the 2011 uprising. (Gleick, 2014; Kelley et al., 2015; Ash & Obradovich, 2020; Daoudy, 2023). Geographically situated within the Fertile Crescent and overlapping the Mediterranean/MENA regions, Syria has been acutely exposed to the tangible symptoms of climate change: rising temperatures, drier winters, reduced precipitation, and elevated evapotranspiration rates (Gleick, 2014, p.338; Kelley et al., 2015, p.3241). Meteorological data between 2007 and 2010 reveal that Syria's rainfall dropped dramatically relative to long-term averages (see Figure 4), while mean annual temperatures rose as it was “the worst drought in the instrumental record” (Kelley et al., 2015, p. 3241).

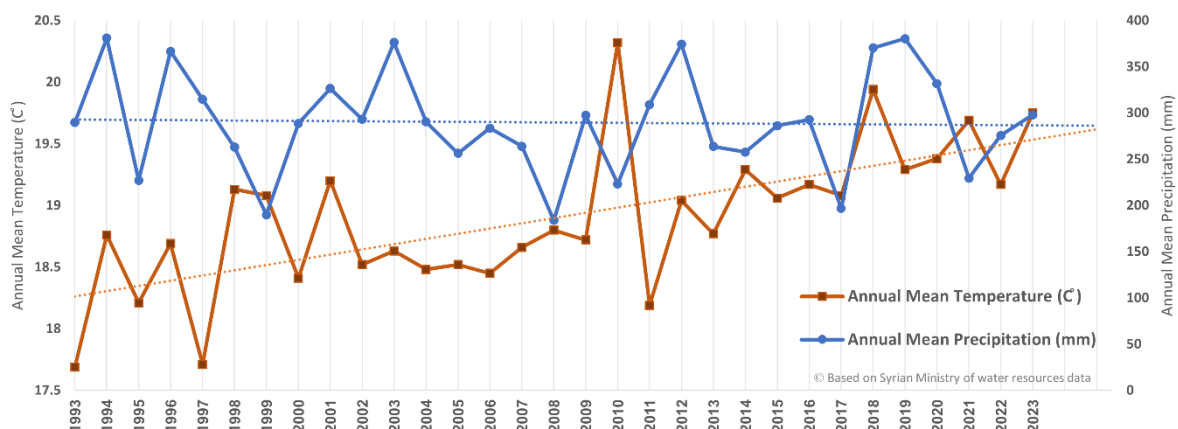


Figure 4: Long-term precipitation and temperature in Syria (based on data from the Syria Ministry of Water)

The Jazira region—historically Syria’s breadbasket—was particularly devastated: wheat and barley yields declined by 47 and 67 per cent respectively (Ababsa, 2015, p. 204; Selby, 2018, p. 262) and experienced a loss of 85 per cent of livestock and three-quarter of crops (Gürçan, 2019, p.9), and dramatic groundwater depletion (Kelley et al., 2015, p. 3241). The socio-economic ramifications of these climatic shifts have been devastating. Agriculture, historically contributing 25–30 per cent of Syria’s GDP and employing a substantial rural workforce, collapsed under drought stress due to mass migration (Kelley et al., 2015, p. 3241; Selby, 2018, p. 266; Barnes, 2009, p. 525; Asfour & Hovil, 2024, p. 11). Rural depopulation was driven not solely by crop failure, but also by rapidly escalating food prices, infrastructural decay, and

systemic socio-political marginalization and corruption (Daoudy, 2023, p. 8; Asfour & Hovil, 2024, p. 11; Kelley et al., 2015, p. 3241-42; Mueller et al., 2021, p. 32). However, it is highly insufficient to explain the socio-economic collapse experienced in Rojava's Cezire region, particularly the breakdown of the agricultural sector and the associated internal displacement, primarily through climate change and meteorological drought. In fact, climatic factors may account for only a minimal part of the crisis. Researchers further critique simplistic degradation narratives, highlighting that structural economic reforms, resource mismanagement, and neo liberalization played a more decisive role (De Châtel, 2014; Selby et al., 2017; Selby, 2018; Daoudy, 2020a; Ababsa, 2015; Mohr, 2021; Fröhlich, 2016; Eklund & Thompson, 2017; Gürcan, 2019; Hoffmann, 2018; Eklund et al., 2022; Dinc & Eklund, 2024; Kelley et al., 2015, p. 3242). Crucially, comparative studies demonstrate that similar climatic pressures in southeastern Türkiye and Iraq did not precipitate comparable socio-economic breakdowns (Eklund & Thompson, 2017; Eklund, 2022), emphasizing the mediating role of governance, resource management, infrastructure and socio-political frameworks.

While the impacts of climate change-induced drought are undeniable, the root causes of the water and agricultural crises in Rojava are more deeply entangled with historical colonial legacies and contemporary Türkiye neo-colonial hydro-hegemony (Zeitoun & Warner, 2006). None of the conventional drought categories—meteorological, hydrological, agricultural, and socio-economic (Wilhite & Glantz, 1985, pp. 113–116)—nor more recently added types such as ecological, flash flooding, or groundwater drought adequately capture the dynamics at play in Rojava. To address this conceptual lacuna, we developed the concept of *colonial drought*, which is the systemic production and intensification of drought conditions through internal colonization and neo-colonial hydro-hegemony, where water resources are controlled and extracted to consolidate power over peripheral regions. It emerges through infrastructure-driven hydrosocial restructuring, thereby transforming environmental design into a strategic instrument of domination. In this context, internal colonization—such as the Ba‘ath regime’s internal colonization policies toward Rojava—and regional hydro-hegemony—epitomized by Türkiye’s Southeastern Anatolia Project (GAP)—operationalize strategies of water control and extractivism to reinforce center–periphery asymmetries. Through large-scale infrastructural projects (dams, irrigation canals), expanded technocratic-bureaucratic apparatuses, and security interventions, these mechanisms reconfigure hydrosocial territories, transforming water from a common ecological resource into a tool of political domination and economic extraction. The impacts of colonial drought manifest along three interwoven dimensions: ecological

degradation, marked by pollution, desertification, aquifer depletion, and ecosystem collapse; socio-economic collapse, characterized by the destruction of subsistence agriculture, rural proletarianization, and forced displacement; and political-cultural marginalization, involving the erosion of local governance structures, the dispossession of indigenous cultural practices, and the suppression of autonomous hydro-social orders (see Figure 5). Colonial drought thus transcends climatic determinism, positioning environmental degradation within the broader structural violences of internal colonialism, resource extraction, and structural socio-political domination and socio-spatial design.

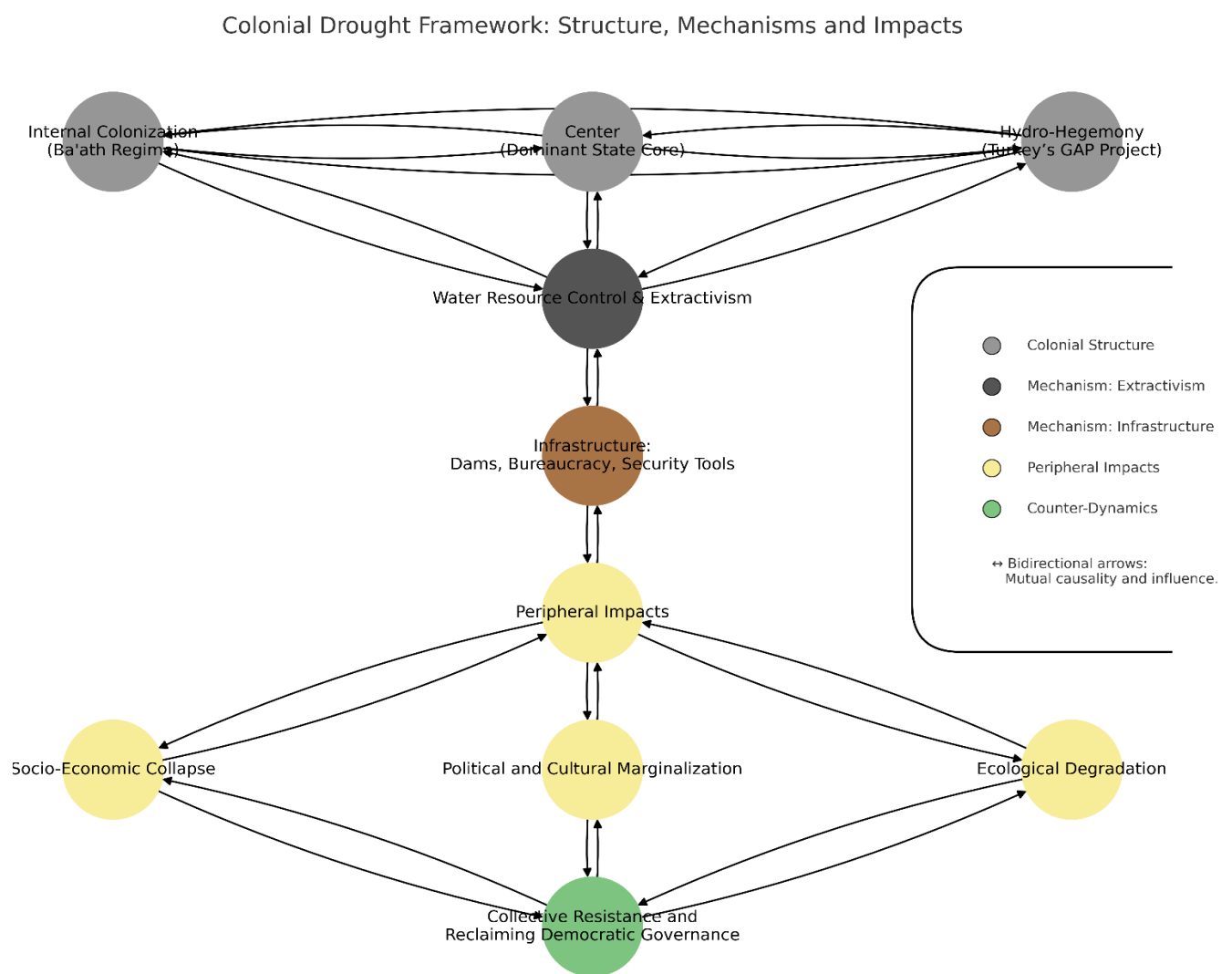


Figure 5: The structure of colonial drought and the dynamics of resistance

5.3.3. The structure of colonial drought: Internal colonization

Dams and related agricultural modernization in Rojava cannot be reduced to a technocratic hydro-economic and development project. Rather, they constitute a central mechanism of state formation and internal colonization, through which the Baathist regime restructured territory, population, and resources to consolidate power (Barnes, 2009; Daoudy, 2020b; Türk & Jongerden, 2024). Thus, water governance and hydraulic infrastructures must be viewed as socio-technical designs that are fundamentally shaped by and constitutive of state power, spatial ordering, and modalities of control (Harris, 2002, 2012; Jongerden 2010; Bilgen, 2017, 2021; Menga, 2015, 2016; Menga & Swyngedouw, 2018; Türk, 2023; Barnes, 2009; Scott, 1998; Mitchell, 2002). Dams, irrigation schemes, and agricultural subsidies under the Ba'ath regime were not merely hydro-developmental interventions; they were instruments of internal colonization that restructured the socio-political and ecological fabric of Rojava, transforming it into a peripheral extractive zone, strategically marginalized from the center (Türk & Jongerden, 2024).

In this regard, the construction of dams in Syria—especially the Euphrates' Tabqa Dam (1973)—exemplifies the authoritarian modernist state's performance of hydraulic sovereignty. The Tabqa Dam and Lake Assad, like Türkiye's Atatürk Dam, must be seen as projects that materialize state sovereignty and redesign socio-spatial relations to make populations legible, governable, and extractable. These monumental structures materialized the state's technological rationality, mobilizing water as an economic extractivism, showpiece of state image and a disciplinary medium. As Timothy Mitchell (2002) states that, "large dams offered a way to build not just irrigation and power systems, but nation-states themselves" (p. 44). These infrastructures reorganized waterscapes, erased existing hydro-social relations, and produced legibility and governability in the Foucauldian sense (Akinci et al., 2020). Much like Türkiye's Southeastern Anatolia Project (GAP), Syria's dam regime symbolized territorial socio-spatial design—drawing not only borders and basins, but boundaries between citizens and subjects, centers and margins. Water thus became a terrain of both state formation and domination (see Türk & Jongerden, 2024). These processes are neither neutral nor benign. This was carried out through a colonial logic that integrated hydraulic infrastructure, demographic engineering, and monocultural agricultural expansion. As Türk and Jongerden (2024) emphasize, this was not simply modernization and development—it was colonization and de-development of Rojava by

agriculture-based nationalist development. In doing so, Rojava's Jazire region—especially Hasakah—was transformed into an agro-extractive periphery dedicated to “strategic crops” such as wheat, barley, and cotton (Hinnebusch et al. 2011; Türk & Jongerden, 2024, p.5). These crops were irrigated largely through unsustainable groundwater extraction: by the early 2000s, 99 per cent of Syria's sustainable water yield was consumed by irrigation, with 85–90 per cent of total water use allocated to agriculture, and 60 per cent of that dependent on groundwater (Varela-Ortega & Sagardoy, 2003, p. 343). In Hasakah, groundwater depletion exceeded 300 per cent of the basin's safe yield by 2001 and surface flows such as the Khabour and Yarmouk Rivers dried up well before the 2007–2009 drought (Selby, 2018, p. 266). Far from being a result of population growth or climatic factors alone, this water crisis was structurally induced by a state-led model of resource overexploitation tied to center-periphery extraction. The conversion of the Jazira into Syria's breadbasket—producing up to 70 per cent of the country's wheat and 69 per cent of its cotton (Ababsa, 2015, p. 201)—was accompanied by severe socio-ecological degradation as an extractive zone. State-supported irrigation expanded from 650,000 ha in 1985 to over 1.4 million ha by 2005 (Barnes, 2009, p. 515). This model did not aim to develop Rojava in any inclusive sense, but to extract raw materials (e.g., cotton, wheat) for processing and consumption in and for “center” (Türk & Jongerden, 2024). As a result of this de-development project, by 2003/04, 58 per cent of Syria's poor resided in the northeast, with Hasakah experiencing the deepest rural poverty and highest unemployment rates (Selby, 2018, p. 266). This reflects what Roy (1995), Yadirgi (2020) and Türk and Jongerden (2024) identify as “de-development” under internal colonialism: the periphery is made to serve the center's developmental needs while itself being impoverished, marginalized, and rendered dependent. This transformation was part and parcel of a broader strategy of demographic engineering, through which the Ba'ath regime's infamous Arab Belt project (1962–1975) sought to dispossess and displace Kurdish populations and Arabization of the region as it is conceptualized as nation-state colonialism, a specific variant of internal colonialism (see Türk & Jongerden, 2024). Rather than being driven by population growth or environmental limits, scarcity was actively constructed through a development model that subsidized large-scale irrigation, prioritized cotton exports, and ignored ecological sustainability. That is why water crisis in Syria was framed as apolitical and ahistorical, masking the role of agricultural policy in generating crisis (Barnes, 2009, p. 519).

In sum, Rojava's water crisis is not reducible to environmental scarcity, nor can it be fully captured by notions such as “internal frontier” (Selby, 2018) which obscure the asymmetric

power and extractive logic at play (Neuburger, 2017). Rather, the region's socio-ecological crisis is best understood as a manifestation of colonial drought: a condition systematically produced by internal colonization through agricultural extractivism under a state-led nationalist agriculture-based development project. This framing reveals how water becomes both a target and a tool of internal colonization, woven into a wider matrix of territorial domination, cultural erasure, and environmental exploitation through socio-spatial design technologies. The drought of 2007–2010 cannot be understood outside this structural context. The crisis was not merely triggered by environmental shock but produced by decades of colonial political economy: the destruction of water resources by extractivist agriculture. As Selby (2018) states that “...vulnerability to drought always has political-economic causes, rooted in the extension of, and contradictions inherent in, capitalist economic relations and associated processes of state-building, colonisation and resistance.” (p. 272).

5.3.4. The structure of colonial drought: Hydro-hegemony and transboundary waters

Water, as a life-sustaining ecological commons, has increasingly been territorialized and politicized through modern nation-state projects. In the contemporary Middle East, no case exemplifies this more clearly than the politics of the Euphrates–Tigris basin (see Figure 6). The nation-state has historically mobilized hydraulic infrastructure as a central instrument of territorial governance, materializing sovereignty through dams, irrigation networks, and large-scale water transfers (Tinti, 2023). Rojava's hydrological reality is dominated by its position downstream in the Euphrates-Tigris basin, one of the most conflict-prone transboundary watersheds in the world. Türkiye controls over 90 per cent of the Euphrates' flow and over 50 per cent of the Tigris (Mueller et al., 2021, p. 12; FAO, 2009). As the upstream power par excellence, Türkiye controls the headwaters of both rivers, granting it a significant geostrategic advantage over Syria and Iraq. The 1987 Türkiye–Syria protocol was an interim and bilateral agreement that has proven ineffective in providing a sustainable solution, as Türkiye has consistently failed to implement its provisions (Tinti, 2023, p.185; Kibaroglu, 2014, p. 3; Daher, 2022, p. 8; Mueller et al., 2021, p. 25). Through its Southeastern Anatolia Project (GAP), launched in 1970s, Türkiye constructed 22 dams and 19 hydroelectric plants (see Figure 6), significantly reducing the downstream flow of the Euphrates (Daoudy, 2023, p. 9). This massive infrastructural expansion, justified as a development initiative, is in reality a textbook case of hydro-hegemony—the consolidation of state power internally (Jongerden, 2010; Türk, 2023)

and control over transboundary waters through asymmetric power, coercive diplomacy, and infrastructural dominance (Zeitoun & Warner, 2006). In this framework, power is exercised not just through infrastructure like dams, but also through legal framing, diplomacy, and strategic inaction. For instance, Türkiye has consistently refused to sign the 1997 UN Convention on the Non-Navigational Uses of International Watercourses, asserting a national-sovereign rights approach and rejecting co-sovereignty claims by downstream riparians (Mueller et al., 2021, p. 55; Kibaroglu, 2014, p. 8; Daoudy, 2023, p. 9). Its official policy treats transboundary rivers not as shared international waters but as resources under exclusive territorial sovereignty.

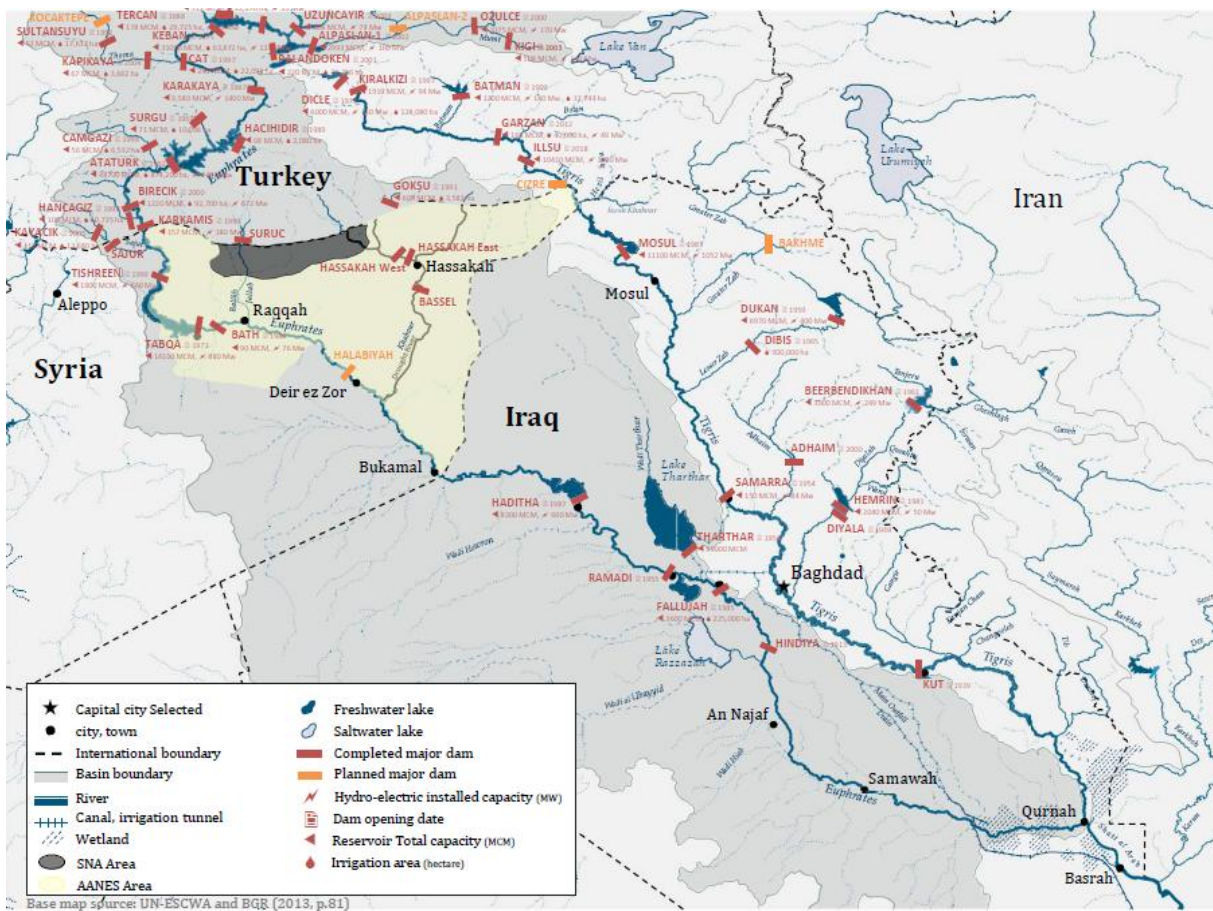


Figure 6: The Euphrates–Tigris Basin and Dams (based on UN-ESCWA & BGR, 2013, p. 81)

The consequences of Türkiye’s hydro-hegemony are acute in Syria, particularly in Rojava’s Cezire (Jazira) region, where water crisis has been systematically exacerbated (Daher, 2022). The GAP has severely curtailed Euphrates flows into Syria—by some estimates, by more than 40 per cent—contributing to aquifer depletion, the desiccation of the Khabur River, and

recurring agricultural collapse². This infrastructural intervention parallels the Ba‘ath regime’s internal colonization of Rojava via large dams, extractive agriculture, and demographic engineering. Both regimes—Ankara and Damascus—have pursued similar strategies of state-building through water control, seeking to assimilate, dispossess, and marginalize Kurds economically and culturally. Hydro-hegemony becomes particularly stark in Türkiye’s use of water weaponization. Weaponized water denial in Rojava mirrors historic practices of water warfare, as detailed by Daoudy (2020b), where hydraulic infrastructure becomes a military and political tool of domination. In this sense, Türkiye’s hydro-hegemony is not limited to upstream control—it extends into geopolitical strategies of subversion against Kurdish self-governance and autonomy. This strategic use of water as a tool of coercion and domination demonstrates the multi-scalar implications of colonial drought. Türkiye’s upstream position, infrastructural capacity, military leverage, and refusal to engage in equitable legal frameworks create a structural vulnerability for downstream communities. Therefore, Türkiye’s control of the Euphrates and Tigris rivers exemplifies the structural dimensions of water scarcity in Rojava. (Daoudy, 2020b; Daher, 2022; Tinti, 2023). Thus, understanding drought in Rojava requires going beyond meteorological explanations to include these political and infrastructural dynamics. As such, the concept of colonial drought becomes essential: it captures the intentional, systemic, and violent production of water scarcity as a tool of colonial domination. In this framework, water ceases to be merely a natural resource and becomes a medium of colonial power, resistance, and socio-ecological struggle. This makes the concept of colonial drought indispensable for understanding water crisis in the region—not as a climate-induced anomaly but as a politically produced condition rooted in state violence, infrastructural power, and hydrological colonialism as implemented by similar policies by Türkiye in Bakur Kurdistan. Drought, in this context, is not merely the absence of water, but the presence of colonial domination.

² Leyla Saruxan and Mehran Ahmed, co-chairs of the Agriculture and Irrigation Council, 21 November 2023; Selah Hemza Mihemed, co-chair of the Agriculture and Water Authority, Cizîrê/Derik, 2 December 2023; Aldar Hesên Xelef, Water & Irrigation Department, Hesekê, 21 May 2024; Fehîm Hüseyn, Public Water Authority, Qamishlo, 28 May 2024; Berivan Silo, co-chair, General director of Water Administration in Qamishlo, 28 May 2024.

5.4. Collective resistance and rhizomatic governance: Envisioning socio-ecological transformation

Unequal access to water and water scarcity are not simply natural conditions; they are politically produced and organised by colonialism and capitalism, operating through the nation state (Boelens et al., 2018; Strakosch & Macoun, 2012; Wolfe, 2006; Berry & Jackson, 2018; Sultana, 2018; Estes, 2019; Gibbs, 2009; Mathur & Mulwafu, 2018; Robison et al., 2018). Indigenous and local communities have historically developed deep, situated knowledge of hydrological cycles, land management, and resource stewardship (Hartwig et al., 2022). Yet colonial and capitalist systems have violently disrupted these practices through commodification, land and water grabbing/dispossession, forced sedentarization, and extractive development projects (Mehta et al., 2012; Austin-Broos, 2013; Berry & Jackson, 2018; Poelina et al., 2019; Hartwig et al., 2022). What Robison et al. (2018) call “water colonialism” refers to this multifaceted process: the dispossession, exclusion, and marginalization of Indigenous peoples and local communities through laws, infrastructures, discourses, and paradigms that assert settler and state control over water (Hartwig et al., 2022, pp. 34–35). This is not merely a historical injustice—it is ongoing and operates through new regimes of transboundary water control, large-scale irrigation schemes, dam construction, and private water grabbing—all of which perpetuate patterns of extraction and marginalization (D’Souza, 2006; Mehta et al., 2012; Mathur & Mulwafu, 2018; Hartwig et al., 2022). In Rojava, such dynamics are visible in Türkiye’s hydro-hegemony and the long legacy of Syria internal colonialism. However, colonial drought is not simply an end condition—it is a terrain of struggle. While its infrastructural violence seeks to reproduce hierarchies of control, silence local knowledges and governance, and deepen peripheral dependency, it simultaneously provokes resistance, solidarity, and reclamation. This chapter situates resistance not as reactive or marginal, but as a constitutive counterforce—emerging from below, from within the ruins of dispossession and destruction. In Rojava, what unfolds is not resilience or adaptation, but an insurgent hydropolitics that challenges the epistemic and material foundations of colonial water regimes through reconstruction of democratic governance and transformation of socio-ecological terrain. To this end, it advocates for a paradigm shift toward socio-ecological transformation, a framework grounded in justice, plurality, and decolonial politics. Emphasizing the significance of alternative ontologies and epistemologies, it highlights the case of Rojava as a pluriverse through rhizomatic democracy (Türk & Jongerden, 2024). To address colonial drought, we

explore how rhizomatic governance in Rojava articulates a political ontology of socio-ecological transformation—a direct challenge to epistemic and ontological structures of colonial drought as a collective resistance.

5.4.1. Challenging the colonial drought: Rhizomatic governance as collective resistance and political ontology of socio-ecological transformation

5.4.1.1. Rhizomatic governance

The socio-ecological transformation in Rojava is underpinned by the principles of democratic confederalism, which are rooted in Abdullah Öcalan's ideological shift towards social ecology and libertarian municipalism, inspired by Bookchin (Gerber & Brincat, 2021; Öcalan 2020; Bookchin 2006). Rojava's praxis embodies a profound rupture from statist, capitalist, and colonialist structures. This transformation refers to a new political ontology what we conceptualize as a rhizomatic governance, which is based on confederal relationality of political nodes (commune, people assemblies), economic nodes (cooperatives, eco-factories), epistemic nodes (academies, jineoloji), territorial nodes (municipalities), gender nodes (Kongra-Star, Jinwar, Eko-jin) and emerging forms of social, cultural and ecological initiatives and movements as new nodes (Figure 7).

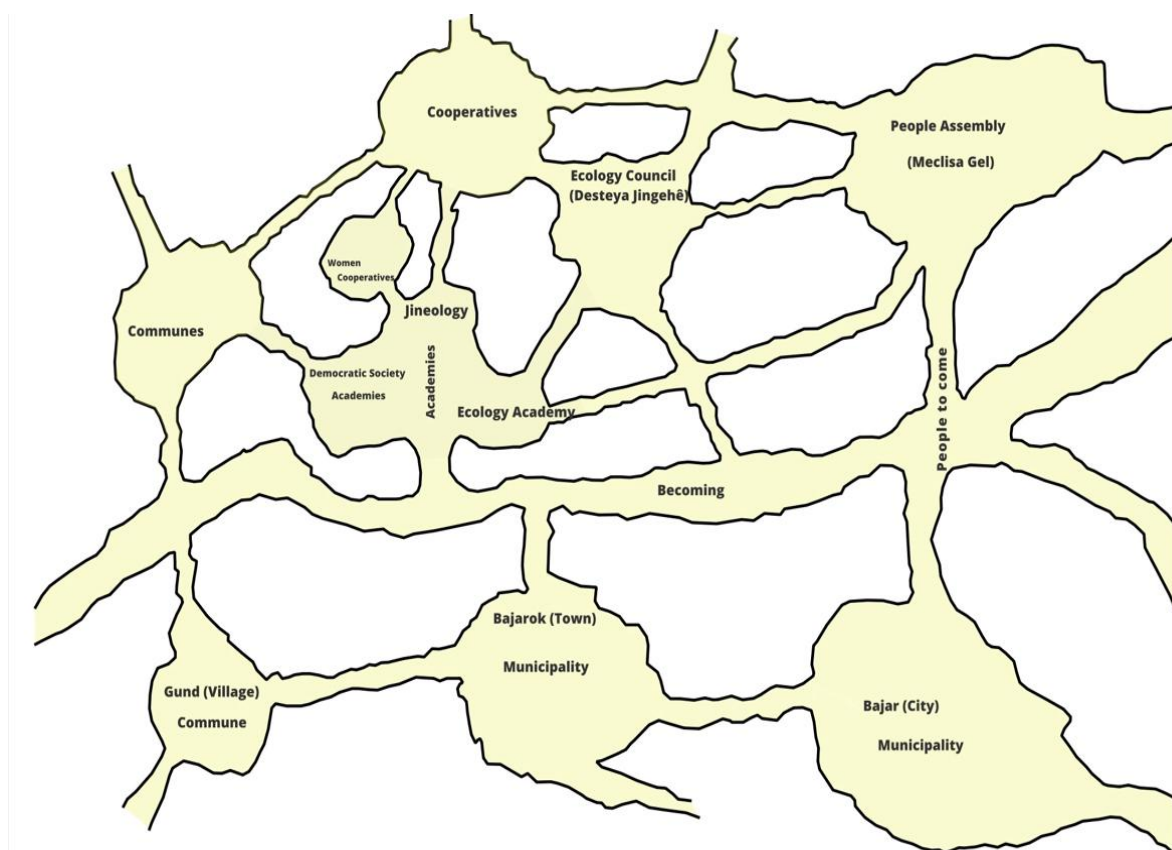


Figure 7: Rhizomatic governance: The political ontology of socio-ecological transformation in Rojava

This organizational modelling refers to the decentralised logic of the rhizome, a concept developed by Deleuze and Guattari (1987) and defined by non-linearity, decentralization, and multiplicity. It serves as a conceptual framework for understanding governance in Rojava. In Deleuze and Guattari's rhizomatic thinking, nodes are dynamic points of connection, continuously emerging and transforming as they interact with other nodes through processes of flow, rupture, and reconnection. Unlike hierarchical structures, rhizomes lack a central organizing authority and are characterized by multiple, non-hierarchical entry and exit points. Applied to governance, the rhizome embodies dynamic, interconnected institutions that resist centralized authority and domination as Deleuze and Guattari state that "a rhizome may be broken, shattered at a given spot, but it will start up again on one of its old lines, or on new lines. You can never get rid of ants because they form an animal rhizome that can rebound time and again after most of it has been destroyed" (Deleuze & Guattari, 1987, p. 9). In the context of Rojava, nodes such as communes, cooperatives, academies, ecological initiatives or women movement emerge and operate within these fluid, interconnected networks, reflecting a dynamic and relational process of socio-ecological transformation and resistance. However,

“although the rhizome metaphor emphasises the ‘scaling out’,” in Rojava’s democratic confederalism formation “there are also forms of ‘scaling up’” (Türk & Jongerden, 2024, p. 8; see also Garcia-Arias & Flores Gutiérrez, 2024). Therefore, rhizomatic governance as a political ontology cannot be explained only in terms of the autonomous units and horizontal interrelationships described above. ‘Scaling up’ forms take shape where these units are coordinated, intersected, separated and reconnected in fluid and dynamic forms (Türk & Jongerden, 2024).

To this end, Ecology Council, functions as a critical and epistemic node, works to ensure the realization of ecology—one of the three fundamental pillars of the Kurdish Freedom Movement paradigm—at the practical, epistemic, and mental levels, making it the primary ethical and aesthetic principle across all areas. It oversees whether all legal, political, economic, and practical regulations and practices of the Rojava administration conform to ecological principles, provides guidance and perspective, and offers training through its affiliated Ecology Academy in response to needs or requests.³ Through its Ecology Academy, the council fosters ecological literacy and epistemic pluralism, integrating ecological consciousness into legal frameworks and educational curricula (Hunt & Türk, 2024 forthcoming). The council also connects directly with local communes, cooperatives, and municipalities, providing training, coordinating initiatives, and ensuring that ecological principles inform governance and practice.

Communes are defined as the stem cells of society and the smallest units of rhizomatic governance, where local assemblies manage water, land distribution, and agriculture, embodying grassroots participation to discuss their problems, make collective decisions, and develop practical solutions. As foundational democratic institutions, communes ensure the political presence and voice of all segments of society—across gender, class, and ethnic lines. Participation is predicated on adherence to ecological and women’s liberation principles.⁴ In essence, the commune is the cellular unit of a democracy in which people can resist, defend themselves, and build dignified, just, and free lives—on the basis of their own language, culture, and identity—against centralized power, capital, and ethno-religious nationalism. Their structure enables flexible, needs-based coordination through sub-committees addressing defence, education, service, agriculture, and other areas (Grasso, 2022; Wimmer, 2024). These

³ Birahîm Hemdî Eshed, AANES cochair of Ecology Council (Desteya Jingehe), 2 June 2024; Eli, consultant at Ecology Committee and Ecology Academy, 30 April 2024, 12 May 2025.

⁴ Rüstem, consultant at Cezire region Municipalities Union, 26 March 24

sub-committees collaborate with cooperatives, the Ecology Council, municipalities and academies to align communal initiatives with broader ecological and social goals. Their co-chair and members are ex officio delegation of bajarok (town) or Bajar (city) people assemblies (city councils). They have organic and ontological relationship with cooperatives since meeting people needs and deconstructing economic dependency and marginalization of colonial legacy is a crucial task.⁵ That is why the slogan, “Ji bo her komûnek kooperatifek” (A cooperative for each commune) encapsulates the organic and structural complementarity between communes and cooperatives. Especially in agriculture, the cooperative model prioritizes ecological principles, self-sufficiency, and social benefit, forming the socio-economic foundation of rhizomatic governance. Therefore, cooperatives emerge as vital economic nodes, disrupting extractive colonial and capitalist models and fostering collective management of resources. Providing access to water, land, and production for historically marginalized communities—including women, Kurds, Yazidis, and Assyrians—they enact a decolonial economic practice.⁶ Therefore, what was required was not just economic reconstruction, but an ontological and epistemic rupture from colonial center–periphery structures—a genuine decolonial economy. The cooperative movement in Rojava was developed precisely with this in mind: to build an economy grounded in anti-colonial, anti-capitalist, and anti-patriarchal principles and rooted in social and ecological parameters (Aslan, 2023; Türk & Jongerden, 2024). Cooperatives coordinate with municipalities to integrate infrastructural development and resource allocation. Building a new ontology; however, requires reclaiming and reconstructing local histories, epistemologies, and knowledge systems (Sultana, 2022; Simpson, 2021; Mignolo & Walsh, 2018; Escobar, 2018). Members and pioneers of communes and cooperatives regularly attend training programs at academies. Academies, particularly the Democratic Society Academies, Jineoloji academies and Ecology Academy, disrupt epistemic hierarchies by integrating feminist, ecological, and communal knowledge. These institutions serve as spaces for the continual transformation of thought and action, embedding critical reflection into daily practice. They aim to dismantle the ideological foundations of capitalist modernity, shaped by colonial, statist, nationalist, and patriarchal mentalities.⁷ Their central mission is to construct a communal, ecological, democratic, and feminist consciousness. The academies play a vital role

⁵ Hesên Koçer, co-chair of Executive Council of the Democratic Autonomous Administration of North-East Syria, 29 March 2024; Rüstem, consultant at Cezire region Municipalities Union, 26 March 24

⁶ Basil Mihemed and Cihan Ferho, co-chairs of the Heskê Canton Cooperatives. 24 May 2024

⁷ Siyar and Medya, co-chairs of the Cizîrê Region Cooperatives. 3 June 2024; Gulistan Murad, spokesperson for the Economics Committee at Kongra Star. 16 December 2023.

in the ideological formation of commune and cooperative members, and in overcoming structural challenges they face in practice. These trainings involve collective reflection on practices, critique and self-critique, and continual transformation of both thought and action.⁸

These three structures—communes, cooperatives, and academies—function as the living and fluid components of rhizomatic governance. However, they require coordination through a “scaling up” node, to which they are vertically affiliated: the people’s assemblies. Each city’s people’s assembly functions as the highest decision-making body and is grounded in the principle of democratic nationhood, allowing for the broadest participation. Through this system, all social classes, genders, ethnic, and religious groups can govern themselves directly as an emerging new subject or becoming as “people to come” (Deleuze & Guattari, 1994, pp. 108-110). People’s Assemblies function as confederal nodes, coordinating decisions across different sectors and scales. The co-chairs of the people’s assembly are ex officio members of the executive council of the canton. People’s Assemblies consist of 60 per cent by elected delegation and 40 per cent by organizations and minority members (Wimmer, 2024, pp. 14-15). All civil and institutional organizations have representatives as members of the People Assembly, including the co-mayors of municipalities. They facilitate communication and integration between communes, cooperatives, academies, and municipalities, ensuring that decision-making reflects diverse community needs and perspectives. They operate as spaces for coordination and collective execution. Municipalities serve as the practical and administrative function of this network—implementing decisions, coordinating services, and materializing the branching and multifaceted structure of the rhizome in everyday life.⁹ Therefore, municipalities, can be conceptualized as territorial nodes, integrate socio-ecological decision-making into material and administrative frameworks. Legal frameworks enable municipalities to coordinate cooperative projects, infrastructure development, and resource management, serving as the physical spaces where different governance nodes converge. Municipalities interact with communes and cooperatives, aligning practical governance with communal and ecological priorities as a space of democratic society.¹⁰ To this end, municipalities operationalize governance by linking local decision-making with resource management, infrastructure, and environmental planning. By defining municipalities as the

⁸ Eli, consultant at Ecology Committee and Ecology Academy, 30 April 2024, 12 May 2025.

⁹ Rüstem, consultant at the Cizîrê Region Municipalities Union. 26 March 2024; Egid, official at the Haseke Municipality. 24 May 2024.

¹⁰ Eli, consultant at Ecology Committee and Ecology Academy, 30 April 2024, 12 May 2025.

embodied spaces where these nodes materialize, these territorial nodes co-produce territorial realities through everyday actions, negotiations, and contestations, transcending the reified categories of state-centric and colonial territorial governance (Deleuze & Guattari, 1987; Delaney 2005; Neuburger, 2017; Escobar, 2018, 2020). Thus, rhizomatic governance in Rojava constitutes a non-statist, decolonial, feminist, and ecological reimagining of political ontology. It offers a radical alternative to climate change adaptation and mitigation policies, colonial water exploitation and water crises, materializing a pluralistic and interconnected model of governance. Having such an ontological formation is crucial for addressing and overcoming colonial drought, climate change, and the water crisis in Rojava, a colonized land devastated by civil war. Co-chair of Ecology Council, Birahîm reflected on this critically:

We have been discussing the term ‘adaptation’ that you mentioned, and the debates surrounding it. The logic of this approach is that if you have the resources, money, and technology, you can achieve results. Since the infrastructure of colonized geographies like ours has already been destroyed, adopting such an approach would be self-denial. We need a new and radical solution and transformation. We need to deconstruct and reconstruct the political, social and economic systems holistically—that is what we are trying to do here.¹¹

Thus, this model embodies a rupture from Western-dominated climate narratives, affirming Kurdish and traditional and liberatory ecological knowledge as central to survival and flourishing. Its resistance capacity and potential lie in its theory and practice—solidarity, ecological care, and democratic autonomy woven into a living, dynamic rhizomatic governance.

5.4.1.2. Praxis and challenges

The heterogeneous assemblages and autonomous rhizomatic relationalities we have outlined may initially appear as mere imagination; however, they in fact reflect material realities on the ground as praxis and challenges of these formations. Grounded in democratic confederalism as operated through rhizomatic governance, this transformation represents not merely ecological regeneration, but a profound socio-cultural and political reconstitution aimed at dismantling historical colonial legacies, patriarchal and capitalist structures through enhancing local

¹¹ Birahîm Hemdî Eshed, AANES cochair of Ecology Council (Desteya Jingehe), 2 June 2024;

communes and assemblies. Decisions regarding land use, resource management, and environmental governance are collectively determined, embedding ecological considerations into everyday practices. Emin, formulate it strikingly,

If you want to communalize water and land, the essence is this—it's about socialization of eco-economy through communes and cooperatives or empowering family farming. It's about making it available for society's benefit, under the condition that it is ecological. It must not be used for gaining power, or for any management body to exploit.¹²

This approach fundamentally contests conventional statist governance by promoting localized autonomy and ecological stewardship. In doing so, the social economy in Rojava constitutes a critical praxis challenging the historical patterns of colonial exploitation and capitalist accumulation. The Arab Belt project, in particular, displaced and dispossessed Kurds, rendering them economically and culturally marginalized (Aslan, 2023). In this context, traditional agricultural methods, local seeds, and traditional knowledge were systematically eroded, and under the Ba'ath regime's monocrop policies, many were reduced to cheap labor for industrial agriculture. As a decolonization process, over half a million hectares of previously regime-controlled land have been redistributed to cooperatives, women-led groups, and agroforestry initiatives. The Autonomous Administration has notably advanced local self-sufficiency by developing 500-600 cooperatives (Türk & Jongerden 2024, p. 12). These cooperatives integrate democratic management with ecological sustainability, laying a tangible foundation for communal production and economic self-sufficiency. Both in agriculture and economy, comprehensive adoption of agroecological practices is central to this transformation, including diversified cropping systems, rainwater harvesting, agroforestry, and drought-resistant seed propagation.¹³ Agricultural diversification initiatives have successfully introduced crops such as chickpeas, lentils, olives, and figs, radically reducing dependency on monoculture practices historically dominant under regime policies (Türk & Jongerden 2024). These efforts signify a crucial transition toward ecological regeneration and food sovereignty, countering previous

¹² Emin, Union of Kurdistan Communities (Koma Civakên Kurdistan, KCK) Economy Committee and Economy Council of AANES, 17 May 2024

¹³ Leyla Saruxan and Mehran Ahmed, co-chairs of the Agriculture and Irrigation Council, 21 November 2023; Karker Ismail, co-chair of the Office of Community Projects and Cooperatives of Cezire Region, 21 November 2023; Delila, agricultural coordinator in the women's village of Jinwar, 27 May 2024.; Siyar and Medya, co-chairs of the Cizîrê Region Cooperatives, 3 June 2024; Basil Mihemed and Cihan Ferho, co-chairs of the Heseke Canton Cooperatives, 24 May 2024; H. Ahîn, coordinator of the Damsal Women's Economy Project, near Tirbespiyê, 4 May 2024; Seed Institution (Saziya Tov), Mihemed Hisên and Şiyar Hesen.

destructive monocultural strategies concerning the de-development of Rojava as a periphery and ecological degradation (Hunt, 2019; Pimbert, 2021; Türk & Jongerden, 2024). Siyar notes, “Crop rotation and diversification have greatly improved our land and economic self-sufficiency”.¹⁴ Such a praxis is inherently part of the decolonization of mind and epistemologies. A critical epistemological transformation is occurring in Rojava through education, embedding ecological consciousness and decolonial thought within new pedagogical frameworks (Hunt & Türk, 2024 forthcoming). As epistemological nodes, academies cultivate ecological literacy and critical pedagogy. This is a crucial intervention as “Kurds have forgotten their own aesthetic, ethical values, and knowledge systems due to occupiers’ education systems”.¹⁵ Central to this educational revolution is *Jineoloji* (science of women), which seeks to reconstruct marginalized knowledge systems, emphasizing women's traditional ecological roles. Gulistan Murad highlights how women economy and women cooperatives interweaving liberation of women and economy through ecological principles.¹⁶ Roj from Eko-Jin states that “Jineoloji integrates nature, women, and life into a holistic educational framework, reclaiming suppressed traditional ecological knowledge”.¹⁷ Jineoloji reinforces this praxis by reviving women's ecological knowledge and practices, essential for rebalancing social and ecological relations, thus restoring integral communal connections long suppressed by patriarchal and colonial systems (Hunt & Türk 2024; Türk 2024 forthcoming).

A field survey conducted among 50 villagers and farmers in Rojava provides insight into evolving perceptions of the ongoing drought and water crisis, as well as responses to socio-ecological transformation efforts. The sample included participants of diverse ethnic, religious, gender, and age groups, with a majority (n=40) actively engaged in communes, cooperatives, and academies. The remaining ten participants, serving as a control group, had limited or no engagement with these institutions and often expressed ideological opposition to the Rojava Autonomous Administration.

Among the politically engaged cohort, there was a notable shift in environmental and political consciousness. When queried on the causes of the drought and water crisis, 70 per cent attributed the crisis to climate change and upstream water restrictions imposed by Türkiye,

¹⁴ Siyar, co-chair of the Cizîrê Region Cooperatives. 3 June 2024

¹⁵ Rüstem, consultant at the Cizîrê Region Municipalities Union. 26 March 2024

¹⁶ Gulistan Murad, spokesperson for the Economics Committee at Kongra Star. 16 December 2023.

¹⁷ Roj, lecturer and Kongra Star Ecology Committee member, 27 June 2024

reflecting a heightened ecological awareness consistent with the political education programs implemented within the communes. 20 per cent interpreted the crisis through a religious lens, as divine punishment, while 10 per cent expressed uncertainty. This contrasts sharply with the traditional belief systems predominant prior to the establishment of the Rojava administration. Participants' perceptions of resource ownership and governance further illustrate this transformation. When asked "Who owns water and land?" 60 per cent of the engaged respondents answered, "the people," 30 per cent "God," and just 10 per cent "the state." Regarding management and distribution of water and land, 40 per cent endorsed collective ownership ("the people"), 55 per cent emphasized local agricultural actors (farmers, peasants, workers), and only 5 per cent favoured state control. This stands in stark contrast to the Ba'athist legacy of centralization and extractivist policies, which emphasized state control over natural resources. Conversely, the control group displayed markedly different interpretations, rooted in traditionalist and religious worldviews. Among this group, 8 attributed the drought to divine punishment linked to moral decay, while only 1 mentioned climate change. Regarding ownership of land and water, 8 insisted on divine ownership, 1 selected "the existing landowners" and 1 chose "the state." Governance preferences were more fragmented: 2 favoured state control, 5 reluctantly supported the Rojava Administration, and 3 abstained. Despite ideological differences, a shared recognition of the groundwater crisis was evident across all groups. Participants consistently reported that groundwater levels had declined precipitously over the past 30–40 years, necessitating deeper and more frequent drilling. Drought has long been cited as a major problem. Therefore, water shortage was cited as the most important problem by 80 per cent. This empirical observation underscores the systemic nature of the crisis, transcending ideological affiliations.

When asked about solutions to the water crisis, engaged participants demonstrated a shift towards rational, ecologically conscious strategies, informed by their involvement in educational programs and communal practices. Proposed solutions included the adoption of water-efficient irrigation techniques, cultivation of drought-resistant crops (such as cumin), expansion of fruit tree planting (with an emphasis on almonds and olives), and engagement in diplomatic initiatives to pressure Türkiye to restore transboundary water flows. However, a residual reliance on deep well drilling as an immediate coping strategy highlight both the severity of the crisis and the constraints of local capacity. Nonetheless, both commune participants and the control group expressed dissatisfaction with current support mechanisms

from the Rojava Autonomous Administration, particularly in relation to diesel fuel, irrigation infrastructure, fertilizers, and pesticides. A 64-year-old farmer expressed it strikingly,

Of course, drought and water shortage affect us. But those with money have drilled new water wells, bought diesel and continue farming. But what about people like us? If God is punishing us, if the *heval* (in Kurdish means friend, refers to Kurdish movement's cadres) have no means and power!¹⁸

A common critique identified the lack of agricultural expertise among leading cadres, and the perception that cooperative structures were often driven more by political commitment than technical competence. The call for trained agronomists and technical specialists was unanimous, signalling a pragmatic recognition of the limits of ideology when confronted with ecological crises. The survey reveals a heterogeneous landscape of perceptions and responses, where engagement in communal structures and political-ecological education appears to foster both critical awareness and constructive strategies for transformation.

Despite this encouraging transformation, Rojava faces substantial structural barriers due to persistent Turkish military aggression, extensive embargoes, and deliberate destruction of essential infrastructure. The governance of the region has increasingly taken on a necropolitical character, that is, it is organised around the power to decide who may live and who must die (Mbembe, 2003). The deliberate targeting of infrastructure by Türkiye and its backed jihadist groups manifests a necropolitical strategy aimed at destabilizing and displacing communities, significantly disrupting socio-ecological stability. This ongoing violence fosters demographic shifts and exacerbates communal insecurity, profoundly weakening the region's ecological and communal existence.¹⁹ The risks, uncertainties, and threats facing Rojava under the Jolani regime—which rose to power following the collapse of the Assad regime in December 2024—remain ongoing. Approximately, 60 per cent of the Autonomous Administration's budget is consumed by defence expenditures, severely constraining resources available for ecological and economic initiatives for last few years.²⁰ Hesên Koçer describes the gravity of this constraint: “Our potential is continuously undermined by external threats; survival has often overshadowed

¹⁸ A farmer. 17 May 2024.

¹⁹ Hesên Koçer, co-chair of Executive Council of the Democratic Autonomous Administration of North-East Syria, 29 March 2024; Bîrahîm Hemdî Eshed, AANES co-chair of Ecology Council (Desteya Jîngehe), 2 June 2024; Basîl Mihemed and Cihan Ferho, co-chairs of the Heskê Canton Cooperatives. 24 May 2024; Eli, consultant at Ecology Committee and Ecology Academy, 30 April 2024, 12 May 2025.

²⁰ Personal conversation with several pioneer cadres, May 2024.

ecological projects and transformation”.²¹ Eli has critically described the situation in response to theory-practice balance:

We are fighting for our existence! We are trying to defend our people and meet their needs under the war and embargo. The current level of realization and practicality is not even a quarter of our potential and capacity. This must be taken into account when evaluating Rojava’s praxis!²²

Internal ideological tensions are another main challenge, significantly hindering the transformative potential of communal practices. Discrepancies between cadre idealism and local socio-economic realities often result in dissatisfaction and limited participation from the populace. Lack of expert labour and institutional inefficiencies remain critical challenges, characterized by bureaucratic overlaps and weak coordination among parallel governance structures. Majority of experts and farmers highlight this institutional fragmentation as a substantial impediment, often leading to competitive dynamics rather than cooperative governance, diluting ecological and communal efforts. The fragility of Rojava’s socio-ecological transformation reflects broader contradictions—between decolonial aspirations and survival imperatives under war conditions, between ideological purity and pragmatic adaptation, and between ecological visions and resource constraints. However, these challenges do not diminish the radical significance of Rojava’s experiment. Rather, they reveal the necessity of continued reflection, transformation, and international solidarity. Strengthening local praxis and fostering international ecological and decolonial alliances are vital to overcoming barriers and deepening the socio-ecological transformation.²³ Rather than responding to ecological crises through technocratic adaptation frameworks, Rojava’s model reorganizes society-nature relationality itself as a new political ontology.

5.5. Conclusion

The protracted water crisis in Rojava cannot be fully grasped through reductionist explanations centered on climate variability, neoliberal policy failure, or poor governance. As this study has

²¹ Hesên Koçer, co-chair of Executive Council of the Democratic Autonomous Administration of North-East Syria, 29 March 2024

²² Eli, consultant at Ecology Committee and Ecology Academy, 30 April 2024, 12 May 2025.

²³ Group discussion with International Commune of Make Rojava Green Again (MRGA), 22 May 2024

shown, it is best understood as a colonial drought: a layered and systemic condition produced through the combined structure of internal colonialism and regional neo-colonial hydro-hegemony. The Ba‘ath regime’s treatment of Rojava as an extractive periphery through monoculture agricultural development—set the stage for profound ecological degradation. Türkiye’s control over transboundary water flows has deepened this devastation, weaponizing water as a tool of dispossession, demographic engineering, and political subjugation. At the same time, this research reveals how Rojava has constructed an alternative pathway through the reconfiguration of governance, ecological and democratic life grounded in new political ontology and epistemology. Framed here as an ontology of socio-ecological transformation, this emerging model challenges hegemonic, technocratic responses to environmental collapse by forging non-extractive, relational, and pluriversal modes of being, knowing and doing. Dominant discourses and policies shaped around adaptation, mitigation, and resilience are inadequate to resolve the escalating ecological crisis marked by climate change and water scarcity. These frameworks fail to address the root causes embedded in colonial and capitalist systems of extractivist destruction and the disintegration of the nature-society relationality. What is urgently needed instead—particularly in colonized geographies of the Global South, such as Rojava—is a new ontological and epistemological foundation that can reconstruct the relationality between nature and society. On this basis, a genuinely transformative socio-ecological reconfiguration becomes not only necessary but imperative. In doing so, rhizomatic governance—rooted in direct democracy, ecological stewardship, and gender liberation through communes, cooperatives, academies, municipalities and People Assemblies—Rojava demonstrates how decolonial, communal and feminist futures are actively being built from below. Communes manage land and irrigation collectively, food sovereignty is enacted through cooperatives, and education is politicized and democratized according to ecological and feminist principles and aesthetics. Communal water committees, women-led cooperatives, and ecological regeneration projects redefine water as commons embedded in democratic decision-making, not as a resource for external extraction or coercive control.

As a result, Rojava teaches us four key lessons: (1) Drought is a hydro-social and political phenomenon structured by power relations, infrastructural violence, and colonial design. (2) Colonial drought mechanisms function through ecocidal development/extractivism by de-development of and control on periphery and epistemicide, undermining livelihoods and knowledge systems to produce dispossession and displacement. (3) Effective responses require ontological transformation, centering democratic autonomy, ecological stewardship, and care

beyond technocratic adaptation. (4) Water justice is inseparable from anti-colonial, anti-capitalist, and anti-patriarchal struggles, as well as communal and democratic management of resources by local communities. Ultimately, this article contributes to rethinking the relationship between nature, society, knowledge, and power under conditions of colonial dispossession and ecological crisis. By centering Rojava as a living site of resistance and decolonization, it affirms the possibility of planetary alternatives—rooted not in abstract utopias, but in concrete, contested, and collective efforts to remake the world otherwise.

This research also acknowledges the alarming levels of land, water and air pollution caused by petroleum extractivism across the region as well as the waste problem. While these are undoubtedly linked to the same colonial infrastructures and logics explored throughout this paper, a full analysis of these issues lies beyond the scope of this article and remains a necessary subject for future research.

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Chapter 6. Article IV: “Education under a tree”: Greening education through ecopedagogy in North and East Syria (Rojava)

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Hunt, S.E., & Türk, N. (under review). “Education under a tree”: Greening education through ecopedagogy in North and East Syria (Rojava). In Türk, N., Jongerden, J., & Hunt, S.E. (eds.). *Ecology and decoloniality in Rojava/North and East Syria: Destruction, reconstruction, and challenges*. London: Bloomsbury.

6.1. Abstract

This research investigates the transformation of educational paradigms in the context of the Autonomous Administration of North and East Syria/Rojava, focusing on the integration of ecopedagogy into the evolving educational system. It traces the shift from a Ba‘athist education model, characterized by patriarchy, centralism, nationalism, sectarianism, and the erasure of local cultures, languages, and knowledge systems, to an alternative framework rooted in ecological principles, democratic governance, and the affirmation of the multiplicity of ethno-religious and socio-cultural structures. Drawing on qualitative fieldwork, interviews, and critical literature, the analysis highlights how ecopedagogy is being harnessed as a transformative tool, not only challenging the colonial, patriarchal, and capitalist legacies of the past but also fostering socio-ecological transformation. By developing traditional and liberatory epistemologies through academies such as those dedicated to *jineology* (women’s science), the study underscores how this paradigm shift supports a broader vision of justice and the decolonization of minds and daily life practices. This approach to education fosters an inclusive, decolonial praxis that redefines the relationship between society, environment, and knowledge, thereby contributing significantly to socio-ecological transformation in the region.

Keywords: Ecopedagogy, Jineology, Democratic Confederalism, Social Ecology, Ecological Crisis, Women’s Liberation, Ecology

6.2. Introduction

Education is at the heart of the socio-cultural transformation blossoming in the Democratic Autonomous Administration of North and East Syria (DAANES). Ongoing since 2012, the precarious Rojava Revolution is founded on the Kurdish freedom movement’s principles of democratic confederalism. This paper considers the place of ecology, one of the major pillars of this paradigm, within the educational system. Education and ecology alike are crucial to survive, thrive, and develop in the existential crisis that threatens this region and the planet. This is first understood in the immediate context of neighbouring colonial power Türkiye’s relentless attacks on civilian infrastructure and policies of ecocide. Second, it is necessary to acknowledge DAANES’ current oil dependency at a time when climate change and other multifaceted aspects of the ecological crisis are all too apparent. As such, DAANES is an extreme example of the situation that the rest of the world faces, at once confronted by hegemonic violence that threatens all communities yet inextricably reliant upon aggressively extractivist systems and infrastructure at the expense of our own ultimate interests. Educational practices, including ecopedagogy, inevitably reflect these tensions. As a priority, education is a confluence of attempts to realise audacious, indeed perhaps utopian desires for personal and communal fulfilment, coupled with the messy realities of this invidious situation. As Elise Boyle Espinosa and Adam Ronan conclude in their study of this under-researched education system, while the outcomes are inspirational, they are often contradictory too (Boyle Espinosa and Ronan 2023).

At the outset, it is necessary to consider ecopedagogy, a concept adapted from the ideas of radical Brazilian educationalist theorist Paulo Freire. In his influential *Pedagogy of the Oppressed*, Freire makes the case for the creative and empowering force of “critical consciousness” (1970, preface 16), intended to encourage participative learning to expand personal agency through enhanced sense-making and deepened understanding, bridging reflection and action to achieve greater social justice through the transformation of oppressive systems. Freire did not coin the term ecopedagogy, but ecological awareness was increasingly integrated into his educational philosophy. This emphasis culminated in his final writing,

published posthumously in *Pedagogy of Indignation*, where he made clear his holistic belief that inclusive concern for other species and the living world is an essential component of revolutionary thought and practice for the planetary citizen: “the notion seems deplorable to me of engaging in progressive, revolutionary discourse while embracing a practice that negates life - that pollutes the air, the waters, the fields, and devastates forests, destroys the trees and threatens the animals.” Ecopedagogy complements the principles of social ecology which hold that environmental degradation has deep roots in dominatory attitudes that must be addressed systemically if multiple ecological crises are to be successfully mitigated, and that ecological thinking must be a part of movements for social justice and human liberation. For these reasons, the Kurdish movement has embraced both Freire’s ecopedagogy and social ecology in its curricula in North-East Syria and throughout the Kurdish region (Platzky Miller 2022, 133; Mashuq 2024).

Necmettin Türk’s fieldwork interviews, conducted in Kurdish and Arabic during the summer of 2024, underpins the current research. We were fortunate that fifteen respondents were able to provide information and share their perspectives on, and experiences of, the democratic society academies. These were a combination of current attendees and lecturers, and included the co-chair of Education Committee of DAANES’ Ecology Council and another representative of the Ecology Council, a Kongra Star ecology committee member, and a lecturer at the Jineology Academy. Such qualitative primary data is becoming increasingly challenging to collect due to the grave security situation. The present study is also informed by the limited literature on educational initiatives available. We will first outline the present mosaic of educational typologies as it has been evolving since 2011, determined as elsewhere by factors such as age, communities of adult learners, literacy, and vocational provision. We will focus upon the inclusion of ecology within NE Syria’s educational academies.

Having contextualised this discussion, it will be necessary to critically examine the theory and practice relating to the appearance of ecological issues within the educational programmes. Several factors distinguish DAANES’s education sharply from the hegemonic system that the Syrian Ba’athist regime formerly imposed. Attempts to make a transition from capitalism to a solidarity economy have implications for what, in Marxian terms, has been conceived as the means of mental production.²⁴ While an emphasis upon critical thinking and reflective practice

²⁴ Marx talks about the means of mental production in “The Illusion of the Epoch” in *The German Ideology*. <https://www.marxists.org/archive/marx/works/1845/german-ideology/ch01b.htm>.

may be shared with much conventional pedagogic practice, the values-based approach takes a different form when developed through the lens of democratic confederalism, rather than the instrumentalist and positivist norms prevailing elsewhere. A further distinguishing feature is the integration of ecological sustainability and women's empowerment, promoted through the Kurdish concept of *jineology*. Internationalist volunteers have reported that this is strongly emphasised from the outset when they undertake training in North and East Syria (NES) (Walker and Chant 2024). Finally, the democratic pillar of democratic confederalism is implemented through the decision-making processes of the committees and regional councils that administer the educational system.

The specific forms of internal colonialism that have prevailed in this predominantly Kurdish area have brought about ambiguous relationships with the land, extractivism and pollution.²⁵ The Ba'athists had historically denied Kurdish communities citizenship and implemented exclusionary patterns of land tenure. Extractivist policies imposed agricultural monoculture, with commodities transferred to enrich other parts of the nation state, leading to underdevelopment which compounded deprivation and alienation in NE Syria (see Türk and Jongerden 2024). This situation recalls Walter Rodney's analysis of colonial powers' deliberate underdevelopment of Africa as a matter of policy and control (Rodney 2018). Such historical lack of agency speaks of the forced severance many Kurds' experience in their relationships to their surroundings and the living world, leaving lands that are psychologically abandoned. This is in part corroborated by a comment from one of our interviewees, Respondent P, who explained that some Kurds had come to regard the lifestyles of those who maintained their livelihoods in rural areas as "backward." At the same time, Respondent P goes on to speak of a reversal of such attitudinal trends, claiming that one of the achievements of revolutionary Rojava has been to reestablish Kurds' "connection with their culture and nature" through education.²⁶ Writer of the Kurdish diaspora, Dilar Dirik, moreover, identifies a countervailing, more redeeming perspective grounded in anti-colonial resistance. She speaks of the powerful concept of *welatparêzî*, a love of homeland that is distinguished from statist notions of patriotism, with its patriarchal connotations, explaining the idea as:

[...] a commitment to liberate the lands from colonization, occupation, and militarism. Women's love for the homeland is read against male-dominated notions

²⁵ The subject of a forthcoming article by Necmettin Türk.

²⁶ Respondent P, 27 May 2024.

of borders, nations, and states, as universal love for humanity and the earth combined with an attachment to one’s ancestral lands and ecology. Protecting cultures, identities, and geographies from genocide and forced assimilation is seen as indivisible from self-defence (Dirik 2022, 66).

The prospectus of the Andrea Wolf Institute (part of the Jineolojî Academy in North-East Syria), *Revolutionary Education*, also makes clear that this intimate love of specific, familiar localities, and care to protect them, is sharply distinct from nationalistic, globalist, and colonial views, affirming that the Kurmanji term *Welatparezî* “is defined interrelated to internationalism - not in detriment of any other place, or excluding any other peoples” (Andrea Wolf Institute 2020, 7). Such divergent attitudes underpin the central function of education in the cultural front to bolster mindsets of resistance and liberation in a context where physical terrain is being despoiled and geographies forcibly reconfigured through airstrikes and the damming of rivers. The final sections of our research will examine how the theoretical underpinnings of ecology is becoming embedded in the evolving educational structures and processes. Informed by analysis of the cadres’ responses, we will consider ecopedagogy as it is being implemented with the wider educational programmes and some of its practical outcomes as its insights are taken up by civil society. Elif Sarican, Nika Dubrovsky, and Elizaveta Mhaili have cited Abdullah Öcalan’s aspiration that every movement home is treated as a revolutionary academy (Sarican, Dubrovsky, and Mhaili 2023, 273). One of our interviewees, Respondent B alludes to Öcalan’s call directly “entering each home and turning the environment into a school is itself resistance”.²⁷ In this context, education is indeed at the heart of the struggle for a liberated society.

6.3. NE Syria’s educational system

Article 62

Education is free at all levels, and primary and intermediate education is compulsory.

(DAANES new social contract 2023)

²⁷ Respondent B, 15 June 2024.

Before evaluating the educational system of the Autonomous Administration's first decade, it is necessary to recognise and reflect upon a situation where even the word "struggle" feels inadequate. For most children, the experience of schooling is likely to be at best stop-go. Occupation and conflict with Islamic State and other Jihadist forces depleted the number of functioning schools in NE Syria during the earlier years of the Civil War. The invasion of Afrin in 2018, and further annexations of areas around Ras al-Ayn/Serê Kaniyê and Tell Abyad/Gire Spi during Türkiye's so-called "Operation Peace Spring" in 2019, led to the closure of progressive educational initiatives in affected settlements. According to deeply critical reports, these were replaced by the imposition of Turkish state doctrine and sharia law, and the relegation and suppression of Kurdish language teaching in Kurdish-majority Afrin (Syrians for Truth and Justice 2023). In 2020-2021, the coronavirus epidemic further disrupted timetables due to school closures in NE Syria, with attempts to deliver education online in a situation where electricity supply was precarious, internet connections intermittent, and access to digital devices was unreliable. Many more schools and supporting infrastructure have been lost to targeted destruction and collateral damage from ongoing Turkish airstrikes.²⁸ Additionally, in the chaos of war, school buildings have been recommissioned for other purposes, often as crowded makeshift shelters for refugees and internally displaced people, sometimes as weapons depots, and even to detain Islamic State affiliates.²⁹ This situation, exacerbated by economic embargoes imposed by hostile neighbours, also accounts for a widespread lack of even the most basic educational resources, including schoolbooks, chairs, and desks. Reportedly, desperate communities have even been forced to burn school furniture as firewood to survive wintry conditions (Stefani and Ruge 2019). Complex economic, social, and psychological factors also impede progress. An educational system based on Abdullah Öcalan's concept of a Democratic Nation was first introduced in 2015. The political task of introducing curricula based on radical pedagogies that are directly counter to preceding policies is formidable. A paradigm shift cannot be seamlessly translated into classroom practice.

²⁸ KCS Research Team in Rojava, interview with Semira Hac Eli, the co-chair of DAANES Ministry of Education, "Rojava Education Co-Chair: Turkey is Bombing Our Schools," *Kurdish Center for Studies* website (15 January 2024), accessed 7 July 2024, <https://nlka.net/eng/rojava-education-co-chair-turkey-is-bombing-our-schools/>.

²⁹ Karlos Zurutuza, "Stuck in limbo: The displaced residents of Syria's northeast," *Middle East Eye*, (10 January 2020, accessed 7 July 2024, <https://www.middleeasteye.net/news/stories-despair-among-displaced-syrians-northeast>; Samer Al Ahmad, "Education Under Strain in Al-Hasakeh Governorate: Disjoined Curricula and Political Antagonism Impact Civilians," *Syria policy briefs*, Friedrich-Ebert Stiftung website (December 2023), <https://library.fes.de/pdf-files/bueros/beirut/20813.pdf>, accessed 21 July 2024; Jeff Seldin, "IS Cells in NE Syria Retaliate for Foiled Prison-Break Plot," *Voice of America News* website (9 November 2021), accessed 11 July 2024, <https://rojavainformationcenter.org/2021/11/the-voice-of-america-is-cells-in-ne-syria-retaliate-for-foiled-prison-break-plot/>.

Improvisation and experiment are inevitably called for in the nascent structures that are evolving. Furthermore, the implementation of aspirations for a programme grounded in principles of democratic confederalism risks alienating families and communities who are politically and religiously antagonistic to the project, or who feel it is against their interests. Alleging fear of indoctrination, children have either been withheld from schools or aspects of democratic-confederal principles have been compromised. For example, in a 2020 interview, Kemal Musa head of the Education Committee in Arab-majority Deir-ez-Zor, reported that key components of the DAANES curriculum were not taught, with jineology excluded to accommodate local sensitivities. At the same time, features such as the male-female co-chair system for school boards and other features of democratic accountability were adopted, and corporal punishment was abolished (Rojava Information Center 2020). Critics who have objected to the dominance of the ideology of democratic confederalism in the educational curriculum have included ecological ideas among their concerns, according to Samer Al Ahmad, who writes:

...its focus on PYD ideology and PKK leader Abdullah Ocalan’s concepts, such as environmental social theory and Jineology, while excluding other viewpoints, has caused discontent among the region’s Arab, Kurdish, and Syriac communities (Al Ahmad 2023).

While the implementation of aspirations for a programme grounded in democratic confederalism met with some initial challenges from families and communities sceptical of such principles, since 2023-2024, the same curriculum has been accepted and applied in all parts of NE Syria, including the Arab region. This is thought to be for pragmatic reasons, so that children can receive a full education within the prevailing administration. Nevertheless, it marks an important extension of the democratic curriculum, beyond its most sympathetic areas. A shortage of qualified and experienced teachers compounds NE Syria’s infrastructural difficulties. The recruitment and training of new teachers is, therefore, urgent. Since the assessment system is not regarded as accredited, there is no international recognition of the qualifications awarded by DAANES educational institutions. Finally, social deprivation makes it necessary for many children to work to contribute to their households, placing further impediments to attendance. In such circumstances, school children are either missing valuable education altogether, studying under conditions of regular disruption and displacement, and frequently experiencing psychological trauma endemic to war zones. Imperfect and contingent as this situation may be, the Autonomous Administration’s educational system is functioning

and under democratic governance. As elsewhere, there is provision by age, spanning early-years care in pre-school nurseries, through to primary and secondary schools. According to the Education Board, there are 4,798 schools in the DAANES area, catering for 860,730 students (figures from October 2022) (Youssef 2022). There are reportedly 41,182 teachers (Mahmud 2023). Multiculturalism is encouraged to help resolve historical social tensions in this polyethnic region. Throughout DAANES, linguistic diversity is protected and valued so pupils are educated on pluralist principles, with school teaching conducted in the Kurdish, Arabic, or Syriac mother tongues.³⁰ The universal curriculum requires that a second regional language is learned as part of secondary education, with a third, international language, typically English, added for later levels. This approach is explicitly intended to enhance co-existence through mutual understanding and to celebrate diversity. In the words of Becet Hussein, a member of the Qamishli Canton School Board, “If we want to live together, we have to learn the languages of others” (Stefani and Ruge 2019). There is also a school in Aleppo that follows the DAANES curriculum and principles, an anomaly explained by its location in the autonomous majority Kurdish enclave of Sheikh Maqsd (Rojava Information Center 2020). Within this educational context, the Education Committee of the Ecology Council (*Desteya Jingeh* in Kurdish) works to embed ecological principles. It has input into curriculum design and development, ensuring that up-to-date ecological content is represented at primary, secondary, and tertiary levels. It undertakes teacher training so that educators are familiar with ecological issues and their connection to the wider principles of democratic confederalism and environmental justice. The Committee is also tasked with engaging with local communities to raise awareness of ecological issues and perhaps helping to make the case for the link to social justice where there is reluctance to adopt democratic-confederal ideas.³¹

At higher educational level, four small universities have opened (one in Afrin was forced to close due to the Turkish invasion of that canton in 2018). The University of Rojava in Qamişlo and the University of Kobanê were joined by a third institution in 2021, when the Al-Sharq University / Zanîngeha Şerq opened in Raqqa. There is an Education and Coordinating Organization of Universities, although higher education is more directly administered by the

³⁰ Mahmud, “A look at northeast Syria's exceptional and diverse education system,” Khabat Abbas, response to question from Stephen Hunt, “Peace in Kurdistan Social Ecology Network on Self-Determination in the 21st Century,” Open Protest Network on *Youtube* (31 July 2024), <https://www.youtube.com/watch?v=S-FfErFHxh0>, accessed 4 September 2024.

³¹ Respondent N, 29 August 2024.

Executive Council.³² While numbers are increasing, these higher education institutions are relatively small. A report by the Rojava Information Center records that “2,482 students attended DAANES universities in the 2021/2022 academic year.” This is in part attributed to exacting standards for admission with less than a third of applicants accepted, and demands for regular attendance, which must not fall below 80 per cent.

Ecological concerns are embedded within the educational systems of NE Syria’s higher education institutions. The philosophical and ethical statement of the University of Kobani, in particular, explicitly includes a commitment to ecological principles. Its Faculty of Natural Sciences includes aspirations to “preserving natural resources and reducing environmental degradation and climate change” and “energy conservation and promotion of renewable energy sources.”³³ A representative confirmed that they regarded it as the University’s duty to ensure that their students are educated in “ethics and environmental protection” as well as science. The reasons are two-fold. First, to promote conservation issues in the Middle East where awareness is lacking, despite the many ecological problems. Second, to integrate ecology as a pillar of democratic education. Ecology is therefore a priority for the future: “By educating students, we will create a generation that will be aware of the environment and that the culture of ecological protection will be at the forefront...”³⁴ There are currently plans to launch a Master of Arts in Political Ecology programme at the University of Rojava in 2025.³⁵ In keeping with other aspects of civil society, all levels of education are administered through a system of committees and councils, facilitated by co-chairs (Rojava Information Center 2022b). The Northern and Eastern Syrian Education Council is the overarching body for coordinating education across DAANES. Immediate governance is devolved to the education committees that are established in each commune, where residents are encouraged to attend and participate in regular meetings (critically, however, participation is often less than would be desirable). Teachers’ unions have an important function in selection for the local committees (Rojava Information Center 2022a). The Autonomous Administration’s Education and Learning Commission coordinates the committees at the inter-regional level. Councils are made up of members of local committees

³² Executive Council for North and East Syria, “Annual Report of the Executive Council for North and East Syria – 2020 (6 February 2021), trans. by Rojava Information Center, *Rojava Information Center* website, accessed 15 July 2024, <https://rojavainformationcenter.org/2021/02/annual-report-of-the-executive-council-for-north-and-east-syria-2020/>.

³³ University of Kobani, “Faculty of Natural Sciences”, University of Kobani home page (English), <https://kobani-uni.ac/en/university-section/9>, accessed 25 June 2024.

³⁴ Representative of the University of Kobani, e-mail message to Stephen Hunt, 30 August 2024.

³⁵ Necmettin Türk is one of the founders of this programme,

with their membership partly elected (60 per cent), with the remainder from quotas to ensure that all groups within society are represented. Such structures are mirrored and complemented by the democratic institutions of the larger ethnic and faith communities, such as the Armenian Education and Training Committee, which operates under the auspices of the Armenian Social Council.³⁶ The Women's Foundation (*Waqfa Jin*), linked to the wider women's movement through Kongra Star, also administers some smaller specialised schools, such as "establishing kindergartens, orphanages and specialized schools for disabled children. For example, there is a school at Jinwar, the women's village, where there is also vocational education for women (Rojava Information Center 2019). Education is free at schools and universities. Education represents a significant part of the DAANES budget, which is challenging since the administration is not prosperous. While attendance at school is compulsory, acceptance to university is dependent upon passing rigorous entrance examinations. Even though all can apply, only approximately one third of those who sit the examinations are successful. However, those who meet this required standard pay no tutorial fees and are also provided with free student accommodation (Rojava Information Center 2022b, 10-17). Shortly, the focus of our study will be chiefly upon the adult education programmes of the academies. Beyond the formal primary, secondary, and tertiary education and the academies of DAANES, there are other institutions for learning and training that should be recognised. There are vocational centres such as those for teacher training, and specialist institutions that cater for Kurdish language teaching (Knapp, Flach, and Ayboğa 2016, 175). Examples are the Celadet Bedirxan Academy for Kurdish Language, History and Literature in Qamişlo, and the Ferzad Kemanger Academy for Kurdish Language and Education in Kobanî (ibid., 175-176). There are also facilities for students with special educational needs and disabilities (ibid., 184). Educational provision within camps for internally displaced people and refugees is a priority, yet limited by stretched resources. This includes, for example, schooling within the camps at Washokani and Serêkaniyê.³⁷ In a 2020 report, the Rojava Information Center investigates initiatives to provide schooling and efforts to "rehabilitate" Islamic State affiliates in detention centres such as Al-Hawl, illustrating some of the unique challenges that the Autonomous Administration faces

³⁶ Rojava Information Center, "An interview with the Armenian Social Council in Qamishlo," *Rojava Information Center* website (21 September 2024), accessed 14 July 2024, <https://rojavainformationcenter.org/2022/09/an-interview-with-the-armenian-social-council-in-qamishlo/>.

³⁷ Rojava Information Center, "Interview [with Berzan Abdullah, Administrator of] – Washokani IDP Camp" (15 October 2023), accessed 25 August 2024, <https://rojavainformationcenter.org/2023/10/interview-washokani-camp/>; ANF, "Third primary school opens in Serêkaniyê Refugee Camp", ANF website (8 March 2023), accessed 25 August 2024, <https://anfenglishmobile.com/rojava-syria/third-primary-school-opens-in-serekaniye-refugees-camp-65911>.

(Rojava Information Center 2020). Over the border in Iraq, the long-running Mekmour Camp, with its strong affinities to the wider Kurdish freedom movement, reportedly has a well-established educational system with 4,000 students and 150 teachers in twelve schools.³⁸ Beyond the secular DAANES provision and curricula, some private educational enterprise continue, including religious schooling. The regime retains administrative jurisdiction over some schools in Qamishlo, Al-Hasakeh, and elsewhere (Al Ahmad 2023). Having completed our survey of educational provision in NE Syria, we will now focus on adult education in the academies and our interviewees’ responses to questions about the integration of ecology within such programmes.

6.3.1. Academies

As for schools and universities, education is also free for cadres and participating adults in the academies of NE Syria. These are an essential part of the eclectic mix of educational opportunities that complements the formal provision. There are educational academies for movement cadres in several major settlements, for example the Mesopotamian Social Sciences Academy, in Qamişlo, with some run by the Kurdish women’s movement, such as the Women’s Education and Science Academy, Kobanî, and the Nuri Dersîmî academies in Rimelan and elsewhere (Knapp, Flach, and Ayboğa 2016, 180-183). Some academies cater for volunteers visiting from the international community, arranging programmes mostly taught in English language. Respondent B notes the influence of pedagogies from radical political traditions such as socialism, anarchism, and feminism, and of alternative educational programmes offered by the liberation theology movement in Latin America, and by the Turkish left.³⁹ Nevertheless, it is accentuated that to be successful it is vital to create educational approaches that are “understanding of integrity to current social problems” and adapted to the specific circumstances in which they are being implemented.⁴⁰ The provision of such education entails not only fiscal but also philosophical and ethical considerations. It is imperative that the

³⁸ Paul Trowbridge, “Joy, Grief and Resilience in Makhmour Refugee Camp,” *Lacuna Magazine* (3 May 2023), accessed 24 August 2024, <https://lacuna.org.uk/migration/joy-grief-and-resilience-in-makhmour-refugee-camp/>; Jinha Women’s News Agency, “Kurdish Education in Makhmour Camp: From Tents to Schools,” Jinha Women’s News Agency website (15 May 2024), accessed 24 August 2024, <https://jinhaagency.com/en/actual/kurdish-education-in-makhmour-camp-from-tents-to-schools-35050>.

³⁹ Respondent B, 15 June 2024.

⁴⁰ Respondent B, 15 June 2024.

academies operate outside of the capitalist system as far as possible so that independent thinking can develop, and knowledge is available to all, regardless of whether they can afford attendance fees. At the academies, everyone is expected to participate not only in seminars but also in maintaining the learning spaces and carrying out tasks such as the preparation of meals. An attendee, Respondent A, explains:

Our academies are open to all citizens living in North-East Syria. Our only criterion is that they believe in freedom, democracy and building on these values in the society of the democratic autonomous administration of North-East Syria. Therefore, regardless of ethnic identity, political party views, religious values, they can participate in training in our academies.

There is perhaps a tension here as the claim that entrants are welcome whatever their political views could surely militate against the expectation that they are sympathetic to the values of democratic autonomy. Comments from Respondent B demonstrate a high-level of consciousness of the value of liberatory education and insists that this should not be based on waged intellectual labour, while acknowledging some of the obstacles and shortcomings in its realization:

Contrary to the mentality of the system, which widens the distance between knowledge and knowing with money, sharing all its knowledge and know-how with society as much as possible is a sine qua non of community academy work. In order to make this principle vital, there is no question of paid or paid-for education. The ideology of democratic modernity is based on non-capitalist social organization.⁴¹

This commentator underscores the significance of free education but readily concedes that this has not been met and is a work in progress. While ideally, Respondent B urges, a cooperative should meet “the necessities of life of intellectual and research labor”, this ideal has not been attained. Most relevant for our present topic is the work of the Ecology Academy, and its role in coordinating ecological issues throughout NE Syria’s educational system. This has been set up under the auspices of the Education Committee of the Ecological Council. Respondent N, its co-chair, itemises the responsibilities of this Committee as:

- Developing educational curricula that address environmental topics
- Organizing workshops and awareness seminars
- Coordinating with educational institutions to implement environmental programs

⁴¹ Respondent B, 15 June 2024.

- Collaborating with agencies to execute environmental education projects⁴²

6.4. Social ecology

*We define the society we envision and struggle for as a democratic ecological society.*⁴³

Let us now undertake a more detailed examination of the perspectives and reflections upon ecology’s place in the educational system that emerge from the fieldwork transcripts. The ideas that inform and activate the curriculum come from both within the Kurdish movement and international expressions of political ecology. Respondent N cites key philosophical inspirations as Abdullah Öcalan’s books, Murray Bookchin’s *The Ecology of Freedom* (1982) and *The Modern Crisis* (1986) and the activist groups such as Fridays for a Future.⁴⁴ Other thinkers known to appear on the reading list for the Ecology Committee, include Paulo Freire, perhaps the most influential theorist of critical pedagogy, Malcom Ferdinand, philosopher of decolonizing ecology, the Marxist geographer, David Harvey, and the Indian writer and activist Vandana Shiva.⁴⁵ As would be expected, several respondents mentioned ecology as a part of their endorsement of democratic confederalism, which prioritises social wellbeing and ecological sustainability. One, Respondent F, reiterates the three pillars which constitute its core principles, namely that “It is based on the democratic, ecological and women's freedom paradigm, which is the key to solving all social problems.” Another, Respondent L, a lecturer and Kongra Star Ecology Committee member, unites this mutually affirming triad of themes, arguing that “women need to develop the science of ecology in order to continue the line of existence that constitutes the communal-democratic essence of society throughout history.”⁴⁶ Such a perspective is representative of the academies’ ethos, which is in direct contrast to capitalist modernity, a condition which Kurdish critics feel has caused the increasing destruction of the living world and the loss of communal and democratic values, thus undermining social solidarity. Respondent L’s claim to historical consciousness here is also indicative of the way that the pedagogy of the academies is founded on an understanding of deep history, in part inspired by Öcalan’s reading of Fernand Braudel and Murray Bookchin. Far from being locked into nostalgia, deep history is conceived as a dynamic strategy to

⁴² Respondent N, 29 August 2024.

⁴³ Respondent E, 15 June 2024.

⁴⁴ Respondent N, 29 August 2024.

⁴⁵ Ali, a cadre of the Ecology Committee, personal communication with Necmettin Türk, 26 September 2024.

⁴⁶ Respondent L, 27 June 2024.

reinvigorate future potentialities. Just as deep time underpins ecology with a sense of almost unimaginable antiquity and complexity with respect to the origins of the physical world and life on earth, so deep history profoundly expands this dimension in the cultural realm. Deep history offers explanatory accounts of the origins of the present day conflictual and unjust societies with their cataclysmic ecological consequences that do not rely upon assumptions of essential and inevitable evil. Instead, it posits a logical wrong turn in humanity's deep past leading to the accumulation and concentration of wealth and monopolization of power that can be critically analysed and, theoretically, dismantled. Faith in this understanding opens space for alternative possibilities. This explains the hopefulness that persists in the Kurdish educational programme, and the movement more broadly, notwithstanding its perilous situation in a violent and disintegrating Middle East. By contrast, therefore, Respondent F claims that to be severed from history and social memory makes it much easier for an individual to be socialised into the service of hegemonic capitalism. Respondent M celebrates connection with the natural world and a sense of social continuity as a source of re-energizing power, writing that "the naturalness of the labor-based lifestyle in non-industrialized, non-urbanized areas, i.e. villages and zozans, brings the vitality of social memory and thus the vitality of ancient knowledge." Again, Respondent L suggests the future-oriented character of the academies of NE Syria with personal training objectives that include:

[...] the ability to recognize oneself first and then the nature and environment (in the context of social relations), the awareness that one is not only a learner but also a good teacher, the ability to synthesize the past and the present.⁴⁷

Reactivating and reconnecting with the past, therefore, is key to replenishing the future through aspirations to bring about the full realization of the condition of democratic modernity, the antithesis of capitalist modernity, grounded in "a new nature-universe perspective and social construction system."⁴⁸ In this context, it is apposite that the land often called the Fertile Crescent, an area substantially contiguous with present-day Kurdish populations, is the site of the Sumerian and Mesopotamian civilizations thought to have been the origins of the earliest agricultural settlements, a shift destined to transform nearly all human relations thereafter. While not contemporary with this transition, the emergence of state formations in earlier civilizational phases in this region also explains the nascent rise of urban societies (Scott 2017,

⁴⁷ Respondent L, 27 June 2024.

⁴⁸ Respondent L, 27 June 2024.

3). Two respondents (B, F) reference the Sumerian period, typically associating it with the development of patriarchal, class society. Bookchin’s *Ecology of Freedom* (1982), a major influence upon Öcalan’s ideas, attributes the inception of hierarchical and dominatory modes of thought regarding society and attitudes towards the natural world, even predating the state and the class system, as the original cause and ongoing source of social injustice, the divergence of interests, and unfreedom. An enduring sense of social solidarity derived from colonial subjugation and traditions of resistance have ensured that the dispersal of power through distributed forms of governance advocated by Bookchin and Öcalan resonated with the Kurdish movement’s communalist sensibilities. It is now endeavouring to put them into practice in NE Syria’s neighbourhood communes, through its committees and councils. Within this current of thought, Respondent L adopts the insight from social ecology that ecological destruction is an inevitable outcome of deep-rooted social dysfunction, insisting that:

We cannot understand society and the domination over nature without understanding the style shaped in the prison of oppression and exploitation of the hierarchical and statist society where the slavery of women is at its peak.⁴⁹

The question of the human domination of the living world and other species is deepened by Respondent F’s reference to a tripartite categorization of nature, an allusion to Bookchin’s dialectical naturalism and core to his philosophy of political ecology: “The nature that includes all beings as the first nature, and people and society as the second nature, consciously bringing people together in a harmony of both natures as the third nature.”⁵⁰ Based on the notion of humanity attaining what Bookchin called “second nature” in *The Ecology of Freedom*, a uniquely heightened although imperfect subjectivity, this perspective articulates progress to a relationship with the living world which is characterised by complementarity over antagonism (Bookchin 1982, 279). In the prospectus of the Andrea Wolf Institute based on jineology, better treatment of other species, would also advance human liberation:

In the field of ecology, the aim is to overcome the existing conflict between nature and society. The environment, seen as the first nature, and the society, seen as the second nature, need to unite for a liberated society, a third nature, or way to connect everything together holisitically. (Andrea Wolf Institute 2020, 40)

⁴⁹ Respondent L, 27 June 2024.

⁵⁰ Respondent F, 15 June 2024.

There is little indication in the Kurdish literature as to how this third nature could be developed in practice. Indeed, cautionary critics have charged that Bookchin himself is unclear about what human-led “third nature” may mean in practice.⁵¹ As a minimum it would likely consist of rewilding and restoration of damaged ecosystems. Might controversial, anthropocentric forms of “geo-engineering” be admissible, and if so, with what consequences? Perhaps, and even more intriguingly, it could involve exploring and celebrating alternative forms of evolutionary consciousness on the part of highly evolved intelligences such as the great apes and even cetaceans and cephalopods, with attempts to communicate more fully across species barriers, and entail the mentoring of other species, by teaching symbolic languages. The insistence on thinking holistically, is another aspect of ecopedagogy that seeks to integrate social and ecological thinking, rather than presenting social and ecological issues as separate fields of study. The pioneering theorist of critical pedagogy, Paulo Freire, believed to be writing about ecopedagogy at the end of his life, asserted that ecology has “gained tremendous importance” and “must be present in any educational practice of a radical, critical, and liberating nature (Freire 2016).” Even within primary and secondary education there is an intention to nurture expanded awareness both of the macro-scale planetary perspective and micro-level detail, leading to knowledge of the interdependency of all organisms within ecosystems. It is hoped that such holistic perspectives may lead to enhanced comprehension of a kind that can navigate the relationships between contexts of multiple scales. Respondent P, for example, comments upon the desire to promote both the social character of the study of the living world and even its intrinsic extra-human dimension and autotelic value. “The system of education”, they argue:

[...] is based on key principles, one of which is the concept of the universe, science, society, and nature conservation. Nature and non-human forms exist independently and predate human existence. They are not mere material or property for us to use and consume. We are dependent on nature, and protecting and preserving nature serves the entire society, not just our needs and existence.⁵²

The imaginative integration of social liberation and ecological concern is further illustrated by the unity of thought regarding women and ecological wellbeing.

⁵¹ A dilemma thoughtfully explored in an account of Kim Stanley Robinson and Robyn Eckersley’s responses to the implications of Bookchin’s concept of “third nature”, Shaun Huston, “Murray Bookchin on Mars! The Production of Nature in the Mars Trilogy”, in *Kim Stanley Robinson Maps the Unimaginable: Critical Essays*, ed. William J. Burling (McFarland, 2009), 231–41, <https://theanarchistlibrary.org/library/shaun-huston-murray-bookchin-on-mars>, accessed 18 August 2024.

⁵² Respondent P, 27 May 2024.

6.4.1. Eko-jin

*Nature, like women, is polluted, battered and mistreated.*⁵³

Ecology and jineology, the science of women, are distinctive areas of study within NE Syria’s educational curricula. There is even a compound Kurdish term, *eko-jin*, that unites these intimately related concepts. Respondent M insisted that it is necessary to integrate women’s liberation and the ethical treatment of the living world as mutually reinforcing ideas that set a standard for a liberated society. As a lecturer at the Jineoloji Academy, this respondent commented, “just as ecology is an agenda inherent to the women's liberation struggle, the women's liberation agenda must also be inherent to the ecological struggle.”⁵⁴ Respondent L further expounded the view that hierarchical systems are at the root of exploitative relations in the present day. As we have noted, in social-ecological thought it is axiomatic that relations of hierarchy account for the appearance and continuation of social injustice and ecological degradation. In interview, Respondent L asserted that: “The oppression and domination experienced by women and nature today is the product of the same hierarchical and power-oriented statist system.”⁵⁵

Essentializing tropes that represent women as nature are problematic as they tend to translate to women’s culturally boundaried positions and reduced agency as natural. In the Middle East and beyond, this attribution of reduced agency and lower status enables ongoing exclusion from full participation in the cultural realm. Respondent M, by contrast, makes a powerful case that women can reclaim their lives and be liberated through the living world. To break from the constraints that have sometimes delimited women’s role to the domestic sphere, the living world can enable self-realization:

She tries to forget the pain of her severed connection with the street and nature by longing for and striving for a beautiful home. It is a prison. This is why it is important for women to connect with nature, society and the street, especially in the process of building their own free identity.⁵⁶

⁵³ Respondent L, 27 June 2024.

⁵⁴ Respondent M, 27 June 2024.

⁵⁵ Respondent L, 27 June 2024.

⁵⁶ Respondent M, 27 June 2024.

This aspiration for women to break out of the imprisoning condition which patriarchal and statist cultures imposed upon them – mindsets that socially conservative women often embraced – is supported with an emancipatory call to expand their spatial geography, in part through being in the “natural” world. Progress in this respect relies upon cultivating understanding through education. In particular, this is conceived of education grounded in the ancient Greek concept of *paideia* translating to a holistic, ethical, education of a kind taken up and reinvigorated in a revolutionary form by twentieth-century thinkers such as Hannah Arendt, Murray Bookchin, and Cornelius Castoriadis looking to bring up new generations that would be at once autonomous and socially oriented in their outlook and practical conduct. Such a commitment to self-development and empowerment within authentically democratic structures is also present within Kurdish culture. Here, similar connotations are conveyed by the rallying concept of *xwebûn*, with its emphasis upon self-knowledge and realization. For example, Respondent L, stresses the importance of women “gaining personality and identity and in pioneering social organization”, in all spaces including “ecological life”.⁵⁷ The case is forcefully made that creating free alternatives to “nation-statism” in cultural and economic life “are important steps women take towards becoming themselves (*xwebûn*)”, a profound form of personal actualization within an emancipated society. It is important to recognise that such notions of self-realization contrast with liberal notions of subjective agency which, as Dirik argues, privilege competitive assertion of the individual subject within hierarchical structures and power systems (Dirik 2022, xxi). Relational self-concepts proposed by theorists such as Nancy Chodorow and, even more pertinent in this context, Freya Mathews’ expanded theory of interconnectivity, the *Ecological Self* are strands of feminist thought that are potentially more complementary (Mathews 1991).

A current syllabus from the Academy of Ecology suggests that within the academies, there is assessment of different, and sometimes conflicting, views within political ecology, enabling for example, opportunities to explore theoretical tensions that often exist between the perspectives of social ecology and deep ecology.⁵⁸ Such debates relating to the degree of identification with, and connection to “nature”, will be played out in discussions within the women’s academies. According to the programme of the Kurdish women’s movement’s Andrea Wolf Institute, jineology works in eight areas, which include both education and ecology. These will, therefore,

⁵⁷ Respondent L, 27 June 2024.

⁵⁸ Respondent D, 27 June 2024.

be priorities for the seven women’s research centres throughout NE Syria.⁵⁹ There is an ethic of self-education and care for the education of others within the women’s academies. Social roles are shared to maintain the administrative and practical operation of the institutions. Ecology and jineology are also embedded across the academies, which include specialist programmes for men. Respondent E, representing the DAANES Ecology Council, confirms that an ecology course features in all “military, social and unique women's academies”.⁶⁰ This is a significant development since researcher Elise Boyle Espinosa, who undertook previous fieldwork on the education system in 2018 found that it “did not seem to feature to the extent that we would have hoped”.⁶¹ In all academies there is an emphasis on lifelong learning to transform mentality through critical reflection on the pathway towards *xwebûn*.

6.4.2. “The struggle for the transformation of mentality”

What Respondent M calls “The struggle for the transformation of mentality” is deeply embedded in the ethos of the academies. Unsurprisingly, education is seen as the foremost area for changing mindsets on the cultural frontline. Respondent C, for example, stated that “With the awareness that life is built through education, we must treat educational activities in all areas of life and in all social spaces as a first and fundamental priority as a work of mentality construction.”⁶² This entails resistance on multiple levels that distinguish the education system in the academies and throughout NE Syria, from prevailing ideologies that are applied not only in neighbouring Türkiye, Syrian Arab Republic, and Iraq, but worldwide. As we have seen, the academies offer jineology rather than patriarchal or liberal feminist approaches and uphold the values of democratic confederalism rejecting capitalist modernity and what Respondent’s C terms “statist mentality”,⁶³ as well as what is deemed to be holistic rather than positivist science. In a particularly apt ecological metaphor of irrigation, Respondent B spoke of the opportunity for an innovative approach previously denied to the peoples of the Middle East and Kurdistan, that “makes each and every individual of the people important and valuable,” and that is counter to “existing calcified mentalities and desertified minds”. In this appraisal, “Leader APO's

⁵⁹ Nüjiyan Adar, “Jineology is Everywhere,” *Co-operation in Mesopotamia* website (30 March 2022), <https://mesopotamia.coop/jineology-is-everywhere/>, accessed 25 August 2024.

⁶⁰ Respondent E, 15 June 2024.

⁶¹ Elise Boyle Espinosa, e-mail message to Stephen Hunt, 17 May 2024.

⁶² Respondent C, 15 June 2024.

⁶³ Respondent C, 15 June 2024.

[Öcalan's] philosophical approach and struggle of "greening the dried tree" has been the life water of the community academies."⁶⁴ In a similar vein, Respondent L speaks of the importance of developing consciousness through the academies and society so that "revitalization of women takes place in the same period of time as the greening of the barren soil."⁶⁵ Finally, however, to be authentic and meaningful changes in mentality and abstract ideas of liberation must be actualised in the ability of women to create and inhabit physical spaces. Respondent M makes the point that it is essential to for a woman to "overcome the fences in her mind but also to meet with the spaces where she can exist."⁶⁶

Several respondents (B, E, F, L, O) identify the need to address alienation as a means to achieving a free mentality. Alienation takes cultural forms, such as the imposition of an educational system that suppresses teaching in the mother tongue, as Respondent F suggests.⁶⁷ Central to the present discussion, moreover, the idea of humanity's alienation from the living world is also prevalent. Respondent B speaks of "repercussions" that include "the incompatibility of nature and balanced life". Taking a social-ecological approach, Respondent E argued that only when a more humanised society that is free from alienation is attained can the conditions be met to resolve the "ecological problems that our world is facing today".⁶⁸ Similarly, Respondent L speaks of the current system's disconnect and alienation from nature and the need for an ecological approach that "deals not only with plants and animals, but also with human-nature relations."⁶⁹ Respondent F stresses the importance of the lens of *eko-jin* for an economic, social, and philosophical understanding to bring about a life that is compatible with the natural environment. Believing that "Ecology is one of the cornerstones of a peaceful and thoughtful society", Respondent F argues a transformation in thinking beyond the "environmentalist movement" is necessary to "overcome the alienation between women and nature, man and nature."⁷⁰ The colonial, extractivist mindsets that foster such alienation from the living world are firmly embedded in the ideology of capitalist accumulation. This is difficult to dislodge since it gives rise to forms of exploitation enacted by local and dominant states and

⁶⁴ Respondent B, 15 June 2024.

⁶⁵ Respondent L, 27 June 2024.

⁶⁶ Respondent M, 27 June 2024.

⁶⁷ Respondent F, 15 June 2024.

⁶⁸ Respondent E, 15 June 2024.

⁶⁹ Respondent L, 27 June 2024.

⁷⁰ Respondent F, 15 June 2024.

corporations alike. Respondent M unequivocally attributes the ecological destruction in the region to his cause:

The main and dominant cause of the ecological destruction we are experiencing in Kurdistan today is occupation, colonialism and war. Our mountains, villages, plains, forests and waters are being deliberately plundered and destroyed.⁷¹

This holds true for the Ba‘athist regime of Syria: “There is an approach that completely uses the accumulation it has obtained for its own benefit and applies the law of accumulation offered by capitalism to the soil, water and air.”⁷² Furthermore, the hegemonic power and hold upon ideas is so great that, unfortunately, it is also internalised, to the regret of Respondent L who observes: “Local people approach nature from a narrow, everyday perspective, without thinking about the soil, not making it barren or calculating the consequences of climate change in the future.”⁷³ Such is the grip of global capital upon commodity exchanges, that even the Autonomous Administration is entrenched, for example, in its reliance for revenues upon the use of oil resources, typically unrefined due to the lack of refineries in NE Syria, a legacy of colonial development structures that benefited the centralised state, while underdeveloping areas considered as peripheral. Several of the respondents from the academies express criticism of positivism and scientism, which they regard as the intellectual cause of social and ecological destruction. A major task is to challenge and reconnect the “fragmented perspective” born of alienating colonial conditions to a more holistic approach. The philosophical and ethical rejection of positivist epistemology distinguishes the academies’ curricula not only from many prevailing liberal outlooks, but also from the paradigm of the Soviet era, which for Respondent B retained a methodology that “could not overcome the crude materialist understanding and the positivist method of investigation.”⁷⁴ The extreme subject-object split particularly as defined and refined since the Scientific Revolution, eroded a sense of the sanctity of life with catastrophic consequences. Respondent L, for example, who explicitly rejects positivist perspectives, calls for “redeveloping” respect for nature so that it is understood to be “alive and sacred” again.⁷⁵ It is argued that positivism’s over reliance upon empirical methods assumes a problematic detachment that severs science from life. For social constructionists, it has long been axiomatic that the conditions in which scientific and technological research and

⁷¹ Respondent M, 27 June 2024.

⁷² Respondent L, 27 June 2024.

⁷³ Respondent L, 27 June 2024.

⁷⁴ Respondent B, 15 June 2024.

⁷⁵ Respondent L, 27 June 2024.

development are conducted are not neutral.⁷⁶ Despite the guise of objectivity, scientific endeavour takes place within a matrix of power relationships that determine what is studied, the means by which its objectives are formulated, the tools conditions and *modus operandi* of scientific research, what research is culturally backed and funded, the language in which its findings and outputs are articulated, and the uses to which it is put. Respondent M, a critic within the Jineoloji Academy, is forthright in identifying this aspect of the “positivist paradigm”:

[...] The main characteristic of the scientific method developed along this axis is the claim to make the unknown known through experimentation and observation. Everything is permissible to realize this claim. The process of making the divide-and-rule policy dominate the scientific approach is an engine of social genocide and memocide as a crime against society. Science has ceased to be a method of making sense of life, of understanding our place in the universe, of systematizing and sharing the knowledge of life, and has entered the service of power.⁷⁷

Similarly, Respondent F argues that the Scientific Revolution has exacerbated dualistic thinking, leading to “the enslavement of nature, then of women and slaves, and finally the objectification of the entire society.” This disconnect with the natural world brought about by the radical subject-object split, has had fatal social impacts and brought about a destructive relationship with a dominated, commodified, exploited natural world. Consequently, science has operated within a paradigm which has been disconnected from social purposes, and “no longer has any field of concern except the goals of capital and the state”. This trajectory has led to negative social and ecological outcomes such as the invention and development of nuclear weapons.⁷⁸ For Respondent L in the Kongra Star ecology committee, the constraints and distortion of existing power structures lead to “unidirectionality” and “pseudoscience” that prevents the flourishing of more holistic science, grounded in ecological thinking, since “ecology comprehends the human-nature relationship in its entirety and develops a life in harmony with nature by taking a comprehensive approach to solve social problems.”⁷⁹

⁷⁶ A problem of epistemology that scientific philosopher Donna Haraway grappled with in the 1980s in her discussion of the dilemmas of objectivity, “Situated Knowledges: The Science Question in Feminism and the Privilege of Partial Perspective,” *Feminist Studies* 14, no. 3 (Autumn, 1988): 575-599, JSTOR.

⁷⁷ Respondent M, 27 June 2024.

⁷⁸ Respondent F, 15 June 2024.

⁷⁹ Respondent L, 27 June 2024.

It is important to accentuate, therefore, that the strong critique of scientism is not a rejection of science and technology, but an attack upon the positivist assumptions with which they are currently understood and developed within the context of capitalist modernity. Citing Öcalan, Respondent L suggests that issues such as climate change are symptomatic of underlying, deep seated errors in the philosophical understanding of society and the living world, "Liberalism, as in every social issue, tries to suppress the structural essence of the ecological problem and blame technology, fossil fuels and consumer society."⁸⁰ For this systemic reason, an intellectual and cultural struggle through “a holistic, multidimensional learning system” is at the heart of the Kurdish project for social transformation.⁸¹ Respondent M concurs that “Overcoming this knowledge-science structuring is essential for the realization of the unity of free society, free individual, free nature.”⁸² So, how is this shift in mentality towards a socially oriented approach, grounded in ideas from ecology and jineology, reflected in the pedagogy of the academies? Several respondents (A, B, C, F, L) emphasise the importance of critical thinking for learning and the co-creation of knowledge. According to the ethos of democratic education, active strategies are favoured over rote-learning and memorizing approaches. Qualities such as creativity and imagination are highly valued. Respondent F advocates “critical, libertarian and radical pedagogy”,⁸³ while Respondent A, describes the adoption of “the method of questioning, interpreting and discussing knowledge through the collective creation and discovery of knowledge, rather than the transmission and teaching of existing knowledge.”⁸⁴ Further noting that “critical-questioning and curiosity are indispensable principles of the mindset,” Respondent A cites the practice of holding monthly debates as structured occasions to put such intellectual skills into practice.⁸⁵ An exploratory, dialogic approach not only constitutes a dynamic experience but also nurtures a culture of inclusivity and egalitarian relationships within learning spaces, deconstructing hierarchical relations between teachers and learners. Breaking down the polarity of teacher and learner in favour of more discursive, reciprocal relationships in classes is important since mutual support raises and enhances the learning environment for all. Ecopedagogy and all lessons and individual contributions are routinely evaluated, within a process of critical reflective practice. This is put into effect through

⁸⁰ Respondent L, 27 June 2024.

⁸¹ Respondent L, 27 June 2024.

⁸² Respondent M, 27 June 2024.

⁸³ Respondent F, 15 June 2024.

⁸⁴ Respondent A, 15 June 2024.

⁸⁵ Respondent A, 15 June 2024.

*tekmi*⁸⁶ sessions, designed to nurture harmony by addressing and resolving “quarrels and resentments” and other interpersonal tensions.⁸⁷ As Respondent B explains, these time slots for collective self-criticism are intended to cultivate confidence in political participation and implement rigorous assessment of the democratic pedagogy to enhance ongoing processual development.⁸⁸ This approach reflects an overarching aim within NE Syria, where education or *perwerde*, is one of the three pillars of the communes and a core objective for “changing mentalities and empowering people” (Rojava Information Center 2019, 22)⁸⁹ Relatedly, Respondent L connects this spirit of autonomous thinking and self-empowerment with ecological awareness and speaks of the culture of active learning:

In the trainings, question-answer, understanding and deepening the speaker, seeing oneself in what is happening outside oneself, feeling that the other is oneself, revealing and completing self-confidence through dialogue methods, encouraging research, using as many techniques as needed [...] ⁹⁰

Respondent N specifically discusses the power of *tekmi*s to enhance the effectiveness of ecological pedagogy. To build time in to summarise and critically reflect helps to ensure that processes are transparent, set objectives and monitor progress, highlight what works well and identify where adaptation may be necessary. For Respondent N:

Providing *tekmi*s in environmental sessions can be a valuable tool for uncovering both strengths and weaknesses. By offering a comprehensive view of achievements and challenges, it allows for effective improvement of environmental performance and adjustment of strategies.⁹¹

The integration of ecology and the development of ecopedagogies into NE Syria’s educational programmes has been gradual and is recent. While ecology as a general principle of democratic confederalism featured in the 2015 charter for democratic education, its inclusion in the developing curriculum has been incremental. Elise Boyle Espinosa and Adam Ronan investigative fieldwork conducted in 2018 found that “environmental conservation” is mentioned in English language textbooks for schoolchildren which feature chapters such as

⁸⁶ A collective debriefing and participation at the beginning and at the end of the training, where the organizational and individual situation of the participants is reported verbally and in writing.

⁸⁷ Respondent A, 15 June 2024.

⁸⁸ Respondent B, 15 June 2024.

⁸⁹ Rojava Information Center, “Beyond the Frontlines”, 22.

⁹⁰ Respondent L, 27 June 2024.

⁹¹ Respondent N, 29 August 2024.

“Keeping the environment clean” and cite a teaching manual where “loves the natural environment and protects it from damage and pollution” is cited as a key quality of the successful teacher (Boyle Espinosa and Ronan 2023, 2291).⁹² However, as previously mentioned, Espinosa commented that ecology did not feature as much as had been anticipated at that time. Nevertheless, it seems that environmental concerns have recently become more prominent and are closely integrated to ideas about regional history and identity, the destructive impact of capitalism on the natural world, and the need to promote ecological awareness for future generations. Respondent P, Co-chair of the educational administration in the Amude area, affirms that “free and green pedagogy has become a fundamental principle within the academy and continues to this day.”⁹³ Ecological issues are therefore more conspicuous in schoolbooks. Respondent P details the content of two standard texts: “The issue of ecology is now included in our books: the first nature, the second nature, and the consequences of disrupting the balance.”

[...] We have a primary book called "Society and Life," where students learn how their connection with the land is formed and how they are committed to their nature. The sacredness of nature is emphasized in that book; everything is provided to help them understand and comprehend.

A second text, which translates as the "Book of Culture and Ethics", includes an ecology section, where:

[...] topics such as ecology, environmental preservation, land and water management, tree planting, climate change, and its impacts on society, pollution, and health are covered. The book also discusses agriculture, detailing how the destruction of nature occurred and how nature can be protected. The connection between children and nature, as well as their commitment to protecting it, is emphasized.⁹⁴

Ecological ideas are now increasingly apparent within adult academy learning programmes too. Respondent E, representing DAANES' Ecology Council, speaks of progress in attaining formal status for the ecology committees as autonomous entities. This has led to successful efforts to include ecological issues:

[...] discussions, researches, international institutions and organizations, trends of thought on the agenda of ecology in all its dimensions have started to be the

⁹² Espinosa and Ronan, “Rojava’s ‘war of education’”, 2291.

⁹³ Respondent P, 27 May 2024.

⁹⁴ Both quotations from Respondent P, 27 May 2024.

subject of more research and examination after this process, and it has become more covered both in schools and in our academies.⁹⁵

As would be expected, the Academy of Ecology offers comprehensive courses based on ecological ideas. Respondent D shared the syllabus for an intensive five-day seminar programme run in early June 2024. Starting with a conceptual framework for understanding ecology outside of the constraints of positivist approaches, the programme incorporates an assessment of the chief dimensions of the ecological crisis within the context of global capitalism and the state, the prominent categories of ideological debate within political ecology, strategies for moving towards an ecological society across key sectors, and an understanding of the ecological crisis within the colonial context of Kurdistan grounded in Öcalan's ideas. Following the study of these four defined areas of content, a final day is reserved for "Discussions, evaluation and planning."⁹⁶ Such a prospectus gives us a strong sense of why ecology is included as an essential pillar of democratic confederalism. For the most part the planetary consciousness of political ecology meshes comfortably with the spirit of international solidarity at the heart of the Kurdish revolutionary project. Environmental destruction has social consequences, as well as ecocidal impacts upon other species, outcomes that are inextricable. Ecopedagogy lends itself to what Greg William Misaszek terms "deepened and widened reflexivity" (Misaszek 2023, 1259). Its integration within the academies stimulates grounded thinking. It invites the deeper contemplation of geographical, ecological, and cultural specificity within the wider macro-context of both geopolitics and the Anthropocene. It analyses the radical causes of present intractable dysfunctional realities, without obfuscating potentiality, the enduring power of alternative social imaginaries for revolutionary transformation. Ecology also appears in more general programmes for cadres in NE Syria's academies. Respondent E confirms that "It can be said that a course titled Ecology has recently been added to the curriculum of all our academies."⁹⁷ This is corroborated by the replies from Respondent C, who outlines an intensive three-stage education programme for full cadre training (25-days basic, 45-days preparatory, and 75-90 days construction), open to "any citizen who accepts the democratic ecological women's liberationist paradigm in principle" and the DAANES social contract. Progress to the construction stage, enabling a more proactive role in the institutions of building democratic autonomy, is conditional upon passing the previous stages. From the synopsis of the cadre training shared by Respondent C, it is evident that the focus upon political

⁹⁵ Respondent E, 15 June 2024.

⁹⁶ Syllabus shared by Respondent D, 15 June 2024.

⁹⁷ Respondent E, 15 June 2024.

ecology appears in the more advanced stage of cadre training. Environmental concerns do not appear explicitly in the Basic programme, although may be included within the two days devoted to “History of Kurdistan (geographical and chronological)”. The syllabus for the preparatory stage includes two days on “natural community”. Three days of study are dedicated to “Social ecology and communal economy” in the construction stage.⁹⁸ According to insights from social ecology, social justice is only possible where there is ecological security and the ability to meet social needs sustainably. At the same time, local and planetary destruction can only be addressed where there is understanding of the deep social causes of the ecological crisis. Respondent H, therefore, makes the case for foregrounding ecology in all education, “in this century, if the environment as a science of ecology is not a main and essential part of education, this education will remain empty and abstract in its content.”⁹⁹ All our interviewees accepted the value of ecological learning, and the cultivation of “ecological consciousness”,¹⁰⁰ or “ecological intelligence.”¹⁰¹ While it may be the case that any learners who did not subscribe to this positive attitude might not have taken the time to respond, it is unsurprising that adult students at the academies are enthusiastic about ecology since it a fundamental pillar of democratic confederalism underpinning the courses that they have proactively signed up to study. Those prompted to verbalise their feelings about studying ecological issues, therefore, used terms such as “motivating”,¹⁰² “stimulating”,¹⁰³ and “enthusiastic”.¹⁰⁴ Respondent K expressed of the value of ecological education as follows:

It alerts learners and trainees to the importance of the environment in all fields, especially health, economic, development, eco-tourism, rational energy management and benefiting from it for current and future generations, and what the environment reflects on climate change.¹⁰⁵

Respondent J emphasises that ecology is paramount for understanding social and economic issues in the present day: “Perhaps no problems were as important and serious as the ecological ones that could put the real face of the business capital systems or organized networks on the agenda of humanity.”¹⁰⁶

⁹⁸ Respondent C, 15 June 2024, discussion and course syllabus for cadre training.

⁹⁹ Respondent H, 18 June 2024.

¹⁰⁰ A phrase used by Respondents H and J, 21 June 2024.

¹⁰¹ Respondent O’s phrase, 21 September 2024.

¹⁰² Respondent G, 18 June 2024.

¹⁰³ Respondent I, 18 June 2024.

¹⁰⁴ Respondent J, 20 June 2024.

¹⁰⁵ Respondent K, 17 June 2024.

¹⁰⁶ Respondent J, 20 June 2024.

6.4.3. Uniting theory and practice

The slogan “the greatest welatparêzî is planting trees and building forests” is based on [ecological science].¹⁰⁷

In addition to theoretical aspects of ecopedagogy expressed in the interviews, several of the academies’ attendees offer observations that relate to the practical application of their education. The Education Committee of the Ecological Council has a coordinating role to ensure that environmental education complements practical sustainability initiatives being promoted within NE Syria, such as renewable energy projects.¹⁰⁸ Several respondents’ comments convey a sense of the multiple imaginative and sensory ways through which the natural world is experienced and interpreted. These draw upon wisdom traditions and ancestral knowledges. Belief systems which revere nature as “sacred”, particularly those of the Yazidis, Alevis, and Zoroastrians, have endured in the region. Active learning includes, therefore, attention to the intimate relationships between human culture and the living environment, and thinking about how the latter can be cared for and enjoyed through other modes of sense-making. The inclusion of alternative attunements to the natural world, consisting of both practical and metaphorical understandings, leads to a profusion of insights beyond the confines of scientism. Deliberation upon diverse lifeways affords a broader vision in the classroom context, breaking down the rigid separation between humanistic and scientific approaches. Respondent L, for example, writes:

It is possible to access a wide range of information on nature in society, including mythological narratives, stories (*çirok*), fairy tales, experiences gained in cultivating large gardens, details of soil care, herbs of women healers, medicines made from herbs, manual examination and treatment methods of women doctors, etc...¹⁰⁹

Similarly, respondent M alludes to other ways of understanding the natural world, as at once living and animated but also metaphorical “*çirok, stranên dengbêjan* (songs of traditional musicians), lullabies, oral literature and culture in general is a rich field of learning.”¹¹⁰ Such anthropological areas of study may appear to be more connected to otherworldly or historical

¹⁰⁷ Respondent F, 15 June 2024.

¹⁰⁸ Respondent N, 29 August 2024.

¹⁰⁹ Respondent L, 27 June 2024.

¹¹⁰ Respondent M, 27 June 2024.

issues. However, reclaiming these knowledges also has direct application to present-day horticulture and medicine. Further comments by Respondent L develop this case:

We are working on everything from the medicinal properties of herbs to the restoration of the soil to its fertility and natural seeds. In the field of health, various trainings are given to the community on the diseases caused by artificial nutrition and the benefits of eating organic foods.¹¹¹

In this way, radical pedagogy relates to wider cultural and political aspirations, sparking continuous conversations to assess its relevance to the progressive remaking of NE Syria. Several respondents referred to the application of practical measures to bring about an ecologically just society. These begin with schoolchildren and are extended into the research undertaken at the academies. Respondent P stresses the practical nature of ecological learning at school where children learn both self-care and earth-care, so that pupils’ horizons are expanded beyond the confines of individualism from an early age:

Our teachers try to teach students through practice. For example, we form a special ecological committee in the school. As for the students themselves, what are their duties? By practically cleaning the school and its yard, planting trees, and taking care of them, students learn these practices firsthand. We plant trees three times a year. Students clean and maintain the school, and we have organized several campaigns where students cleaned the streets around the school, teaching them the culture of cleanliness. They learn through hands-on experience, including self-care, maintaining their living spaces (rooms, classrooms, gardens, etc.), and separating waste. In collaboration with environmental NGOs or associations, we have built gardens in many schools where students plant trees and care for them. Similar efforts are made in other areas where we have cleaned and prepared spaces for tree planting.¹¹²

This vivid account of practical ecopedagogy recalls the cultivation of the five skins of the ecological aware person expounded by Austrian artist and activist Friedensreich Hundertwasser (1928-2000), namely epidermis, clothes, houses, identity, and earth. Straightforward and concrete activities such as inculcating responsibility for the immediate school environment and beyond help to cultivate ecological intelligence through the everyday recognition of

¹¹¹ Respondent L, 27 June 2024.

¹¹² Respondent P, 27 May 2024.

interconnectivity between the planetary, the interpersonal, and the individual, rendering such potentially abstract concepts easily relatable. Respondent P, acknowledges, however, that such efforts do not always meet with instant success in single generation, conceding that “students, despite our teachings, sometimes still burn a tree, break a table, or cause damage to the school.”¹¹³ This intention to bring the work of education into the wider social arena occurs in a context where climate change and water shortages are an increasing challenge for agriculture, and there is a public health crisis due to air pollution and other carcinogens. Respondent O, for example, expressed the importance of analysing environmental threats, reflecting that “Our connection to nature is always present in our minds,” since the area is already experiencing crises such as mass fatalities caused by air pollution, and extensive flooding and the loss of farmed animals due to adverse climatic conditions.¹¹⁴ Consequently, the need to understand ecological issues to inform practical responses is immediate and urgent. With embargoes and the lack of support from international agencies, it is essential that expertise is nurtured within domestic universities and academies (Rojava Information Center 2022b, 30).¹¹⁵ Research and knowledge sharing about eco-agriculture and ecotechnologies will be key to enhancing livelihoods by sustainable means in a solidarity economy. Respondent F, for example, states that “[...] the technological revolution of anti-industrialization is an urgent necessity in the Middle East. These whole ecological methods can be discussed and projected in academies.”¹¹⁶ Respondent K also speaks to the importance of educating citizens on issues such as climate change, pollution and biodiversity to promote an ecological society. It is argued that ecological education “alerts learners and trainees to the importance of the environment in all fields, especially health, economic, development, eco-tourism, rational energy management and benefiting from it for current and future generations, and what the environment reflects on climate change.”¹¹⁷

Respondent K also outlines the practical know-how content of the academy course that can be applied to transforming NE Syria into an ecological society. Practical means to achieve ecological sustainability are itemised as tree planting, recycling and other ecotechnologies, research centres for strategic planning for “sustainable environmental solutions,” exchanges

¹¹³ Respondent P, 27 May 2024.

¹¹⁴ Respondent O, 21 September 2024.

¹¹⁵ Rojava Information Center, *Young and Promising*, 30.

¹¹⁶ Respondent F, 15 June 2024.

¹¹⁷ Respondent K, 17 June 2024.

with international organizations, agricultural improvement using organic methods and good water management practices, and public information about ecological issues.¹¹⁸ Relatedly, Respondent N stresses the value of case studies as a means to consider the application of ecological knowledge in specific contexts so that “tailored solutions” can be adopted.¹¹⁹ The unity of theory and practice is also beautifully illustrated by the approach to food and cooking at the academies. Conscious communal planning leads to a synthesis of pedagogy and matters concerning social justice, autonomy, and ecological practice. Several respondents (A, B, G, H, I, K, O) replied to a question about communal food preparation. Responsibility for cooking and washing dishes is spread between all attendees on a rotational basis. Working together in this practical way, helps to generate an ethos of collective participation. In the words of Respondent A, it raises “labor consciousness and the will to be self-sufficient”.¹²⁰ Preparing meals together challenges traditional gender roles and breaks down barriers between servers and consumers, extending the learning space by creating opportunities for mutual skill sharing. This communal practice not only makes for efficient use of time, but reduces the energy costs of cooking and reduces waste, minimizing the ecological impacts of food preparation. Respondent K for example speaks of learning “how to dispose of excess food in a way that does not harm the environment” as a part of this process. This spirit of commensality is carried over into the custom of picnicking, a perhaps unexpected theme mentioned by Respondents A and C. These expeditions involve travelling beyond the academy and urban areas, creating, for Respondent C “a relaxation in the relationships in the environment.”¹²¹ Furthermore, not only is food prepared by the attendees, Respondent I explains that an element of self-sufficiency is even carried over to growing food at the academies:

most teaching places and academies have sufficient space for agriculture, so this space must be exploited for agriculture and the joint work of the trainees in it, as it is part of the training in harmony with the environment, and therefore we have reaped good results from this training and meeting the trainees’ need for food.¹²²

Kurdish commentators have noted the close etymological affinity between the Kurdish word for education, *perwerde* and the verb to fly.¹²³ So, it is fitting that liberatory pedagogies are

¹¹⁸ Respondent K, 17 June 2024.

¹¹⁹ Respondent N, 29 August 2024.

¹²⁰ Respondent A, 15 June 2024.

¹²¹ Respondent C, 15 June 2024.

¹²² Respondent I, 18 June 2024.

¹²³ Zozan Sima, “The Women’s Emancipatory Pedagogical Approach of Jineoloji, *Jineoloji* website (21 December 2023), <https://jineoloji.eu/en/2023/08/13/the-womens-emancipatory-pedagogical-approach-of-jineoloji/>, accessed 3 September 2024.

about taking flight and not being caged within delimited physical or intellectual spaces. In the words of Respondent M:

Contrary to stereotypical practices that squeeze knowledge between walls and in books, it is based on vitalizing knowledge, sociologizing historical knowledge, that is, integrating it into today's analysis and developing the power of thought.¹²⁴

Ecological learning in NE Syria, therefore, is embedded within a combination of radical pedagogies and practical organization of educational life within democratic structures. We have seen that this approach is sharply at odds with educational systems prevailing elsewhere, and rests within a paradigm that would demand a transformational shift to be translated and adopted more extensively. Nevertheless, its previously under-researched ecopedagogy, offers invaluable insight into the potential for alternative education. If hegemonic systems continue to fail socially and ecologically, the need for a paradigmatic shift towards such utopian aspirations and efforts may yet appear proactive, realistic, and effective. They can teach us much.

6.5. Conclusion

A long-term commitment to radical education in fact requires less belief in the stories we tell ourselves and others at the outset, and a greater attention to what we learn along the way: unexpected successes and failures. (Laurence Cox)

Education is a priority for the Autonomous Administration in NE Syria. Schools, universities, academies, and even homes as centres of learning are regarded as the chief front for defending the cultural gains of the 2012 Rojava Revolution. Efforts to provide universal and liberatory education are taking place in a precarious situation, due to both ongoing aggression from the Turkish military and its Islamist affiliates, and economic embargoes imposed by hostile neighbours.¹²⁵ Antipathy and scepticism towards its democratic educational curricula from some of its own population also represent a challenge that should be acknowledged. Notwithstanding such difficulties, attendees at the adult education academies express refreshing passion for contextualised learning and knowledge. The present study has examined the

¹²⁴ Respondent M, 27 June 2024.

¹²⁵ Turkish drones murdered two Kurdish journalists, Gulistan Tara and Hero Bahadin, in northern Iraq, at the time of writing in August 2024. <https://cpj.org/2024/08/2-journalists-killed-1-injured-in-drone-strike-in-iraqi-kurdistan/>

incremental integration of a commitment to ecological principles into the educational programmes of DAANES. The focus has been upon the place of ecological principles within the academies’ curricula and radical pedagogies. We have seen that this nascent development complements and is of a piece with the other pillars of democratic confederalism, women’s liberation or jineology, and participatory democracy. This creative synthesis is apparent, for example in a remark by Respondent L who asserted that “it is necessary to deal with women and ecological life in a dialectical connection.”¹²⁶ The greening of the curriculum reflects the insight from social ecology that there can be no long-term social or economic justice without addressing the ecological crisis, but also that the latter cannot be achieved without attention to the deep social dysfunction that is causing planetary boundaries to be catastrophically breached. The interview responses have been invaluable in helping to identify what distinguishes the democratic society academies from the educational systems guided by capitalist and statist ideologies. For example, the respondents consistently critique and reject positivism when researching and interpreting the physical world. Democratic education is associated in the minds of the respondents with secular ethics in contrast to religious morality and amoral capitalism. Our interviewees have revealed this ethical dimension through their claims for social justice, pedagogy, and in ecological principles. Respondent L referred to the “ethics and aesthetics of freedom”,¹²⁷ while Respondent M remarked that “moral-political society”¹²⁸ is their reference for “knowledge structuring”, and Respondent K held that “our society’s goals are moral, political, and ecological.”¹²⁹ Respondent O states that “A free and ethical relationship within the nature-society represents a democratic life in its essence,” in contrast with Liberalism where capitalist industrial practices prevail which, they argue, represent a fracturing of the “collective memory” of human societies which maintained intimate bonds with the living world.¹³⁰ Respondent M, in particular, defines the “paradigm of democratic modernity” in a way extends respect for others beyond humans to other species arguing that it includes consciousness of “the sinfulness of harming trees and animals, the culture of helping those in need, the resolution of disputes through social consultation, or the knowledge that one should not hurt the earth...”¹³¹ Finally, the fieldwork has indicated that the inclusion of ecopedagogy within the learning strategies of democratic society academies is meaningful to respondents

¹²⁶ Respondent L, 27 June 2024.

¹²⁷ Respondent L, 27 June 2024.

¹²⁸ Respondent M, 27 June 2024.

¹²⁹ Respondent K, 17 June 2024.

¹³⁰ Respondent O, 21 September 2024.

¹³¹ Respondent M, 27 June 2024.

because it powerfully enables thinking about place-making in a non-statist way, since ecosystems are not defined and constrained by national borders. There are parallels between the celebration of the cultural and linguistic diversity of NE Syria's multiethnic communities and the need to move away from agricultural monocultures and to defend biodiversity. We have seen that the democratic confederalist cadres of the academics understand ecological resilience in the context of colonial history and contemporary geopolitics. The impetus to defend the cultural specificity and integrity of Kurdistan by resisting the homogenizing impact of globalization, coexists and complements internationalist and planetary sensibilities. In addition to the theoretical rethinking of NE Syria's unique educational system, respondents have also indicated the way that the practical application of ecological measures are helping to build social resilience. In the words of Respondent B, "Community academies strive to play a core role in the reorganization of life."¹³² As we have seen, therefore, the radical pedagogies of the academies have real-world applications. The work of the Education Committee of the of the Ecological Council is tasked with ensuring that ecological education at all levels is aligned with the principles and objectives of DAANES. They intended to spiral out to enhance and remake society, rather than circulate exclusively within a closed circuit of academia. Respondent M speaks of education as working with a paradigm that seeks to activate learning in a way that perpetuates social change.¹³³ This is a work in progress and a work of progress.

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¹³² Respondent B, 15 June 2024.

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Chapter 7. Article V: Ecological struggle as ethos of decolonization: The emergence of a decolonial ecology in Northern (Bakur) and Western (Rojava) Kurdistan

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7.1. Abstract

This article examines why Kurdish Freedom Movement adopted ecological struggle as a foundational pillar in its politics of decolonization and the reclamation of autonomy. Framed within the context of colonial and ecocidal developmentalism, the research analyses how policies initiated during the Ottoman Empire and perpetuated by Turkish and Syrian nation-states systematically dismantled Kurdish economic and ecological foundations to entrench colonial dominance. Conceptualizing these dynamics as an ecocidal developmentalism, the article elucidates how ecocidal socio-spatial designs have functioned as mechanisms of elimination, targeting the socio-political, economic, and cultural fabric of Kurdish society under the guise of development in *Bakur* (Northern Kurdistan) and *Rojava* (Western Kurdistan). In response, the Kurdish Freedom Movement embraces an ecological struggle rooted in communal ownership, ecological stewardship, and traditional and liberatory epistemologies through the political ontology of Democratic Autonomy. This ethos challenges colonial and capitalist structures by advancing grassroots governance, gender liberation, social justice and eco-economy, signifies the emergence of a decolonial ecology.

Keywords: ecocidal developmentalism, de-development, colonialism, decolonial ecology, democratic autonomy

7.2. Introduction

This article investigates why and how the Kurdish Freedom Movement (KFM) has embraced ecological struggle as a foundational principle for fostering autonomy and advancing decolonization, with a particular focus on developing eco-economy as a structural mechanism to counter systematic ecocidal development policies imposed by colonial powers. It analyses how colonial and ecocidal development strategies in Kurdistan systematically targeted natural resources and geographic assets to dismantle Kurdish economic structures and entrench colonial dominance. These policies, initiated by the Ottoman Empire in the 1830s, were perpetuated by its successors, Türkiye and Syria. By conceptualizing this dynamic as colonial and later ecocidal developmentalism, the study elucidates how such socio-spatial designs were deliberately employed by colonial powers as instruments of elimination against Kurds' essential socio-economic and cultural elements.

The existing literature on this subject remains both limited and fragmented. Some studies explore colonial dynamics within the Kurdish Freedom Movement's (KFM) ideological and political framework (Akkaya and Jongerden 2013; Knapp et al. 2016; Matin 2021; Sunca 2023a, 2023b; Holmes 2024; Jongerden 2024). Others address the colonial underdevelopment of Northern Kurdistan (East and Southeast of Türkiye) as a peripheral region (Yadirgi 2020), KFM's alternative economic model in Bakur (Aslan 2022), and conceptualizing ecological destruction and development projects through conflict and counterinsurgency dynamics (Harris 2002, 2012; Van Etten et al 2008; Jongerden 2010; Türk 2023). The struggles in Rojava/the North and East Syria has attracted considerable attention, particularly concerning ecological and feminist movements (Piccardi 2021; Dirik 2022; Piccardi and Barca 2022) and the anti-capitalist dimensions of Rojava's economy (Aslan 2023). Other works explore the intersections of ecocide, ethnic cleansing, and neo-colonial dynamics of settler colonialism in Rojava (Schmidinger 2019; Grasso 2024; Dinc & Türk, 2025). Further studies highlight the autonomous agricultural system (Jongerden 2021) and reconstruction of ecological agriculture as a decolonial endeavour (Türk and Jongerden 2024). Ecology's role in Rojava's educational framework has also been explored (Espinosa and Ronan 2023; Hunt and Türk 2024 forthcoming). There are few works focused on the role of ecological struggle in Kurdish liberation process often framed through an anti-capitalist rather than anti-colonialist lens (Aslan 2022; Hunt 2019, 2021; Pimbert 2021). A couple of works create link between ecological struggle and decoloniality without delving into colonial economic marginalization/de-

development structure (Sustam 2021, 2022; Yıldırım 2024). This research aims to contribute to filling this gap.

Drawing on my extensive engagement as an activist-scholar in ecology movements across Kurdistan and Europe, including participation in tens of local and international conferences, seminars, and workshops, this study benefits from a nuanced understanding of the philosophical and practical evolution of KFM's ecological struggle. As a pioneer and founding activist of the Kurdistan Ecology Movement-Europe (*Tevgera Ekolojîya Kurdîstanê-Ewrûpa*, TEV-EKO), I have organised and participated in significant events related to this topic in Bakur, Rojava, and Europe in 2024¹³⁴. The methodology adopts a qualitative approach, incorporating online interviews, surveys, fieldwork and group discussions across Bakur, Europe, and Rojava. In Bakur, three online interviews with key figures and a group discussion with Mesopotamia Ecology Movement (MEM) board members provided collective insights into the region's ecological activism. In Europe, data collection included two in-depth interviews with key figures. In Rojava, data collection was extensive, combining online interviews and fieldwork as part of my PhD research. In 2024, 17 interviews and 40 surveys were conducted in three provinces/towns (Dêrik, Tirbespiyê, and Hesekê, 12 villages). Fieldwork from April to June 2024 deepened the understanding of Rojava's case, supplemented by 21 online expert interviews in 2023 and group discussions in *Jinwar*¹³⁵ and Dêrik with leading local figures.

¹³⁴ 1) Ecology Conference: "Ecology in Democratic Modernity - From Resistance to Construction: Perspectives, Possibilities & Challenges" held on April 26-27, 2024, Qamiso/Rojava-Syria

2) Agriculture Symposium by Democracy and Equality Party (DEM Party): "Building the Future in Agriculture and Food" held on September 6-7, 2024, Wan/Turkey

3) "The Global Gathering for Climate and Life – Anti-COP 2024", November 4–9, 2024, Oaxaca/Mexico.

4) Ecology Symposium: Kurdistan Ecology Movement-Europe (*Tevgera Ekolojî ya Kurdîstanê-Ewrûpa*, TEV-EKO) was declared, held on 21-22 December, 2024, Freiburg/Germany.

¹³⁵ Existing literature identifies *Jinwar* as a "place of women" (Cioni and Patassini 2021, 133) or an "eco-village of women" (Dirik 2022; Piccardi 2021). However, both definitions are inadequate and lead to a superficial understanding of the concept. In Kurdish, *cîh* translates to "place," while *war* refers to a place where one feels safe, experiences belonging, and finds meaning in identity and existence—a space that is marked, adopted, designed, and recognised as familiar and peaceful. This concept has no English equivalent and is more akin to the German concept of *Heimat*. The notion of *Heimat* encompasses a sense of spatial and social familiarity that fosters safety, trust, and stability. It is intrinsically tied to community, space, and tradition, serving as the foundation of identity (Greverus 1979). In this regard, *jînwar* signifies the making-world or the ontological design of a world where women dismantle the codes of patriarchy, colonialism, and capitalist exploitation through *xwe afirandin* (self-creation or realisation). In a group discussion with four pioneering women in *Jinwar*, they emphasised that the concept should not be confined to a single location (referring to its current physical setting). On the contrary, they argued that it embodies a perspective advocating for the *jînwarisation* of all cities and villages.

The article unfolds in three analytical sections. The first section examines how the Ottoman Empire's centralization in the 19th century transformed Kurdistan into an underdeveloped periphery through the liquidation of Kurdish political, demographic and economic foundations as a colonial structure. This chapter conceptualises ecocidal developmentalism as the nation-state colonial structures of the Turkish and Syrian states as a continuation of the Ottomans' periphery-centre developmental structure. To this end, it argues that ecocidal developmentalism has become the main methods of the colonial structure in terms of designing the socio-spatiality and liquidating the economic foundations of the Kurds. This strategy seeks to eliminate Kurdish culture and resistance while consolidating state exploitation and dominance.

Second section delves into the emergence of a decolonial ecology in response to these colonial structures, framing ecological resistance as ethos of decolonization. Drawing on Öcalan's philosophical shift and the KFM's unique political framework, this chapter explores the role of eco-economy and ecological stewardship in building democratic autonomy to challenge colonial structures through indigenous epistemologies and ontologies.

The discussion synthesises these findings by connecting colonial policies, ecological destruction, and the KFM's ecological resistance strategies. To this end, it reveals how KFM put great efforts to bring together Kurdish ecology movements from four parts of Kurdistan and across the world while addressing the challenges KFM faces in operationalizing its ecological vision amid external threats and internal obstacles.

7.3. Recolonization of Kurdistan as a periphery

Historically, Kurdistan has been invaded by external empires for its abundant resources and strategic location. The most dramatic phase of re-colonization began with the Ottoman Empire's centralization during the *Tanzimat* (Imperial Edict of Reorganization, 1839) (Yadirgi 2020; Kaymak and Teoman 2016, 13), dismantling Kurdish socio-economic (agriculture and nomadic) and political structures (confederal tribal system). Between the sixteenth and nineteenth centuries until the 1830s, Kurdistan exercised near-complete autonomy over land, taxation, law, and military matters (Kaymak and Teoman 2016, 10-11; Tejel 2009, 65; Bozkurt and Akpınar 2011, 172-74; Yadirgi 2020, 64-94). However, the Ottoman Empire reasserted direct control in the 1830s and 1840s as it was identified "fully reconquering of the country of

Kurdistan” (Bozkurt and Akpınar 2011, 98). This liquidation of Kurdish autonomy was formalised by founding the Kurdistan Province (*Eyalet-i Kurdistan*), administered by a colonial governor (Bozkurt and Akpınar 2011, 28; Yadirgi 2020, 98). Thereafter, the *Hamidiye* Regiments under Abdülhamid II gained territory and power, seizing Armenian and Kurdish lands, while Kurdish communities faced forced displacement (Yadirgi 2020, 159). Demographic engineering became key, echoing Wolfe’s assertion that “invasion is a structure not an event” (Wolfe 2006, 388), and continued through the 1857 settlement regulation and the 1858 Land Code, which dismantled Kurdish communal production (Fratantuono 2018 6; Kaymak and Teoman 2016, 15) through settling “mass Muslim migration from the Crimean Peninsula, Caucasus, and Balkans” in Kurdistan (Fratantuono 2018, 1). During the First World War, genocidal strategies intensified; after Armenian genocide (1915), over one million Kurds were displaced to western Anatolia (Üngör 2011, 117; Yadirgi 2020, 159; Jongerden 2007, 178), and properties were seized under “abandoned property laws” (Adalet 2022, 7; Yadirgi 2020, 160-61).

After the Committee of Union and Progress (CUP, 1913–1918) took power in 1913, these policies were extended to Kurds and non-Turks/Muslims, featuring “mass murder, deportations, expropriation and dispossession of economic resources...” (Yadirgi 2020, 58-59). Following the Ottoman Empire’s collapse, Britain and France partitioned Kurdistan via the Sykes-Picot Agreement (1916). Under the Treaty of Lausanne (1923), Kurds lost self-determination rights enshrined in the Treaty of Sevres (1920), leading to official denial of their existence without any status (Beşikçi 1990; Bozarslan 1996; McDowall 2004; Öktem 2005; Fernandes 2012; Duruiz 2020; Jongerden 2024).

Today, *Bakur* and Kurdistan’s other parts remain among the least developed regions in their respective states, with chronic deficits in per capita income, healthcare, education, and employment due to this colonial legacy (McDowall 2004; Harris 2008; Knapp et al. 2016; Yadirgi 2020; Pimbert 2021; Aslan 2023). Yet, from the seventeenth to nineteenth centuries, these territories supported prosperous agriculture, manufacturing, and trade. Veli Yadirgi (2020) attributes *de-development* to deliberate colonial economic policies. Ottoman Kurdistan’s agriculture, generating more than half of total tax revenues (108), and its robust industrial sector (144) were systematically eroded by centralization, land dispossession, and the 1858 Land Code, concentrating land in the hands of a small elite “not interested in increasing productivity” (Heper 2007, 16; Yadirgi 2020, 40-41). According to Yadirgi, underdevelopment was not

caused by geographic isolation and lack of infrastructure but by *colonial de-development* (Yadirgi 2020, 264-65). These historical disruptions continue to shape Kurdistan's socio-economic landscape, highlighting the enduring impact of colonial and centralised statecraft.

7.3.1. Ecocidal developmentalism as a colonial continuum in nation-state colonialism

Although classical forms of colonialism may no longer exist in their historical guise, colonialism perpetuates itself through renewed methods of exploitation, subjugation, and systemic violence, concealed by ideological and discursive technologies as well as epistemic and scientific violence (Quijano, 2007; Moses 2008; Ferdinand 2022). This colonial continuity becomes manifest in the concept of “development,” which Escobar (2018) calls “the most portentous social experiments of the past seventy years—a grand design gone sour” (xiii). “Development” encompasses modern scientific techniques, technocratic rituals, practices, and discourses as a laboratory project applied to cultures, socio-economic modes of production, and diverse identities deemed backward savages (Visvanathan 1988; Escobar 1995, 2018; Scott 1998; Hinton 2002; Mitchell 2002; Dunlap 2018; Crook and Short 2020; Wise 2021). Such a process has led to the annihilation of both nature and communities, identified by some scholars as “developmental genocide” (coined by Helen Fein) (Dunlap 2021, 219) and “ecocidal development,” (Wise 2021, 195) reflecting a genocide-ecocide nexus rooted in colonial development. To this end, de-development, as a product of colonial development policies, aims “to ensure that there will be no economic base, even one that is malformed, to support an independent indigenous existence” (Roy 1995, 4).

Damien Short (2016) significantly unveiled that “genocide’s inherently colonial nature” (3) is often enacted through environmental destruction/ecocide, which can result in destruction of *essential foundations* of local communities. Ecocide involves extensive destruction of ecosystems to such a degree that the communities reliant on them face existential threats. Crucially, Crook and Short argue that ecocide can be a method of genocide if “environmental destruction results in conditions of life that fundamentally threaten a social group’s cultural and/or physical existence” (Crook and Short 2014, 298). Many state-led or corporate ecocidal projects are justified under economic “intent” or the guise of “economic development” and often result in the forced displacement of local peoples who have lived sustainably on these lands for centuries (Dunlap 2018, 557). By displacing these communities, states and

corporations undermine their way of life, culture, and ultimately their survival. Such displacement is not only material dispossession, but also spiritual dispossession. This represents a genocidal act, not in the form of direct killing but through the erasure of a way of life that defines the group (Short 2010, 2016; Crook and Short 2014; Lindgren 2018; Dunlap 2018, 2021, 2023; Jaber, 2019; Priestley 2021; Wise, 2021; Lynch et al., 2021; Dinc & Türk, 2025). Thus, ecocide and cultural genocide, two sides of the colonial coin, are historically exercised and legitimised through development designs. Turkish and Syrian regimes exemplify this nation-state colonialism and employ internal colonization, embodied by ecocidal developmentalism, whereby ecocide functions not merely as an economic exploitative practice but also as part of the state's "logic of elimination," completing cultural genocide as stated by article 11 of the Ecology Conference, "...ecocide in Kurdistan does not only demonstrate typical nation-state developmentalism or exploitation by capitalist corporations, but is also enacted as a technique of elimination/cultural genocide."¹³⁶

Such policies prompt demographic engineering, extreme land concentration, economic marginalization/poverty, and ecological destructions. Examining ecological destruction in Bakur and Rojava Kurdistan as a colonial continuity anchored in the eliminatory logic of ecocidal developmentalism clarifies why the KFM regards ecological struggle as existential within its ontology, as stated by article 5 of Ecology Symposium:

Ecocide and the liquidation of the economic structure are intertwined and function as the main tools of cultural genocide. Therefore, the ecological struggle in Kurdistan must be anti-colonialist and anti-capitalist and is one of the most fundamental ethics and duties of *welatparêzî*.¹³⁷

This article does not aim to debate the existence of ecocide in Kurdistan as discussed somewhere else (see Türk 2023); rather, it seeks to analyse its implications and strategic deployment. A review of state discourse, institutional reports, and practices—including the GAP, Arab Belt project, extractivism, forest burning and deforestation, and the construction of security dams—reveals numerous instances where systematic ecological destruction has been adopted as a military and political strategy to suppress the Kurdish struggle (Harris 2002, 2008, 2012; McDowall, 2004; Van Etten et al 2008; Tejel 2009; Jongerden 2010; Bilgen 2017; Akıncı

¹³⁶ "Ecology in Democratic Modernity - From Resistance to Construction: Perspectives, Possibilities & Challenges" held on April 26-27, 2024, Qamislo/Rojava-Syria

¹³⁷ *Welatparêzî*, a Kurdish term, means to love, serve and defend one's homeland.

et al. 2020; Dinc et al. 2021; Adalet 2022; Türk 2023; Türk and Jongerden 2024; Dinc & Türk, 2025). The concept of *ecocidal developmentalism* is employed here to describe how large-scale ecological destruction, with long-term and often irreversible impacts on local ecosystems and populations, is deliberately designed and framed as development. This approach is utilised by Turkish and Syrian regimes as a tool to assimilate the Kurdish population socially and culturally by dismantling their economic and ecological foundations, while simultaneously building national and colonial economies through the exploitation and plundering of Bakur and Rojava Kurdistan under the guise of development.

7.3.2. *Bakur* Kurdistan

Bakur has long been treated as a periphery by the Turkish state, subjected to deliberate underdevelopment, repression, systematic exploitation, cultural genocide, and ecological destruction, all intended to integrate *Bakur* into the Turkish nation-state and suppress Kurdish autonomy. Turkification policies, framed as Western-style nation-building, form a core feature of the modern Turkish state (Bozarslan, 1996; Öktem 2005, 2008; Zeydanlıoğlu 2008; Fernandes 2010, 2012; Yadirgi 2020; Jongerden 2007, 2024). Turkish nationalists increasingly focused on restructuring the countryside, treating the drive to “nationalize” rural Türkiye as a project of “internal colonization”. They cast state agents as (cultural) missionaries, implying that Anatolia’s rural populations needed to be “converted” or “civilized” and “become Turks”. Their “nation-building” mindset was not a mere metaphor; they believed the new built environments they created could in fact transform villagers into Turks (Jongerden, 2007, p.198). Under the discourse of “development,” Kurdish geography and culture are labelled “backwards,” “feudal,” “tribal,” “traditional,” or “underdeveloped,” necessitating socio-economic and cultural interventions; indeed, “the southeast’s atavism is necessarily articulated through notions of Kurdishness.” (Harris 2008, 1704). As a result, the *underdevelopment of Kurdishness* has become a key narrative justifying the destruction of *essential foundations* of Kurds.

Drawing on the blueprint of the Armenian genocide by CUP; its successor Kemalist regime designed elimination policies targeting Kurds, culminating in measures such as the Plan for the Reformation of the East (1925) and the Resettlement Law (1934). Under these laws, Kurdish tribal assets were seized, and Kurdish-majority regions depopulated, with inhabitants forcibly

relocated to western Türkiye (Üngör 2011, 149-162). In 1927, Foreign Minister Tevfik Rüstü Aras openly endorsed dismantling Kurdish territorial unity, population, and socio-economic structures to ensure Kurds remained “incapable of self-government” by “clearing out the Kurds of their valleys ... and settl[ing] Turkish peasants there” (Yadirgi 2020, 167). Over half a million Kurds were thus displaced, while thousands of Muslim/Turkish families were settled in *Bakur*, stifling Kurdish nationalism and entrenching reliance on the central state. Marshal Fevzi Çakmak, an early key Republican architect, believed “economic development and wealth would ... lead to the development of nationalism among the Kurds” (Kılıç 1998, 97), prompting systematic underdevelopment through minimal investment in Bakur within Five-Year Development Plans of the 1930s (Yadirgi 2020, 168-170). Turkish state combined this colonial development structure with ecocidal developmentalist policies. To this end, the Southeast Anatolia Project (GAP), launched in the 1970s (expanding in the 1980s and 1990s), is one of Türkiye’s largest regional “development” projects, encompassing 22 dams, 19 hydroelectric plants on the Euphrates and Tigris, and irrigation for 1.8 million hectares. While this initiative provides energy, irrigation, and export crops, it also serves as an instrument of assimilation/suppression by redesigning Kurdish socio-spatiality (Harris 2002, 2006, 2008, 2012; Jongerden 2010; Bilgen 2017, Akinci et al. 2020, Türk 2023). Some see GAP as “a new phase of exploitation and colonization of both natural and human resources in the region” (Aytar 1991, 7). Observers note that GAP overlooks “cultural sustainability,” radically altering or erasing the language, lifestyle, and social organization of nomadic populations (Harris 2002, 753). Consequently, subsistence-based peasant and nomadic livelihoods have been decimated, exacerbating dispossession, inequality, and economic marginalization. Agricultural policies replaced drought-resistant crops with export-oriented varieties, and denied most Kurdish-majority areas water resources due to colonial extreme land inequality. In addition to soil salinization, erosion, waterlogging, and deforestation, GAP prompted displacement and shifting patterns of land ownership (Harris 2002, 2006, 2008, 2012; Jongerden 2010; Bilgen 2017, Akinci et al. 2020; Adalet 2022; Türk 2023). In contrast to the state’s “development” discourse to justify GAP economic, social, cultural and ecological destructions, the reality shows an opposite image: By 2006, 74 per cent of hydroelectric projects were complete, but only 14 per cent of irrigation was implemented (USARM 2009, 42). Most energy generated left the region, with 69 per cent of electricity exported elsewhere (Yadirgi 2020, 253). By 2006, Bakur’s per capita income was 54 per cent below the national average, and all twenty-one provinces of Bakur there received fewer state subsidies combined than a single province in western Türkiye: Bursa (USARM 2009, 35-36). Between 2002 and 2007, 69 per cent of public investment in

Bakur targeted energy extraction, with only 28 per cent for health and education (USARM 2009, 43).

Additional *ecocidal extractivism* compounds these issues. In 2021 alone, “7 per cent of forests” in Besta, Cudi, and Şırnak were cut down, “hundreds of thousands of hectares,” visible via Google Earth.¹³⁸ Hundreds of small Hydroelectric Power Plant Projects (HEPPs) and more than 50 dams, particularly in Dersim and Botan, serve “security reasons.”¹³⁹ Dersim is invaded by 145 mining projects over 143,000 hectares, while Botan suffers from rampant dam-building, deforestation, and illicit mining¹⁴⁰.

From the CUP to the Kemalist era (1919-1950), Turkish policies integrated colonial demographic and economic engineering, systematically producing *de-development* in Bakur (Yadirgi 2020, pp. 265-66). Since the 1960s, these strategies have become more sophisticated through ecocidal developmentalism and extractivism, functioning as new methods of de-development and cultural genocide that perpetuate the subjugation of the Kurdish people.

7.3.3. Rojava

After independence from French colonial rule in 1946, Syria adopted a nation-state model enforcing Arabization to consolidate state power and homogenise its population. The Ba‘ath Party’s rise in 1963 intensified these policies, subjecting Kurds to forced assimilation, dispossession, and territorial fragmentation under a national developmentalist agenda (McDowall 2004; Tejel 2009; Aslan 2023; Türk and Jongerden 2024). A cornerstone of this approach was the *Arab Belt* policy (1960s–1975), which displaced tens of thousands of Kurds and seized fertile lands along the northern border with Türkiye (McDowall 2004). It disrupted Kurdish territorial unity, severed links to Türkiye and Iraq, and created an Arab settler to buffer Syria from broader Kurdish regions (McDowall, 2004; Türk and Jongerden 2024). As a result,

¹³⁸ Haber Merkezi, “Şırnak’taki ağaç kıyımı uydudan da görüntülendi”, Yeşil Gazete, August 11, 2022, <https://yesilgazete.org/sirnaktaki-agac-kiyimi-uydudan-da-goruntulendi/>.

Ecohub, “Deforestation in Kurdistan due to the war”, 2023, <https://www.ecohubmap.com/hot-spot/kurdistan-areas-deforested-by-turkey-used-to-be-so-dense-one-could-barely-see-the-sky/11yxis0ml7u76ki1>.

¹³⁹ <https://cdniys.tarimorman.gov.tr/api/File/GetFile/425/KonuIcerik/759/1107/DosyaGaleri/dsi-2007-faaliyet-raporu.pdf>, 142.

¹⁴⁰ <https://yesilgazete.org/dersim-dosyasi-hes-baraj-ve-maden-ocagi-kiskacinda-bir-kent/>

Rojava faced colonial de-development through the statist central agricultural modernization and socio-spatial designs (Türk and Jongerden 2024). To this end, Rojava's de-development was amplified by extractivist *ecocidal developmentalism*, often mirroring Türkiye's GAP. Under the Ba'ath regime, large-scale irrigation and monocrop production, managed by the General Authority for Development of the Euphrates Basin (GADEB) (Springborg 1981, 192-193), centred on the Tabqa Dam (completed in the 1970s) to boost wheat, barley, and cotton yields (ibid., 192). However, this export-oriented production mainly benefited areas outside Rojava, leaving it an "ecologically degraded periphery" (Türk and Jongerden 2024, 15). For instance, Cezire region of Rojava covers 40 per cent of Syria fertile land and contributing significantly to agriculture and energy production. The region produces two-thirds of Syria's cereals, including 70 per cent of its wheat, and three-quarters of its hydrocarbons. Despite its economic role, Cezire has been underdeveloped as it is the poorest region of Syria, with high illiteracy and unemployment rates. By 2010, poverty had risen to 80 per cent, and unemployment stood at 25 per cent in 2004, more than double the national average (Ababsa 2015, 201).

Türkiye's 2018 invasion of Afrin, was one of the three cantons of Rojava before the invasion, advanced this colonial continuum, displacing over 300,000 Kurds in favour of Arab and Turkmen settlers—reducing Afrin's Kurdish population from about 90s to 20s per cent by 2023 (Schmidinger 2019; SOHR 2021; Human Rights Organisation Afrin 2023; Dinc & Türk, 2025). This ethnic cleansing was coupled with ecological devastation, notably the uprooting of more than 350,000 olive trees, with some reports suggesting even higher figures (SOHR 2021; Human Rights Organisation Afrin, 2023; Dinc & Türk, 2025). Such actions underscore how *ecocide and genocide* intersect, a point echoed by the Ecology Conference, which states that "ecocide and genocide are intertwined in Kurdistan as fundamental policies of the region's colonial states" (Article 6).

Consequently, the systematic demographic engineering, economic and cultural marginalization, resource extraction, and ecological destruction enacted by both Syria and Türkiye constitute systematic colonial economic resources control mechanisms and socio-spatial designs that debilitate Kurdish communities and constrains any future for Kurdish self-determination.

7.4. Emergence of a decolonial ecology

The ongoing structural crisis of the existing capitalist modernity is deeply rooted in the intertwined legacies of colonialism, capitalism, and patriarchal systems. These legacies have contributed to a vast range of interconnected crises, including climate crisis, ecological degradation, extreme social inequality, structural violence and exploitation, loss of biodiversity and cultures through genocidal and ecocidal practices and policies at global level. These interconnected crises are produced by what Ferdinand calls “colonial ecology,” a term refers to “an ecology whose function it is to preserve colonial inhabitation and the forms of human and non-human domination that come with it.” (Ferdinand 2022, 115). In colonial ecology, the ecological crisis, often portrayed as a purely environmental issue, must be understood as intrinsically linked to historical processes of colonization and capitalist exploitation. The colonial roots of the ecological crisis, which Ferdinand terms the “colonial fracture,” continue to be obscured by environmentalist movements that fail to confront the historical exploitation of both human and non-human life as a second fracture: the environmental fracture (ibid., 175). In response to these challenges, decolonial ecology offers a framework that directly confronts the colonial and environmental fractures that define capitalist modernity. Decolonial ecology seeks to dismantle the colonial structures that have shaped not only human relationships but also relationships with the non-human forms. Ferdinand describes decolonial ecology as “an ecology of struggle” that is intimately linked to demands for political emancipation, social justice, and equity (Ferdinand 2022, 175). This approach is rooted in the recognition that colonialism is not merely a historical phenomenon but an ongoing process that continues to shape contemporary ecological and social conditions. In contrast to modernity’s one-world ontology, which promotes a singular, universal worldview dominated by capitalist and colonialist logic, the concept of the pluriverse, central to Escobar’s work, offers a vision of a world where multiple ontologies and ways of knowing, being and doing coexist. The pluriverse, as Escobar (2018) explains, “is a world where many worlds fit” through Zapatista’s formulation (xvi), and it is through this framework that decolonial ecology seeks to reimagine the relationship between humans and non-humans. World-making, as Escobar describes it, involves a relational ontology that recognizes the interconnectedness of all beings, both human and non-human (Escobar 2018, 230). To this end, “designs for the pluriverse becomes a tool for reimagining and reconstructing local worlds.” (ibid., 4). Reconstructing Kurdistan as a new world through decolonial ecologies has become ontological ethos of KFM as discussed below.

7.4.1. Ecological struggle as ethos of decolonization

The transformation of the Kurdish movement into a socio-ecological revolutionary force is deeply rooted in Abdullah Öcalan's ideological shift, particularly influenced by Murray Bookchin's philosophy of social ecology and libertarian municipalism. Central to this shift is the recognition that the ecological crisis is inseparable from the social problems of domination and exploitation (Bookchin 2006; Öcalan 2020; Gerber and Brincat 2021). According to Öcalan, solving the ecological crisis required overcoming all forms of hierarchy, including capitalist, colonialist, and patriarchal structures (Öcalan 2020). By integrating social ecology into its political framework, the Kurdish movement positions a radical rupture with the statist, capitalist and colonialist epistemology and ontology of Europe and the West. To this end, it adopts ontology of multiplicity that affirms its own difference through reconstructing Kurdish historical forms of knowledge and socio-political forms according to contemporary needs (Akkaya and Jongerden 2013; Öcalan 2020; Cemgil 2021; Matin 2021; Goner 2023; Sunca, 2023a; Hunt and Türk 2024 forthcoming). To this end, it aims at the construction of an ecological society based on direct democracy through the political forms of democratic autonomy and libertarian municipalism as a new constitutive and creative design (*afîrîner*)¹⁴¹ as well as based on ecological agriculture and eco-economy (Knapp et al. 2016; Jongerden 2021; Hunt, 2019, 2021; Aslan 2022, 2023; Jongerden 2024; Türk and Jongerden 2024). This ideological transformation corresponds to the socialization of this ideological transformation, as many ecological movements, which include both political and social dynamics, aim at a socio-ecological transformation by struggling against the genocide-ecocide nexus in Kurdistan.

¹⁴¹ *Afirandin* (verb) in Kurdish refers to bring into existence, to construct, to constitute, to build, to invent, to create, to generate, to found, to design (artwork, idea, process) and to make. The *afîrîner* (subject/noun) can be a god, an artist, an artisan, body or nature, as well as time, thought/thinking and ideology. *Xwe afirandin* refers to self-creation and self-affirmation, and not necessarily refers any external references. The concept of *Xwebûn* (being oneself), for example, is an *afîrîner* ontologically, as a praxis of moral affirmation and decolonisation that involves processes of knowing, understanding and loving oneself.

I offer this concept to identify Öcalan philosophy as an *afîrîner* regarding how it transforms and designs dynamics of KFM and Kurdish society ontologically. This perspective allows us to see and read Öcalan's texts as a design of political ontology rather than epistemic sources.

7.4.2. Decolonization of epistemology and ontology

The traditional socio-political fabric of Kurdish society, particularly within *Asiret* confederations or autonomous tribal structures, has long been defined by a deep connection between land, culture, and spiritual beliefs. The land-based economy of peasants, pastoralists, and nomads was not merely an economic structure but also a cultural one, with profound ties to nature as a sacred entity. Belief systems such as Yezidism, Zoroastrianism, and *Rêya Heqî* (the path of truth, in Kurdish) place nature and land at the core of their cosmologies, which inform the socio-political structures in Kurdistan¹⁴². This sacred relationship is based on mutual interdependence between humans, nature, and all beings. Rituals at sacred sites like trees, mountains, and rivers further reflect the community's reverence for nature through the concept of *jiare*, that function as the focal points of religious and cultural practices (Gültekin 2021, 227). In contemporary times, these deep-rooted values have been infused with new meaning through ecological struggles in Dersim province (center of *Rêya Heqî*/Kurdish Alevi), such as protests against dam projects, mining and deforestation as well as commodification, making environmental activism central to their identity and political movements (Gültekin 2019, 2021; Deniz 2019, 2021). The natural cores of this ecological society continue to shape and enrich the ecological policies of the KFM. In interviews conducted during the fieldwork in Rojava, locals emphasized the profound connection between land, identity, and history. A striking statement from a Yezidi interviewee reflects this sentiment: “The land is not just soil—it is the memory of our people, our sacred mother, and our dignity”¹⁴³. This connection is echoed in responses from the survey within 40 participants who described land as “sacred, memory, homeland, mother, dignity, and bread.” This is crucial ground where KFM weaving national liberation and ecological struggle together as Berivan Yunis points out, “the ecological struggle is necessary to build an ecological consciousness, which will generate *welatparêzî* inherently”¹⁴⁴. These land-based political, economic and socio-cultural structures have produced its own epistemology and relational ontology with nature, which have become foundational of

¹⁴² Interview: Menal, Zoroastrian community representer and lecturer, 06 January 2024; Nesim Shemo (Yezidi community representer), lecturer at University of Rojava, 17 May 2024

¹⁴³ Nesim Shemo (Yezidi community representer), lecturer at University of Rojava, 17 May 2024

¹⁴⁴ Berivan Yunis, Kongra Star's academies, 20 May 2024

*Kurdayeti*¹⁴⁵. Not surprisingly Turkish and Syrian colonial rules have long targeted this interconnectedness. Through violent mechanisms of ecocide and genocide in order seeks to sever this bond between the Kurdish people and their land, thereby dismantling their cultural identity. The construction of massive dams, mining, and deforestation projects disrupt the Kurdish way of life and destroy ecosystems, making the ecological struggle a cornerstone of the KFM¹⁴⁶. Firat Anli states this relationality concerning thousands of village burnings and forced displacement in 1990s as, "... thousands of years of land-based knowledge, agricultural production techniques and inter-generational transmission of land-based culture were prevented or disrupted. Many indigenous seeds, plant varieties and even animal breeds have become extinct."¹⁴⁷ In response to the ongoing colonial destruction of Kurdish land and culture, Öcalan identified ecological destruction as a "de-Kurdification policy" (Bilgen 2017, 7). Zilan argues this point as:

Ecological struggle has to be anti-colonial as well as women's liberationist. The main and dominant cause of the ecological destruction we are experiencing in Kurdistan is occupation, colonialism and war. Our mountains, villages, plains, forests and waters are being deliberately plundered and destroyed¹⁴⁸.

In doing so, as part of the broader anti-colonial movement, KFM advocates for the reconstruction and affirmation of indigenous epistemologies and ontologies as states by the article 3 of the Ecology Conference,

The Conference embraces the religious, philosophical and intellectual heritage of the peoples of Mesopotamia and Kurdistan and their belief systems, which originate from the natural society where nature is sacred and the relationship with it are sustainable and respectful. It harnesses these ecological values in the contemporary freedom struggle and opens the space for their reinvigoration as the core nodes of democratic modernity. Within this context, the Conference rejects colonial, Eurocentric and statist epistemologies and scientism¹⁴⁹.

¹⁴⁵ *Kurdayeti* represents what Hobsbawm refers to as "proto-national bonds" or aligns with Smith's concept of "ethnie." Therefore, *Kurdayeti* should not be considered synonymous with Kurdish nationalism and should not be used interchangeably (see Tunc 2018).

¹⁴⁶ Birahîm Hemdî Eshed, AANES cochair of Ecology Council (Desteya Jingehe), 2 June 2024; Ercan Ayboga, co-founder of MEM and activist in TEV-EKO, 12 March 2024; Dr. Fatih Cicek (Sociology), consultant for agriculture policy in DEM Party, 02 October 2024; Group discussion with MEM board, 28 October 2024; Egid Özdemir, ecology activist in Northern Kurdistan, 21 October 2024

¹⁴⁷ Firat Anli, former co-mayor of Diyarbakır Metropolitan Municipality, 03 March 2024

¹⁴⁸ Zilan, lecturer at the Jineoloji Academy, 27 June 2024

¹⁴⁹ "Ecology in Democratic Modernity - From Resistance to Construction: Perspectives, Possibilities & Challenges" held on April 26-27, 2024 in Qamislo/Rojava-Syria

This concept includes a return to traditional knowledge systems related to land, agriculture, and community governance. For example, the use of traditional seeds and farming techniques—practices that had sustained Kurdish communities for centuries—are being revitalized as part of the ecological struggle¹⁵⁰. This revival is not only about environmental sustainability but also about resisting the imposition of monocultural, extractive colonialist farming methods (Türk and Jongerden 2024). *Jineoloji* (science of women) plays a central role in this epistemic reconstruction through seeking, restoring and creating knowledge systems that were erased or marginalized under colonial and patriarchal rule (Piccardi 2021). As Zilan stated,

Jineology seeks to overcome the epistemic violence of positivist science through integrating nature, women, and life into a holistic framework. This knowledge was once a part of our daily life, through traditional medicine, stories, and myth, but was destroyed by modern science¹⁵¹.

Parallely, Roj explained that “*Jineoloji* aims to restore the lost balance between society and nature, between men and women, and between the spiritual and material worlds”¹⁵². This perspective challenges the positivist science that has been used to justify the exploitation of both women and nature. One critical element of *Jineoloji* is the emphasis on women’s traditional knowledge of nature and healing. For example, in *Jinwar*, women not only work the land using traditional farming methods but also pass down their knowledge of herbal medicine and healing practices, which have been marginalized by modern, male-dominated science. Roj notes that “returning to these knowledge systems is not just about reclaiming our history; it is about building a future where humans and nature live in harmony”¹⁵³. This is nexus of liberation of three colonies (women-nature-Kurdistan) as negation of time is one of the main instruments of colonizers to de-historicise and dehumanise the colonized people (Mbembe 2015). Escobar points out “...the realm of myth and story in shaping our worlds—is part and parcel of design’s coming to terms with the very historicity of the worlds and things of human creation in the current tumultuous age.” (Escobar 2018, 15). The use of mythology, oral histories, and cultural narratives in daily life, as well as ecological education, forms the backbone of this knowledge system, demonstrating how Kurdish women are both agents and preservers of ecological

¹⁵⁰ Saziya Tov (Seed Institution) of AANES, 9 May 2024; Firat Anli, former co-mayor of Diyarbakır Metropolitan Municipality, 03 March 2024; Group discussion with MEM board, 28 October 2024; Egid Özdemir, ecology activist in Northern Kurdistan, 21 October 2024

¹⁵¹ Zilan, lecturer at the Jineoloji Academy, 27 June 2024

¹⁵² Roj, lecturer and Kongra Star Ecology Committee member, 27 June 2024

¹⁵³ Ibid.

wisdom and defenders of homeland and culture (Piccardi 2021; Dirik 2022; Hunt and Türk 2024 forthcoming). Parallel to these autonomous areas, ANNES/Democratic Autonomy of Rojava is greening its pedagogy in whole education system. To this end, the *Jineoloji* academies, democratic society academies, and ecology academies, a new ecopedagogy is emerging that teaches not only to deal with the ecological crisis but also how to integrate ecological thinking into governance, municipalities, health, and daily life¹⁵⁴. This pedagogy challenges the statist and patriarchal mindsets that have been dominant in existing societal and educational system (Hunt and Türk 2024 forthcoming).

This epistemic shift associated with a new ontological design, based on democratic autonomy, which reject the centralized power of the nation-state and embraces a pluralistic, non-hierarchical system based on grassroots democracy (Ocalan 2020; Akkaya and Jongerden 2013; Jongerden 2016; Dirik 2022; Sunca 2023b). This relational ontology or ontology of multiplicity decentralizes decision-making and brings governance closer to the land and the people who rely on it. In Rojava, democratic autonomy has been implemented through a series of local councils and communes in rhizomatic sense, where decisions about land use, resource management, and environmental protection are made collectively. Deconstruction of colonial states and capitalist corporations allows for a more sustainable relationship with nature, as local communities are better equipped to manage their resources in a way that meets their needs without depleting them (Türk and Jongerden 2024).

Through the revival of traditional knowledge systems, the construction of eco-pedagogies, and the implementation of democratic autonomy, the Kurdish people are actively building a decolonial ecology that offers an alternative to the destructive forces of capitalist modernity. KFM ecologic struggle as an ontological design or *afrîner* is not only challenging the structures of capitalist modernity and its colonial legacy but making a world where multiple ways of knowing and being can coexist with “its own materiality, epistemology, power relations, aesthetics, ethics, and more...” (Sunca, 2023a, 76) as a decolonial ecology.

¹⁵⁴ Berivan Omari, lecturer at University of Rojava (Social ecology), co-chairperson of Cezire region municipalities union, 26 May 24

7.4.3. Challenging ecocidal and colonial developmentalism by reclaiming autonomy and land

The KFM addresses colonial de-development and ecocide by advocating for democratic autonomy and emphasizing the role of municipalities and self-sufficient eco-economy as foundational elements in the liberation process¹⁵⁵. The concept of autonomy, central to the struggle of indigenous and colonized peoples worldwide as a “cultural, ecological, and political process” (Escobar 2018, 173). Colonized communities resist the destructions of colonial state or corporations’ land dispossession, extractivism, dams, deforestation, racist economic marginalization, loss of culture and biodiversity (Escobar 2018; Ferdinand 2022). Instead of state-driven ecocidal development based on assumed needs and market-based solutions, *autonomía* emphasizes forms of learning, healing, living, producing, and more that are less subject to external control and regulation (Escobar 2018, 181). Autonomy is, therefore, a collective resistance rooted in the community’s creation of non-liberal, non-state, and non-capitalist ways of being, doing, and knowing, forming a political ontology of “collective biographies of micro-revolutions for self-determination.” (Botero 2015, 17-19 cited in Escobar 2018, 182). The Kurdish people’s struggle for autonomy in the liberation process, especially within the framework of democratic confederalism, is remarkable in that it constitutes a rejection of nation-state colonialism and capitalist exploitation. In this context, democratic autonomy is more than a political framework of self-governance; it embodies an ontological stance as a holistic restructuring of society along communal, feminist, and ecological lines (Akkaya and Jongerden 2013; Hunt, 2021; Sustam, 2021; Pimbert, 2021; Aslan, 2023; Jongerden 2024; Türk and Jongerden 2024). In this regard, the KFM seeks to build an economic autonomy according to eco-economy principles that directly counters the colonial and ecocidal developmentalist agenda and capitalist exploitation. The eco-economy concept integrates ecological values into economic activities, including sustainable agriculture, democratic resource management, and communal ownership of land. This approach has been central to the

¹⁵⁵ Hesên Koçer, Co-Chair of Executive Council of the Democratic Autonomous Administration of North-East Syria, 29 March 2024; Firat Anli, former co-mayor of Diyarbakır Metropolitan Municipality, 03 March 2024; Birahe Hemdî Eshed, AANES cochair of Ecology Council (Desteya Jingehe), 2 June 2024; Berivan Omari, lecturer at University of Rojava (Social ecology), co-chairperson of Cezire region municipalities union, 26 May 24; Rûstem, consultant at Cezire region Municipalities Union, 26 March 24

Kurdish effort to build a self-sufficient economy through agricultural production¹⁵⁶. This reality was stated in Article 12 of the Conference declaration as well, “Agriculture is the main socio-economic element of the Kurds, especially in Rojava. Therefore, policies on land and water are strategic elements of the decolonization and/or liberation process.”¹⁵⁷

One of the key challenges; however, in both Bakur and Rojava has been extreme land inequality and landlessness issue, a legacy of colonial policies, to build ecological agro economy. In Rojava, the collapse of the Syrian state’s authority allowed the Autonomous Administration to control large swaths of land, redistributing it to cooperatives and local communities. Agricultural cooperatives have been formed, focusing on food autonomy and the communal management of resources. These cooperatives include peasants, women, and marginalized groups, embodying the KFM’s inclusive vision for overcoming the nation-state’s ecocidal developmentalism and extractivist capitalist model (Aslan 2023; Türk and Jongerden 2024) .

In contrast, Bakur faces greater challenges in reclaiming land, although land reform was one of the main promises of the GAP, it has never been implemented. Since this colonial structure is the economic pillar of the state’s colonial continuum, it is only possible for the state to dismantle this structure through land reform if it agrees to recognise Kurdish autonomy. Despite this, the Kurdish movement in Bakur has developed alternative strategies for communalizing land through local governance and municipal projects. These efforts highlight the role of agriculture in resisting colonial economic marginalization and capitalist exploitation and promoting economic autonomy through cooperatives as a core of their reconstruction of economy program and activities¹⁵⁸. Anli stated that they had conducted similar projects under the economy department of the Municipality including “agricultural infrastructure and equipment needs of the villages, providing animal, fertiliser and seed support, founding seed bank, building

¹⁵⁶Leyla Saruxan and Mehran Ahmed, Co-Chairs of the Agriculture and Irrigation Council, 21 November 2023; Karker Ismail, Co-Chair of the Office of Community Projects and Cooperatives of Cezire Region, 21 November 2023; Emin, Union of Kurdistan Communities (Koma Civakên Kurdistan, KCK) Economy Committee and Economy Council of AANES, 17 May 2024; Birahim Hemdi Eshed, AANES cochair of Ecology Council (Desteya Jingehe), 2 June 2024; Ercan Ayboga, co-founder of MEM and activist in TEV-EKO, 12 March 2024; Dr. Fatih Cicek (Sociology), consultant for agriculture policy in DEM Party, 02 October 2024; Group discussion with MEM board, 28 October 2024; Egid Özdemir, ecology activist in Northern Kurdistan, 21 October 2024

¹⁵⁷ "Ecology in Democratic Modernity - From Resistance to Construction: Perspectives, Possibilities & Challenges" held on April 26-27, 2024 in Qamislo/Rojava-Syria

¹⁵⁸ Firat Anli, former co-mayor of Diyarbakır Metropolitan Municipality, 03 March 2024; Ercan Ayboga, co-founder of MEM and activist in TEV-EKO, 12 March 2024; Dr. Fatih Cicek (Sociology), consultant for agriculture policy in DEM Party, 02 October 2024; Group discussion with MEM board, 28 October 2024

irrigation canals and ponds.” However, between 2016-2024 the Turkish state occupied the hundreds pro-Kurdish municipalities in Bakur, many of their initiatives—such as agricultural projects and eco-conscious urban planning—were dismantled by Turkish-appointed trustees after local Kurdish mayors were removed from office.¹⁵⁹

Reconstruction efforts in Rojava and Bakur are deeply informed by the Kurdish movement’s commitment to ecological principles and traditional practices. Conferences and symposiums held by the Kurdish movement have highlighted the importance of applying traditional and ecological agricultural methods, as well as integrating eco-economy principles to combat the environmental destruction wrought by colonial and ecocidal developmentalism. The reconstruction process is not just physical but also cultural and ideological, with a focus on creating a democratic space for ethno-religious and economically marginalised groups, ecologic culture, and women’s liberation.¹⁶⁰

7.4.4. Discussion

The findings indicate that ecology—one of the three principal pillars of the KFM—has historically lagged behind women’s liberation and democratic autonomy in both theory and practice.¹⁶¹ Recent internal critiques, however, have prompted structural transformations and heightened attention to ecology, most visibly in Rojava’s agriculture, education and economy¹⁶². In *Bakur* and *Rojava Kurdistan* alike, ecological resistance has emerged as a countermeasure to neo-colonial and ecocidal state policies, including extensive dam projects, deforestation, monoculture agriculture, and resource extraction. Rather than coincidental

¹⁵⁹ Firat Anli, former co-mayor of Diyarbakır Metropolitan Municipality, 03 March 2024

¹⁶⁰ Birahîm Hemdî Eshed, AANES cochair of Ecology Council (Desteya Jingehe), 2 June 2024; Berivan Omari, lecturer at University of Rojava (Social ecology), co-chairperson of Cezire region municipalities union, 26 May 2024; Rûstem, consultant at Cezire region Municipalities Union, 26 March 24; Hesên Koçer, Co-Chair of Executive Council of the Democratic Autonomous Administration of North-East Syria, 29 March 2024; Firat Anli, former co-mayor of Diyarbakır Metropolitan Municipality, 03 March 2024; Ercan Ayboga, co-founder of MEM and activist in TEV-EKO, 12 March 2024; Dr. Fatih Cicek (Sociology), consultant for agriculture policy in DEM Party, 02 October 2024; Group discussion with MEM board, 28 October 2024

¹⁶¹ KCK Ecology Committee, briefing report, January 2024

¹⁶² Ali, AANES Ecology Committee, 2 June 2024; Birahîm Hemdî Eshed, AANES cochair of Ecology Council (Desteya Jingehe), 2 June 2024; Berivan Omari, lecturer at University of Rojava (Social ecology), co-chairperson of Cezire region municipalities union, 26 May 24; Ercan Ayboga, co-founder of MEM and activist in TEV-EKO, 12 March 2024

outcomes of “development,” these destructive tactics function as deliberate tools for displacing Kurdish communities, appropriating their lands, degrading the environment, and eroding cultural identity¹⁶³. As the Kurdistan Communities Union (KCK) Ecology Committee emphasises, “for the Kurdish people, the ecological struggle is also a struggle for the defence and survival of their country, land, history and all national values,”¹⁶⁴ underscoring how ecological activism directly intersects with broader aspirations for decolonization.

In *Bakur*, Kurdish municipalities and ecological movements endeavour to reclaim communal land and resources despite harsh governmental repression. The Mesopotamia Ecology Movement (*Tevgera Ekoloji ya Mezopotamyayê*), founded in 2011, leads initiatives aimed at safeguarding Kurdish territories from state-driven environmental damage, notably by restoring ancestral seeds and reviving traditional farming approaches¹⁶⁵. The persistent persecution of pro-Kurdish municipalities, destruction of collective agricultural projects, and incarceration of prominent political figures reveal the Turkish state’s apprehension that ecological and communal self-organization could facilitate Kurdish autonomy¹⁶⁶.

By contrast, *Rojava* has made more tangible strides, in part due to the power vacuum resulting from the Syrian state’s collapse in 2012. The Autonomous Administration seized this opportunity to pilot democratic autonomy on a broader scale, creating agricultural cooperatives, decentralised production facilities, and other essential industries, thereby advancing local self-sufficiency. Central to this process is the “(re)diversification of agriculture” and “rhizomatic governance structure,” encompassing drought-resistant crops, seed heritage, agroforestry, organic composting, crop rotation, and water-harvesting techniques (Türk and Jongerden 2024, 11-15). Nevertheless, the fall of the Assad regime on 8 December 2024 has endangered

¹⁶³ Birahîm Hemdî Eshed, AANES cochair of Ecology Council (Desteya Jingehe), 2 June 2024

Firat Anli, former co-mayor of Diyarbakır Metropolitan Municipality, 03 March 2024

Ercan Ayboga, co-founder of MEM and activist in TEV-EKO, 12 March 2024

Dr. Fatih Cicek (Sociology), consultant for agriculture policy in DEM Party, 02 October 2024

Group discussion with MEM board, 28 October 2024

Egid Özdemir, ecology activist in Northern Kurdistan, 21 October 2024

¹⁶⁴ KCK Ecology Committee, briefing report, January 2024

¹⁶⁵ Ercan Ayboga, co-founder of MEM and activist in TEV-EKO, 12 March 2024; Group discussion with MEM board, 28 October 2024

¹⁶⁶ Firat Anli, former co-mayor of Diyarbakır Metropolitan Municipality, 03 March 2024; Ercan Ayboga, co-founder of MEM and activist in TEV-EKO, 12 March 2024; Dr. Fatih Cicek (Sociology), consultant for agriculture policy in DEM Party, 02 October 2024; Group discussion with MEM board, 28 October 2024; Egid Özdemir, ecology activist in Northern Kurdistan, 21 October 2024

Rojava's de-facto status, as Türkiye's neo-colonial invasions and jihadist proxies pose an existential threat¹⁶⁷.

Despite these upheavals, the Kurdish movement's unwavering commitment to forging an ecological society remains integral to its decolonizing project. Throughout Bakur and Rojava, the KFM aims to reconstruct society via ecological and women's liberation principles, democratic governance, and communal economy—placing democratic autonomy at the heart of decolonization process. The KFM has established or collaborated with several key ecology organizational bodies, including the Kurdistan Communities Union (KCK) Ecology Committee, the Peoples' Equality and Democracy Party (DEM Party) Ecology and Agriculture Commission, the European Kurdish Democratic Societies Congress (KCD-E) Ecology Committee, the Mesopotamia Ecology Movement (MEM), the Democratic Society Congress (DTK) Ecology Committee, the AANES's Ecology Council, Ecology Committee and Academy of Ecology, newly formed Kurdistan Ecology Movement-Europe (*Tevgera Ekolojî ya Kurdîstan a Ewrûpa, TEV-EKO*) and hundreds local ecology movements, associations and initiatives across Kurdistan. These entities coordinate councils, assemblies, and public campaigns to confront ecocidal and genocidal state projects from an anti-colonial, anti-capitalist, and anti-patriarchal standpoint. Consequently, ecology becomes not merely an environmental concern but also a defence of Kurdish cultural existence, resisting exploitation by Türkiye and Syria's nation-state frameworks. The KCK Ecology Committee reinforces this: "The geography of Kurdistan, which was made uninhabitable in the past years by the forced displacement and massacres, is now being evacuated through the policy of ecological destruction as a war strategy."¹⁶⁸

Moreover, KFM's ecological activism transcends regional boundaries, aligning with international movements that question the devastations of neo-colonial extractivism and ecocidal developmentalism. The proposed *Kurdistan Ecology Platform and International Solidarity Network*, outlined the Conference and the Symposium, aims to consolidate ecological forces across the four parts of Kurdistan, the Middle East, Europe, and the Global South¹⁶⁹.

¹⁶⁷ https://www.democracynow.org/2024/12/24/syria_kurds

¹⁶⁸ KCK Ecology Committee, briefing report, January 2024

¹⁶⁹ "Ecology in Democratic Modernity - From Resistance to Construction: Perspectives, Possibilities & Challenges" held on April 26-27, 2024 in Qamislo/Rojava-Syria; Ecology Symposium: Kurdistan Ecology Movement-Europe (*Tevgera Ekolojî ya Kurdîstanê-Ewrûpa, TEV-EKO*) was declared, held on 21-22 December, 2024 in Freiburg/Germany

This unification is crucial given persistent challenges such as extractivism, dam projects, deforestation, land inequality, drought, biodiversity loss and pollution.

7.5. Conclusion

The historical analysis demonstrates how the Ottomans centralization policies redefined Kurdistan as an underdeveloped periphery, effectively liquidating Kurdish autonomy and imposing economic and political subjugation. This foundation of colonial de-development has evolved into modern state practices in Bakur and Rojava, where ecocidal extractivism and developmentalism have systematically displaced and dispossessed Kurdish people. Therefore, ecocidal developmentalism of Turkish and Syrian regimes serves not only as a mechanism of resource extraction and exploitation but as a tool for dismantling the essential socio-political, cultural and economic foundations of Kurdish society as a method of cultural genocide. This nation-state colonial structure rooted in three primary colonial continuums—de-development, ecocide, and cultural genocide—these practices have systematically undermined the socio-political, economic, cultural and environmental foundations of Kurdish society. To this end, the article reveals that KFM's ecological struggle is presented as a decolonial ethos that challenges not only the socio-economic and ecological destructions but also the reclamation of political, economic, cultural and spiritual autonomy. This movement articulates an alternative vision of ecology, ethics and epistemology as a multiplicity, rooted in Democratic Autonomy as a political ontology. This new ontological design for autonomy emerges as an innovative counter-structure, positioning ecological resistance as foundational to Kurdish autonomy, economic and ecologic regenerating and cultural survival. KFM's ecological framework, thus, is not simply a response to environmental destructions; it is a deliberate, strategic reassertion of traditional Kurdish knowledge, governance structures, and economic systems through the creation of ecological councils, communal resources ownership, eco-economy based agricultural cooperatives and promoting diverse ecologies within local knowledge. These efforts embody a practical decolonial ethos that seeks to challenge the epistemic and material structures of colonial and capitalist powers. By emphasizing communal and democratic governance, sustainability, and women liberation, the KFM articulates a decolonial ecology that inherently resists the capitalist commodification of nature and the homogenizing forces of nation-state colonialism. Through an ethos of communal stewardship and the decolonization of knowledge systems, KFM underlines the significance of decolonial ecology as a framework for addressing the interconnected crises of ecological degradation, colonialism, capitalism, racism, patriarchy,

and social injustice. To this end, the KFM's ecological struggle provides a transformative model of decolonial ecology, deeply rooted in Kurdish historical practices and reimagined through democratic autonomy, suggest that decolonization is not only a political but also an ecological imperative.

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SECTION III. CONCEPTUAL INTERVENTIONS, SYNTHESIS AND DISCUSSION

Thinking is neither abstract nor detached from its geo-historical locations and conditions, from materials, events, and affects; it becomes a political act when it composes new ways of being, doing, and knowing, emerging within its own territorial relationalities.

After Deleuze

Introduction to Section III

Decolonization is not a simple, finite undertaking. Due to its epistemic and ontological structures, it demands a complex and holistic transformation sustained by ongoing, intertwined processes of deconstruction and reconstruction. Decoloniality as praxis thus requires “ongoing processes of thought-analysis-reflection-action, processes that encourage a theorizing from practice and a practicing of theory” – that is, a continual movement in which practice informs theory and theory is tested and reshaped through practice (Walsh, 2020, p. 607). This is especially crucial at the epistemic level. To this end, the present research not only adopts decolonial methodologies, but also undertakes a deconstruction of colonial conceptual structures and designs. Conceptual intervention here is not a matter of importing novel terminology or poetic turns of phrase; it is a political project of knowledge decolonization and liberation. This raises a set of guiding questions: how do the processes and designs that shape our thinking actually operate? What roles do concepts play in configuring a mindset? Which kinds of concepts are at stake, and what renders a concept emancipatory and anti-colonial?

The signature of colonial epistemologies and their conceptual architectures is a universalist, abstract regime of knowledge that presumes to speak for the whole. As a dispositive (an apparatus), this regime furnished Europe with the knowledge, cultural and moral rationalities needed to design and regulate regions elsewhere, legitimating epistemic and material violence while operationalizing exploitation (Mignolo & Walsh, 2018). Coupled with hierarchical orders of knowing and relating, it functioned as a weaponized structure: enabling Europe to deterritorialize from its own locus, extend itself spatially, colonize and assimilate other

geographies, and reterritorialize its norms as durable and self-evident. The disciplinary apparatus of (colonial) geography has been central to this process, and remains so. Accordingly, any decolonization that confronts the prevailing colonial order and its statist, power-centric formations cannot stop at critique or deconstruction. It must also engage in concept creation and forge counter-epistemologies as part of an affirmative and anti-colonial ontological and epistemological reconstruction (Escobar 2018; Ferdinand, 2022).

To this end, “Thinking is neither a line drawn between subject and object nor a revolving of one around the other. Rather, thinking takes place in the relationship of territory and the earth” (Deleuze & Guattari, 1994, p. 85). This geophilosophical provocation ruptures the inherited image of thought that has dominated Western philosophy since Descartes. Where the colonial episteme universalized abstraction and claimed to ground truth in a view from nowhere (Haraway, 1988), Deleuze and Guattari insist that thought is always embedded, territorialized and immanent. Their intervention resonates with, and is deepened by, decolonial scholarship. As Aníbal Quijano (2007, pp. 168–169) argues, the modern/colonial world system is structured by a “coloniality of power” that naturalizes Eurocentric epistemologies and presents them as universal reason. The violence of colonialism lies not only in conquest and dispossession, but also in the imposition of an abstract, hierarchical knowledge system that disqualifies other ways of knowing. Walter D. Mignolo (2009) therefore calls for “epistemic disobedience”: a refusal of the epistemic privilege of the West, and an insistence on the legitimacy of independent, situated thought. For Mignolo and Walsh (2018, pp. 2–5), decoloniality requires not just critique but the construction of alternative conceptual horizons, the creation of categories and sensibilities that emerge from subaltern struggles and worlds.

It is here that Arturo Escobar’s (2018) call for a pluriverse becomes especially salient. Against the “one-world world” of Eurocentric universalism, Escobar affirms a multiplicity of worlds, ontologies and epistemologies, “a world where many worlds fit” (p. 16). The pluriverse does not merely affirm difference; it insists upon the irreducible multiplicity of worlds and the conceptual structures they generate. Traditional and liberatory epistemologies are not minor variants within a dominant paradigm, but enact different ontological grounds altogether. Arising from intimate relations with territory – plateaus, rivers, forests, soils, ancestors—these are situated and embodied knowledges that refuse the abstraction of colonial science. In this sense, they converge with Deleuze and Guattari’s (1994) insistence that concepts must be

territorial, while extending the critique by grounding it explicitly in the lived experience of coloniality and the demand for decolonization.

Geophilosophy, accordingly, does not simply relocate philosophy into geography; it transforms the very conditions of thinking. To be geophilosophical is to affirm that concepts are not timeless essences awaiting discovery, but creations tied to earth, milieu and stratum. “Concepts are not waiting for us ready-made, like heavenly bodies. There is no heaven for concepts. They must be invented, fabricated, or rather created and would be nothing without their creator’s signature” (Deleuze & Guattari, 1994, p. 5). Philosophy is not the revelation of eternal truths, but the practice of concept creation, inventions inseparable from the territories and struggles in which they emerge. In this sense, concept creation, in a geophilosophical register, addresses the function of inventing “concepts of events that summon us to new opportunities, or that make us aware of events that promise a ‘new earth’ through constructing their concepts” (Bonta & Protevi, 2004, p. 30). For Deleuze and Guattari, territory is not a bounded land or static container. In *A Thousand Plateaus*, they show that territory is produced through expressive acts, through refrains that carve out zones of habitation. The song of a bird, the rhythm of a chant, the ritual of planting or harvest – these are territorializing refrains (Deleuze & Guattari, 1987, pp. 312–321). “The territory is the first assemblage, the first thing to constitute an assemblage; the assemblage is fundamentally territorial” (ibid., p. 323). Territory is made and remade through rhythms and affects; it is a living assemblage, porous to deterritorializations and reterritorializations. This territorial ontology stands in stark contrast to colonial epistemologies. Where colonial sciences impose striated spaces – gridded, hierarchical, centrally coded – territorial thinking reveals smooth and holey spaces where multiplicities coexist, deterritorialize and connect (Bonta & Protevi, 2004, p. 9).

The rhizome becomes the privileged figure of this form of thought. Against the arborescent logic of colonial hierarchies, the rhizome is flat, immanent and connective. It grows underground, branching in multiple directions, irreducible to a single origin or telos. It resists eradication because it proliferates through lines of flight, with budded fragments capable of re-emerging elsewhere. Rhizomatic thought privileges connection, heterogeneity and multiplicity rather than hierarchical bifurcation (ibid., pp. 136–137). This is not merely metaphorical, but is ontological. In political terms, it signals a mode of organization that resists centralization and over-coding by the state. Rhizomatic politics, like rhizomatic thought, multiplies connections and sustains heterogeneity.

Geophilosophy, then, is inseparable from territorial thinking. Both insist that thought is an immanent process within territories and milieus, produced in encounters with earth, land and environment. For decolonial thought, this orientation is pivotal. Colonial epistemologies sought to erase territorial multiplicities by imposing universalist abstractions. Territorial thinking and geophilosophy reopen the possibility of situated concept creation. They invite us to ask: what kinds of concepts emerge from the mountains and plateaus of Kurdistan, from the deserts of Syria, from the lands of Mesopotamia? In this sense, the revolutionary experiment in Rojava is not an application of external ideology, but a geophilosophical event: a becoming-territorial in which democratic autonomy, women's liberation and ecology are continually composed, tested and reterritorialized in relation to the living dynamics of a geography. Deconstructing colonial epistemologies through geophilosophy and territorial thinking does not simply invert hierarchies or replace one universalism with another. It loosens the binaries that sustain colonial abstraction: replacing fixed essences with becomings, scalar containers with relational fields, and representational distance with immanent co-composition. It cultivates sensitivity to multiplicity, to the assemblages and affects that compose place through continuous (de)territorializations.

The significance of these debates for Kurdistan and Rojava is profound. To struggle against ecocidal coloniality requires more than resistance; it requires the invention of concepts adequate to the lived realities of oppression and the emergent practices of autonomy. Such concepts are geophilosophical in Deleuze and Guattari's sense, and decolonial in Quijano's and Mignolo's: they refuse abstract universalism and articulate situated, insurgent knowledges. Through these lenses, it becomes clear that the received lexicon is insufficient to grasp the emergent forms of governance in Rojava, and the political, historical and structural conditions of destruction they seek to undo. The existing literature, terminology and conceptual frameworks are not only inadequate; they also obscure the historical and colonial background of the crisis. Socio-ecological breakdown is too often narrated through depoliticizing, neoliberal and neocolonial lenses, such as 'climate change', corruption, conflict, neoliberalization or resource mismanagement, thereby deflecting attention from the structures of colonial domination that produce it.

Bringing these structures to light necessitates new conceptual tools. That is why, in this thesis, I advance original concepts, including ecocidal coloniality, colonial drought, rhizomatic governance and rhizomatic democracy, to name specifically Kurdish modalities through which

scarcity is produced, authority is distributed, and political life proliferates laterally across communes, cooperatives and academies. In parallel, I co-developed the concept of nation-state colonialism to clarify how modern state formation in the region operates as a colonial structure across political, demographic and socio-ecological registers. At the same time, I rework and localize existing frameworks by giving them a Kurdish inflection: I adopted the notion of decolonial ecology (Ferdinand, 2022) as an ethos and praxis of liberation grounded in the region's socio-ecological relations; and I extend the genocide-ecocide nexus to show how environmental destruction functions as a method of cultural and physical elimination in Afrin and beyond (Short, 2010; Crook & Short, 2014; Short, 2016). I also formulate ecocidal developmentalism to capture how "development" projects operate as mechanisms of ecological and social erasure, and specify ecocide as counterinsurgency to name the operational coupling of environmental devastation with security doctrine.

Importantly, these interventions did not precede the field; they emerged through encounters with the territory, extended dialogue with local actors and collaborative critique with colleagues. Their purpose is not to impose a template, but to articulate, as precisely as possible, the sociological, historical and contemporary dynamics of Rojava and Kurdistan within a decolonizing process, that is, concepts created with the earth and from territory rather than abstracted from it. Thus, the conceptual labour of this dissertation is not simply to apply existing theories to a case; it is to engage in the geophilosophical and decolonial practice of creating concepts within a territory. These concepts were necessitated by the realities of Rojava and Kurdistan, yet they resonate beyond, contributing to broader struggles for epistemic justice and decolonization. They exemplify how philosophy, in Deleuze and Guattari's sense, can become geophilosophy: a practice of concept creation in and with the earth, against the abstractions of colonial universalism, and towards the affirmation of a pluriverse of worlds.

The conceptual synthesis developed above grounds the cumulative design of this dissertation. Its five constituent articles can best be described, in Deleuzian terms, as rhizomatic: connected, symbiotic and semiotic, yet retaining their autonomy. Each article stands as a plateau, able to be read independently, but together they compose a heterogeneous assemblage, a multiplicity whose meaning is intensified by their interrelations. The dissertation thus refuses a linear architecture; instead, it is composed as a rhizome, with transversal connections, resonances and divergences between texts. This form mirrors both the objects of study, Rojava's rhizomatic governance and socio-ecological experimentation, and the epistemological stance of the

research, which affirms multiplicity, heterogeneity and situated concept creation. Within this rhizomatic constellation, two dynamics emerge that traverse and connect the articles. The first dynamic traces the ecocidal coloniality and its operational dimensions that have shaped Kurdistan as a periphery: the genocidal and ecocidal logics of nation-state colonialism, the socio-spatial technologies of ecocidal developmentalism, and systemic underdevelopment imposed through agricultural modernization, hydro-hegemonic control, extractive infrastructures, and colonial pedagogy and epistemic erasure.

Together, the five articles expose main dimensions of ecocidal coloniality. Articles I and V analyse how Ba'athist agricultural modernization functioned as a vehicle of internal colonization, with ecocide producing a de-developed periphery in Rojava, and how its dismantling opened pathways for a decolonial reconstruction of agriculture. Article III similarly introduces the concept of colonial drought, demonstrating how Syrian monocrop agriculture and Turkish hydro-hegemony intersect to produce water scarcity and drought as an ecocidal legacy of the colonial structure. Article II exposes additional facets of ecocidal coloniality by clarifying why ecocide is intrinsic to nation-state colonialism. To this end, it reveals the genocide-ecocide nexus through a comparative analysis of olive tree destruction in Palestine and Afrin, highlighting how environmental violence functions as a deliberate strategy of cultural and physical extermination. Accordingly, Article IV demonstrates that colonial elimination is not only material, but also epistemic, psychosocial, mental and spiritual, which results in a degeneration of societal ecology. Colonial pedagogy works to manufacture a 'slave mentality' and a subjugated society. Under Ba'athist rule, schooling served as a primary apparatus of assimilation, nationalism and patriarchal control: Kurdish language, history and culture were effaced, while a centralized, Arabization-oriented curriculum was imposed as a tool of epistemicide.

The second dynamic addresses the responses to ecocidal coloniality: the emergence of democratic autonomy and rhizomatic governance as a political ontology of socio-ecological transformation as a decolonial ecology. Here, the focus shifts from domination to resistance, from analysis of destruction to the conceptualization of alternatives. Article I already prefigures this by exploring the decolonization of agriculture as both a material and epistemic practice. Article III extends this trajectory by showing how rhizomatic governance in Rojava, as a political ontology of socio-ecological transformation, reimagines socio-ecological relations in response to colonial drought. Article IV elaborates this line further by examining ecopedagogy

and the transformation of education in Rojava as a crucial dimension of socio-ecological transformation, linking ecology, *jineoloji* and democratic confederalism. Finally, Article V brings these threads together, situating ecological struggle as the ethos of the KFM's decolonial praxis, articulated through eco-economy, grassroots governance, and traditional and liberatory epistemologies. These articles collectively demonstrate that ecological struggle is not a peripheral dimension of Kurdish politics, but a constitutive axis of decolonization itself, giving rise to a decolonial ecology that affirms multiplicity and enacts a pluriverse.

The two dynamics, colonial domination and decolonial transformation, thus provide coherence to the dissertation without reducing its heterogeneity. They function as transversal axes through which the five articles can be read together. In the following two chapters, these axes are developed explicitly. Chapter 8 interrogates vectors of ecocidal coloniality within its operational dimensions. Chapter 9 turns to socio-ecological transformation as a decolonization process, analysing the emergence of democratic autonomy and its operational formation, rhizomatic governance, as political ontologies of resistance and transformation, culminating in the articulation of decolonial ecology. In this way, the cumulative project of the dissertation unfolds: a geophilosophical and decolonial cartography of Rojava and Kurdistan that not only diagnoses colonial destruction, but also traces the experimental construction of emancipatory alternatives.

Chapter 8. The vectors and operational dimensions of ecocidal coloniality: Ecocide, socio-spatial engineering and developmentalism

This chapter examines the vectors and dimensions of ecocidal coloniality in Kurdistan, and sets out the conceptual scaffolding for the thesis. It asks how nation-state colonialism has been materialized through socio-spatial design, and why its operational logics are inherently ecocidal. Working across historical, ecological and epistemic registers, it shows that what is often named ‘development’ has functioned as an architecture of elimination and de-development. Chapter section 8.1 reconstructs the colonial continuum linking Ottoman centralization to the modern nation-state projects of Türkiye and Syria. It traces how land, water and forests were reorganized through land codes, demographic engineering, dams, irrigation grids and monocrop agriculture, dissolving autonomous Kurdish formations and producing extractive margins. The argument is that ecological degradation was not collateral; it was constitutive of ecocidal coloniality. Chapter section 8.2 analyses the impacts of this ecocidal coloniality on the essential foundations of the Kurdish people. It maps how de-development, water control and the weaponization of water, ecocide as counterinsurgency, and assimilationist pedagogy combine to dismantle livelihoods, territory, language, memory, and ethical and aesthetic worlds. Two compact tables organize the evidence and detail the techniques and mechanisms through which control is exercised in North and East Syria, from hydraulic megaprojects and groundwater overextraction to curricular assimilation and border militarization, together with their observed outcomes. Taken together, these sections demonstrate operational dimensions of ecocidal coloniality, and that ecological destruction, scarcity and underdevelopment are politically produced.

8.1. The colonial continuum of socio-spatial design: The roots of ecocidal coloniality

The colonial status of Kurdistan constitutes one of the most systematically denied realities of the modern world. While the Ottoman and Iranian Empires once treated Kurdistan as a frontier

zone valued for resources and strategic location in the sixteenth and seventeenth centuries (McDowall, 2004; Kaya, 2020; Atmaca, 2021; Dehqan & Genç, 2022), the nineteenth and twentieth centuries witnessed a more insidious form of domination through Ottoman centralization and the rule of Western colonial powers, namely Great Britain and France. The twentieth century was probably the most dramatic phase, involving the recolonization of Kurdistan as a de-developed periphery through the genocidal and ecocidal policies of the four newly founded nation-states between which Kurdistan was divided. This transformation was not merely political or military, and was a project of socio-spatial engineering designed to dismantle the essential economic, cultural, socio-ecological and political foundations of the Kurdish people. From the Ottoman Tanzimat reforms of the mid-nineteenth century to the postcolonial nation-state projects of Türkiye, Syria, Iraq and Iran, Kurdistan was subjected to a logic of elimination by de-development: prosperity and autonomy were deliberately liquidated, and the region was rendered dependent and fragmented, governed through extractive logics that drained resources while systematically impoverishing its socio-ecologies. These practices mimicked the European nation-state form, while adapting it to local geopolitical conditions (Beşikçi, 1990; Beşikçi, 1991; Bozarslan, 1996; McDowall, 2004; Öktem, 2005; Öktem, 2008; Tejel, 2009; Fernandes, 2012; Beşikçi, 2014; Duruiz, 2020; Yadirgi, 2020; Goner, 2023; Türk, 2023; Jongerden, 2024).

The colonial condition of Kurdistan cannot be fully captured by either the framework of franchise colonialism, concerned primarily with the extraction of wealth, or that of settler colonialism, oriented towards permanent demographic replacement. Rather, the Kurdish regions have been subjected to a layered process of a specific form of internal colonialism: nation-state colonialism, whereby homogenizing national projects aim to dissolve Kurdish geographies, cultures, socio-ecologies and political formations into statist identities (Türk & Jongerden, 2024). This colonial logic has relied on a relation between centre and periphery that echoes Friedrich List's nationalist political economy: the centre flourishes precisely by transforming its periphery into a zone of raw material extraction and demographic engineering (Yadirgi, 2020; Türk & Jongerden, 2024). In Kurdistan, monocrop agriculture and hydraulic projects were the primary vehicles of this transformation. The result was a dual process: the state secured political consolidation and infrastructural sovereignty, while Kurdish regions were forcibly integrated into national economies as extractive margins. This is what I conceptualize in Article V as ecocidal developmentalism: a socio-spatial design in which 'development' operates as a colonial weapon, liquidating the ecological and economic foundations of the

Kurdish people while legitimizing colonial dispossession under the brand of modernization and progress. The Ba‘athist regime’s massive irrigation schemes in Rojava managed by the General Authority for Development of the Euphrates Basin (GADEB) (Springborg, 1981), as well as the Turkish GAP project in Bakur, exemplify this dynamic (Article V).

The genealogy of ecocidal coloniality; however, stretches back to Ottoman centralization. During the Tanzimat era (1839 onwards), the Empire moved to dismantle Kurdish confederal autonomy and tribal governance, recolonizing Kurdistan as a governable province under direct administration (Bozkurt & Akpınar, 2011; Yadirgi, 2020). The Hamidiye Regiments and the 1858 Land Code intensified this process, redistributing communal lands to elites and settlers while targeting Armenians and Kurds with dispossession and displacement (Heper, 2007; Fratantuono, 2018; Yadirgi, 2020). This liquidation of Kurdish autonomy was not an isolated episode, but the onset of a long trajectory in which demographic engineering, forced migration and land seizure became structural techniques of domination. This trajectory crystallized with the partition of Kurdistan after the First World War: a territory simultaneously subjected to the colonizing projects of multiple states. Each state mobilized nationalism as its vehicle of colonization, imposing cultural assimilation and economic dependency while denying the very existence of Kurdistan.

Yet paradoxically, much academic and political discourse has refused to recognize this colonial condition. The result was what Goner (2023) calls an “invisible international colony”: Kurdistan’s colonized reality obscured not only by the violence of states, but also by epistemic blind spots that naturalized its nonexistence. This epistemic denial is not incidental. It reflects the coloniality of knowledge identified by Aníbal Quijano (2007): the imposition of universalist categories that render local experiences illegible. Here, the refusal to recognize Kurdistan as a colony functions as a form of epistemic counterinsurgency. It denies the naming of colonial domination and, in so doing, obstructs the articulation of decolonial alternatives. The intervention of this thesis, through the concepts of ecocidal coloniality, nation-state colonialism, colonial drought and ecocidal developmentalism, is precisely to rupture these barriers by creating a conceptual vocabulary rooted in Kurdistan’s experience. These concepts are not imported descriptors, but insurgent categories born of and with the territory, advancing what Mignolo (2009) terms epistemic disobedience. To this end, understanding ecocidal coloniality in Kurdistan thus requires holding together its political, socio-spatial, ecological and epistemic dimensions: legal and political engineering, ecocidal developmentalism, socio-spatial

and hydraulic design, militarized ecological violence, epistemic domination and erasure, and biopolitical marginalization.

To this end, Article V demonstrates that Ottoman centralization set in motion a pattern of demographic engineering and dispossession that would later be refined by the Turkish state and Ba‘athist Syria. In both cases, the state sought not only to assimilate Kurds culturally, but also to dissolve Kurdistan ecologically and economically – transforming it into an underdeveloped or dependent periphery while eliminating its autonomous foundations. Article I extends this argument by showing how Ba‘athist agricultural modernization was not a neutral development programme, but a vehicle of internal colonization. Through massive irrigation projects, monocropping of strategic crops, and the Arab Belt policy, the Syrian state reconfigured Rojava’s socio-ecological landscape to assert control, displace Kurdish populations and embed the region in a system of extractive dependency (McDowall, 2004; Tejel, 2009; Hinnebusch et al., 2011). These practices reveal the intimate linkage between nation-state colonialism and ecocide. The large-scale ecological degradation resulting from monocrop agriculture, soil exhaustion and groundwater depletion was not accidental, but constitutive of agricultural policies (Barnes, 2009; Selby, 2018; Türk & Jongerden, 2024).

Ecocide also operates as a counterinsurgency strategy: by destroying the ecological bases of Kurdish survival and autonomy, the state has pursued elimination through environmental means. The case of Afrin, analysed in Article II, illustrates this nexus with brutal clarity. Here, the Turkish invasion of 2018 not only imposed military occupation and ethnic cleansing; it also unleashed systematic destruction of olive trees, land appropriation and environmental degradation. Such acts cannot be dismissed as collateral damage. They demonstrate how ecocide and genocide intertwine in ecocidal coloniality: ecological destruction functions as a deliberate method of cultural elimination. Therefore, colonial state consolidation in Kurdistan has unfolded through socio-spatial design and resource control: the deliberate reorganization of land, water and populations to entrench domination. This design, especially visible in Türkiye and Syria, shows how colonial power embeds itself in material infrastructures such as dams, irrigation canals and monocrop fields, as well as in demographic and legal regimes (Articles III and V). These infrastructures did not only supply energy or agricultural output; they functioned as technologies of control. They rendered Kurdish lands extractable for the benefit of the centre, while impoverishing and stripping Kurdish communities of autonomy over their territory. Article I demonstrates that these measures turned Rojava into the breadbasket of Syria, yet in a

manner that simultaneously underdeveloped the region. Food and fibre were extracted to supply processing industries in Aleppo, Hama and Damascus, while Kurdish farmers remained dependent on external markets for flour, textiles and manufactured goods (Türk & Jongerden, 2024). This de-development (Roy, 1995; Soleimani & Mohammadpour, 2020; Yadirgi, 2020) was not simply an unintended outcome, but a deliberate strategy to tether Kurdistan to the centre as a subordinate periphery, while weakening the socio-economic bases of Kurdish political dynamics and cultural life.

The historical trajectory, from Ottoman centralization to Ba‘athist agricultural modernization, from the Turkish GAP to the Syrian Arab Belt, shows that Kurdistan’s colonization is inseparable from socio-spatial design. Land codes, irrigation schemes, dams and monocultures have been as decisive as armies and decrees in dismantling Kurdish autonomy. Article III adds a critical dimension of ecocidal coloniality (socio-spatial and hydraulic design) by showing how this model of agrarian modernization was built upon the overexploitation of water resources, both surface and underground. The Tabqa Dam, inaugurated in 1973, has embodied a technopolitical rationality in which hydraulic sovereignty was materialized in concrete, canals and bureaucratic regulation (Springborg, 1981). Through such infrastructures, water was deterritorialized from local social worlds and reterritorialized as a resource of the state. Conceived as a symbol of hydraulic sovereignty and national modernization, it simultaneously recolonized the socio-ecological fabric of Rojava: transforming diverse farming systems into monocultures irrigated through unsustainable groundwater extraction, depleting aquifers and salinizing soils (Barnes, 2009; Hinnebusch et al. 2011; Ababsa, 2015; Selby, 2018; Daher, 2022). Far from being an innocent technocratic project, the dam materialized the state’s colonial rationality (Scott, 1998; Mitchell, 2002; Harris, 2012; Daoudy, 2020b). It reorganized territory in Foucauldian terms, making populations legible, governable, and extractable, while producing ecological crisis as a side effect of domination (Schmitt, 2017; Akıncı et al., 2020).

Agricultural modernization under the Ba‘ath regime, exemplified by its socio-spatial and hydraulic design as one of the main dimensions of ecocidal coloniality, makes visible the full force of what I conceptualize as *colonial drought*. Drought is usually cast as a climatic phenomenon, but as Article III demonstrates, in Rojava, scarcity has been systematically engineered through state-led monocrop agricultural and hydrological policies. This is drought not as natural calamity, but as colonial product. By transforming the Jazira region into Syria’s breadbasket for ‘strategic crops’, the Ba‘ath regime turned water into a tool of extraction.

Intensive irrigation, monocropping and unsustainable groundwater pumping produced a situation where, by the early 2000s, 85–90 per cent of Syria's water use was for agriculture, much of it in Rojava (Varela-Ortega & Sagardoy, 2003, p. 343). Rivers such as the Khabour and the Yarmouk had already dried up well before the 2007–2010 drought, which was the most severe on record, while aquifers were being depleted at more than three times their safe yield (Selby, 2018, p. 266). If the Ba'ath regime's hydraulic projects restructured Rojava into an extractive zone, Türkiye's control of the Euphrates and Tigris reinforced scarcity downstream. Through its GAP, Türkiye constructed 22 dams and 19 hydropower plants, capturing over 90 per cent of the Euphrates' flow (Mueller et al., 2021; Daoudy, 2023). This infrastructural dominance allowed Ankara to regulate the lifeblood of Syria and Iraq, transforming water into an instrument of geopolitical leverage (Zeitoun & Warner, 2006). For Rojava, positioned downstream and already subjected to internal colonization, Türkiye's hydro-hegemony compounded vulnerability: reducing water flows, destroying ecological balances and weaponizing scarcity against Kurdish autonomy (Barnes, 2009; Mueller et al., 2021; Daher, 2022; Daoudy, 2023; Tinti, 2023).

Across these contexts, drought reveals itself not as the absence of water but as the presence of colonial power. Colonial drought emerges at the intersection of internal colonization and regional hydro-hegemony, as the crystallization of ecocidal developmentalism in hydrological form. Colonial drought thus reframes drought as a political and colonial condition, revealing the socio-spatial design of scarcity itself. Here, the line between nation-state colonialism and ecocide becomes clear: ecological degradation is not an accidental failure of state power but a constitutive element of ecocidal coloniality in Kurdistan. This is why Afrin provides a paradigmatic case where these logics culminate, and are conceptualized through the genocide-ecocide nexus. As Article II details, Afrin's olive groves were not merely sources of subsistence and livelihood, but anchors of cultural memory and territorial identity. By cutting down and appropriating olive trees, Turkish-backed militias sought to dismantle the socio-ecological fabric that tied Kurds to their land, while simultaneously enriching themselves through plunder. This militarized ecological violence as dimension of ecocidal coloniality exemplifies how ecocide functions as counterinsurgency: a method to sever communities from their socio-ecological and cultural foundations and thereby undermine their capacity for resistance.

The concept of the genocide-ecocide nexus captures this dimension of ecocidal coloniality and its continuum best: ecocide is not a by-product but a method of genocide, a technology of

erasure that operates by destroying the material bases of existence. From the burning of forests in Bakur to the uprooting of olive groves in Afrin, from the drying of the Khabour to the weaponization of the Euphrates, the destruction of ecological systems has served as a means of cultural and physical extermination and exploitation. To ignore this is to reproduce the epistemic violence that denies Kurdistan's colonial status. Seen through this lens, nation-state colonialism in Kurdistan is inseparable from ecocidal practices. The homogenizing projects of Arabization in Syria and Turkification in Türkiye mobilized ecological destruction as a weapon through mega development projects and ecocidal policies: forests burned, rivers diverted, fields poisoned, olive groves uprooted. Under the guise of modernization, the state advances not life but death: displacing populations, dispossessing land and dismantling ecological systems in ways that weaken the Kurdish people's capacity to endure and reproduce themselves.

As a crucial complementary element of these policies, Article IV brings another dimension into focus: the epistemic and affective consequences of these colonial projects. By denying the cultural and linguistic existence of Kurds, dispossessing them of land and imposing monocultural agriculture, colonial rule produced what interviewees described as a psychological severance from the land, a condition in which rural livelihoods were disparaged as 'backward'. Such alienation illustrates that internal colonialism operates not only through material dispossession, but also through epistemic erasure, disqualifying Kurdish ways of knowing and relating to land while imposing extractivist rationalities. In this sense, colonial infrastructures do not simply transform economies and ecologies; they also reconfigure subjectivities, reproducing what Steve Biko called "the greatest weapon in the hand of the oppressor: the mind of the oppressed" (Wilson, 2011, p. 152). Article IV, in this regard, demonstrates that epistemicide is a crucial dimension of ecocidal coloniality, and foregrounds the necessity of ecopedagogy as a decolonial response.

When read together, these five articles show how ecocidal coloniality operates through intertwined dimensions of ecocidal coloniality: legal and political engineering, ecocidal developmentalism, socio-spatial and hydraulic design, militarized ecological violence, epistemic domination and erasure. These are not parallel processes but mutually reinforcing logics, forming a colonial assemblage that has persistently targeted Kurdish existence. This synthesis also clarifies why new conceptual interventions are necessary. Existing frameworks, including climate determinism, corruption discourse and neoliberal governance failure given as structural reasons behind the Syrian uprising in 2011 (Trigo et al., 2010; Gleick, 2014; Kelley

et al., 2015; Werrell et al., 2015; Ash & Obradovich, 2020), and scholars who adopted more critical lenses and focused on structural and infrastructural elements (Ababsa, 2015; Fröhlich, 2016; Eklund & Thompson, 2017; Selby et al., 2017; Hoffmann, 2018; Selby, 2018; Daoudy, 2020a; Mohr, 2021; Eklund et al., 2022; Dinc & Eklund, 2024), fail to capture the colonial production of scarcity, dispossession and erasure. By introducing concepts like nation-state colonialism, colonial drought, the genocide-ecocide nexus and ecocidal developmentalism, the present research names the specific modality of colonial structure: ecocidal coloniality. These concepts do not replace one another, but resonate rhizomatically: each illuminates a dimension of the colonial assemblage, while their interconnections reveal a broader structure of domination. Together, they articulate a geophilosophical and decolonial analysis adequate to the lived realities of Rojava, demonstrating that colonialism here is not a matter of past occupation but an ongoing structure, continuously reterritorialized through infrastructures, policies and epistemologies. Thus, the concepts created and developed here constitute not only analytic tools, but also acts of epistemic disobedience and decolonization. They name what has been systematically denied, revealing the colonial and genocidal structure of the nation-state as it materializes in Kurdistan.

In sum, the historic structure in Rojava is a colonial formation that operates simultaneously on ecological, demographic, economic, infrastructural and epistemic levels. By weaving together, the insights of the five articles, this synthesis demonstrates that agricultural modernization, demographic engineering and hydraulic infrastructures have not been isolated policies, but mutually reinforcing technologies of colonial domination. They have transformed Rojava into a periphery, dispossessed its people, eroded its ecologies and sought to assimilate its culture. The result was not development but de-development, not inclusion but elimination. This is the modality of ecocidal coloniality, a structure whose logics continue to shape the region, with impacts that remain devastating, as discussed below.

8.2. Ecocidal coloniality and its impacts on the foundations of Kurdish society

Ecocidal coloniality is not merely an ordering of power over territory; it is an apparatus for undoing the material, ecological, and epistemic grounds of Kurdish collective existence. Its

dimensions are diverse (agrarian modernization, hydraulic megaprojects, settlement and demographic engineering, curricular control), but their effects converge (see Tables 3 and 4)

Table 3: Mapping the structure of ecocidal coloniality, conceptual framework and impacts

Ecocidal coloniality	Concept(s)	Article(s)	Ecocidal and genocidal impacts
1) Legal-political engineering	Nation-state colonialism: demographic, economic and ecological engineering, Arabization or Turkification	I, III, V	• Ethnic cleansing/forced displacement • Cultural genocide • Denialism of fundamental rights • Criminalization and systemic discrimination (Arab Belt policies)
2) Ecocidal developmentalism	State-led agrarian modernization: internal colonization and value capture/de-development	I, III, V	• Poverty • Economic dependency • Dispossession/economic marginalization/landlessness • Systemic ecological degradation (salinization, exhausted land, chemical pollution, loss of seeds and biodiversity)
3) Socio-spatial and hydraulic designs	Colonial drought: dams and hydro infrastructure, transboundary water politics; groundwater overextraction; weaponization of water	I, III, V	• Water crisis • Public health crisis • Agricultural economic collapse • Undermining AANES
4) Militarized ecological violence	Genocide-ecocide nexus: deliberate environmental destruction as a technique of elimination; ecocide as counterinsurgency; settler-colonial demographic engineering	II, V	• Demographic change: destruction of the identity of specific socio-ecologies • Social death and cultural genocide via destruction of the link between people and land: their land-based culture, socio-economic formation and social fabric

			• The dismantling of knowledge systems related to agriculture and the environment and the disruption of intergenerational knowledge flows
5) Epistemic domination and erasure	Epistemicide: assimilatory curricula; language suppression; erasing traditional knowledge systems; symbolic displacement in public space	I, IV, V	• The erasure of knowledge systems • The dependent and enslaving mentality • The reactionary patriarchal, statist and capitalist ideologies • Attempts to degrade, erode and eliminate Kurdish culture, language and identity • Attempts to eradicate the ethical and aesthetic norms of the Kurds
6) Biopolitical marginalization	—	—	• Widespread pollution • Unequal distributions of disease and toxic exposure along lines of ethnicity, gender, class and peripheral location

To this end, Article V traces a line from the Ottoman roots through the Turkish state in Bakur and the Syrian Ba‘ath regime in Rojava: what is presented as development becomes ecocidal developmentalism, a programme that reorganizes production for the centre, corrodes local ecologies, fabricates economic dependency and gradually liquidates the economic base required for autonomous existence (Roy, 1995; Yadirgi, 2020). In Syria, as Article I demonstrates, the Ba‘athist transformation of agriculture exemplified this logic. Crop calendars, inputs and marketing were folded into centrally mandated plans, with annual targets and crop prescriptions that reorient land use to ‘strategic crops’ (Springborg, 1981; Hinnebusch, 2011). The result was not agrarian uplift but de-development: Rojava became an ecologically degraded periphery, a provider of primary commodities for processing in the Arab core (Springborg, 1981; Türk & Jongerden, 2024). The socio-economic outcomes were stark: despite Cizîre covering 40 per cent of Syria’s fertile land and producing two-thirds of cereals (including about 70 per cent of wheat) and three-quarters of hydrocarbons, it remained Syria’s poorest region, with high illiteracy, unemployment at 25 per cent in 2004 (more than double the national rate) and poverty

reaching about 80 per cent by 2010 (Ababsa, 2015, p. 201). This structured underdevelopment forced a shift from smallholder autonomy to wage dependence in subordinated employment circuits, that is, cheap-labour incorporation rather than local industrial upgrading, precisely the outcome Roy (1995: 4) identifies as de-development: ensuring “no economic base ... to support an independent indigenous existence.” In short, by exporting grains and fibres to the centre for value-adding, and by withholding the industrial, technological and knowledge infrastructures required for transformation in situ, the regime reproduced an economic dependency trap that precluded an autonomous economy (Soleimani & Mohammadpour, 2020; Yadirgi, 2020).

The same periphery-centre design appears across the border in Türkiye’s GAP, which promised irrigation over 1.8 million hectares, but delivered a different arithmetic: by 2006, 74 per cent of hydropower capacity was completed while only 14 per cent of planned irrigation was implemented; 69 per cent of the generated electricity left the region; and per capita income in Bakur stood 54 per cent below the national average (USARM, 2009, pp. 35–36, 42–43; Yadirgi, 2020, p. 253). Between 2002 and 2007, public investment overwhelmingly favoured extraction (69 per cent) over health and education (28 per cent) (USARM, 2009, p. 43). As in Syria, the discourse of development provided the alibi for an extractive redesign of Kurdish socio-spatiality (Harris, 2002, Harris, 2006, Harris, 2008; Jongerden, 2010; Harris, 2012; Bilgen, 2017; Akıncı et al., 2020; Bilgen, 2021; Türk, 2023). In both cases, economic marginalization and underdevelopment are not inadvertent residues, but engineered conditions of planned de-development and dependency as a colonial continuum. Hence the claim: ecocidal developmentalism is not an exception but a rule of domination, elimination and exploitation in Kurdistan. Layered upon this colonial economic architecture is demographic engineering, most notably the Ba’athist settler-colonial Arab Belt (McDowall, 2004; Tejel, 2009; Aslan, 2023). Here the agrarian, the hydraulic and the demographic meet: irrigation grids, land expropriation and settler agriculture co-produce dispossession. Predictably, the ecological balance sheet was disastrous, with salinization, soil exhaustion and water pollution intensified by monocropping and fertilizer regimes, compounded by climate change-induced drought and upstream damming in Türkiye (Chatel, 2014; Gleick, 2014; Kelley et al., 2015; Daoudy, 2020a; Daher, 2022; Tinti, 2023). De-development and ecological degradation are thus two faces of the same colonial calculus: ecocidal developmentalism.

The starkest expression of this convergence is monocrop-agriculture-based development, overwhelmingly reliant on non-ecological irrigation and overextraction of water, one of the

main drivers of colonial drought. Article III details the multilayered structure of this colonial destruction and its *longue durée* impacts. Rojava lies largely within semi-arid and arid zones, conditions already marginal for rainfed agriculture (FAO, 2018; Baba et al., 2021, p. 1; Aydin-Kandemir & Yıldız, 2022, p. 3). Syria's hydrology is overwhelmingly Euphrates-centred (approximately 85 per cent of surface supply), while approximately 60 per cent of irrigated lands nationwide depend on groundwater (Baba et al., 2021, pp. 4, 7). After decades of over-pumping and weak regulation, aquifers in North and East Syria have fallen to 20-year lows; in Jazira and Hasakah, approximately 75 per cent of irrigation relies on groundwater, with over 200,000 largely unregulated private wells (Baba et al., 2021, p. 14; Aydin-Kandemir & Yıldız, 2022, pp. 5–6; REACH, 2023, p. 1). Since 2019, Türkiye's crossborder releases have dropped far below the 1987 understanding of 500 m³/s, falling to 160–214 m³/s, and driving Tabqa and Tishrin towards dead levels, compromising hydroelectricity and municipal supply for roughly 5.5 million people in North and East Syria (Water Directory of Hasaka, 2024: 2; Daher, 2022: 13). The Alouk station, serving nearly 1 million inhabitants of Heseke, has been repeatedly closed since the occupation of Serêkaniyê in 2019 (Daoudy, 2020b, p. 1362; Daher, 2022, p. 13; Daoudy, 2023, p. 15; REACH, 2023, p. 13). Families resort to trucked water that consumes 7–10 per cent of household income (REACH, 2023, p. 13; Water Directory of Hasaka, 2024, p. 10). Predictably, public health collapses along hydrological lines: diarrhoea, leishmaniasis, hepatitis A and cholera flare, with internally displaced persons camps acutely affected (Mueller et al., 2021, p. 18; Daher, 2022, pp. 16–17; Water Directory of Hasaka, 2024, p. 13). Scarcity here is not meteorological fate but a politically produced condition, an infrastructural choke on electricity, irrigation and potable supply that erodes the infrastructural sovereignty upon which democratic autonomy depends (Daoudy, 2020b; Daher, 2022; Daoudy, 2023).

Taken together, these destructions and policies expose another dimension of ecocidal coloniality, namely colonizers' intent and the use of ecocide as a technique of counterinsurgency, as discussed in Article II. The same colonial grammar targets the living archive through which people sustain memory and livelihood: the grove, the seed, the tree. A re-centred genocide scholarship, attentive to Lemkin's emphasis on the "essential foundations", makes plain that the destruction of land-bound social worlds can be genocidal even in the absence of mass killing (Lemkin, 1944; Wolfe, 2006; Moses, 2010; Mennecke, 2012, p. 145). Building on this, the genocide-ecocide nexus shows how environmental destruction imposes conditions of life that imperil a group's cultural or physical existence (Short, 2010; Crook & Short, 2014; Short, 2017; Crook et al., 2018).

In Afrin after 2018, ecocide has functioned as counterinsurgency. The region's economy and culture have historically revolved around approximately 18 million olive trees, yielding approximately 270,000 tons of olives and approximately 55,000 tons of oil annually (Human Rights Organisation Afrin, 2023, p. 36). Following occupation, more than 300,000 trees have been uprooted by Türkiye-backed militias, who reportedly appropriated groves and imposed 'taxes' (North Press Agency, 2020; Human Rights Organisation Afrin, 2023). In tandem, demographic engineering has remade the population map: the Kurdish share dropped from over 90 per cent to the low 20s, with reports indicating more than 130,000 settlers installed between 2018 and 2021; widespread killing, enforced disappearances and torture of remaining Kurdish residents; suppression of Kurdish language and toponyms; Turkification of public space; and 'model settlements' built with transnational associations, with some sources noting the settlement of Hamas-affiliated militias (Syrian Observatory for Human Rights, 2021; Human Rights Organisation Afrin, 2022; Human Rights Organisation Afrin, 2023; North Press Agency, 2023; Human Rights Watch, 2024).

As a constitutive part of this ethnic cleansing, the destruction and seizure of Afrin's olive groves operates as ecocide as counterinsurgency: it collapses economic autonomy by dismantling olive-based livelihoods and women-led and cooperative networks, forcing households into remittances, precarious trade and aid; it silences ritual and seasonal knowledge, including harvests, oral histories and place-based calendars, turning olive oil into a militarized commodity; and it creates a generational rupture, as uprooted trees and militarized spaces block the embodied transmission of cultivation, care and stewardship. This is not collateral damage, but ecological warfare that produces social death and functions as cultural genocide by severing people from the landscapes that sustain identity and resistance. In this sense, ecocide doubles as epistemicide, the silencing of embodied knowledge that presupposes the continued presence of more-than-human teachers. The logic of elimination is thus simultaneously material, symbolic and pedagogical.

Article IV brings to light the continuity of this epistemic erasure and the suppression of traditional ecological knowledge through education systems and colonial pedagogy. Education in Rojava exposes how the colonial formation operates as epistemicide: the systematic dismantling and replacement of knowledge systems that sustain Kurdish collective life. Anchored in a longer Ba'athist education mentality, with Arabized curricula, the downgrading or erasure of Kurdish language and history, and a centralized, submissive pedagogy, this project

is intensified under occupation through the Turkification of curricula and public space, the removal or marginalization of Kurdish instruction, and toponymic erasure in Afrin and Serêkaniyê (Human Rights Organisation Afrin, 2023; Human Rights Watch, 2024). Its reactionary patriarchal, statist and capitalist orientation redirects schooling away from ecological stewardship and communal care towards subordinated, dependency-bound employment, cultivating dependency rather than democratic capacity. By stripping curricula, denying credential recognition, displacing symbols in public space, and suppressing festivals, arts and everyday cultural practices, it targets the ethical and aesthetic norms through which communities reproduce themselves. In aggregate, education is re-coded from a vehicle of collective flourishing into an instrument for unmaking, weakening the intergenerational relay of language, memory and practice, and curtailing a people's capacity to know, remember and act (Boyle Espinosa & Ronan, 2023; Syrians for Truth and Justice, 2023). Colonial pedagogy aims to re-code subjectivity, producing a submissive mentality attuned to patriarchal, statist and capitalist hierarchies.

Seen together, these dimensions of ecocidal coloniality are not separate sectors of destruction, but interwoven vectors of elimination. The dam that prescribes a crop turns, in drought, into a lever of discipline, as control over water flows becomes a means of enforcing obedience and punishing those who step out of line; the settlement plan that displaces farmers also converts groves into militia revenue streams; the school whose desks are burned as firewood under bombardment teaches, of necessity, the ethics of survival rather than flourishing. Water becomes toxic and unpotable; soils whiten with salt; trees, once the scaffolds of memory, are uprooted; language is exiled from classrooms and streets. The foundational elements of Kurdish existence, including livelihoods, territory, language, memory, and ethical and aesthetic worlds, are targeted simultaneously (see Table 4).

Table 4: Techniques, mechanisms, and observed outcomes of nation-state colonial control over socio-ecologies in North and East Syria

Technique	Domain	Policy	Mechanism	Examples in North and East Syria	Observed outcomes
Hydraulic megaprojects	Water	Dams, canals,	Upstream capture and flow regulation	Euphrates control; Tishrin and	Scarcity; seasonal volatility;

Section III: Conceptual interventions, synthesis and discussion

		interbasin transfers	restructure hydro-social relations	Tabqa regime legacies; upstream restriction	irrigation cuts; dependency
Weaponization of water	Water	Crossborder valve control; station shutdowns	Periodic reduction or interruption of water supply	Alouk pumping interruptions; grid switching manoeuvres	Dry taps; tanker reliance; waterborne disease risk
Groundwater overextraction	Water/land	Subsidized monocrop; unregulated wells	Pumping beyond recharge; cone-of-depression effects	Hasakah plain wells; emergency boreholes	Aquifer decline; salinization; land subsidence
Monocropping	Agriculture	Wheat and cotton mandates; input chains	Simplifies agroecosystems; input lock-in	State wheat procurement; cotton corridors	Yield volatility; pest pressure; biodiversity loss
Input dependency	Agriculture	Fertilizer and seed regimes	Externalize fertility; suppress landraces	Hybrid wheat; imported fertilizer cycles	Soil fatigue; cost shocks; debt exposure
Expropriation and enclosure	Land	Confiscation ; military zones	Remove communal rights; repurpose land use	Afrin olive groves seizure; buffer zones	Dispossession ; income loss; displacement
Arab Belt resettlement	Demography/land	Engineered settlement belts	Demographic re-engineering of territory	Settlement of Arabs in Cezîrê region	Territorial fragmentation; cultural erasure

Chapter 8: The vectors and operational dimensions of ecocidal coloniality

Energy grid leverage	Energy/water	Targeted outages; selective feeds	Disable pumps, cold-chain and clinics	Regional blackouts after strikes	Service collapse; generator reliance; diesel pollution
Targeting civilian infrastructure	Infrastructure	Strikes on water, power and grain facilities	Destroy life-supporting systems	Strikes on stations, mills and depots	Service disruption; forced migration; morbidity
Crop burning and sabotage	Agriculture	Covert arson; border flashpoints	Destroy harvests; deter cultivation	Wheat field fires in front-line zones	Food insecurity; livelihood shock
Curricular assimilation	Knowledge	Centralized curricula; language bans	Decouple youth from local ecologies	Pre-2012 school programmes	Epistemic erosion; language loss
Border militarization	Mobility/economy	Walls; lethal zones; tariff choke points	Sever trade, labour and seed flows	Northern wall segments	Market isolation; price spikes

This is why, for the KFM, ecological struggle is not a sectoral policy, but an ethos of decolonization: an ontological defence of life against ecocidal coloniality that names exploitation and destruction, indeed annihilation, as ‘progress’ and ‘development’ (Harris, 2002; McDowall, 2004; Harris, 2006; Harris, 2008; Van Etten et al., 2008; Tejel, 2009; Jongerden, 2010; Harris, 2012; Bilgen, 2017; Akıncı et al., 2020; Bilgen, 2021; Dinc et al., 2021; Adalet, 2022; Türk, 2023; Türk & Jongerden, 2024; Dinc & Türk, 2025).

Chapter 8 has shown that ecocidal coloniality in Kurdistan is not merely a political order, but a socio-ecological formation that organizes territory, resources and subjectivities through a coherent repertoire of dimensions. What is branded as ‘development’ operates as ecocidal developmentalism: agrarian modernization, hydraulic megaprojects and demographic engineering that integrate Kurdish regions as extractive margins while dissolving their ecological and economic bases. This design generates colonial drought, where scarcity is

produced by infrastructural capture and hydro-hegemony, rather than by climate alone. It also forges a genocide-ecocide nexus, most visible in Afrin, where the destruction and appropriation of olive groves functions as counterinsurgency, collapsing livelihoods, severing land-bound culture and engineering social death. In parallel, epistemicide targets the conditions of collective endurance: schooling and public discourse disqualify Kurdish languages, knowledges and ecological practices, recoding dependency as common sense.

Taken together, dimensions and vectors of ecocidal coloniality, namely legal and political engineering, ecocidal developmentalism, spatial and hydraulic design, militarized ecological violence, epistemic domination, and biopolitical marginalization, compose a single apparatus that de-develops Kurdistan, while rendering it governable and extractable. If Kurdistan and Rojava's present is shaped by the ecocidal coloniality whose impacts dismantle the foundational elements of the Kurdish people, then anti-colonial praxis must be ecological in means and ontological in horizon. The next chapter turns to those insurgent reconstructions: rhizomatic democracy, governance and resistance, the eco-agrarian economy, and ecopedagogy as socio-ecological transformation.

Chapter 9. Democratic resistance and socio-ecological transformation: The emergence of a decolonial ecology

This chapter advances the thesis's central claim that decolonization in Rojava unfolds along two intertwined lines that are simultaneously political and ecological. First, it examines the construction of democratic governance against the nation-state's ecocidal, genocidal, centralizing, monopolistic and homogenizing form of rule. The turn to democratic autonomy relocates authority from the state to organized society, and generates a distinctive, locally grounded practice of decentralized, horizontal, non-statist and non-hierarchical self-rule, conceptualized as rhizomatic democracy, operating through semiotic and symbiotic confederal clusters. From within these clusters, rhizomatic governance emerges as a political ontology that materializes decoloniality across institutions, infrastructures and everyday practice as a rhizomatic resistance. Second, the chapter argues that ecological struggle against ecocidal coloniality functions as the ethos of decolonization and the principal driver of socio-ecological transformation, giving rise to a decolonial ecology. Empirically, the analysis follows three interlinked domains where this transformation is most legible: ecological agriculture and the communal governance of land, water, energy and seeds; the social and communal economy, including cooperatives, eco-factories and the women's economy; and epistemic transformation through *jineoloji* and ecopedagogy. Finally, the chapter registers limits and contradictions produced by permanent coercion and material scarcity, including war, embargo, weaponized water, capacity gaps, land justice disputes, cadre-peasant tensions, risks of bureaucratic drift and uneven institutionalization. Read together, these dynamics show how democratic resistance, embodied in rhizomatic governance as the political ontology of socio-ecological transformation, constitutes decolonization at once ontologically and epistemologically through ecological struggle, as well as through its associated ethics and aesthetics.

9.1 Democratic resistance: Rhizomatic governance as a political ontology

After the collapse of the Ba'ath regime's colonial order in Rojava on 19 July 2012, political institutions underwent several stages as an experimental process. In 2018, AANES was declared

as a new political formation and system of autonomous governance. A substantial scholarly literature has examined this revolutionary reconfiguration, situating the radical transformation of governance within the Kurdish movement's ideological and political turn inspired by the thought of Abdullah Öcalan (Sunca, 2020; Cemgil, 2021; Gerber & Brincat, 2021; Matin, 2021). Against an authoritarian, centralist, monist, antidemocratic nation-state paradigm, what emerged in Rojava was a radical democratic rupture, not merely an administrative shift but a transformation at the level of political ontology. The post-2012 reconfiguration of Rojava marks an ontological rupture with colonial governance in Rojava/Kurdistan. The most marginalized and excluded Kurds dismantled core elements of the colonial order and asserted self-governance, refusing to replicate the colonial nation-state form and advancing instead a model grounded in democratic autonomy and confederal self-administration. Decentring the state as the metric of self-determination, theorists of the movement have elaborated an ideological and political framework of self-government without a state (Knapp et al., 2016; Gunes, 2019; Jongerden, 2019; Matin, 2021; Türk & Jongerden, 2024). Rather than pursuing state formation and reproducing a new elite, the party and movement reoriented towards strengthening society through self-administration. This radical move resonates with Fanon's insistence that postcolonial polities should not imitate European state forms; instead, they should build decolonial alternatives (Fanon, 1963), notably democratic autonomy and democratic confederalism, proposed as institutional cores of any democratic nation or community (Öcalan, 2020; Sunca, 2023; Jongerden, 2024).

In this sense, radical democracy names a current of thought that relocates the site of democratic power from state institutions to the organized capacities of society itself. Its genealogy crosses several lines. Gramsci locates democratization in civil society, conceived as a dense bulwark that can both sustain and contest state power (Gramsci, 1971). Laclau and Mouffe radicalize democracy by articulating heterogeneous social demands into a shared political space, combining different voices through processes of articulation rather than presuming a single subject (Laclau & Mouffe, 1985). Taken together, these lines refuse the nation-state as democracy's necessary form, and open a horizon in which democratic practice becomes the organized power of those who act.

Within this horizon, the Kurdish movement's ideological evolution is decisive. After the collapse of the Soviet Union, the movement continually reassessed global transformations and sought a new strategic horizon. This reassessment accelerated with Öcalan's imprisonment in

1999, when the movement undertook a marked turn to radical democracy, with Öcalan rejecting both the “classical Kurdish nationalist line” and “a leftist interpretation of a similar tendency” (Akkaya & Jongerden, 2013, p. 186). In his subsequent writings, Öcalan translated this reorientation into an ideological programme, democratic confederalism and democratic autonomy, which pursue politics beyond and without the state. As complementary dimensions of post-statist self-rule, they refer to “the right of people to determine their economic, cultural, and social affairs” (Akkaya & Jongerden, 2013, p. 187). Democratic autonomy is not a devolved function of a nation-state. In Escobar’s terms, autonomy is a cultural, ecological, and political process: it delinks everyday life from state and corporate control and marketized “solutions,” privileging situated practices of learning, healing, living, and producing that are less exposed to external regulation (Escobar, 2018, pp. 173, 181). This orientation directly counters the canonical repertoire of colonial and ecocidal destructions, including land dispossession, extractivism such as dams and deforestation, racialized economic marginalization, pollution, and the erosion of culture and biodiversity, as documented across Indigenous and colonized contexts (Escobar, 2018; Ferdinand, 2022). Autonomy, accordingly, functions as collective resistance and world-making: a political ontology in which communities craft non-liberal, non-state, non-capitalist modes of being/doing/knowing, what Botero terms “collective biographies of micro-revolutions for self-determination” (Botero, 2015, pp. 17–19, cited in Escobar, 2018, p. 182).

The genealogy of non-statist governance runs through Murray Bookchin’s municipalist-confederal tradition, which Öcalan explicitly draws upon. For Bookchin, communalism/libertarian municipalism rests on direct democratic assemblies (community, town, neighbourhood) confederated by mandated, recallable deputies: an alternative to the nation-state (Bookchin, 1991, p. 7). Confederalism is “a network of administrative councils” elected from face-to-face democratic alliances, whose remit is administrative, not legislative, with decision-making power retained in the assemblies (Bookchin, 1990, pp. 9–10). This municipalist-confederal architecture underwrites a “radically new configuration of society” (ibid., p. 4). In the Kurdish context, it provided Öcalan with both a critique of statist horizons and a positive grammar of non-state democracy. This also helps explain why council-based architectures and assemblies matter: they are not merely deliberative, but also institutions of collective capacity, through which diverse communities relate horizontally and speak for themselves, aligning with “politics beyond the state, political organization beyond party, and

political subjectivity beyond class” (Badiou, 2002, pp. 95–97, cited in Akkaya & Jongerden, 2013).

The multitude concept sits in this orbit, foregrounding popular power against separations of people from government and authority (Negri & Hardt, 2004). Finally, this repertoire is prefigurative rather than deferential: democracy is practised in the present through the very institutions that embody it (Maeckelbergh, 2009, p. 67). In this sense, democratic autonomy and democratic confederalism are not merely descriptive labels for the Rojava experience. They are the conceptual and practical means by which the Kurdish movement’s post-1999 paradigm makes radical and direct democracy material – extending the political field into civil society (Gramsci, 1971), radicalizing its scope and composition (Laclau & Mouffe, 1985), and unbinding it from the state form (Bookchin, 1990, 1991). Yet even as elements of these perspectives have been officially adopted and partially implemented, no single framework fully captures the emergent experience in Rojava. What is at stake is an ongoing effort to re-found a multiethnic and multireligious society in a new ontological register, organized around democracy, ecology and women’s liberation, and pursued beyond the nation-state form as part of a sustained anti-colonial and anti-capitalist autonomy. For this reason, I develop the conceptual framework of rhizomatic democracy and rhizomatic governance which, I argue, best captures these dynamics, in Articles I and III.

9.1.1. Rhizomatic democracy

Following Deleuze and Guattari (1987), the rhizome names a mode of organization in which any point can connect to any other; it traverses heterogeneous terrains and operates across both regimes of signs and non-signifying states. It is irreducible to a unified One or to a discrete, countable multiple. Rather than a vertical, transversal branching form, rhizomes spread through horizontally oriented stems without a single organizing centre. For Deleuze and Guattari, this decentralized, non-hierarchical arrangement is preferable because it enables heterogeneous linkages among parts (Somers-Hall, 2013, p. 2). Unlike a structure defined by fixed positions and binary relations, a rhizome is composed of lines, lines of segmentarity and stratification and, at the limit, a line of flight or deterritorialization along which a multiplicity metamorphoses and changes in nature. The distinction between arborescent and rhizomatic formations functions as an analytic method rather than an ontological either/or. In *A Thousand Plateaus*,

arborescence figures hierarchy, centralization and vertical organization, a root–tree schema embodying order, unity and transcendence. The rhizome maps a plane of immanence, a network of proliferating connections, and transversal and horizontal logics of growth and survival (Deleuze & Guattari, 1987, p. 21). No system is purely one or the other: trees are always subject to rhizomatic intrusions, while rhizomes are prone to arborescent capture and centralization. The opposition is therefore not absolute but methodological: the tree is a figure of tracing and transcendence; the rhizome, a figure of mapping and immanence (Deleuze & Guattari, 1987, pp. 20–21; Bonta & Protevi, 2004, p. 52).

To make this analytic operational, I emphasize five rhizomatic principles and their Rojava correlates: connection/heterogeneity (cross-linkages among communes, academies, women’s structures); multiplicity (parallel nodes performing similar functions), asignifying rupture (reconstitution after attacks, such as energy/water decentralization), and cartography over tracing (iterative, situated ‘mapping’ via commune practice rather than reproducing fixed blueprints). This highlights not only the primacy of horizontal relations, but also the rhizome’s immanent capacity to regenerate from any part and the fact that every social arrangement inevitably produces some vertical forms. Thus, while the rhizome metaphor stresses “scaling out,” we must also attend to necessary modes of “scaling up” (García-Arias et al., 2024). In Rojava, scaling out refers to lateral proliferation and redundancy across communes and committees; scaling up indicates confederal clustering that coordinates across nodes without acquiring superior sovereignty. This tendency is evident in Rojava’s evolving organizational practices. For instance, each commune has a coordinating executive committee and a set of open working committees. The commune serves simultaneously as a deliberative arena and an implementing body, led by two co-chairs with representatives from the committees. Communes federate upward into neighbourhood councils and then into district, city and regional councils/cantons, which together constitute AANES. While the names of these councils have shifted over time, the underlying principle is unchanged: a confederal network that devolves decision-making to communities rather than concentrating it in a central bureaucracy, ensuring active participation across the community. Here, confederal nodes are mandated and recallable coordinators; they do not monopolize legislative authority (Bookchin, 1990, pp. 9–10).

This is the logic of democratic confederalism: interlinked, self-administered communes constitute a horizontal mesh that enables joint decision-making through confederal clustering, whereby village assemblies compose new assemblies for deliberation and coordination (Türk

& Jongerden, 2024). Crucially, such assemblies do not acquire higher authority – they aggregate, and coordinate mandates issued below. To grasp the intermediate strata between politically driven coordinating bodies and on-the-ground organizational formations through which everyday resistance, transformation and becoming take shape, the concepts of rhizomatic democracy and rhizomatic governance are more precise and useful. This claim rests on three core field-based observations and empirical findings, as well as more than a decade of Rojava's organizational transformation and experimentation.

First, rhizomatic democracy, while it may incorporate the mechanisms of direct democracy and the organizational forms of radical democracy, differs from both in crucial ways. Rather than defining itself solely in opposition to state and power (a purely negative, non-state, non-hierarchical stance), it persists by building spaces without them and, in the face of assault, tracing lines of flight that open new terrains on which to constitute itself. This does not mean that rhizomatic democracy and thinking do not contain anti-capitalist, anti-colonial and anti-power/anti-statist elements. But rhizomatic democracy does not function as a counterprogramme aimed at eliminating them. Rather, it creates terrains that contain the possibilities, contingencies and potentialities of life and forms without and beyond them. Accordingly, its causal logic is affirmative–compositional: if coercive capture intensifies, then nodes redirect energies along new lines (withdraw, recombine, reappear), sustaining democratic practice without mirroring statist forms. By contrast, opposition-centred organizing inevitably co-produces its adversary: as antagonists risk becoming one another, settling in non-state plateaus and territories enables new forms of life (Checchi, 2021). This is vital, because it allows organization and imagination without mimicking colonial and statist formations contingently. In short, rhizomatic democracy privileges composition over reactive negation, and endures by inventing spaces where domination cannot easily take hold.

Second, whereas classical debates on radical and direct democracy focus on dismantling representation and related forms of domination so that society governs itself, rhizomatic democracy carries no fixed ideal or telos. It has no blueprint or utopia. Its manifesto of freedom is a field of possibility and contingencies in which non-state and non-power forces can actualize themselves in territories that are not 'ideal', but that nonetheless make escape from domination and exploitation possible. This is an affirmative, creative ontology of multiplicity and becoming. For that reason, it allows unforeseen forms, ruptures and inventive modes of existence: it dynamically defends itself against incursions of power and, when life is targeted,

it can withdraw, flee and reappear elsewhere, in new territories and newly configured forms. This distinguishes rhizomatic governance from polycentric approaches (Ostrom, 2010) and from standard network-governance models: beyond coordination, it prefiguratively remakes socio-ecologies and subjectivities, and includes lines of flight that standard polycentric models do not theorize. In other words, in conventional democratic accounts, communes, assemblies and councils are assumed to coordinate and administer all public affairs, political, defence, economic, social, cultural, legal, educational and health matters, and more. In rhizomatic democracy, however, these give way to each field's own organizational forms, institutional designs and repertoires of resistance: a living current of resistance that continually maps new lines.

For example, the Kurdish guerrilla movement organizes in autonomous units in a rhizomatic way; facing attacks by centralized state and army, it constantly shifts ground, retreats, escapes, re-establishes territory, strikes, defends, and withdraws. Following the same logic, when the Turkish state systematically targeted and destroyed Rojava's infrastructure (especially in energy, economy and water), the revolution's pioneers and responsible cadres (some of them former guerrillas), together with the local population, turned to decentralized energy and water systems. This was not a single master plan, but a distributed, adaptive repertoire oriented to resistance, endurance and survival. In practice, the turn to solar panels in Rojava began precisely in this way. In principle, a state could strike every rooftop panel or wellhead installation, but doing so is economically and militarily irrational: destroying assets costing hundreds of dollars would require munitions worth tens of thousands. In the civil sphere, energy and water systems have been decentralized under bombardment. Parts can be destroyed, but the network re-forms and rebuilds; like an ant nest, it is extraordinarily resilient and resists eradication. Similar logics animate other areas: communes, cooperatives, factories and academies adjust functions and connections to sustain provisioning and learning under duress. Article I shows that the agricultural domain has been organized through this formation, from the village level to cantons and AANES, from commune agriculture committees to the AANES Agriculture and Irrigation Council. It also demonstrates symbiotic relationality among the Economy Council, the Ecology Council, and the Agriculture and Irrigation Council.

Third, although AANES and its multiple organizational forms are dominant in the region, there are heterogeneous formations in a rhizomatic sense. Different Kurdish political parties, tribes and individuals, NGOs, and private/capitalist enterprises exist and do not join the Rojava model.

Even in the economy, compared with communal cooperative systems and AANES-led factories, capitalist enterprises are often the dominant economic form. These tendencies and emergences, together with their contradictions and splits, show that the revolution in the making in Rojava is a dynamic process, interlaced with capitalist and statist systems through continual movements of deterritorialization and reterritorialization. Deleuze and Guattari (1987) warn that rhizomes can be captured (pp. 224–226) and become arborescent (pp. 20–21); acknowledging this, Rojava's anti-capture designs (co-chair system, rotation/recall, autonomous women's nodes, academy training, normative-legal constraints) work to prevent vertical consolidation while maintaining operational effectiveness. In zones where counter-revolutionary tendencies gain traction, rhizomatic forms of organization and resistance come into play, recomposing themselves through new organizational forms and lines of action. For example, the women's movement has positioned itself against such counter-revolutionary waves as an autonomous structure (a node) within the broader AANES assemblage. While the education system and the universities articulate to the whole as academic nodes, *jineoloji* and the Democratic Society Academies continue to situate themselves as distinct nodes.

These heterogeneous forms are, precisely, the privileged expression of the rhizomatic character. These tendencies are territorial, contrasted with the merely functional orientations often found in radical and direct democratic experiments: as constituent and creative movements, they choose to organize deterritorialization and trace lines of flight against capitalist and statist tendencies. Rojava undoubtedly comprises diverse formations and elements, yet these three dynamics most clearly disclose the Rojava Revolution's distinctive rhizomatic configuration. They point to a formation that departs from prevailing democratic formulations, debates and models. It is therefore apt to describe what is being lived and shaped in Rojava as rhizomatic democracy, and to recognize the presence of a rhizomatic governance network that affirms and defends life and the earth, as discussed below.

9.1.2. Rhizomatic governance

Article III presents Rojava's autonomy formation as the construction of rhizomatic governance: a mode of life and territorial organization that refuses the modernist grammar of submission to a central state, and instead composes conditions for living beyond the nation-state as the vehicle of a decolonial shift (Paradies 2020; Walsh 2020). Democratic autonomy is the political horizon

and institutional grammar; rhizomatic governance is the modal logic and lived dynamic that make that horizon material. What appears here is not a technical redesign of administration, but a political ontology, namely a confederal clustering that breaks with statist, capitalist and colonial arrangements by multiplying points of initiative, linking them transversally and keeping them capable of flight when confronted by capture. By political ontology is meant the patterned relations and practices that compose socio-natures and subjects, tracked here through institutional rules, infrastructures and training, so that democratic autonomy is not merely designed but enacted through rhizomatic governance. These relational and horizontal clusters contain diverse nodes which organize life across multiple spheres. Nodes are political (communes, people's assemblies), economic (cooperatives, eco-factories), epistemic (academies, *jineoloji*), territorial (municipalities) and gendered (Kongra Star, Jinwar, Eko-jin), together with proliferating social, cultural and ecological initiatives that continually become new nodes. These rhizomatic governance networks have become a political ontology of socio-ecological transformation through their semiotic, symbiotic and transversal relationality, as discussed in Article III.

To this end, as a critical and epistemic node, academies (Democratic Society Academies, Jineoloji Academies and the Ecology Academy) unsettle epistemic hierarchies by weaving together feminist, ecological and communal knowledges (Escobar, 2018; Mignolo & Walsh, 2018; Simpson, 2021; Sultana, 2022). They serve as spaces for the co-transformation of thought and action, enabling collective reflection, critique and self-critique aimed at disarticulating the ideological bases of capitalist modernity and cultivating communal, ecological, democratic and feminist consciousness. Members of communes and cooperatives circulate through these programmes; training addresses practical obstacles while consolidating ethos.

In this regard, the Ecology Council works to materialize ecology, one of the KFM's three pillars, practically, epistemically and affectively as an ethical and aesthetic principle across domains. It reviews legal and administrative practice for ecological consonance, provides guidance, and conducts training through the Ecology Academy, cultivating ecological literacy and epistemic pluralism and integrating ecological criteria into legal frameworks and curricula (Article IV). The Ecology Council connects directly with communes, cooperatives and municipalities, coordinating initiatives and embedding ecological commitments in everyday life, particularly through cooperation between this epistemic node and the communes.

The commune functions as the stem cell of society and the smallest node of rhizomatic governance: local assemblies that manage water, land and agriculture; deliberate problems; make decisions collectively; and implement solutions across gender, class and ethno-religious lines. Participation is tethered to ecological and women's liberation principles. Flexible subcommittees (defence, education, services, agriculture) coordinate with cooperatives, the Ecology Council, municipalities and academies, so that communal initiatives align with broader socio-ecological aims (Grasso, 2022; Wimmer, 2024). Co-chairs and committee representatives serve as delegates to people's assemblies (city assemblies), maintaining continuous feedback between primary and confederal scales. Thus, governance 'scales' through confederal clustering rather than vertical sovereignty. This is the organization of politics and socio-economic fields autonomously through confederated clusters.

As economic nodes in this formation, cooperatives condense the thesis of a decolonial economy. The slogan "*Ji bo her komûnek kooperatîfek*" ("A cooperative for each commune") identifies the organic complementarity between political and economic organization: cooperatives prioritize ecological principles, self-sufficiency and social benefit; they disrupt extractive colonial and capitalist circuits; and they collectivize access to land, water and production for historically marginalized groups, including landless peasants, workers, women, Kurds, Yazidis and Assyrians. This is not merely an economic reconstruction but an ontological and epistemic rupture with centre-periphery coloniality, building an anti-colonial, anti-capitalist and antipatriarchal economy rooted in social and ecological parameters. Cooperatives coordinate with municipalities to integrate infrastructure and resource allocation.

As territorial nodes, municipalities materialize decisions: they coordinate services, implement infrastructure and embed socio-ecological criteria in administration. They are the embodied spaces where governance nodes converge; legal frameworks enable cooperative projects, infrastructure development and resource management; and territorial realities are co-produced through everyday action, negotiation and contestation, which exceeds state-centric imaginaries (Deleuze & Guattari, 1987; Delaney, 2005; Escobar, 2018; Escobar, 2020).

Coordinating all these autonomous nodes requires confederated clusters and carefully delimited scaling-up. Thus, rhizomatic governance cannot be reduced to autonomous units and horizontal ties alone. Scaling-up emerges where units are coordinated, intersected, separated and reconnected in fluid, dynamic forms. People's Assemblies act as such confederal nodes,

coordinating decisions across sectors and scales. Co-chairs of city assemblies sit *ex officio* on canton executive councils; composition typically mixes 60 per cent elected delegates and 40 per cent organizations and minorities (Wimmer, 2024, pp. 14–15). All civil bodies, including municipal co-mayors, are represented, which facilitates bidirectional integration among communes, cooperatives, academies and municipalities. In contrast to the nation-state's monist identity politics, this formation enables diverse ethnic, religious and cultural groups, and individuals, to coexist under the rubric of the Democratic Nation.

Together, these symbiotic and semiotic relations constitute rhizomatic governance: a non-statist, decolonial, feminist and ecological reimagining of political ontology, offering a substantive alternative to technocratic adaptation and mitigation scripts that leave intact the drivers of collapse. By re-relating society and nature, it addresses the roots of colonial dispossession, displacement, ecocidal developmentalism, ecocide, colonial drought, climate breakdown, and water crisis on a war-torn, colonized land. Communes collectivize land and irrigation, cooperatives enact food autonomy, education is politicized and democratized along ecological and feminist lines, and women-led cooperatives, communal water committees and ecological regeneration projects refigure land and water as commons rather than resources for external extraction or coercive control. The resulting configuration is neither a return to the state form nor a mere aggregation of localities, but a relational, pluriversal infrastructure of autonomy, capable of deterring pressure through dispersal and recomposing itself along new lines of common life.

On this basis, the democratic and horizontal institutions in Rojava are not romanticized abstractions but practices formed over years as strategies of resistance and survival, that is, instances of rhizomatic governance. The Rojava Revolution, led by veteran guerrillas and political cadres with roughly 20–30 years of experience of struggle, draws almost instinctively on accumulated practices and lessons. The Kurdish guerrilla movement offers a distinctive instantiation of a rhizome: although a central headquarters exists, strategy and tactics are enacted through autonomous units capable of operating independently. Relational linkages and mutual support bind units together, yet the destruction of any single unit is offset through replacement and recomposition. This form, in many respects, resembles the irrepressible and inextinguishable formation of a rhizome: “You can never get rid of ants because they form an animal rhizome that can rebound time and again after most of it has been destroyed” (Deleuze & Guattari, 1987, p. 9). Enacted through communes, cooperatives, academies, municipalities

and People's Assemblies, where rhizomatic democracy, ecological stewardship and women's liberation reorganize land, water, and social reproduction from below, rhizomatic governance renders decolonial, communal and feminist futures materially thinkable and buildable.

Yet, as Deleuze and Guattari insist, rhizomes are not inherently liberatory; they can congeal into hierarchies. The very capacities that enable distributed endurance can tip towards centralization and antidemocratic consolidation. Elements of this hybridization already appear in Rojava's economic and military spheres. These formations pose a serious internal risk, namely a drift that can degrade the revolution's democratic ethos. Consequently, while the rhizome enhances resilience, a tendency towards centralization in Rojava's rhizomatic organization would undermine both its democratic-autonomous principles and its capacity for resistance and renewal. To overcome such risks, the political formation of Rojava, which is grounded in principles of rhizomatic democracy and rhizomatic governance, advances a novel form of resistance, as elaborated below.

9.1.3. Rhizomatic resistances: Resisting mimicry, affirming difference and transversality

This subchapter reframes resistance line in Rojava beyond its familiar portrayal as a rare rupture. Much of the classic literature equates resistance with dramatic moments of confrontation that interrupt the ordinary stability of power. Checchi (2021) shows how this framing remains tied to an event-based imaginary in which resistance is circumscribed by barricades, clashes and short-lived uprisings that begin and end with the enemy they oppose. This traditional understanding (oppositional tradition) treats resistance as inseparable from an enemy and measures its life by the duration of struggle: resistance to what, against whom. This view renders resistance derivative and finite. Once the adversary falls, resistance either dissolves or is captured by the new order. A second axis, the "reactive tradition," gives power conceptual primacy and reduces resistance to response (ibid., p. 6). This reading draws on Nietzsche's (1989) distinction between noble (active forces) and slave (reactive forces) moralities, considered alongside colonizer-colonized dynamics. Active forces have greater intensity; they set the terms of relation, command reactive forces, and realize their capacities in affirmative and value-creating ways. Reactive forces have lesser intensity; they are subordinated to active forces and remain separated from what they can do, which yields life-negating, value-conserving orientations. Crucially, their moral world and value schema are

formed reactively in relation to the master's values (Woodward, 2013, p. 118). They are, in this sense, products of power.

To overcome this trap, Foucault's assertion that "resistance comes first, and resistance remains superior to the forces of the process" points in a different direction: resistance is primary because it is creative and affirmative before it is oppositional, with power responding to it rather than the other way around (Foucault, 1997, p. 167). Since power and resistance are coextensive and immanent in this register, their relation is not a singular duel, but a shifting "multiplicity of points of resistance" that reconfigure the field itself (Foucault, 1978, p. 95). That is why Caygill (2013, p. 5) insists there is never a pure moment of resistance, only "clusters or sequences of resistance and counter-resistance" that seize or surrender initiative. In this register, opposition is not the essence of resistance, but an accident that appears when creative trajectories are blocked by reactive forces, hence Checchi's provocation: "If resistance needs to be against something, it is primarily against the against that follows it" (Checchi, 2021, p. 6).

Transitioning from this theoretical ground to anti-colonial resistances clarifies why many movements faltered or reproduced what they opposed. Mbembe (2015), re-reading Fanon, insists that decolonization is a project of self-ownership/self-affirmation and world-making in time. It turns away from mimicry and "imitation," rejects Europe as model, and affirms the capacity to craft new forms of life and new temporalities in which the "not yet" becomes thinkable and liveable. For Fanon (1963), the colonial order negates time by confining the colonized to repetition. Decolonization must break this law of repetition by creating new subjects who recover authorship over image and essence, refusing the gap imposed by colonial representation of the time. Anti-colonial praxis fails when it mirrors the state form, colonial epistemology, or remains captive to reactive resentment. It succeeds when it affirms its own difference, invents and creates different way being, knowing and doing, and refuses the binary of either seizing the state or withdrawing into purity (Quijano, 2007; Mignolo, 2009; Sousa Santos, 2016; Acosta & Brand, 2018; Blaser & de la Cadena, 2018; Escobar, 2018; Mignolo & Walsh, 2018; Tello Valle Hiriart, 2024). Otherwise, resistance movements risk becoming little more than waves breaking upon the shore of the colonial ocean, condemned to a tragic, nihilistic repetition.

The intellectual effort to escape this nihilistic cycle and the servile mentality that sustains it has a long genealogy. Among the most creative interventions is Deleuze's reading of Nietzsche.

Deleuze concludes *Nietzsche and Philosophy* by claiming that human history largely unfolds as nihilism, negation, and reaction, sustained by the repetitive reign of reactive forces. At the point where this sequence exhausts itself, what persists is affirmation as the sole quality of the will to power, action as the distinctive quality of force, and becoming active force as the creator of its own difference (Deleuze, 1983). Since, for Deleuze, difference is not opposition, affirmation can be sustained without reinstalling negation. Hence his formulation: “Negation is *opposed* to affirmation but affirmation *differs* from negation” (Deleuze, 1983, p. 188). To cast affirmation as “opposed” to negation would reintroduce the negative and mistake opposition for the essence of the negative. On this basis, Deleuze’s account of overcoming nihilism rests on the affinity between affirmation and a concept of difference that is non-oppositional and non-negative. In *Difference and Repetition*, Deleuze develops difference in itself as irreducible to identity or opposition, and articulates it through multiplicity, temporal syntheses (which underpin becoming) and objective chance (the role of unpredictable connections in the flow of time). Multiplicity is not a mere difference between fixed things, but internal relational lines; becoming is the process through which differences are synthesized across time; chance intervenes in this synthesis by disrupting deterministic continuity. Against this, a nihilistic metaphysics would attempt to reabsorb becoming into stable Being and judge the multiple by the One (Deleuze, 1994).

By affirming difference rather than combating negation, Deleuze figures the overcoming of nihilism as the becoming active forces and the affirmation of life (Patton, 2000). To this end, Deleuze’s interpretation of Foucault provides a significant ontological reformulation of resistance in relation to power and force. Beyond the dual capacities “to affect” and “to be affected”, Deleuze identifies a “third power,” which he describes as the possibility of “resistance” (Deleuze, 1988, p. 89). This “third power” is not a reactive counterforce within existing power relations, but arises from the “outside”, a plane of potentiality that exceeds any given diagram of forces. Resistance, therefore, is the creative capacity that continually redraws the boundaries of power by generating new relations and possibilities for transformation. On this register, the problem is not to negate negation, which reinstalls dialectic. It is to practice a politics in which affirmation does not derive its meaning from what it denies. Hence Checchi’s formulation that affirmation sets trajectories on which power later installs itself: resistance “always finds new routes, new paths, new trajectories, new openings”, and persists as “resistant re-existences” rather than a single decisive act (Checchi, 2021, p. 19). Contemporary practices that craft alternative socio-ecologies across human and more-than-human alliances illustrate

this logic as an “insurgent posthumanism” (Papadopoulos, 2018). They prefigure different ways of living, and opposition appears only when power stumbles upon their paths (Greyl et al., 2012; Checchi, 2021, p. 13).

With this groundwork, I introduce rhizomatic resistance to extend the earlier account of rhizomatic democracy and governance by naming a mode of struggle that is ontologically transversal (Guattari, 2015; Guattari et al., 2015). In the terms advanced by Deleuze and Guattari, a rhizome functions through connection and heterogeneity, multiplicity and lines of flight, and by mapping rather than tracing: “any point of a rhizome can be connected to any other” and “it is composed not of units but of dimensions” (Deleuze & Guattari, 1987, pp. 7–8). In this schema, transversality is not a mere tactic; it becomes the mode by which struggle traverses domains, scales and subject positions to produce new consistencies. It cuts across the molar partitions of economy, ecology, pedagogy and polity, while weaving micropolitical revolutions into collective forms without recentring them in a unitary sovereign identity. This approach evokes Guattari’s notion of “a molecular revolution” (Sustam, 2021, p. 42). In this sense, rhizomatic resistance is transversal because it forges lateral and oblique connections and treats differences as relational resources for creation rather than identities to be defended.

Materially, rhizomatic resistance is visible when communes, cooperatives, assemblies, academies and municipalities function as distributed nodes that coordinate without reproducing state logics. Epistemically, it works through liberatory and decolonial knowledges that provincialize European paradigms and reactivate local archives without enclosing them. Ethically and aesthetically, it affirms practices that bind care, economy, pedagogy and ecology in its own differences. Strategically, it treats opposition as an accident. The primary work is to build and link socio-ecological forms that prefigure the future in the present, to maintain redundancy and dispersion so that no single point of failure can disable the whole, and to keep organizational forms porous enough to sustain their own transformation. In Foucault’s language, it multiplies “points of resistance” so thoroughly within the network that attempts at capture must continually chase a moving field rather than a central target (Foucault, 1978, p. 96).

Rojava’s experience helps specify the stakes. Rojava’s refusal to mirror the nation-state or to pursue sovereignty as identity is not a sign of weakness, but a condition of autonomy. By rejecting mimicry and affirming its own historical, epistemological and geographical resources,

Rojava holds open a politics of difference rather than opposition (Silva Fobister, 2022). Democratic autonomy is the political ontology that carries this refusal. Rhizomatic governance is its expression and operational logic. The eco-agrarian economy and ecopedagogy are its socio-ecological drivers. Together they enact an ethos in which ecological struggle is constitutive rather than supplementary, and in which resistance is measured by the proliferation of life-supporting relations rather than by victories over enemies. In this sense, resistance subordinates its oppositional moment to creation and transformation. It is ‘against’ only when creation is obstructed, and even then, it remains oriented to openings rather than closure (Checchi, 2021, pp. 8, 12).

This conceptualization also guards against two recurrent traps. First, it resists the centralization of resistances that would translate multiplicity into a single and fixed identity, which would reproduce the unity of sovereign power it contests. Second, it guards against reactive tradition by affirming difference as such, rather than defining identity through the negation of an Other. The criterion is neither purity nor permanence. It is endurance in becoming, the capacity to turn every closure into a relay for new openings and to sustain “the real urgency of the inexistent”, the future already at work in the present (Negri, 1991, p. 160).

In sum, rhizomatic resistance names a decolonial praxis that is ontologically transversal. It avoids mimicry, affirms differences and becomes otherwise. It is primary rather than derivative, creative rather than merely oppositional, multiple rather than centralized. It does not seek to seize the state in order to install identity. It seeks to compose a pluriverse of socio-ecologies in which life can continue and transform through rhizomatic governance. In Rojava, this is what has made resistance durable under existential pressure. It generates the trajectories on which power is forced to react, and it continually reopens what closure attempts to seal. Rhizomatic governance as a political ontology of socio-ecological transformation is the pivot of this analysis. It conceptualizes Rojava’s alternative pathway: a reconfiguration of governance, ecological practice and democratic life grounded in new ontological and epistemic commitments that refuse technocratic ‘adaptation, mitigation, resilience’ scripts. Instead, it forges non-extractive, relational, and pluriversal modes of being, knowing and doing that confront the colonial-capitalist roots of interlocking crises: genocidal and ecocidal destruction and armed conflict, dispossession and poverty, the climate crisis, and the collapse of the international political and legal orders. Enacted through communes, cooperatives, academies, municipalities and People’s Assemblies, where rhizomatic democracy, ecological stewardship

and women's liberation reorganize land, water and social reproduction from below, rhizomatic governance renders decolonial, communal and feminist futures materially thinkable and buildable. The next section of this chapter traces the praxis of this socio-ecological transformation.

9.2. Socio-ecological transformation as a decolonial praxis: The emergence of a decolonial ecology

Rojava is subjected to layered colonial and neocolonial violence of ecocidal coloniality: ecocidal developmentalism, deliberate de-development, dispossession and demographic engineering, epistemic and biodiversity erasure, assimilation, armed conflict, drought and water crisis, and pollution, now intensified by climate change (Articles I to V). Confronting this convergent structural crisis requires transformation rather than adaptation and mitigation. Because these destructions are embedded in socio-spatial design, environmental engineering, and regimes of resource control, the response must itself be ecological, namely a programme of socio-ecological transformation (Articles III and V). I advance rhizomatic governance as a political ontology of socio-ecological transformation. Its operative aspects and features do not merely decentralize administration; they remake socio-ecologies and political subjects, prefiguratively shifting everyday life beyond statist, extractive and patriarchal logics.

To this end, I argue for a paradigm shift towards socio-ecological transformation, grounded in justice, plurality and decolonial politics, and present Rojava as a pluriversal formation where alternative ontologies and epistemologies are actively enacted. In Rojava, collective responses to hydro-social enclosure, the collapse of monocrop agriculture and ecological degradation have manifested in re-appropriated water infrastructures, communal water and land governance, and the politicization of ecological struggle. Because colonial drought and climate change have produced the water crisis (Article III) alongside a wider spectrum of ecological degradation, which includes the overexploitation of groundwater linked to monocrop agriculture (Articles I, III and V), the initial axes of socio-ecological transformation centre on water and agriculture without being confined to them. Article V shows that, in the face of ecocidal coloniality, ecological struggle has become an essential element of this transformation. In parallel, decolonizing education and designing an ecological curriculum through ecopedagogy have been pursued (Article IV), aligning learning processes with the political and ecological aims of

the movement. What unfolds is not resilience or adaptation, but an insurgent ecological struggle that challenges the epistemic and material foundations of colonial water and agricultural regimes through the reconstruction of democratic governance and the transformation of socio-ecological terrain.

This approach marks a radical rupture with dominant capitalist, statist and market-oriented solutions anchored in technocratic and positivist epistemologies. A striking example is the globalized praxis of climate change adaptation, mitigation and resilience. The concept of climate change adaptation has become the hegemonic mantra of contemporary climate discourse, international development discourse and environmental policy (Taylor, 2014). Defined by the Intergovernmental Panel on Climate Change (IPCC, 2014) as “the process of adjustment to actual or expected climate and its effects” (p. 1758), adaptation is presented as a rational response to climate change rather than, as Taylor (2014, p. 52) argues, “a set of relations between forms of knowledge, structures of power, institutional practices and prevailing technologies that delineate ways of thinking about and acting upon processes of social and ecological change.” Adaptation is deeply embedded in a Cartesian worldview that separates society from nature and is infused with the technocratic rationality of capitalist modernity. It promotes a mode of intervention aligned with neoliberal governance, security discourse and risk management regimes. It does not confront the systemic crises at the heart of climate change; it manages them, defers them and governs their consequences (Swyngedouw, 2010; Taylor, 2014; Benjaminsen, 2015; Moore, 2015; Watts, 2015). From the outset, adaptation reproduces the structural logics of colonialism and capitalism (Vaughn, 2022). These regimes universalize Eurocentric knowledge while marginalizing traditional epistemologies that have long sustained ecological balance (Taylor, 2014, p. 48; Watts, 2015, p. 21; Sultana, 2022a, p. 9; Hartwig et al., 2022, pp. 34–35). This epistemic violence masks a deeper political violence: it shifts attention away from structural causes such as colonial land dispossession, capitalist extraction and settler hydrology, and instead asks communities to become more ‘resilient’, more ‘adaptive’ and more governable. In doing so, adaptation depoliticizes the climate and water crisis. It fragments the systemic into the technical, reducing questions of justice to questions of capacity, and makes it nearly impossible to grasp the composition and dynamics of the socio-ecological crisis (Beck, 2010; Shove, 2010; Swyngedouw, 2010; Sultana, 2022a; Sultana, 2022b). These policies do not transform; they legitimize, pacify and manage. Adaptation, therefore, obscures structural violence by promoting individual- or community-level coping strategies rather than addressing systemic root causes (Taylor, 2014; Watts, 2015).

The widely accepted ‘holy trinity’ of adaptation, namely vulnerability, adaptive capacity and resilience, further entrenches this logic. Each of these seemingly neutral concepts conceals political decisions about who is made vulnerable, whose capacity is built and who is expected to be resilient (Taylor, 2014, pp. 53–60). Vulnerability is treated as an inherent condition rather than a historically and politically produced reality. Adaptive capacity privileges access to resources and technologies, thereby reinforcing structural inequalities. Resilience becomes an imperative to bend without breaking, a call for flexibility that displaces structural transformation with individualized survival. These terms, ostensibly value-neutral, actually reproduce existing inequalities by isolating climate impacts from their socio-economic and political contexts (Eriksen & O’Brien, 2007; St. Clair, 2010; Taylor, 2014; Hackfort & Burchardt, 2016; Vaughn, 2022). They mask the historical and ongoing roles of capitalism and colonialism in producing ecological degradation and socio-economic marginalization (Hornborg, 2001; Malm & Hornborg, 2014; Taylor, 2014; Benjaminsen, 2015; Moore, 2015; Watts, 2015; Hartwig et al., 2022; Sultana, 2022a; Sultana, 2022b). Resilience is a tool of governance, a technology for rendering populations governable through the language of flexibility and self-organization. It facilitates the colonization of political imagination by the state and converts systemic instability into an opportunity for neoliberal ordering (Taylor, 2014). Resilience policies administer life itself, echoing Foucault’s biopolitics (Foucault, 2008), but now with a distinctly neoliberal inflection that valorizes markets as adaptive and self-correcting systems. As such, adaptation, security, risk management and resilience together form the contemporary hegemony through which neoliberal rule is exercised (Lemke, 2011; Taylor, 2014; Watts, 2015, pp. 40–41).

To be sure, certain critical strands within the adaptation literature push back against technocratic interpretations, emphasizing how pre-existing social inequalities shape vulnerability to climate change (O’Brien et al., 2010; Pelling, 2011; Taylor, 2014). Yet in effect, adaptation functions as a discursive and material extension of colonial intervention, underwritten by Northern epistemologies and institutional dominance. Indigenous and local communities have historically developed deep, situated knowledge of hydrological cycles, land management and resource stewardship (Hartwig et al., 2022). Colonial and capitalist systems have violently disrupted these practices through commodification, land- and water-grabbing and dispossession, forced sedentarization, and extractive development projects (Mehta et al., 2012; Austin-Brooks, 2013; Berry & Jackson, 2018; Poelina et al., 2019; Hartwig et al., 2022). Adaptation discourse, however, is far from addressing the root causes of dispossession and

instead emphasizes coping strategies. A major shortcoming of adaptation, therefore, is its temporal myopia: it fails to account for historical responsibility of colonialism and the *longue durée* of ecological degradation. Oscillations in climate have always existed, but current crises are embedded in centuries of colonial expansion, extractive development and capitalist accumulation by dispossession (Harvey, 2003).

To respond meaningfully to these structural forms of ecological degradation, including drought and water crisis, we must move beyond adaptation and towards a framework of socio-ecological transformation. This concept refers to comprehensive, far-reaching and systemic changes in the metabolic, symbolic and institutional relations between society and nature. It recognizes that ecological questions are inherently social questions, intertwined with power, domination and inequality. Rather than offering technical fixes, it envisions a democratic, inclusive and participatory renegotiation of how societies live with and within nature (Hackfort & Burchardt, 2016; Görg et al., 2017; Danso-Dahmen & Degenhardt, 2018; Brand et al., 2020; Kister et al., 2024). Socio-ecological transformation challenges the primacy of capitalist development, fossil-fuel dependency and extractivist economies (Brand & Wissen, 2017; Malm, 2020; Wainwright, 2020). It recognizes that the multiple crises we face – ecological, economic, political – cannot be resolved within the existing system’s technocratic approaches (Brand, 2016; Görg et al., 2017; Svampa, 2019; Brand, 2022; Dorn et al., 2022). In doing so, transformation must confront how and why resources are produced, consumed and distributed under conditions of global inequality (Chen, 2010; Hackfort & Burchardt, 2016; Görg et al., 2017; Kister et al., 2024). It must also embrace alternative worldviews, ontologies and governance forms that enable plural ways of being, knowing and doing (Sousa Santos, 2016; Acosta & Brand, 2018; Tello Valle Hiriart, 2024). This is the promise of the pluriverse: a world in which many worlds can coexist (Blaser & de la Cadena, 2018; Escobar, 2018). Decolonial and Indigenous epistemologies insist on the affirmation of diverse cosmologies, modes of governance and relations to nature.

Rojava, despite its limitations and ongoing challenges, offers a compelling instance of such transformation. It represents an emergent pluriverse, attempting a radical reorganization of society through communal governance, women’s liberation, cooperative economies and ecological agriculture. Through the creation of communes, cooperatives and people’s assemblies, Rojava has sought to enact a political ontology rooted in care, collectivity, and autonomy from capitalist and statist paradigms. In this context, transformation entails

dismantling the colonial-capitalist world system that produces climate crisis, water scarcity and ecological degradation; it rejects the instrumental logic of adaptation and advances a new design rooted in justice, plurality and rhizomatic democracy. This ontological transformation is associated with holistic, multiple ontic regulations and construction, as well as deconstruction processes. In this regard, socio-ecological transformation in Rojava operates through concurrent regenerative transformations in ecological agriculture and the communal governance of land, water, energy and seeds; the social and communal eco-economy, including cooperatives, eco-factories and the women's economy; and epistemic transformation through academies, ecopedagogy and *jineoloji*.

9.2.1. Agriculture and communal governance of land, water, energy and seeds

Article I explored the reconstruction of agriculture as decolonization. Since 2012, the praxis in Rojava has been marked by a profound restructuring of the agrarian landscape, representing one of the most significant socio-ecological transformations in the region. A central element of this transformation has been the communalization of land. Approximately 500,000 hectares of land that were previously under regime control have been designated as *erdên giştî* (public land). Around 80 per cent of these lands have been allocated to communal purposes, including agricultural cooperatives and the women's economy with its associated cooperatives. Such measures not only dismantle the regime's legacy of centralized, monopolistic land management, but also establish the material basis for a system of communal governance rooted in collective ownership and ecological principles. Agriculture has been reoriented away from the monocropping model that previously dominated the region under Ba'athist colonial rule. Instead, diversification has become a cornerstone of the new policy framework. A wide variety of vegetables, pulses and agroforestry crops are now cultivated. Complementing this diversification, several hundred thousand fruit-bearing trees have been planted in recent years. Such diversification strategies not only strengthen food autonomy but also contribute to ecological resilience, by regenerating soils and reducing vulnerability to drought and crop failure. The Seed Institution of AANES plays a pivotal role in this ecological regeneration. Its activities focus on developing and distributing drought-resistant and locally adapted seed varieties.

In parallel, substantial resources have been invested in water-efficient irrigation systems. Particularly notable is the introduction of rainwater harvesting and drip irrigation, which marks a sharp departure from the large-scale hydraulic infrastructure and state-driven water politics of Syria and Türkiye. Together, these transformative practices provide the basis for localized food systems that break with dependency on external markets and state-controlled supply chains. Thus, agroecological practices are not supplementary but integral to the economic and ecological reorganization of the region. Diversified cropping systems, crop rotation, rainwater harvesting, agroforestry and the propagation of drought-resistant seeds form the backbone of this new approach. Such practices are simultaneously ecological and political, undermining the monocultural strategies that were imposed to de-develop Rojava as a periphery and to degrade its socio-ecologies as discussed in Article I. Thus, ecological farming practices are deeply intertwined with economic autonomy, soil regeneration and the political ethos of self-governance. Taken together, these developments constitute more than just technical adjustments to agriculture. They embody a radical reconfiguration of the relationship between society, land and ecology. The shift from monocropping and centralized state control to diversified agroecological practices and communal governance of resources represents a profound departure from the logics of domination and extraction. This praxis not only addresses the immediate material challenges of drought, soil degradation and food insecurity, but also exemplifies how ecological agriculture can become the foundation for broader socio-ecological transformation as decolonization.

9.2.2. The social and communal eco-economy (cooperatives, eco-factories, women's economy)

Articles I and V have comprehensively shown that ecocidal coloniality in Rojava and Bakur Kurdistan were based on de-development, dispossession and ecocidal developmentalism. The social and communal eco-economy in North and East Syria emerged as a deliberate counter to the Ba'ath regime's peripheralization of Rojava and the dispossession produced by the Arab Belt policy. The latter displaced and dispossessed Kurdish communities, pushing them to the economic and cultural margins, and eroding customary agricultural practices, local seed cultures and traditional knowledge. Under monocrop directives, many became a reserve of low-wage labour for industrial agriculture.

Against this background, the current transformation is organized as decolonization in practice: construction of democratic, ecological forms of production embedded in communal institutions. AANES has advanced local self-provisioning through the development of roughly 500–600 cooperatives, which link democratic management with ecological sustainability and generate a material base for collective production and economic self-reliance. This cooperative surge is not limited to cultivation; it reorganizes entire circuits of production, circulation and consumption. Ecological regeneration and agroeconomic reconstruction are reinforced through crop rotation, organic composting, recycling and reuse, and the localization of processing and supply chains. The cooperative sphere functions as both an economic node and a governance architecture. Decisions over inputs, pricing for the public and reinvestment priorities are discussed within assemblies that articulate social benefit and use value, rather than market profit, as guiding principles. In this sense, the eco-economy is simultaneously an economic strategy and a political pedagogy oriented to collective welfare.

A core legacy of colonial de-development was the strategic absence of processing capacity in Kurdish cities, an engineered dependency that funnelled raw produce outward while value was captured in the centre, as discussed in Articles I and V. Decolonizing this structure underpins an explicit eco-industrial strategy: build processing capacity close to production, shorten commodity chains and retain value locally. Over the past decade, this approach has yielded concrete infrastructures. AANES has established a wide range of processing factories approaching self-sufficiency in several staples. Together, these eco-factories anchor a provisioning model aimed at local markets, stabilizing prices and buffering communities from external shocks.

They also materialize the social and communal economy's orientation to use value and social benefit and its commitment to democratic and ecological principles, including the dismantling of hierarchical domination along the lines of gender, class and nature. The women's economy is integral rather than ancillary to this reorganization. Women-led cooperatives and production units are both sites of material provisioning and spaces where decision-making, skill formation and income are collectively redistributed. In practice, this means that questions of product mix, pricing and reinvestment are not treated as technical matters, but as social choices negotiated through assemblies in which women's bodies, labour and knowledge are positioned as central to economic life (Aslan, 2023). Taken together, the cooperative system, the build-out of eco-factories and the institutional role of the women's economy recompose social economy around

collective needs, ecological regeneration and democratic governance as one of the main pillars of socio-ecological transformation.

9.2.3. Epistemic transformation (academies, ecopedagogy, *jineoloji*)

Articles IV and V have examined the epistemicide in ecocidal coloniality and the development of liberatory epistemologies in Kurdistan. The epistemic transformation underway in Rojava is anchored in a return to territory-based knowledge systems, while building new institutions that can sustain a pluralistic, democratic and ecological society. At its core is the revitalization of practices long suppressed by colonial and patriarchal orders, including traditional seeds, farming techniques and communal governance repertoires that historically sustained the essential socio-economic foundations of the Kurds. In this horizon, *jineoloji* assumes a constitutive role by recovering, reconstructing and producing knowledge that had been erased or marginalized, and by reframing knowing as a relational, life-centred practice (Piccardi, 2021; Dirik, 2022). From this vantage, *jineoloji* seeks to re-balance relations between society and nature, women and men, and spiritual and material worlds. The project is thus at once ecological and decolonial: a refusal of temporal erasure (Mbembe, 2015) and an insistence that myth, oral histories and cultural narrative are part of the design grammar for world-making in turbulent times (Escobar, 2018, p. 15).

Running parallel to these autonomous spaces, AANES has begun to green pedagogy across the education system. Through *jineoloji* academies, Democratic Society Academies and ecology academies, an ecopedagogy is taking shape that does not only teach about ecological crisis, but weaves ecological thinking into governance, municipal practice, health and daily life, confronting statist and patriarchal mentalities embedded in prior schooling. This epistemic and pedagogical turn is inseparable from the ontological redesign associated with rhizomatic governance and the decolonial project of democratic autonomy. This epistemic shift is also an educational reconstruction. The move away from the Ba'athist paradigm, which was marked by patriarchy, centralism, nationalism, sectarianism, and the erasure of Kurdish and other local cultures, towards a curriculum that affirms ecological principles, democratic governance and ethnoreligious plurality, is visible across levels. Schools now prioritize mother-tongue education (Kurdish, Arabic, Syriac) with mandatory second, and later third languages, to support coexistence; administrators emphasize that linguistic learning is an instrument of social

peace (Stefani & Ruge, 2019). Content on ecology appears in core texts, covering first and second nature, land and water stewardship, planting, pollution, climate change, and the ethics of care. In classrooms, ecopedagogy is deliberately practical: pupils form ecology committees, clean and maintain shared spaces, separate waste, and plant trees in schoolyards and nearby streets, learning the sociality of care alongside technical skills, as discussed comprehensively in Article IV.

The higher education landscape remains small but symbolically significant, as these institutions articulate ecology as an ethical pillar and frame ecological literacy as part of democratic education. Beyond formal schooling, adult academies constitute the crucible of epistemic change. Their pedagogy is dialogic and collective: seminars are coupled with shared reproductive labour, enacting non-hierarchical social relations in the learning space. The academies advocate a holistic epistemology in which ecology frames inquiry, ethics and method. In this sense, education is not an adjunct to ecological struggle; it is the struggle's medium: turning homes into academies, classrooms into commons, and knowledge into a practice of collective world-making. Such a transformation is a radical departure from colonial epistemologies.

The preceding discussion has demonstrated that the socio-ecological transformation unfolding in Rojava is not a rhetorical claim, but a lived practice. Through ecological agriculture and communal governance of resources, the material infrastructures of ecocidal coloniality are being dismantled. By establishing a social and communal eco-economy through cooperatives, women's initiatives and eco-factories, the economic circuits of dependency and de-development are being repaired and redirected towards collective sufficiency. Finally, through epistemic transformation, the epistemic erasure of the colonial structure is being critically deconstructed and resituated within plural, relational ontologies, and traditional and liberatory knowledge systems.

These practices reveal a coherent and multilayered project of socio-ecological transformation. They confront the metabolic rifts created by ecocidal colonial structure, redistribute stewardship through commoning, prioritize use value and social benefit over profit, and reclaim suppressed epistemologies that affirm women's knowledge and ecological interdependence. What emerges is not simply an ecological policy agenda but the outlines of a decolonial ecology: a praxis in which ecology becomes inseparable from autonomy, decolonization and liberation. It is on this

basis that the ecological struggle has assumed a central place within the KFM. No longer a secondary concern, it has become an ethos of decolonization, interwoven with women's liberation and democratic principles as constitutive pillars of a new political ontology, rhizomatic governance, as discussed in Articles III and V. The following section develops this argument further by examining how the socio-ecological transformation in Rojava crystallizes into the emergence of a decolonial ecology, and why ecological struggle itself must be understood as the ethos of decolonization in Rojava and Kurdistan.

9.3. The emergence of a decolonial ecology: Ecological struggle as ethos of decolonization

The present structural crisis of capitalist modernity is not accidental, but the cumulative outcome of colonialism, capitalism, racism and patriarchy. Their entwined legacies generate the interlocking crises of our time: climate breakdown, ecological degradation, biodiversity loss, entrenched inequality, and cultural erasure through policies with genocidal and ecocidal effects. As Malcolm Ferdinand (2022, p. 115) argues, these are not parallel tracks, but expressions of a colonial ecology, an ecology arranged to secure colonial inhabitation and the domination of human and non-human life. What appears in environmental discourse as a purely 'ecological' problem is inseparable from the colonial fracture, that is, the appropriation of land, water and labour through colonial violence. When this history is bracketed, mainstream environmentalism reproduces a second rupture, an environmental fracture, treating devastation in abstraction from its colonial and capitalist roots through the dualism between nature and society (ibid., 175). Within this frame, ecocidal coloniality reveals inherent logic of ecocide and colonial domination: ecocidal developmentalism, socio-spatial engineering, and legal and administrative control that convert territories into extractive peripheries and weaponize environments against insurgent populations. Dams and diversion schemes, forced monocultures, mining and deforestation, enclosure of commons, demographic engineering, and infrastructural reordering are not neutral projects of development, but techniques of colonial elimination and exploitation. Their rationality is proprietorial and militarized, their epistemology erases local ecologies, and their effect is the dismantling of livelihoods, habitats, and the cultural and political conditions of collective life.

By contrast, decolonial ecology is both diagnosis and praxis. It is, in Ferdinand's terms, an "ecology of struggle", inseparable from emancipation, justice and equity (Ferdinand, 2022, p. 175). Ontologically, it rejects the one-world universalism of capitalist modernity and resonates with the pluriverse, a world where many worlds fit, in which heterogeneous ways of knowing, being and doing coexist without subordination (Escobar, 2018, p. xvi). For Escobar, relational ontologies and designs for the pluriverse become tools for reconstructing local worlds otherwise (ibid., pp. 4, 230).

In this sense, the socio-ecological transformation underway in Rojava/North and East Syria is a concrete instantiation of decolonial ecology. The Rojava Revolution dismantles dimensions of ecocidal coloniality and composes alternatives across three interlinked domains: ecological agriculture and communal governance, the communal eco-economy, and epistemic transformation. Communes deliberate and manage land and water; cooperatives enact social provisioning and food sovereignty; ecological councils and academies embed ecological principles into training, regulation and everyday practice; municipalities materialize decisions in infrastructure and services. Together, these nodes build a non-extractive, relational, pluriversal arrangement oriented to democratic autonomy, operating through the assemblages and nodes of rhizomatic governance. This orientation is not novel or ex nihilo; it resonates with long-standing land-based epistemic and ontological formations. The socio-political fabric of Kurdish society, especially tribe (*aşîret*) confederations and autonomous tribal structures, braids economy, culture and spirituality around land (van Bruinessen, 1992; McDowall, 2004; Jongerden, 2007). Peasant, pastoralist and nomadic livelihoods have been anchored in sacralized relations with nature; belief systems centre land and the more-than-human world in their cosmologies (Gültekin 2021, p. 227). My fieldwork in Rojava confirms this continuity: interlocutors repeatedly tied land to identity and memory. As one Yazidi interviewee put it, "The land is not just soil, it is the memory of our people, our sacred mother, and our dignity."¹⁷⁰ Survey responses echoed this: participants described land as sacred, memory, homeland, mother, dignity and bread. These land-based political, economic and cultural structures have generated their own epistemology and relational ontology, which constitute the historically essential foundations of *Kurdayetî*¹⁷¹ (Article V).

¹⁷⁰ Nesim Shemo, Yazidi community representative and lecturer at University of Rojava, 17 May 2024.

¹⁷¹ *Kurdayetî* represents what Hobsbawm refers to as "proto-national bonds", or Smith's concept of "ethnie". Therefore, *Kurdayetî* should not be considered synonymous with Kurdish nationalism, and should not be used interchangeably with this term or concept (Tunc, 2018).

It is therefore not a coincidence, but a deliberate colonial strategy, that the Ottoman centralization, extended by the Turkish and Syrian nation-states, has targeted the nexus of nature-related belief, traditional knowledge systems, and land, water and agriculture-based socio-economies together with their political and cultural forms. Policies of ecocidal coloniality (dams, mining, deforestation, forced displacement) have been instruments to sever Kurds from their land and to destroy the land-based cultural and ecological fabric of *Kurdeyatî*. As Öcalan has put it, this has amounted to a policy of “de-Kurdification” (Bilgen, 2017, p. 7). When colonial rule attacks ecology to eliminate a people’s material and symbolic grounds of existence, anti-colonial struggle must be ecological. This is why the KFM treats ecology as an ethos of decolonization. Ecology is not a subsidiary policy lane but, alongside women’s liberation and democratic autonomy, a central pillar of the paradigm. Even if ecological practice initially lagged behind the other two pillars, internal critique and institutional innovation have pushed it to the forefront of agriculture, economy and education. Across Bakur and Rojava, ecological resistance directly counters state-driven ecocidal coloniality: displacements induced by dam systems, the imposition of monocultures, deforestation, resource extraction and the administrative enclosures that follow. Resistance thus confronts not only degradation, but the epistemic and material architectures of internal colonialism, reclaiming traditional knowledge, relational ontologies, communal governance and alternative economies as the foundations of autonomy. As the Union of Kurdistan Communities (KCK) Ecology Committee insists, “for the Kurdish people, the ecological struggle is also a struggle for the defence and survival of their country, land, history and all national values.”¹⁷² Thus, decolonial ecology names the politics of life that counters ecocidal coloniality.

Situating this in the broader historical arc clarifies the stakes. A colonial continuum that includes de-development, ecocide and cultural genocide has structured Kurdish subordination. In the present, that continuum takes the form of ecocidal coloniality, displacing and dispossessing communities while degrading ecosystems. Against this, the KFM’s ecological struggle offers not only mitigation but counter-design, a deliberate reassertion of Kurdish traditional knowledge, governance forms and economic systems. Through ecological councils, communal resource ownership, eco-economy-based agricultural cooperatives and ecopedagogies, it advances a decolonial ethos aimed at dismantling the epistemic and material architectures of colonial and capitalist power. In emphasizing rhizomatic governance,

¹⁷² KCK Ecology Committee, briefing report, January 2024.

democratic autonomy and women's liberation, the movement refuses commodification and the homogenizing violence of nation-state colonialism, and recomposes relations between society and nature on terms compatible with autonomy and care. Rojava's praxis shows how ecological struggle can recompose ecological relations and secure autonomy from below as an ethos of decolonization. This is the central argument of the thesis, which is why Article V is devoted entirely to this issue. It is a strategy of life-making that binds nature and community in a single horizon; a political ontology (rhizomatic governance) that treats decolonization as simultaneously political and ecological; and a practical design that translates that horizon into institutions, infrastructures and subjectivities capable of sustaining life otherwise through socio-ecological transformation.

For the Kurdish movement, decolonization is therefore not only a political imperative, but also an ecological one. The next section examines the challenges, deadlocks and limits of this transformation in Rojava, and how these challenges condition the horizons of decolonial ecology in practice. In this regard, the Rojava experience emphasizes several key insights. First, colonial power in Kurdistan is ecocidal in its essence: it is a project that targets ecological systems as a means of dismantling social, cultural and political foundations. Second, ecological struggle is not only about the environment, but also concerns survival, autonomy and identity: it is an essential element of political ontology. Third, the KFM's ecological paradigm represents a significant contribution to global decolonial thought and praxis, integrating eco-economy, gender liberation, and rhizomatic democracy and governance into a coherent ethos of democratic autonomy. Finally, the Rojava case offers a comparative vantage point to think across struggles, showing how decolonial ecology can serve as a framework for resisting colonial ecocide globally.

9.4. Challenges and limits of socio-ecological transformation

Rojava's socio-ecological transformation and decolonization operate under continuous coercion and neocolonial attacks and invasions. Turkish airstrikes and proxy militias repeatedly target energy plants, water installations and productive sites, generating rolling blackouts, pumping failures and factory shutdowns. Local actors describe this as a deliberate necropolitical strategy intended to destabilize communities and force displacement. In late 2024, after 13 years of civil war, HTS, led by Ahmed al-Shara'a, toppled the Assad regime. This new Syrian regime

now controls nearly all Syrian territory, outside North and East Syria which remains Kurdish-controlled, and Suwayda Governorate led by its Druze population. HTS rule has already included massacres targeting the Alawi and Druze communities, and created acute risks for Kurds and other minorities, including Yazidis, Assyrians and Armenians.

It is thus understandable, in budgetary terms, that approximately 60 per cent of recent AANES expenditure has been consumed by defence, leaving little fiscal space for long-horizon ecological investment or basic capital replacement.¹⁷³ The external embargo compounds these constraints by blocking spare parts, capital equipment and trade, thereby collapsing margins in agriculture and the energy sector (Articles I, III and V). As Hesên Koçer puts it: “Our potential is continuously undermined by external threats; survival has often overshadowed ecological projects and transformation.”¹⁷⁴ Eli also underscored the triage logic of governance: “We are fighting for our existence. We are trying to defend our people and meet their needs under the war and embargo. The current level of realization and practicality is not even a quarter of our potential and capacity.”¹⁷⁵ This situation has forced the AANES into a war economy built on oil dependency. Because a colonial relation between centre and periphery left the region without modern refining capacity, crude oil is processed with rudimentary methods that harm both people and ecosystems. The result is a triple risk: economic fragility, serious public health impacts, and extensive land and water contamination. These security dynamics and fragile infrastructures intersect with the weaponization of water by Türkiye. Upstream restrictions and attacks on installations have converted a long drought into what Article III conceptualizes as colonial drought, with cascading effects across farming, municipal supply, public health and electricity generation.

A mixed-methods field survey with 50 villagers and farmers illuminates how these pressures are perceived and navigated (see Appendix A for sampling, districts, languages and consent procedures). Among participants engaged in communes, cooperatives and academies ($n = 40$), 70 per cent attributed the water crisis to climate change and upstream restrictions by Türkiye, 20 per cent framed it in religious terms as divine punishment, and 10 per cent were uncertain. Views of ownership and governance also reflected the political education embedded in communal structures: to the question “Who owns water and land?”, 60 per cent answered “the

¹⁷³ Eli, consultant at Ecology Committee and Ecology Academy, 30 April 2024, 12 May 2025 (online).

¹⁷⁴ Hesên Koçer, co-chair of Executive Council of AANES, 29 March 2024.

¹⁷⁵ Eli, consultant at Ecology Committee and Ecology Academy, 30 April 2024, 12 May 2025 (online).

people”, 30 per cent “God” and 10 per cent “the state”. On management and distribution, 40 per cent endorsed collective ownership, 55 per cent emphasized local producers and only 5 per cent preferred state control. The control group of 10, characterized by minimal engagement and frequent ideological opposition to the AANES, displayed markedly different interpretations: eight described the drought as divine punishment, one cited climate change and one was unsure; on ownership, eight chose “God”, one “current landowners” and one “the state”; on governance, two preferred state control, five reluctantly supported the AANES and three abstained. Despite these divergences, there was consensus on severity: 80 per cent of the whole sample cited water shortage as the most important problem, and respondents across cohorts reported precipitous groundwater declines over the last 30–40 years that now require deeper and more frequent drilling (Article III).

Proposed solutions among engaged participants reveal an incremental but significant shift towards ecologically informed strategies: water-efficient irrigation, drought-resistant crops such as cumin, expansion of fruit trees, especially almonds and olives, and diplomatic initiatives to press for restoration of transboundary flows. Participants trained in the academies, as well as those active in communes and cooperatives, have developed a robust ecological consciousness and are driving socio-ecological transformation. However, AANES still faces major challenges in extending this work to broader segments of society. At the same time, continued reliance on deep well drilling persists as an emergency coping strategy and a marker of constrained capacity. Dissatisfaction with practical support was widely expressed, particularly regarding diesel, irrigation infrastructure, fertilizers and pesticides. A cross-cutting critique across interviews and group discussions concerned the shortage of agricultural expertise among leading cadres, and hence the urgent need for trained agronomists, irrigation engineers, cooperative accountants and technicians.

War, embargo and water scarcity have also reshaped demography. Emigration drivers cited by 65 per cent of survey respondents included worsening economic conditions from the Syrian war, Turkish interventions, embargoes imposed by Türkiye, and drought. Outflows spiked after the occupations of Afrin and Serêkaniyê. Among Kurds, Assyrians and Yazidis, nearly every household reported at least one emigrant; among Arabs, one in ten households did so (Article I). Juridical lack of recognition further limits access to multilateral funds, spare parts and credit lines, locking the economy into cash transactions and informal procurement channels. The loss

of skilled workers amplifies capacity gaps in agriculture, manufacturing, education and health, and weakens the very institutions needed to sustain eco-economy initiatives.

Land justice remains an unresolved fault line. In response to the Arab Belt, AANES has avoided large-scale reallocation, to prevent ethnic polarization and in line with the Democratic Nation principle of multi-identity governance. Policy has focused instead on redistributing state-owned land to those in need. Three-quarters of Kurdish survey respondents strongly criticized this stance, often describing the old Arab Belt policy as de-Kurdification that should be reversed; one fifth judged it a difficult political problem without easy resolution; 5 per cent did not answer. All agreed that those dispossessed should be compensated. Arabs settled under the Arab Belt generally acknowledged the policy's injustice, but argued that responsibility lay with the Ba'ath regime rather than with them, and that they should remain on the land.

In practice, communalizing land and scaling cooperative agriculture have encountered both structural and cultural obstacles. Cadres often attribute slow expansion to statist and individualist habits, whereas many peasants emphasize freeriding, corruption and nepotism, and risk aversion. When asked whether communal use through cooperatives or individual distribution would be preferable, nearly 70 per cent of respondents chose individual distribution, while 24 per cent supported a well-established cooperative system. Under conditions of war and embargo, such risk aversion is rational, which helps to explain preferences for individual distribution even among those sympathetic to collective aims. This mirrors long-standing debates in agrarian thought on the relation between peasant production and socialization, from Leninist nationalization and cooperative cultivation to Alexander Chayanov's emphasis on emancipation through democratically grounded forms that need not dissolve peasant production (Ploeg, 2013; Articles I and V).

Education faces its own layered constraints. Schooling across North and East Syria has been repeatedly interrupted by occupation, blackouts, and the repurposing of buildings as shelters, depots or detention sites. Turkish occupation in Afrin, and the 2019 incursions around Serêkaniyê and Girê Spî, shuttered progressive initiatives and replaced them with Turkish state doctrine and Islamic sharia-based curricula, while Kurdish language instruction was suppressed in Kurdish-majority Afrin (Syrians for Truth and Justice, 2023). The AANES curriculum grounded in democratic confederalism was introduced in 2015 and, after initial resistance, has been applied across the region in 2023/4, including in Arab-majority areas, sometimes with

adjustments for local sensitivities, for example the temporary exclusion of *jineoloji* in Deir ez-Zor, while adopting co-chair system at all administrative levels and banning corporal punishment (Article IV). Chronic shortages of qualified teachers, lack of accreditation, and child labour driven by poverty, further constrain outcomes. Nonetheless, the education system functions under democratic governance, and has begun to institutionalize ecological and feminist content within severe material limits.

These external and internal pressures do not negate the achievements documented elsewhere in the thesis. They explain their unevenness and fragility. Rhizomatic governance, eco-economy, and ecopedagogy have advanced in the teeth of infrastructural attacks, liquidity shortages and skills deficits. The core risk remains escalation. Renewed offensives by Türkiye or expanded proxy operations would be devastating for human security, ecosystems and institutional continuity. A realistic assessment must therefore hold two truths together. First, Rojava has generated significant socio-ecological transformation that materializes a decolonial ethos in practice. Second, the same project is structurally constrained by war, embargo, the weaponization of water, demographic pressures, land conflicts, capacity gaps and juridical lack of recognition, all of which fragment coordination and continually threaten reversal. The analysis that follows in the conclusion of this thesis reads accomplishments and setbacks together, treating transformation not as a linear arc but as a contested process. It is an ongoing revolutionary struggle, in which construction and deconstruction are interwoven.

SECTION IV. Conclusion

Chapter 10. Conclusion

Colonialism, together with its mindsets and material designs globalized through capitalist modernity, has produced perhaps the most dramatic overturning in the history of civilization. There is scarcely a single region of the globe that has not either been a colonial power or been occupied by one in modern history. Peoples, nature and ecosystems, especially in the Global South, have been commodified and exploited; everything judged incompatible with European rational, scientific, philosophical and cultural norms was categorically rejected or destroyed at the epistemic and ontological levels. The power of what was called ‘progress’ resided in the manufacture of mentality: it persuaded peoples, in the name of civilization, to believe in the mythologies of ‘progress’ and ‘development’. This mental conquest was of extraordinary scale, comparable to the self-deification of pharaohs and kings in ancient empires and the monumental symbolism of the pyramids. Epistemic violence fused with unbounded ecocidal and genocidal violence, and its outcomes were catastrophic: the annihilation of languages, cultures, beliefs and knowledge systems, the exhaustion of nature, and a present marked by climate crisis, wars, massacres, hunger and poverty.

Kurdistan functioned as a laboratory in which nearly every modality of this violence was applied. From the late Ottoman period through the post-First World War partitions designed by France and Britain, its Indigenous peoples were denied, its essential foundations were targeted, its country was fragmented, and it was left to the genocide and ecocide policies of four nation-states founded on colonial codes. Against this attempted erasure, Kurds endured terrible massacres and destruction for more than a century and, despite temporary interruptions, have continued to resist. Rojava was born as a pluriverse in the very midst of this storm of colonial devastation.

In this regard, for Rojava and partly for Bakur, this thesis has investigated how colonialism persists as a structuring modality in the contemporary world, no longer through overseas empires, but through the nation-state form itself. It has argued that nation-state colonialism is

intrinsically ecocidal, producing environmental degradation, displacement and dependency as deliberate techniques of governance. Far from being developmental anomalies or policy failures, these are structural outcomes of a colonial logic internalized within state institutions, infrastructures and political economy. The thesis therefore conceptualizes ecocidal coloniality as a framework to understand how new colonial modalities and structural ecological destruction remain co-constitutive under nation-state colonialism. This thesis has also analysed the decolonization process and its dynamics that correspond to Rojava's emergence as a pluriverse (Article V). The five articles together pursue this inquiry at the ontological and epistemic levels, investigating these two intertwined trajectories. In what follows, I make explicit how each research question is answered, and where in the thesis those answers are developed.

The first line of inquiry examines the colonial structure, its mechanisms and its effects. The colonial status of Kurdistan is one of the most systematically denied realities of the modern world. Partitioned among four states, Kurdish regions have been subjected to a layered process of internal colonialism in a specific form: nation-state colonialism. Homogenizing national projects sought to dissolve Kurdish geographies, cultures, socio-ecologies and political formations into statist identities. Understanding, ecocidal coloniality in Kurdistan therefore requires holding together its socio-spatial, ecological and epistemic dimensions.

As Article V shows, Ottoman centralization set in motion patterns of demographic engineering and dispossession later refined by the Turkish state and Ba'athist Syria. The goal was not only cultural assimilation but also the dissolution of Kurdistan as an economic and ecological entity, transforming it into a dependent periphery while eliminating its autonomous bases. This logic has rested on a centre-periphery relation which, in Kurdistan, is based on monocrop agriculture and hydraulic megaprojects, key vehicles of this transformation. As conceptualized in Article V as ecocidal developmentalism, 'development' has operated as a colonial weapon that liquidated the ecological and economic foundations of the Kurdish people, while legitimizing dispossession under the banner of modernization and progress. Article I extends this argument by demonstrating that Ba'athist agricultural modernization was a vehicle of internal colonization rather than a neutral development programme. Through massive irrigation projects, the monocropping of 'strategic crops' and the Arab Belt policy, the Syrian state reconfigured Rojava's socio-ecological landscape to assert control, displace Kurdish populations and embed the region in extractive dependency. The consequent ecological degradation such as soil exhaustion, groundwater depletion and loss of agrobiodiversity was

not accidental. This directly answers Question 1 (Articles I, III and V) by showing how Ba'athist agrarian policy and Turkish hydraulic governance have restructured land and water to dismantle ecological, economic and cultural integrity.

Article III adds the hydrological dimension. Agrarian modernization has rested on overexploitation of surface and groundwater. Conceived as national modernization, the dam at the same time recolonized Rojava's socio-ecologies: diverse farming systems were transformed into monocultures dependent on unsustainable pumping, aquifers were depleted, and soils salinized. Far from an innocent technocratic project, the hydraulic state made populations legible and governable, and produced ecological crisis as an intrinsic effect of domination. This is the backdrop for what is conceptualized as colonial drought. Drought is typically cast as climatic, yet in Rojava scarcity has been systematically engineered through colonial agricultural and hydrological policies. By turning the Cezîrê/Jazira into Syria's breadbasket for 'strategic crops', the Ba'ath regime used water as a tool of extraction. Colonial drought emerges at the intersection of internal colonization and regional hydro-hegemony. Similarly, Türkiye's control of the Euphrates and Tigris through the GAP system captured the great majority of Euphrates flow and enabled Ankara to regulate the lifeblood of Syria and Iraq. For Rojava, already restructured as an extractive zone, this compounded vulnerability: reducing flows, destabilizing ecosystems and weaponizing scarcity against Kurdish autonomy. In short, drought here is not merely the absence of water, but the presence of colonial power. This develops Question 1a (Article III) by explaining how engineered hydro-social systems aggravate drought and agricultural collapse, which climate variability and conflict then intensify.

As a result, ecocide functioned as counterinsurgency: by destroying the ecological bases of Kurdish survival and autonomy, the state pursued elimination through environmental means. Article II offers a stark illustration in Afrin in this regard. The 2018 Turkish invasion imposed military occupation and ethnic cleansing, while unleashing systematic destruction of olive trees and appropriation of olive groves. These acts were not collateral damage; they exemplified how ecocide and genocide intertwine within nation-state colonialism. Ecological destruction became a method of cultural and physical elimination, severing communities from land-based social fabric, knowledge, livelihood and identity. Ecocide doubles as epistemicide. When groves, seeds and rivers are destroyed or commandeered, the embodied knowledge they carry is silenced. Ecocidal coloniality in Kurdistan has therefore unfolded through socio-spatial design and resource control: the deliberate reorganization of land, water and populations to entrench

domination. Dams, irrigation canals, monocrop fields and cadastral regimes have functioned as technologies of colonial rule. They rendered Kurdish lands extractable for the benefit of the centre while impoverishing communities and stripping them of territorial autonomy. This development was a deliberate strategy to tether Kurdistan to the centre as a subordinate periphery, and to weaken the socio-economic bases of Kurdish political and cultural life. However, colonial subjugation and elimination is not only material, but also epistemic and cognitive. These findings set up Question 2 (Articles I and V) by showing why struggles over land and water are also struggles over power and knowledge.

Article IV brings the epistemic and affective consequences into view. Denial of citizenship, dispossession of land and imposition of monocultures produced a psychological severance from place. Systematic denialism and erasure of Kurdish identity, language, culture and epistemologies were the main strategy to consolidate the colonial rule. Rural livelihoods were disparaged as ‘backward’, and Kurdish ways of knowing and relating to land were disqualified while extractivist rationalities were normalized. Colonial infrastructures therefore reconfigured not only economies and ecologies, but also subjectivities and minds. Here the thesis addresses Question 4 (Articles I, III, IV and V) by tracing an ontological and epistemological rupture required to undo statist, capitalist and patriarchal designs.

Read together, the five articles show how dimensions of ecocidal coloniality operate through mutually reinforcing practices of demographic engineering, socio-spatial design, infrastructural domination, ecological destruction and epistemic erasure. Therefore, its instruments vary across time and borders, yet their effects converge: agrarian modernization that reorganizes land use for the benefit of the centre; hydraulic infrastructures that re-code rivers and aquifers into levers of control; demographic engineering that severs people from place; and pedagogies that erode languages, memories and ways of knowing. The effect was not only dispossession, but a re-arrangement of social and ecological life.

Ecocidal coloniality is thus not a defective development model that occasionally goes wrong. It is a structure that secures political consolidation by engineering ecological degradation, economic dependency and epistemic erasure. Ecocidal developmentalism is its operational mode. Colonial drought is its hydrological expression. The genocide-ecocide nexus is its method of counterinsurgency and cultural elimination. Together, they attack the essential foundations of Kurdish existence: the ability to reproduce life on the land, sustain hydro-social

relations, transmit language and knowledge, and practice collective autonomy. This consolidation makes explicit the thesis contribution to debates on colonialism today: it identifies nation-state colonialism as a living structure whose operational logic is intrinsically ecocidal, thereby clarifying why contemporary coloniality must be theorized as ecocidal coloniality.

If the colonial present dismantles the essential foundations of Kurdish people through ecocidal coloniality, then anti-colonial praxis must be ecological in means and ontological in horizon, which is the second line of the thesis. Since colonialism is a structure and its functions are based on ontological level design, decolonization is inherently an ontological issue (Article V). To this end, Rojava rejects colonial nation-state ontology, and instead constructs democratic autonomy as an ontological transformation. It is embodied in rhizomatic democracy and rhizomatic governance. In this regard, rhizomatic governance within transversal resistance in Rojava is not a metaphor laid over familiar institutions; it is an operative political ontology that reorganizes power, knowledge and material life away from the nation-state form. Its distinctiveness lies in how it couples horizontal proliferation with confederal coordination, allowing decision-making to remain anchored in communes while scaling up through mandated, recallable nodes that do not accumulate sovereign authority. This arrangement has proved capable of recomposition under attack, as seen in the decentralization of energy and water systems as a rhizomatic resistance. The capacity to withdraw, reassemble and continue governing from multiple points is not an emergency workaround; it is an immanent property of this resistance. In that sense, the revolution's endurance is inseparable from the rhizome's ability to generate new lines of life when older lines are cut (Articles I and III).

At the same time, a rhizome is not inherently emancipatory. Rojava's experience confirms Deleuze and Guattari's caution that rhizomes can be captured and become arborescent. The very features that enable distributed resilience can harden into centralizing routines, especially in domains like military coordination, large-scale procurement or crisis logistics. The rhizomatic resistance repertoire developed in Rojava is therefore not ornamental. Rotation and recallability, co-chair systems, autonomous women's nodes, academies that institutionalize critique and self-critique, and the embedding of ecological and feminist criteria in everyday administration function as continual checks on vertical drift. Here the density of circulation between communes, cooperatives, academies, municipalities and people's assemblies is decisive. Where those circuits are thick, democratic practice becomes habitual, women's

authority is normalized, and ecological rules for how land and water are used become common sense. Where the circuits thin out, capitalist firms, patronage and older hierarchies fill the space.

This suggests concrete criteria for evaluating depth rather than appearance: the share of provisioning mediated through cooperatives aligned with communes; the extent of women's autonomous decision power across nodes; the degree to which ecological constraints shape budgetary and legal decisions; the frequency with which academics cycle practitioners back into practice with revised methods; and the ability of confederal clusters to coordinate without converting into superior authorities. Rhizomatic governance gains its political force by organizing across this plurality without seeking to eliminate it, while insisting on non-negotiables such as women's liberation, collective decision rights and ecological limits. The price of that openness is permanent contest. The benefit is a form that can hold together diverse material and symbolic worlds without subsuming them under a single identity or a state. Taken together, these features mark a passage from democratic resistance to world-making. Rhizomatic governance as a political ontology does not simply defend communities from the violence of ecocidal coloniality; it composes an infrastructure of autonomy that is capable of transforming relations between society and nature from below. Its value therefore cannot be measured only by institutional diagrams or moments of military success, but by the slow construction of commons, by the production of new ethical and aesthetic norms, and by the emergence of subjects who can act together without taking power. On this basis, it turns to praxis concerning how this political ontology becomes socio-ecological transformation in agriculture, economy, education and everyday life (Article III). This section answers Question 3 (Articles I, II and III) by specifying how AANES develops organizational and administrative responses, and Question 3b (Articles I, III and V) by detailing the democratic governance mechanisms that integrate diverse groups and dismantle centralized logics.

Since Rojava confronts layered crisis and destructions of ecocidal coloniality, a politics of adaptation and mitigation cannot address their causes. What is required is transformation that is ecological in substance and scope. I therefore centre rhizomatic governance as a political ontology of socio-ecological transformation: not administrative decentralization but the remaking of socio-ecologies and political subjects in ways that displace statist, extractive and patriarchal logics and that materialize justice, plurality and decolonial politics in everyday life. Read through this lens, Rojava is a pluriversal formation in which alternative ontologies and epistemologies are enacted rather than merely asserted. Collective responses to hydro-social

enclosure, the collapse of monocrop agriculture and wider ecological degradation have taken form as re-appropriated water infrastructures, communal governance of water and land, and the politicization of ecological struggle. Because colonial drought and climate change have produced a water crisis (Article III), transformation has understandably centred on water and agriculture, while extending into economy and knowledge. What unfolds is an insurgent ecological praxis that contests the epistemic and material foundations of colonial water and agricultural regimes by reconstructing democratic governance and transforming the socio-ecological terrain.

To this end, three interlinked fields crystallize the praxis of socio-ecological transformation that has taken root. Ecological agrarian restructuring has reoriented land from monoculture and hydraulic command towards diversified, place-attuned cultivation; communal allocation and the women's economy anchor this shift, so food production becomes the material spine of autonomy rather than a lever of de-development (Article I). A social and communal eco-economy complements this transformation by reorganizing production, circulation and consumption around use value, collective need and ecological repair; cooperative networks and locally embedded processing re-code the economy away from extractive peripherality and towards democratic stewardship (Articles I and V). An epistemic transformation completes the triad by confronting epistemicide directly: *jineoloji* and ecopedagogy recover and regenerate knowledge systems erased by colonial and patriarchal rule, reframing knowing as relational, life-centred and accountable to the socio-ecological milieu (Articles IV and V). Together, these fields compose a rhizomatic governance of land, economy and knowledge in which communes, cooperatives, academies, municipalities and people's assemblies act as mutually reinforcing nodes. This directly addresses Question 2 (Articles I and V) by showing how land and water struggles are contested through autonomy-building practices, and Question 3a (Articles II, III and V) by clarifying the structural relationality between coloniality and ecocide, and why ecology becomes an ethical and aesthetic ground of the struggle.

Taken together, these fields show that Rojava's project is no mere environmental policy bundle, but the outline of a decolonial ecology. Decolonial ecology is diagnosis and praxis together. It is an ecology of struggle inseparable from emancipation, justice and equity, and it resonates with the pluriverse where heterogeneous ways of knowing, being and doing coexist without subordination (Escobar, 2018; Ferdinand, 2022). Rojava gives this horizon material shape. Communes manage land and water, cooperatives enact social provisioning and food autonomy,

ecological councils and academies embed ecological principles in regulation and everyday practice, and municipalities materialize decisions in infrastructure and services. These nodes generate a non-extractive, relational and plural arrangement oriented to democratic autonomy and operationalized through rhizomatic governance. This configuration also clarifies why ecological struggle has become the ethos of decolonization in the KFM.

In this regard, the thesis therefore advances a consolidated claim. Socio-ecological transformation in Rojava dismantles dimensions of ecocidal coloniality in agriculture, land and water, repairs dependency circuits through cooperatives, eco-factories and the women's economy, and counters epistemicide through academies, ecopedagogy and *jineoloji*. These practices confront metabolic rifts created by ecocidal coloniality, redistribute stewardship through commoning, prioritize social benefit over profit, and affirm relational ontologies that bind community and nature within a single horizon of life-making. In doing so, they render decolonial, communal and feminist futures both thinkable and buildable. This is the core argument of Article V, and the analytic pivot of the thesis: rhizomatic governance is the political ontology through which decolonial ecology emerges as praxis. Therefore, ecological struggle in Rojava is not a sectoral niche, but is the organizing ethic of decolonization, because colonial power in Kurdistan is ecocidal in its essence and targets ecological systems to dismantle social, cultural and political foundations. That is why the main lesson we can learn from Rojava is that the KFM contributes a generative framework to global decolonial praxis by integrating ecological agriculture, eco-economy, ecopedagogy, gender liberation, and rhizomatic democracy and governance into a coherent ethos of democratic autonomy. Just as Rojava learned a great deal from the Zapatistas during its formative years, its interaction, solidarity and reciprocal exchange with current anti-colonial and anti-capitalist revolutionary movements will continue and deepen. In sum, Question 4a (Articles I, III, IV and V) is answered by demonstrating how the model rejects mimicry, and reconfigures socio-ecological terrains through methodological and epistemic frameworks that are relational, feminist and ecological, thereby contributing to current debates by naming and analysing ecocidal coloniality.

Taken together, these findings reveal that the KFM's ecological struggle is not a peripheral or environmental reform project but the ethical and ontological core of decolonization. It exposes that modern statehood itself operates through ecocidal coloniality. Conceptualizing this phenomenon as ecocidal coloniality within its vectors and dimensions, the thesis contributes to contemporary debates in postcolonial studies, political ecology, genocide-ecocide studies and

decolonial thought, by identifying how ecocidal and colonial structures interwoven and have been reconfigured through the nation-state system. However, the lessons drawn from Rojava and from this research comprise rich and diverse experiences that extend beyond this theoretical contribution. Rojava's experience, rooted in its distinctive historical, social and ecological-geographical dynamics, and informed by the accumulated praxis and debates of centuries of anti-colonial struggle across geographies of resistance, offers valuable lessons. The points that follow are illustrative rather than exhaustive.

10.1. Ecocidal coloniality and its operational dimensions

Colonial invasion is a structure, not an event (Wolfe, 2006), and it materializes through ecocidal socio-spatial design. To analyse this structure, this research outlines the concept of ecocidal coloniality and its dimensions, including legal-political engineering, ecocidal developmentalism, socio-spatial-hydraulic design, militarized ecological violence, epistemic domination and biopolitical marginalization. It demonstrates the structural and intrinsic link between colonialism and ecocide, and argues that any decolonization struggle must take this into account.

- Legal-political engineering (Articles I, III and V)
 - Colonial policies were continually legitimized through legal decrees and state discourse. The Ba'ath regime's 1962 special decree, which classified Kurds as "criminal refugees" and enabled their dispossession, exemplifies this political engineering. Türkiye's recent incursions likewise criminalize Kurdish autonomy in Rojava by equating it with terrorism, and seek legitimacy under the rubrics of national security and counterterrorism. The core logic of these policies is to produce the most 'manageable' and 'disciplined' forms: standardized and assimilated individuals and societies.
 - Many colonial wounds and legacies, including land expropriation, do not have simple solutions. Their complexity requires compromise, negotiation, compensation and the honouring of victims. The Arab Belt legacy in Rojava remains an open colonial wound and a trigger of conflict dynamics. Decolonization is therefore a process, not a destination.

- Ecocidal developmentalism (Articles I, III and V)
 - For colonized peoples, ecocidal coloniality has updated classical colonial methods through the discourse and practice of ‘development’. In Kurdistan/Rojava, development policies have produced ecocidal and culturally genocidal outcomes. The colonizer’s development has functioned as a regime of de-development for local peoples. Although Bakur and Rojava are among the richest geographies in water, land and resources, they have become the poorest and most neglected regions of the states that rule them. This is not fate or an internal deficiency; it is the result of colonial design and violence, closely paralleling patterns in parts of Africa and Latin America. Methods vary locally; the consequences are similar.
 - The systematic ethnic cleansing and demographic engineering carried out against the Kurds, particularly in the Arab Belt and Afrin, together with property dispossession, the channelling of economic resources to Arab communities perceived as loyal to the state, and the systematic impoverishment of the Kurds, constitute structural articulations of ecocidal coloniality.
- Socio-spatial and hydraulic design (Articles I, III and V)
 - Drought is not only meteorological, hydrological, agricultural or socio-economic – it is also colonial. In Rojava, colonial drought is a specific, multilayered form produced by internal or nation-state colonialism through monoculture agriculture, dam building and hydro-hegemony over transboundary waters by the upstream state, acting together or separately.
 - Kurdistan was reoccupied through dams, irrigation infrastructures, monoculture agriculture and accompanying security-bureaucratic architectures; land and water were appropriated, and traditional socio-economic elements dismantled. Thus, ecocidal coloniality functions by socio-spatial design.
- Militarized ecological violence (Articles II and V)
 - Colonial socio-spatial designs centred on land, water and forests are ecocidal. Consequently, in Kurdistan/Rojava, ecocide has become a specific technique of cultural genocide and a tool of counterinsurgency. Afrin is the most dramatic example. Hence, the operational reality of colonialism in Kurdistan is a genocide-ecocide nexus.

- Epistemic domination (Articles I, IV and V)
 - Existing colonial terminology and conceptual frames are insufficient to name the violence imposed on colonized peoples. As seen in Kurdistan's Rojava and Bakur, internal colonialism, in its specific variant as nation-state colonialism, can threaten local peoples and ecosystems through ecocidal and genocidal policies in ways that remain obscured.
 - One way to starve and break a land-based people, and thereby assimilate them, is to destroy the knowledge systems that make their land-based socio-economy work. Colonial states in Kurdistan/Rojava have severed people from land, cut new generations off from older ones, and removed women from their roles as holders and transmitters of knowledge. Epistemicide thus enables and perpetuates both cultural genocide and ecocide.
 - The outcomes of colonial violence, such as loss of identity, belonging and dignity, enforced ignorance, and poverty, are often internalized by colonized peoples. This internalization is one of the most effective means of sustaining colonial rule. In Rojava, for decades the Ba'ath regime institutionalized this through schooling and curriculum content. Notably, while the regime invested little in economic, health, social and industrial infrastructure, it invested significantly in education in order to reproduce domination and to enslave minds.
- Biopolitical marginalization
 - Rojava is rich in resources, yet it remains an impoverished region and the site of the most severe ecological pollution and destruction. The Kurds are not only exploited for these resources; they are also slowly killed by the resulting toxic burdens and heightened exposure to disease. Although aspects of these issues are addressed in Articles I, II and III, a full analysis of the relationship between colonialism, extractivism, waste and pollution lies beyond the scope of this research.

10.2. Decolonial ecology and its transformative aspects

Since ecocidal coloniality has multilayered epistemic and ontological structures, and operational mechanisms, technologies and configurations, decolonization must be holistic and structural, including ontological and epistemic reconstruction. Rojava's pathway reflects this.

Ontologically, the nation-state's centralized, homogenizing and domineering form is the institutional and ideological embodiment of capitalist modernity and colonial mentality. It stands in stark contrast to Kurdish historical confederations of autonomous tribes, and to Kurdistan's plural religious and ethnic composition. In Rojava, democratic autonomy and rhizomatic governance can be read as a modern synthesis of that historical ontology with contemporary revolutionary experience.

Epistemologically, Rojava embraces the religious, philosophical, and intellectual heritage of Kurdistan. Reinterpreting this heritage, and integrating women's liberation with democratic and ecological principles in scientific practice, enables decolonial epistemologies. *Jineoloji* and the academies actively advance this aim, rejecting patriarchal, colonial and positivist epistemologies and scientism.

This holistic response contains several key lessons, including the following.

- The most vulnerable and fragile link can catalyse the most effective breakthroughs and revolutions, transforming the whole. Compared to Bakur (north), Başûr (south) and Rojhilat (east) Kurdistan, in terms of the heritage of uprisings, organizational capacity, population, resources and geopolitical attention, historically, Rojava had neither a sustained armed nor long-standing civilian resistance tradition before 2012. It is the smallest and weakest part of Kurdistan in terms of territory, population and resources, yet it became the driver of decolonization and socio-ecological transformation.
- The exploitation and extractivism of colonial structures are ecocidal. Anti-colonial struggle must therefore be ecological. This is why ecological struggle has become the ethos of decolonization in Rojava/Kurdistan, and the principal driver of socio-ecological transformation. In contrast to climate-centred adaptation, resilience and mitigation policy packages, Rojava has adopted a deep and holistic socio-ecological transformation directed against the producer of the climate crisis, capitalist modernity.

- A social economy that prioritizes communal land, ecological agriculture and the communal use of water can form a foundation for economic and socio-ecological transformation for colonized peoples. Crop rotation and diversification, agroforestry, and the adoption of drought-resistant local seeds are not only agronomic and environmental strategies; they regenerate socio-ecologies and build eco-economies.
- Despite shortages and attacks, Rojava's communal or social economy based on cooperatives and eco-factories is moving towards economic autonomy and self-sufficiency. *Aborîya jin*, the autonomously organized women's economy, stands among the most successful formations of the cooperative system, and has catalysed some of the region's most far-reaching socio-economic and cultural transformations.
- Diversify water and energy sources, reduce dependence on single upstream controls, and adopt allocation rules that prioritize aquifer recharge, equity and care. Decentralization of critical infrastructures, especially energy, water and electricity, is key to resisting Türkiye's deliberate attacks on infrastructure. Decentralized, autonomous and relational organizational forms, grounded in symbiotic relationality, create redundancy and flexibility across communes, cooperatives and municipal nodes, allowing systems to reconfigure and self-repair when single points of failure are struck. In the face of the state's centralized and asymmetric power, this sociotechnical dispersal is not merely administrative reform; it is an essential strategy of defence and survival akin to a rhizome.
- Transformation of mentality is the most profound and durable form of anti-colonial struggle. Rebuilding education on democratic, ecological and women's liberation principles is vital. Rojava's ecopedagogy should be read as a crucial instrument of social transformation and the decolonization of the mind.
- When society governs its daily life, long-term planning and strategy through communes, academies, cooperatives, assemblies and democratic municipalities, it becomes a constitutive, transforming subject, and an ontological multiplicity, producing spaces beyond domination. Rotation and recallability, limits on accumulation, participatory budgeting, and open assemblies should be embedded as routine safeguards. This is a living experiment in changing the world without taking power (Holloway, 2002).
- Rojava is neither an ideal model nor an island utopia. It is an anti-colonial, anti-capitalist and antipatriarchal struggle that creates and sustains itself daily against colonial and statist domination and violence. Learning through failure, it treats deconstruction or decolonization and reconstruction as co-constitutive, mutually imbricated practices.

Since 2012, Rojava has undergone genuinely transformative and successful phases while simultaneously confronting existential threats and devastating losses, including the invasions of Afrin and Serêkaniyê by Türkiye. Its present position remains precarious, undermined by Türkiye's neocolonial agenda, and the operations of allied jihadist proxies that strain institutions, erode security and destabilize the region.

- In Rojava, a research method is not an external lens applied to a field; it is a relationship that changes both sides. Entering the research as a Kurdish activist-scholar meant that access, trust and obligation were co-produced with communes, cooperatives, academies and municipal actors. The organic structure of the Rojava Revolution does not simply host research; it transforms the researcher's stance, questions and outputs. Knowledge emerges through encounters that are simultaneously ethical, political and affective. In this sense, positionality is not a fixed attribute, but a process of becoming in relation, and method is a practice of situated accountability that prioritizes reciprocity, safety and usefulness to those who live with the consequences of what is written.
- In both the war against ISIS and the work of revolutionary construction and ongoing socio-ecological transformation, international solidarity has been decisive. Rojava demonstrates what international solidarity networks can achieve, and shows that isolated utopias cannot endure in this century.

Despite its unique transformation and decolonization processes, Rojava operates in a permanent emergency. Turkish strikes and proxy warfare target energy, water and agro-industry; upstream hydro-hegemony turns drought into engineered scarcity; embargoes enforce material depletion and a war economy reliant on rudimentary oil refining with heavy environmental and public health costs. War and scarcity drive displacement and skill loss, while land justice, accreditation and institutional capacity remain unresolved. The project has proven viability, yet faces structural and existential threats: invasion risk by Türkiye and its jihadist proxies, demographic pressure, and ongoing violence and economic instability.

Pollution constitutes a structural, enduring crisis. Ba'athist internal colonialism has converted Rojava into an extractive zone, where decades of monoculture have entrenched heavy chemical and pesticide use that has contaminated soils and waterways. Oil extraction has likewise generated severe air, soil and water pollution. Colonial rule systematically alienates people from land and memory; communities are reduced to a labour reserve and a captive consumer market, as value is extracted from territory while toxicity is offloaded onto society. These dynamics are

poorly monitored, and reliable data sets remain scarce; beyond scattered reports by civil society organizations, the pollution-colonialism inherent relationality in Rojava has largely been neglected in scholarship. A priority for future research, and a natural extension of this thesis, is a rigorous, multiscalar mapping of how extractivism, contaminants and colonial regimes co-produce biopolitical marginalization of ecocidal coloniality.

It is difficult to foresee whether Rojava will overcome these structural crises; the risks and threats are considerable. Whatever the outcome, Rojava will remain an invaluable experiment that delineates the parameters of exit from enslavement, poverty, ecological destruction, identity erasure and indignity for Kurds and other colonized peoples. This experience is an ensemble of practices that materializes the horizon of a moral-political society beyond capitalist modernity: it reconstitutes the political through rhizomatic democracy and autonomy; socializes life through an eco-economy, common ownership and cooperatives; reverses epistemic destruction through *jineoloji* and ecopedagogy; and revives the bond between land, water and social fabric through ecological repair and justice. Even under war, embargo and the weaponization of water, Rojava is less a model than a process, a line of flight, a method, an orientation and an invitation. It is a journey of affirming the earth, believing in it and struggling for it, of dignifying life and peoples. It challenges us to prefigure the future in the present. It is up to us to find the courage to do so.

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APPENDICES

Appendix A. Master of Arts in Political Ecology, Institute of Social Science, University of Rojava, 2025

A.1. General overview

The MA in Political Ecology is an interdisciplinary programme that explores the relationships between politics, society, science and the environment. The programme consists of two years and four semesters. Approximately 15 students will be admitted to the programme. The programme is a multidisciplinary one, which trains students in historical and contemporary ecological and social problems, while giving them the ability to discuss potential solutions to these problems through practical engagement.

The programme aims to prepare students for PhD studies, as well as for professional careers in other areas such as politics and diplomacy. Upon completing this programme, the students shall be able to demonstrate the knowledge, judgment, competence and skills to critically and systematically apply scientific methodology and knowledge to carry out, on a professional level, the institutional functioning and work of the agricultural, economic and ecological pillars. The global climate crisis, and the ongoing capitalist and colonialist extractivist agricultural, environmental and economic policies, have resulted in severe ecological destruction and permanent damage. Additionally, unplanned and unorganized urbanization and environmental pollution make it imperative to implement sustainable ecological urban policies in North and East Syria. Such deep-rooted and structural problems require competence in ecological consciousness, pedagogy and methodology. The participating students will pioneer this ecological consciousness and institutionalization in their respective fields (ecology, agriculture, economy, education and municipalities). It is also a strategic goal to prepare academic staff for the University of Rojava in the field of ecology, by arranging a doctoral programme in Europe every two years for one or two students with academic capacity and ability who successfully complete the programme.

A.2. Programme structure: MA in Political Ecology with 90 ECTS

The MA programme follows the European Credit Transfer and Accumulation System (ECTS), where one ECTS credit corresponds to 20–25 hours of total student workload. Each 6 ECTS course therefore requires approximately 120–150 hours of work over the semester, including seminar attendance, readings, assignments, projects and independent study. Most courses

consist of one 90-minute seminar per week for 14 weeks, or alternatively can be done as a double session of 180 minutes per week for seven weeks, with the remaining hours fulfilled through structured self-study and coursework.

Attendance for both the midterm exam and the final exam is obligatory for the first semester. For the second semester, a paper will be written for each seminar, and assignments and projects will be given if the lecturer wishes. Attendance for lectures is also obligatory. Deadlines for course requirements and exam papers (assignments, projects etc.) must be strictly followed.

A.2.1. First semester (24 ECTS)

(C) = Core course

(E) = Elective course

The first semester introduces the structural and historical causes of the current ecological crisis, including the climate crisis, and the main conceptual and methodological developments and debates in this field. Students choose four courses (24 ECTS).

- Introduction to ecology: Concepts and theoretical frameworks (C) (6 ECTS)
- Social ecology and women's liberation (C) (6 ECTS)
- Ecological struggle as decolonial practice: Socio-ecological transformation in Rojava/North and East Syria (C) (6 ECTS) (taught by Necmettin Türk, University of Hamburg)
- Biogeography and spatial analysis/GIS (C) (6 ECTS)

A.2.2. Second semester (24 ECTS)

The second semester offers students the opportunity to apply the basic knowledge acquired in the first semester to their areas of interest and specialization. The depth in these areas will also help students in the process of identifying their thesis topics. The semester will use different pedagogies, and aims to provide an interactive learning and exchange environment. On a practical level, it motivates societal engagement towards solving ecological problems. Students choose four courses (24 ECTS).

- Environmental rights, ecocide and the ecocide-genocide nexus in Kurdistan (C) (6 ECTS) (taught by Necmettin Türk, University of Hamburg)
- Soil, water and forestry (C) (6 ECTS)
- Climate change and towards a socio-ecological transformation (C) (6 ECTS) (taught by Necmettin Türk, University of Hamburg)
- Pollution, colonialism and necropolitics: War and ecological destruction in Rojava/North and East Syria (C) (6 ECTS)

A.2.3. Third semester (18 ECTS)

During the third semester, the students take practical responsibilities in the institutions and organizations of AANES, in the domains of ecology, agriculture, economy, education councils and municipalities. In cases requiring specific technical and engineering skills, the internship period could also be carried out abroad in eligible institutions in the public and private sectors.

At the end of the third semester, students are required to submit their research proposal to the Programme Director. Students will be informed about their thesis supervisors at the beginning of the fourth semester.

- Internship in AANES or other affiliated institutions (12 ECTS)
- Methodology and research design (C) (6 ECTS)
- Fourth semester (24 ECTS)

A.2.4. Fourth semester (24 ECTS)

In the final semester, students are expected to synthesize and analyse in depth the knowledge and skills acquired throughout the programme at the theoretical and practical level. This semester develops the student's capacity for carrying out independent research and writing an original research-based master's thesis that critically engages with ecological, political and societal dimensions.

- Master's thesis (24 ECTS)

A.3. Admission requirements for the Master of Arts in Ecology

Curriculum vitae and bachelor grades 30 per cent

Interview 30 per cent

Entrance exam 40 per cent

Essential: Language proficiency

Additional: relevant experience

Applicants must have completed a bachelor's degree from an accredited institution. Preferred fields of study include, but are not limited to, environmental studies/science, geography, political science, sociology, anthropology, development studies, international relations, economics, public policy and law.

While not mandatory, relevant experience in environmental organizations, NGOs, AANES institutions and organizations, research institutions, or related fields will strengthen an application. Experience working on issues related to environmental justice, conservation,

Appendix A. Master of Arts in Political Ecology

sustainable development or social movements is particularly valuable. Social and environmental engagement as volunteers or professionals in relevant fields or organizations of AANES will be highly prioritized.

English language proficiency is essential, as the programme is fully taught in English.

Appendix B. Questionnaires

B.1. Questionnaire on livelihood (farmers/peasants, workers) and intersectional categories (ethnic-religious groups, gender, age)

1. Region – Village: _____ Date: _____ / _____ / _____
2. Name: _____
3. Nationality: _____ Ethnicity: _____ Religion: _____ Age: _____
4. Occupation: _____
5. Number of family members – Males: _____ Females: _____
6. Do you own agricultural land? _____ If yes, area in acres: _____
7. What is the type of ownership: inherited, state, administration?

 Who inherits – men only, or also women? _____
8. How did you receive the land? _____
 Date of your investment and acquisition of the land: _____
 Who do you think owns water and land? _____
9. For what purpose is this land used: _____
10. What crops do you grow: _____
 How do you access markets, how do you sell your goods? _____
 How do you set the price? _____
11. What is the type of agriculture? Irrigated / Rainfed
12. If it is irrigated, what is the source of water: River / Spring / Well
13. What are the most significant problems you have faced in agriculture in recent years?
14. Compared to the past, has there been a decrease in rainfall and groundwater?
 Compared to previous years, how has the rate of access to well water changed?
15. What are the effects of drought on agriculture?

Appendix B. Questionnaires

What measures are you taking against drought?

If drought continues in the area, is it possible to obtain irrigation water? If yes, how?

16. What are the main causes of the drought in the region?

If you compare the current situation with _____ years ago (e.g. 20 years ago)

17. Regarding water resources management (rivers / dams), who should manage water resources? How?

18. Has there been a change in agriculture in terms of type of crops / productivity / costs, if compared to the last 10–20 years?

19. Have you changed your farming pattern in the face of drought and extreme heat? If yes, how?

Have you introduced new varieties of seeds that tolerate drought and high temperature?

20. If the land is irrigated, what type of irrigation is used: *al-masakab* (traditional) / Sprinkler irrigation (spray) / Drip irrigation.

21. What type of seeds are used in agriculture and what animal breeds are raised? Local / Foreign.

22. When practicing agriculture, do you think we should preserve the environment? Yes / No.

23. Is AANES fair when distributing seeds, diesel, pesticides, fertilizers and other agricultural subsidies, i.e. does not discriminate between religious and ethnic groups?

How could it be better?

24. There are families in the region who do not own agricultural land. Are you in favour of distributing agricultural land to them? Yes / No.

25. Are you a member of an agricultural cooperative, commune, political party or local council? Yes / No.

If yes, what is your role in it?

26. Are agricultural institutions, organizations or farmer unions and agricultural bodies doing their work properly in serving farmers?

Can they solve the problems farmers face? Explain how/why.

27. Do you receive any support from AANES (diesel, land tillage, seeds, irrigation tools)?

What do you think about agricultural and environmental policies in Rojava?

Did you communicate with AANES about your problems? If so, what problems? How did AANES act?

28. Does AANES intervene as required in cases of drought and disasters?

Is there any educational/informative attempt to prepare farmers against possible problems?

29. Do agricultural engineers in AANES perform their required tasks in the field of support and guidance for farmers?

30. Are there adequate living conditions (health/economic/educational/social) in villages?

If no, what are the reasons?

How should the reality be in the future?

31. Do you have family members, relatives or people from your village who have emigrated to other cities or abroad?

If so, what was the main reason?

If you are unemployed in the village and find a job opportunity in the city, do you plan to move there and work? Yes / No.

32. How do you see the future of agriculture in the region?

Does it cover your standard of living?

Are you optimistic about the development of agriculture in the region?

33. When you face an agricultural problem, do you solve it on your own, or do you turn to AANES institutions?

Do they solve your problem?

34. Have you heard about Arab Belt land policies and their consequences?

If yes, what do you think about it?

If you find it problematic, how should be solved?

Do you think land reform can be solution, and if yes, how?

35. What is the source of your agricultural knowledge?

Is there any sharing of knowledge or other solidarity between villages or communes of villages?

Have you ever learned anything about agriculture from a different ethnic or religious community?

36. Are there tensions in your village between different groups (religious and ethnic-based) or between landless peasants, herders and landlords?

How do you find solutions when you have problems?

37. Would you like to add anything else?

Thank you for your time.

B.2. Rojava rural household survey

Enumerator guidance

- Introduce the study and seek verbal informed consent before proceeding.
- Safety first: Avoid sensitive questions if risk outweighs benefit; do not record names.
- Language: Administer primarily in Kurdish or Arabic; translate terms locally (commune, people's assembly, cooperative).
- Skips: If household reports rainfed-only cultivation in Section D, skip irrigation follow-ups (D2–D3).

Consent script

Thank you for meeting with us. We are conducting a non-profit academic study about livelihoods, agriculture, water and local governance in Rojava. Your participation is voluntary; you may refuse any question or stop at any time. We do not record names. We will keep your information confidential. All materials are handled under EU data-protection principles: explicit consent; data minimization; pseudonymization; encryption at rest and in transit; access controls; and limited retention with separate storage of reidentification keys.

Do you consent to participate? ☐ Yes ☐ No

A. Identification and interview logistics

- A1. Date: ____ / ____ / ____ A2. Start time: ____ A3. End time: ____
 A4. Location: Village ____ District: ☐ Dêrik ☐ Tirbespiyê ☐ Hesekê
 A5. Interview mode: ☐ In-person ☐ Phone/online
 A6. Enumerator initials: ____ A7. Respondent code (no names): ____

B. Household and livelihoods

B1. Household size: Total: ____ Men: ____ Women: ____

B2. Primary occupation (self-described): _____

Secondary: _____

B3. Livelihood category (tick all that apply):

☐ Farmer – landowner

☐ Tenant / sharecropper

☐ Landless agricultural worker

☐ Pastoralist / herder

☐ Other rural work: _____

B4. Household roster (no names; use initials or codes)

Record only members who slept here most nights in the past six months.

Member ID	Gender (M/F/Other)	Age band*	Education**	Primary role***	labour	Seasonal migration (Y/N)

* Age bands: 1 = 17–29; 2 = 30–49; 3 = 50–64; 4 = 65+

** Education: 0 = None; 1 = Primary; 2 = Lower secondary; 3 = Upper secondary; 4 = Vocational/technical; 5 = University +

*** Primary labour role: 1 = Farmer/self-cultivation/landowner; 2 = Farmer; 3 = Wage worker (agri); 4 = Wage worker (non-agri); 5 = Pastoralist/herder; 6 = Student; 7 = Homemaker/care; 8 = Unemployed/job-seeking; 9 = Retired; 10 = Unable to work/disabled; 11 = Other: _____

Do not record names or dates of birth; store any separate consent logs apart from this form.

C. Land, tenure and inheritance

C1. Do you (or your household) have access to agricultural land? ☐ Yes ☐ No

If No, go to section D.

C2. Area (note unit): _____ Unit: ☐ dunam ☐ hectare ☐ acre

C3. Basis of access (tick all that apply):

☐ Ownership (registered/*tapu*)

☐ Customary/inherited

☐ Lease/rent-in

☐ Sharecropping

☐ Communal/collective

☐ State/AANES allocation

☐ Other: _____

C4. How did you acquire access? (open response)

Year of first cultivation here: _____

Inheritance norms: ☐ Men only ☐ Men and women ☐ Depends/other: _____

Enumerator: Emphasize that questions C5/C6 ask for political visions and ethical principles, not documents or titles.

C5. In your view, who is the rightful holder/owner of land and water in this area? (open response)

C6. Who should be entrusted with the governance and stewardship of land and water? (e.g., communes, people's assemblies, cooperatives, AANES bodies, state, elders, etc.) (open response)

D. Water access and irrigation

D1. Main water source for agriculture (tick all that apply):

☐ Rainfed only

☐ Well/groundwater

Appendix B. Questionnaires

- ☐ River/canal
- ☐ Reservoir
- ☐ Trucked water
- ☐ Other: _____

If household reports rainfed-only in D1, skip questions D2/D3.

D2. If irrigated: irrigation method (tick all that apply):

- ☐ Flood/surface
- ☐ Furrow
- ☐ Sprinkler
- ☐ Drip
- ☐ Other: _____

D3. If irrigated: main constraints (Open response):

E. Agriculture, shocks and adaptation

E1. Main crops/livestock (last 12 months): _____

E2. Have you experienced drought or water scarcity impacts in the last 5 years? ☐ Yes ☐ No

E3. If yes, describe the impacts on yields, costs, labour or migration: _____

E4. Adaptation strategies used (tick all that apply):

- ☐ Changing crop mix/varieties
- ☐ Adjusting planting dates
- ☐ Reducing cultivated area
- ☐ Increasing off-farm work
- ☐ Using savings/loans
- ☐ Joining a cooperative
- ☐ Collective water management
- ☐ Other: _____

F. Cooperatives, communes and collective action

F1. Are you a member of a cooperative? ☐ Yes ☐ No If yes, which type? _____

F2. Commune participation: ☐ Regular ☐ Occasional ☐ Never

F3. People's assembly participation: ☐ Regular ☐ Occasional ☐ Never

F4. What benefits or challenges have you experienced from cooperatives/communes? (open)

G. Governance, rights and services

G1. Access to services in the last 12 months (tick all that improved):

- ☐ Seeds/inputs
- ☐ Extension/academy
- ☐ Irrigation maintenance

☐ Market access

☐ Credit

☐ Social support

☐ Other: _____

G2. Perceptions of fairness in land and water governance:

☐ Very fair

☐ Mostly fair

☐ Mixed

☐ Mostly unfair

☐ Very unfair

G3. If unfair/mixed, why? (open response) _____

H. Security and risks

H1. Have security events (e.g. attacks, infrastructure strikes) affected your farming or water access in the last five years? ☐ Yes ☐ No

H2. If yes, please describe (open response): _____

I. Overall perceptions and closing

I1. What are your top three priorities for improving livelihoods and ecological wellbeing here? (open response)

1) _____

2) _____

3) _____

I2. Is there anything else you would like to add? (open response)

Thank you for your time.

Appendix C. Organizational-level expert/key informant interviews by date, with lettered repeats

ID rules: earliest dated interview = number 1; ties on the same date sorted alphabetically by participant/pseudonym. Repeat interviews with the same person/entity receive a single base number with lettered suffixes (a, b, c) in chronological order. Group discussions and email-written respondents are excluded.

ID	Date	Level	Region	Interviewee/role	Mode
1a	2023-11-18	Org.	Rojava	Azad	Online
2a	2023-11-19	Org.	Rojava	Dr Ahmed Al-Mohamed – Livestock Department, Office of Agricultural Affairs (AANES)	Online
3a	2023-11-21	Org.	Rojava	Karker Ismail – co-chair, Office of Community Projects and Cooperatives (Cizîrê)	Online
4a	2023-11-21	Org.	Rojava	Leyla Saruxan and Mehran Ahmed – co-chairs, Agriculture and Irrigation Council (joint)	Online
5a	2023-11-23	Org.	Rojava	Ibrahim – Afrin Human Rights Organization	Online
6a	2023-11-24	Org.	Rojava	Eli – Ecology Committee/Academy consultant	Online
7a	2023-12-02	Comm.	Rojava	Narîman Hisên Hesên – commune co-chair (Gundê Carudiyê)	Online
8a	2023-12-02	Org.	Rojava	Selah Hemza Mihemed – agriculture and water	Online

9a	2023-12-06	Org.	Rojava	Savaş – general economic policy consultant	Online
10a	2023-12-10	Org.	Rojava	Bîrahîm Hemdî Êşed – AANES Environmental Council co-chair	Online
11a	2023-12-15	Comm.	Rojava	Alevi community representative	Online
12a	2023-12-15	Comm.	Rojava	Sunni/Muslim community representative	Online
13a	2023-12-16	Org.	Rojava	Gulistan Murad – spokesperson, Economics Committee (Kongra Star)	Online
14a	2023-12-18	Org.	Rojava	Dilser – international activist	Online
15a	2024-01-06	Comm.	Rojava	Hena – Armenian/Christian representative	Online
16a	2024-01-06	Comm.	Rojava	Menal Şêxmûs – Zoroastrian (Zerdestî) representative; lecturer	Online
17a	2024-01-06	Comm.	Rojava	Sebrî Sewme – Assyrian/Christian representative	Online
18a	2024-02-24	Org.	Rojava	Mihemed Hisên Mamê / Şiyar Hesên Mio – Saziya Tov (Seed Institution)	Online
19a	2024-02-28	Org.	Rojava	Mansur Mahmoud – eco-factories (Kargehên Berhemdarî)	Online
20a	2024-03-03	Org.	Europe	Fîrat Anlı – former co-mayor, Diyarbakır Metropolitan Municipality	Online

Appendix C. Organizational-level expert/key informant interviews

21a	2024-03-12	Org.	Europe/Bakur	Ercan Ayboğa – co-founder Mesopotamia Ecology Movement	Online
22a	2024-03-17	Org.	Rojava	Ismail Ebidelexelîl Sûlêman – General Director of Scientific Agricultural Research	Online
23a	2024-03-26	Org.	Rojava	Rûstem – consultant, Municipalities Union	Online
24a	2024-03-29	Org.	Rojava	Hesen Koçer – Executive Council co-chair (AANES)	Online
25a	2024-04-05	Comm.	Rojava	Gulbahar – commune co-chair	Online
6b	2024-04-30	Org.	Rojava	Eli – Ecology Committee/Academy consultant	In-person
26a	2024-05-04	Org.	Rojava	H. Ahîn – coordinator, Demsal Women’s Economy Project	In-person
18b	2024-05-09	Org.	Rojava	Saziya Tov – leadership follow-up	In-person
27a	2024-05-17	Org.	Rojava	Emin – KCK Economy Committee and AANES Economy Council	In-person
28a	2024-05-17	Comm.	Rojava	Nesîm Şemo – Yazidi representative; University of Rojava	In-person
29a	2024-05-20	Org.	Rojava	Berivan Yunis – Kongra Star academies	In-person
30a	2024-05-24	Org.	Rojava	Basil Mihemed and Cîhan Ferho – co-chairs, Heseke canton cooperatives	In-person
31a	2024-05-26	Org.	Rojava	Berivan Omari — lecturer (social ecology), co-mayor of Qamishli	In-person

32a	2024-05-27	Org.	Rojava	Delîla – agricultural coordinator, Jinwar	In-person
10b	2024-06-02	Org.	Rojava	Bîrahîm Hemdî Êşed – AANES Ecology Council co-chair	In-person
33a	2024-06-03	Org.	Rojava	Siyar – co-chair, Cizîrê region cooperatives	In-person
34a	2024-06-27	Org.	Rojava	Roj – lecturer; Kongra Star Ecology Committee	In-person
35a	2024-06-27	Org.	Rojava	Zîlan – lecturer, Jineolojî Academy	In-person
36a	2024-10-02	Org.	Türkiye	Dr. Fatih Çiçek – DEM Party agricultural policy consultant	Online
37a	2024-10-21	Org.	Türkiye / Bakur / Northern Kurdistan	Egid Özdemir – ecology activist	Online
6c	2025-05-12	Org.	Rojava	Eli – Ecology Committee/Academy consultant	Online

Notes

- Levels: Org = organizational; Comm = community.
- Modes: In-person = conducted during April–June 2024 fieldwork; Online = remote interviews in Online W1 (October 2023–March 2024) or Online W2 (October 2024–March 2025).
- Acronyms: KCK = Union of Kurdistan Communities.
- ID policy: one base number per interviewee/entity across time; lettered suffixes indicate subsequent sessions (e.g. 7a, 7b). Same-day ties sorted alphabetically (see Appendix C).

Appendix D. Ethical guidance for research in Rojava/North and East Syria

This research is conducted in a conflict-affected and politically sensitive context, requiring heightened ethical sensitivity and accountability. The framework follows EU research ethics standards, while adapting them to the situated realities of Rojava.

Informed consent and voluntariness

Participation is strictly voluntary and based on informed consent. Consent is sought orally where literacy, security or cultural context makes written consent inappropriate. Participants are fully informed about the aims, risks and potential uses of the research, with clear emphasis on their right to withdraw at any time without consequence.

Protection of vulnerable groups

Research involves populations affected by war, displacement and socio-economic precarity, who must be treated as vulnerable participants under EU standards. Safeguards include avoiding retraumatization, ensuring that no individual is coerced into participation, and carefully considering power dynamics between researcher and participants.

Privacy, anonymity and data protection

No personally identifiable information is collected unless explicitly authorized. All names and identifiers are anonymized using codes, unless they explicitly request the use of their official names and positions. Data are stored securely with encryption, and access is restricted to the researcher. EU General Data Protection Regulation (GDPR) principles are observed: data are collected for explicit purposes, minimized, and deleted after project completion, unless long-term archiving is consented.

Data sovereignty and reciprocity

Data belongs first to the communities who generate it. Local actors are offered access to transcripts, findings and outputs in accessible formats. Findings are shared back with communes, cooperatives and academies in Kurdish and Arabic, ensuring reciprocity and avoiding extractivism.

Avoidance of harm and risk mitigation

Locations, identities and sensitive information are anonymized to avoid exposing participants to state or militia retaliation. Fieldwork strategies are designed to minimize physical and emotional risk, including flexible interview settings, the option to decline any question, and immediate cessation of interviews if distress arises.

Reflexivity and positionality

The researcher explicitly acknowledges their positionality as a Kurdish activist-scholar in exile. This positionality brings both proximity (shared struggle, language, trust) and responsibility (to

avoid reproducing power hierarchies). Reflexive practices include continuous dialogue with participants and community organizations to assess risks, benefits and responsibilities.

Compliance with EU ethics frameworks

Research aligns with the European Code of Conduct for Research Integrity and Horizon Europe Ethics Appraisal requirements. Key principles include respect for human dignity, avoidance of harm, proportionality, informed consent and fairness. Particular care is taken to respect cultural diversity and Indigenous epistemologies, in line with Article 31 of the United Nations Declaration on the Rights of Indigenous Peoples.

Dissemination and accountability

Outputs will be published in academic venues but also translated and returned to local institutions. Sensitive details that could endanger participants or institutions are omitted from publications. Accountability is understood relationally – not only to EU research bodies, but also to the communities who host and sustain the research.